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Holomorphic Curves and Global Questions in Contact Geometry



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Holomorphic Curves and Global Questions in Contact Geometry

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Dedicated to the Memory of Andreas Floer and Kris Wysocki



Andreas Floer
Author: George M. Bergman.
Source: Archives of the Mathematisches Forschungsinstitut Oberwolfach



Kris Wysocki
Author: Jürgen Pöschel.
Source: Archives of the Mathematisches Forschungsinstitut Oberwolfach

Introduction

Historical Background

In 1976, in his paper [91], Jürgen Moser writes about the classical action principle the following: "However, this variational principle is very degenerate, for example even the Legendre condition is violated, and is certainly not suitable for an existence proof." If this statement would be true, modern symplectic geometry would not exist. In essence, Moser claimed (for what he thought were obvious reasons) that a certain formal variational principle cannot be used to find global periodic orbits for finite-dimensional Hamiltonian systems. In today's language, it was an argument against the existence of Floer theory. Moser's statement was well-founded in what was known at the time. Morse theory in finite dimensions was well-understood, and starting with the work of Palais and Smale, the ideas were applied to infinite-dimensional problems. However, a basic assumption, besides what is called now the Palais-Smale condition, was that Morse indices of a Morse function f were finite, whereas the co-index was allowed to be infinite (in the reversed case one can study $-f$). From Morse theory, one knew that passing a non-degenerate critical point, the homotopy type of the sub-level sets changed by adding a finite-dimensional handle of the appropriate dimension. In the case that the Morse index was infinite, the homotopy type would not change due to the fact that the unit sphere in an infinite-dimensional Hilbert space is contractible. As a consequence, the global existence mechanism for finding critical points by studying the change of homotopy type would not work. This, in a nutshell, was Moser's argument: if the topology of the sub-level sets would never change, how could there be a global mechanism to find them?

The argument is undoubtedly correct for general functionals, but it does not apply to the classical action functional. Indeed, shortly afterward, Moser's former student P. Rabinowitz showed in his paper [95] that a nonlinear wave equation problem, an infinite-dimensional Hamiltonian system, has in fact periodic orbits, by using a generalization of the classical action principle. As J. Moser once told the second author (HH), this paper clearly proved that he was mistaken and he asked

P. Rabinowitz if he could use his methods to prove the existence of periodic orbits in finite-dimensional Hamiltonian systems. This resulted in Rabinowitz' celebrated papers [96] and [97] where the latter prompted the influential Weinstein conjecture [114]. A first breakthrough on the Weinstein conjecture was Viterbo's result that every regular compact energy surface in $\boldsymbol{R}^{2n}$ carries a periodic orbit (see [112]). This led HH and Zehnder to exhibit an important phenomenon in Hamiltonian dynamics called *almost existence* (see [65], and for a sharpening of the result, Struwe's contribution [107], and more generally [66] for a "symplectic capacity viewpoint"). The notion of symplectic capacity was introduced by Ekeland and HH in their papers [27, 28]. The basic observation was that Gromov's non-squeezing result, which was proved by pseudoholomorphic methods, is related to an infinite class of new monotonic symplectic obstructions which are related to Hamiltonian dynamics and associated spectral properties. These papers pave the way for symplectic homology [20, 21, 48, 49].

Rabinowitz's papers are the starting point for considerable research activities. Particularly important are [9] which, together with Conley's index theory [23], lays the foundation for the breakthrough result [24], establishing the Arnold conjecture for fixed points of Hamiltonian diffeomorphisms on standard symplectic tori and the significant contribution by Ekeland and Lasry on the number of periodic orbits on a convex energy surface [29] (see also [26] for some of the subsequent developments). A preprint by Eliashberg also addressing the abovementioned Arnold conjecture [30] was unknown to most researchers in the West.

Another important development was Bennequin's Ph.D. thesis [13], in which among other things, an exotic contact structure on $\boldsymbol{R}^3$ is constructed. This is the first example of what is later called an overtwisted contact structure. Bennequin's work can also be seen as the starting point of Eliashberg's striking work on the subject [31–33].

In 1985, [60], Gromov introduces the theory of pseudoholomorphic curves. Soon thereafter, the pseudoholomorphic curve idea is combined with the Conley-Zehnder variational viewpoint by A. Floer in a series of seminal papers [43–47], which subsequently opens up different areas of mathematics.

Relevant for this book is the following development. In [64], HH develops a version of Gromov's pseudoholomorphic curve theory [60], applicable to contact manifolds and as an application that solves many cases of the Weinstein conjecture. The paper [64] makes a considerable use of two of the before-mentioned papers by Eliashberg, namely, [31, 32].

About This Book and Its Context

In some sense, this book is a "classic." It grew out of a Nachdiplom Vorlesung at the ETH Zürich given by HH during the academic year 1993/1994. The lecture was built around the paper [64] and was almost in definite form in 1995. The key result proved in this text shows that every Reeb vector field on S^3 has at least one period

orbit, establishing the Weinstein conjecture for the three-sphere. Unfortunately, the authors never found the 2 months to finish the project until recently. However, quite a number of students learned the material through the preliminary drafts floating around. The material provides an interesting entry point to the pseudoholomorphic curve theory by proving some of the deeper results in the subject. It has been used by the authors for graduate courses on the subject, and the assumed prerequisites are modest, and the given proofs are very detailed.

The book can be viewed as an entry point for the study of finite energy foliations (see [68, 69]), which initially was used to study low-dimensional dynamics but later evolved as a tool for studying low-dimensional symplectic problems (see [115]). The book is also an entry point to symplectic field theory (SFT) [36]. Results about the Weinstein conjecture for planar contact structures as proved in [3] arise from these two viewpoints. The results by Hryniewicz and collaborators (see [70–73]) are fundamental contributions to the theory of global surfaces of section, which have a precision which 20 years ago would have been unimaginable. Due to a recent result of Joel Fish and HH [39], the methods gain additional importance since they can be used in a suitable modification to prove the existence of so-called feral curves. These can be viewed as a strong generalization of pseudoholomorphic curves and can be used to derive deep results in symplectic dynamics (see [18]). Symplectic dynamics is an emerging field with an already impressive number of important results (see, e.g., [4, 16, 17, 56, 75]). The recent development of feral curves allows to answer a 20-year-old question by the dynamicist Michel Herman, raised at his 1998 ICM address, in the special case of dimension four [61]. More precisely, an application of the theory of feral curves confirms that a compact regular energy surface in the standard symplectic $\boldsymbol{R}^4$ cannot harbor a minimal flow, i.e., a flow for which every orbit is dense. The result strongly hints toward a theory of algebraic invariants for closed invariant subsets of Hamiltonian flows substituting for invariants build on periodic orbits. Of course, this is a question for future research, and the outcome will have ramifications for the development of the field.

What to Do Next?

In the last 40 years, symplectic geometry developed into a huge field, and it is surely challenging for the novice to find her/his way. Assuming the reader has digested the current text, there are several directions for further study. Since we only provided a minimum of background material, it is clearly advantageous to learn more about contact and symplectic geometry. A very useful reference for contact geometry is Geiges' book [54]. However, there have been recent developments around the so-called *hard vs soft dichotomy*, and we refer the reader to the survey article by Eliashberg [35] and the references discussed in this survey. We also recommend the important book by Cieliebak and Eliashberg [19] which is concerned with the symplectic geometry of affine complex manifolds and in which contact geometry plays an important role. These references provide the geometric backbone of a lot

of current research, provided one would like to venture into the geometric realm. Many of the geometric problems require some hard technology, i.e., knowledge of pseudoholomorphic curve theory, particularly when they use symplectic or contact invariants. Another important source of background material are the books by McDuff and Salamon [86, 87].

Focusing back on pseudoholomorphic curve theory, there are many different directions which lead to get a deeper understanding of this theory. Compactness results are an important stepping stone (see [2, 15] and [22]). The approaches differ somewhat and are in line with the seminal work of Gromov [60] but study the more complicated case of symplectic cobordisms with contact type or stable Hamiltonian ends. Joel Fish [40, 41] takes a quite different viewpoint in his foundational papers on compactness properties of pseudoholomorphic curves. His work provides a very powerful new take on the original compactness ideas, and the two papers are the obvious starting point for further developments. In particular, this work prepares very well for a study of [39, 42]. These two papers are the starting point for the new class of "feral curves" and their applications in symplectic dynamics.

Another important basic topic is Siefring's intersection theory (see [102, 103]). After this, one can study finite energy foliations (see [68, 69]). On the more geometric side, there are many applications aiming at the study of low-dimensional symplectic and contact manifolds, and we refer the reader for the further developments to [115]. If one wants to focus on the dynamical applications, they are plentiful as well. Particularly, the classical restricted three-body problem is a very interesting subject of study, and we refer the reader to [50] and its numerous references.

Another interesting direction is concerned with embedded contact homology (ECH), and there are still many open problems. That this theory is very powerful can be easily inferred from the results which have been obtained. ECH is closely related to Seiberg-Witten-Floer theory, and many of its properties can only be derived by this connection. Historically, motivated by the results of Taubes about Seiberg-Witten theory and the event of SFT, M. Hutchings [74] predicted a relationship between Seiberg-Witten-Floer theory and a version of SFT called embedded contact homology (ECH). This theory has striking applications in low-dimensional symplectic geometry and symplectic dynamics, i.e., Irie used it in [75] to prove a smooth closing lemma.

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Chapter 1
An Introduction to Contact Geometry



1.1 Basic Notions of Contact Geometry

A *contact form* on an odd-dimensional manifold M of dimension $2n+1$ is a one-form λ such that the $(2n+1)$-form Ω given by

$$\Omega = \lambda \wedge (d\lambda)^n$$

defines a volume form on M. We observe that any manifold admitting a contact form is necessarily orientable, and that a contact form defines a natural orientation.

Assume now that (M, λ) is a manifold together with a given contact form. First of all we note that λ defines a $2n$-dimensional vector bundle over M. Indeed, consider $\xi \to M$, where ξ is given by

$$\xi_m = \ker(\lambda_m).$$

The linear functional $\lambda_m : T_m M \to \mathbf{R}$ is non-zero since $\lambda \wedge (d\lambda)^n$ defines a volume form. We have the following lemma:

Lemma 1.1.1

- *The 2–form $\omega := d\lambda|_{\xi \oplus \xi}$ is nondegenerate on each fiber, i.e. if there is $v_p \in \xi_p$ for some $p \in M$ so that $\omega_p(v_p, w_p) = 0$ for all $w_p \in \xi_p$ then $v_p = 0$.*
- *There exists a unique vector field X_λ which satisfies*

$$i_{X_\lambda} d\lambda \equiv 0 \quad \textit{and} \quad i_{X_\lambda}\lambda \equiv 1.$$

Proof In order to prove the first statement, we argue by contradiction and assume that there are $p \in M$ and $0 \neq v_1 \in \xi_p$ so that $\omega_p(v_1, w) = 0$ for all $w \in \xi_p$. We can find a basis $\{v_1, \ldots, v_{2n+1}\}$ of $T_p M$ so that the vectors $v_1, \ldots, v_{2n}$ generate the

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subspace ξ_p. Then

$$(\lambda \wedge (d\lambda)^n)(v_1, \ldots, v_{2n+1}) = \lambda(v_{2n+1}) \cdot (d\lambda)^n(v_1, \ldots, v_{2n}) = 0$$

by our assumption on v_1, which contradicts the fact that $\lambda \wedge (d\lambda)^n$ is a volume form.

Let us now prove the second assertion. Pick a point $p \in M$ and an open neighborhood U of p so that there is a local trivialization of $TM|_U$, i.e. there are $2n+1$ linearly independent vector fields $v_1, \ldots, v_{2n+1}$. Since $\lambda \wedge (d\lambda)^n$ is a volume form, we can find $v, w \in \{v_1, \ldots, v_{2n+1}\}$ so that $d\lambda(v, w) \neq 0$ on some open neighborhood $U' \subset U$ of p. We may assume that $d\lambda(v, w) = 1$ by multiplying v with a suitable non vanishing function. We define the following subbundles of $TM|_{U'}$:

$$P_1 := \mathbf{R}v \oplus \mathbf{R}w$$

and

$$E_2 := \{z \in TM|_{U'} \mid d\lambda(z, z_1) = 0 \text{ for all } z_1 \in P_1\}.$$

We have $E_2 \cap P_1 = \{0\}$ and $TM|_{U'} = E_2 + P_1$. Indeed, if $z \in TM|_{U'}$ then $z - d\lambda(z, w)v + d\lambda(z, v)w \in E_2$, hence we obtain

$$TM|_{U'} = P_1 \oplus E_2.$$

We repeat this process now inductively with E_2, i.e. we find $U'' \subset U'$ so that $E_2|_{U''} = P_2 \oplus E_3$, where P_2 is a two–dimensional subbundle of $E_2|_{U''}$ generated by two vector fields, say v' and w', so that $d\lambda(v', w') = 1$. Finally, we end up with $2n+1$ linearly independent vector fields $\{v_1, w_1, \ldots, v_n, w_n, e\}$ defined on some neighborhood of p, so that

- $d\lambda(v_k, w_k) = 1$ for all k,
- $d\lambda(v_k, w_l) = d\lambda(v_k, v_l) = d\lambda(w_k, w_l) = 0$ for all $k \neq l$,
- $d\lambda(e, v_k) = d\lambda(e, w_k) = 0$ for all k.

We have

$$\begin{aligned} 0 \neq\ & (\lambda \wedge (d\lambda)^n)(v_1, w_1, \ldots, v_n, w_n, e) \\ =\ & \lambda(e) d\lambda(v_1, w_1) \ldots d\lambda(v_n, w_n) \\ =\ & \lambda(e), \end{aligned}$$

hence

$$X_\lambda := \frac{e}{\lambda(e)}$$

satisfies both $i_{X_\lambda} d\lambda \equiv 0$ and $i_{X_\lambda}\lambda \equiv 1$. Moreover, this vector field is unique. Indeed, if we had another one, say Y, different from X_λ, then

$$i_{X_\lambda - Y} d\lambda \equiv i_{X_\lambda - Y}\lambda \equiv 0$$

and $X_\lambda(q) - Y(q) \neq 0$ at some point $q \in M$, which is impossible since $\lambda \wedge (d\lambda)^n$ is a volume form. The vector field X_λ that we constructed locally near some point $p \in M$ is actually globally defined since it is unique. □

The 2–form $d\lambda|_{\xi\oplus\xi}$ is, by definition, a *symplectic product* on ξ_m and (ξ, ω) is a *symplectic vector bundle*. We define a line bundle ℓ over M as follows:

$$\ell_m = \{p \in T_m M \mid d\lambda_m(p, q) = 0 \; for \; all \; q \in \xi_m\} = \mathbf{R}X_\lambda.$$

Summing up, a contact form λ on an odd-dimensional manifold M of dimension $2n + 1$ defines a natural splitting of the tangent bundle TM of M into a line bundle $\ell \to M$ with a preferred section X_λ and a symplectic vector bundle (ξ, ω):

$$TM \; = \; (\ell, X_\lambda) \; \oplus \; (\xi, \omega).$$

In the following we shall always denote the data associated to a pair (M, λ) by $\xi, \; \omega, \; \ell$ and X_λ. The vector bundle $\xi \; \to \; M$ without its symplectic structure is called a *contact structure* on M. The vector field X_λ is called the *Reeb vectorfield* associated to λ. We observe that given the contact structure ξ we can re-discover the conformal class of the symplectic structure. Namely take any nowhere vanishing one-form τ with $\ker(\tau) \; = \; \xi$. Then

$$\lambda \; = \; f \cdot \tau$$

for some non vanishing smooth function $f : M \to \mathbf{R}$. We observe that

$$\omega \; = \; d\lambda|(\xi \oplus \xi) \; = \; f \cdot d\tau|(\xi \oplus \xi).$$

We always assume that contact structures are induced by a global 1-form. In the literature they are usually defined as codimension 1 plane field distributions, which are given locally as the kernel of a 1-form such that $\lambda \wedge (d\lambda)^n$ is a volume form. The existence of a global form is then equivalent to the orientability of the line bundle $(TM)/\xi \; \to \; M$. The standard example for a manifold with a contact form is $(\mathbf{R}^{2n+1}, \lambda)$, where

$$\lambda \; := dz \; + \; \sum_{i=1}^{n} x_i dy_i.$$

Here $(x_1, \ldots, x_n, y_1, \ldots, y_n, z)$ are the coordinates in $\mathbf{R}^{2n+1}$. The Reeb vector field is the unit vector field $(0, \ldots, 0, 1)$ into the z-direction. The contact plane ξ at the point $(x_1, \ldots, x_n, y_1, \ldots, y_n, z)$ is spanned by the vectors

$$e_1, \ldots, e_n, f_1, \ldots, f_n$$

where the e_i denote the standard basis vectors of $\mathbf{R}^{2n+1}$ for $(1 \leq i \leq n)$ and the f_i are defined by

$$f_i \;=\; e_{n+i} - x_i \, e_{2n+1}.$$

We also observe that $d\lambda \;=\; \sum_{i=1}^{n} \, dx_i \wedge dy_i$. Hence

$$\omega(e_i, e_j) \;=\; 0, \;\; \omega(f_i, f_j) \;=\; 0 \text{ and } \omega(e_i, f_j) \;=\; \delta_{ij}.$$

Therefore, $(e_1, \ldots, e_n, f_1, \ldots, f_n)$ is a symplectic basis of $\xi_{(x_1,\ldots,x_n,y_1,\ldots,y_n,z)}$. For further reference we call λ the *standard contact form* on $\mathbf{R}^{2n+1}$ (Fig. 1.1). For the following we note two important results which will be proved later in Chap. 2. The first result is concerned with local properties of contact forms and is called Darboux's theorem. We will prove it in Sect. 2.1.

Theorem 1.1.2 *Let (M, λ) and (N, τ) be two manifolds of dimension $2n + 1$ equipped with a contact form. Suppose $m \in M$ and $n \in N$ are given. Then there exists a local diffeomorphism $\psi : U \to V$ defined on an open neighborhood U of m onto an open neighborhood V of n such that*

$$\psi(m) \;=\; n, \; \textit{and} \; \psi^{*}\tau \;=\; \lambda.$$

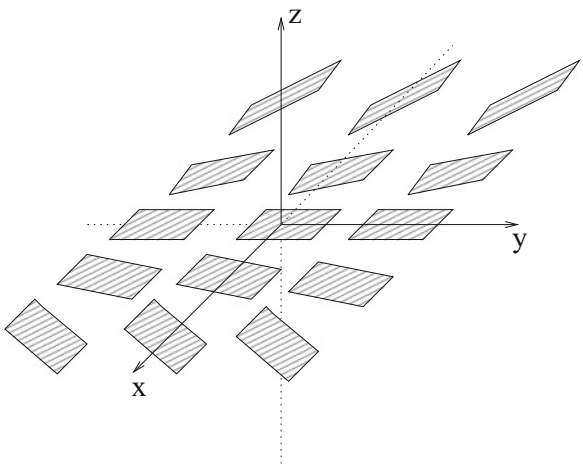


Fig. 1.1 The standard contact structure $\ker(dz + xdy)$ on $\mathbf{R}^3$

The next result studies global properties of contact structures and is referred to as Gray's stability result. We will give a proof in Sect. 2.3:

Theorem 1.1.3 *Assume* $(\xi_t)_{t\in[0,1]}$ *is a smooth family of contact structures on a compact manifold* M *without boundary. Then there exists a smooth arc of diffeomorphisms* $(\phi_t)_{t\in[0,1]}$ *such that* $\phi_0 = Id$ *and*

$$T\phi_t(\xi_0) = \xi_t$$

for all $t \in [0, 1]$.

J. Martinet showed in 1971 that every closed orientable three dimensional manifold M admits a contact form [83]. We will give a proof of this result in Sect. 2.5. Assume ξ is a contact structure on a three-dimensional manifold M induced by a contact form λ. Following Y. Eliashberg, [31], we call ξ *overtwisted* if there exists an embedded closed disk $\mathcal{D}$ in M such that $T\partial\mathcal{D} \subset \xi|_{\mathcal{D}}$ and $\partial\mathcal{D}$ does not contain any point m, where $T_m\mathcal{D} = \xi_m$. We call such a disk $\mathcal{D}$ an *overtwisted disk*. In the literature another definition of overtwisted disk sometimes appears. An additional condition is imposed: There is precisely one point in the interior of the disk where the tangent space to the disk agrees with the contact plane. We will prove in Theorem 3.5.7 that such a nice overtwisted disk always exists on an overtwisted contact manifold, so these two definitions are not different at all.

If such a disk does not exist we call the contact structure *tight*. A contact form is called similarly *overtwisted* or *tight* if it induces a contact structure with the corresponding property. Let f be a non-vanishing function on M. Then λ is overtwisted (tight) if and only if $f\lambda$ is overtwisted (tight).

A fundamental result due to D. Bennequin [13] is the following:

Theorem 1.1.4 *The standard contact structure on* $\mathbf{R}^3$ *is tight.*

We will give a proof of this result in Sect. 9.4. It is quite easy to give an example of an overtwisted contact structure on $\mathbf{R}^3$ (Fig. 1.2). Let (r, ϕ, z) be cylindrical coordinates on $\mathbf{R}^3$. Define μ by

$$\mu = \cos(r)dz + r\sin(r)d\phi. \tag{1.1}$$

Then let $\mathcal{D}$ be the graph of the function f defined on the closed disk of radius π in the x-y-plane by

$$f(x, y) = \varepsilon(\pi^2 - (x^2 + y^2))$$

where ε is sufficiently small. One easily verifies that $\mathcal{D}$ is an overtwisted disk. Hence, Bennequin's Theorem 1.1.4 implies the following:

Theorem 1.1.5 *On* $\mathbf{R}^3$ *there exist two non isomorphic contact structures.*

A diffeomorphism of $\mathbf{R}^3$ mapping one contact structure to the other would preserve overtwisted disks. However due to Theorem 1.1.4 there are no overtwisted

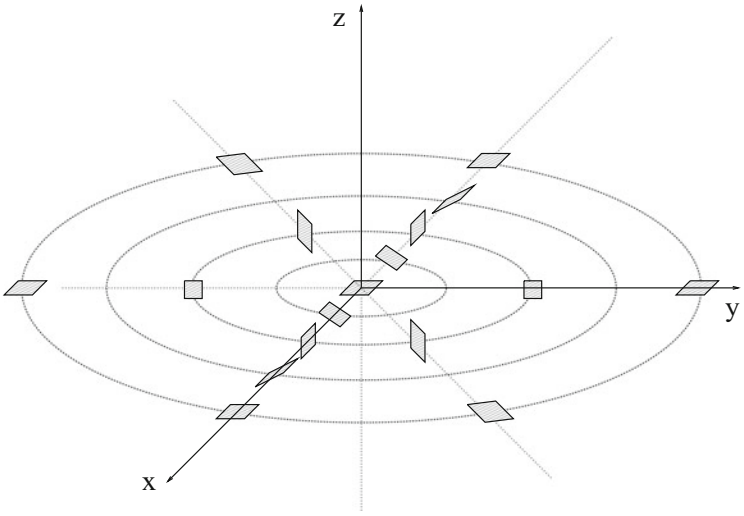


Fig. 1.2 The overtwisted contact structure $\lambda = \cos r dz + r \sin r d\phi$ on $\mathbf{R}^3$: In contrast to the standard tight structure the planes make a full turn

disks for the standard structure, but there are for the overtwisted structure (1.1). We leave it as an exercise to the reader to show that there exists a smooth arc of contact structures connecting the standard one to the overtwisted one in (1.1), just push the twist to ∞. Hence the hypotheses of Theorem 1.1.3 are met except the compactness. However the conclusion cannot hold due to Theorem 1.1.4. This shows that the compactness hypothesis in Gray's theorem is crucial.

We remark that there is a procedure, called Lutz-twist, which makes any contact structure overtwisted by a local modification (see [81, 82] or the monograph by Hansjörg Geiges [52]). So there are lots of overtwisted contact structures.

Next we give some criteria to distinguish tight contact structures on compact three-manifolds without boundary. We have the following result by H. Hofer, [64], which we will prove in Sect. 9.1:

Theorem 1.1.6 *Let M be a compact three-manifold without boundary equipped with an overtwisted contact form λ. Then the associated Reeb vector field has a contractible periodic orbit.*

It is surprising that no topological assumption on M is needed. There is a criterion for tightness due to M. Gromov, see [32, 60]:

Theorem 1.1.7 *Let ξ be a contact structure on the compact three-manifold M without boundary. Assume there exists a compact symplectic four-manifold (W, σ) such that $M = \partial W$ and $\sigma|_M = d\lambda$ for a contact form λ inducing ξ. Then ξ is tight.*

The proofs of all these theorems are a combination of hard analysis as well as subtle geometric arguments concerning the geometry of contact forms. The geometric methods are based on deep result due to Y. Eliashberg [31–33] and E. Giroux [58]. A manifold M satisfying the conditions in Theorem 1.1.7 is called *(symplectically) fillable*. It turns out that the classification of overtwisted contact structures is a purely homotopy-theoretic problem (see Theorem 1.1.8 below). Not every compact orientable three-manifold without boundary admits a tight contact structure. This result is due to J. Etnyre and K. Honda, [38].

In contrast to the intricacies one faces in studying tight contact structures the situation is clear for overtwisted structures due to the pioneering work of Y. Eliashberg. Let us introduce on the set of nowhere vanishing one-forms the equivalence relation $\sim$ by saying that two such forms are equivalent if they can be connected by a smooth arc of nowhere vanishing one-forms. We introduce the equivalence relation $\approx$ for two contact forms if they can be connected by a smooth arc of contact forms. We formulate Y. Eliashberg's result as follows, see [31]:

Theorem 1.1.8 *Let M be a compact three-manifold without boundary and λ_0 and λ_1 two overtwisted contact forms satisfying $\lambda_0 \sim \lambda_1$. Then $\lambda_0 \approx \lambda_1$. Moreover for every nowhere vanishing one-form τ there exists a overtwisted contact form λ satisfying $\tau \sim \lambda$.*

We call a codimension one subbundle E' of a vector bundle E *co-oriented* if and only if E/E' is orientable. Then we may phrase the result above as follows. Every co-oriented plane field is homotopic to an overtwisted contact structure. Two overtwisted contact structures homotopic as plane fields are homotopic as overtwisted contact structures. In view of the Gray's stability result (Theorem 2.3.1) this implies that overtwisted contact structures modulo diffeomorphism isotopic to the identity are in bijective correspondence to homotopy classes of co-oriented plane field distributions in TM.

1.2 Relationship with Hamiltonian Dynamics

Reeb flows are very closely related to Hamiltonian dynamics. For example any isoenergetic flow of a classical Hamiltonian system on a compact regular energy surface is up to a parametrisation of the flow lines a Reeb flow.

Let (M, ω) be a *symplectic manifold* of dimension $2n$, i.e. ω is a closed and nondegenerate 2-form on M. Moreover let $S \subset M$ be a *regular compact energy surface* , i.e. there is a smooth function $H : M \longrightarrow \mathbf{R}$ such that $S = H^{-1}(0)$ and $dH(x) \neq 0$ for all $x \in S$. Recall that the *Hamiltonian vector field* X_H is defined by $dH = i_{X_H}\omega$.

Proposition 1.2.1 *With S as just described assume further that there exists a 1-form λ on S such that $d\lambda = \omega \mid S$ and $\lambda \wedge (d\lambda)^{n-1}$ is a volume form on S. Then the orbits of the Reeb vector field X_λ of λ and the Hamiltonian vector field X_H on S coincide up to parametrisation.*

Proof Consider the characteristic line bundle $\mathcal{L}_S$ over S which is defined as follows:

$$\mathcal{L}_S := \bigcup_{x\in S}(\mathcal{L}_S)_x := \bigcup_{x\in S}\{\xi \in T_xS \mid \omega_x(\xi,\eta) = 0 \text{ for all } \eta \in T_xS\}.$$

We calculate $\omega_x(X_H(x),\eta) = dH(x)(\eta) = 0$ for $\eta \in T_xS$, i.e. X_H is a section in $\mathcal{L}_S$ which does not vanish because $dH(x) \neq 0$ for all $x \in S$. Now $0 = i_{X_\lambda}d\lambda = i_{X_\lambda}(\omega\mid_S)$ on TS, so X_λ is also a non vanishing section in $\mathcal{L}_S$. Therefore we can find a non vanishing smooth function f on S such that $X_\lambda(x) = f(x)X_H(x)$. Suppose that $x : \mathbf{R} \longrightarrow S \subset M$ is a solution of $\dot{x}(t) = X_H(x(t))$ and $\alpha : \mathbf{R} \longrightarrow \mathbf{R}$ solves the ordinary differential equation

$$\dot{\alpha}(t) = f((x\circ\alpha)(t)).$$

Then

$$\begin{aligned}\frac{d}{dt}(x\circ\alpha)(t) &= \dot{x}(\alpha(t))\dot{\alpha}(t)\\ &= X_H((x\circ\alpha)(t))f((x\circ\alpha)(t))\\ &= X_\lambda((x\circ\alpha)(t))\end{aligned}$$

which proves the proposition. □

If we consider Hamiltonian systems of classical mechanics then the symplectic manifold is the cotangent bundle T^*M of some n-dimensional manifold M (this is the phase space of the configuration manifold M). The cotangent bundle carries a canonical symplectic structure ω defined as follows:

Denote by $\tau^* : T^*M \longrightarrow M$, $\tau_1 : TT^*M \longrightarrow T^*M$ and $\tau : TM \longrightarrow M$ the projections. Then the following diagram commutes:

$$\begin{array}{ccc} & & T\tau^* \\ & TT^*M & \longrightarrow TM \\ \tau_1 & \downarrow & \downarrow \ \tau\,. \\ & T^*M & \longrightarrow M \\ & & \tau^* \end{array}$$

Define the canonical 1-form θ on T^*M by

$$\theta(\xi) := (\tau_1\xi)(T\tau^*\xi) \ ; \xi \in TT^*M$$

and a two-form by $\omega := -d\theta$. If we take coordinates $(q_1,\ldots,q_n,p_1,\ldots,p_n)$ on T^*M where $p_1,\ldots,p_n$ describe the fiber then

$$\theta = \sum_{i=1}^n p_i dq_i \quad \text{and} \quad \omega = \sum_{i=1}^n dq_i \wedge dp_i,$$

so ω is symplectic. Now take a compact regular hypersurface $S \subset T^*M$ that can be written as $H^{-1}(0)$ for some classical Hamiltonian

$$H : T^*M \longrightarrow \mathbf{R}$$

$$H(\eta) := \frac{1}{2} g_x^*(\eta, \eta) + V(x) \ , \ x = \tau^*\eta \in M.$$

Here g^* is the dual metric on T^*M. The following theorem implies that Hamiltonian dynamics on a compact regular energy hypersurface is the same as Reeb dynamics for the case of classical Hamiltonians.

Theorem 1.2.2 *Let $S \subset T^*M$ be a compact regular energy hypersurface of a classical Hamiltonian as described above. Then there exists a one form β on S such that $d\beta = \omega \mid S$ and $\beta \wedge (d\beta)^{n-1}$ is a volume form on S.*

Proof The canonical 1-form θ on T^*M fails to be a contact form on S if S intersects the zero section of the cotangent bundle. We will define a function $F : T^*M \longrightarrow \mathbf{R}$ such that $-\theta + \varepsilon \, dF$ is a contact form on S for some suitable $\varepsilon = \varepsilon(S) > 0$. Define

$$F : T^*M \longrightarrow \mathbf{R}$$

$$F(\eta) := dV(x)(\gamma_x^{-1}\eta)$$

where $\gamma_x : T_xM \to T_x^*M$ is the isomorphism given by the metric g on TM. Now let us investigate F and dF in local coordinates

$$(q, p) = (q_1, \ldots, q_n, p_1, \ldots, p_n) \in \mathbf{R}^{2n}$$

on T^*M such that θ and ω are given by

$$\theta = \sum_{i=1}^{n} p_i dq_i \ ; \ \ \omega = \sum_{i=1}^{n} dq_i \wedge dp_i.$$

In local coordinates TM and T^*M are of course both given by $\mathbf{R}^{2n}$ but we identify them via the metric g. We can describe the metric g in local coordinates by

$$g_q(. \ , \ .) = \langle A(q) \ . \ , \ . \ \rangle$$

with some symmetric positive definite matrix

$$A(q) = (a_{ij}(q))_{1 \le i,j \le n}.$$

If $p = (p_1, \ldots, p_n)$ is in T_q^*M then $\gamma_q^{-1} p$ is given by $A(q)^{-1} p$, so we obtain

$$g_q^*(p, p') = \sum_{i,j} \tilde{a}_{ij}(q) p_i p'_j$$

where $\tilde{a}_{ij}$ are the entries of the matrix A^{-1}. Then the Hamiltonian vector field is given by

$$\begin{aligned} X_H(q,\,p) &= \sum_k \left(H_{p_k} \frac{\partial}{\partial q_k} - H_{q_k} \frac{\partial}{\partial p_k} \right) \\ &= \sum_{i \le k} \tilde{a}_{ik}(q)\, p_i \frac{\partial}{\partial q_k} - \sum_k \left(\sum_{i \le l} \frac{\partial \tilde{a}_{il}(q)}{\partial q_k} p_i\, p_l + \frac{\partial V}{\partial q_k}(q) \right) \frac{\partial}{\partial p_k}. \end{aligned}$$

Finally F is given by

$$F(q,\,p) = \sum_{k,l} \frac{\partial V}{\partial q_k}(q) \tilde{a}_{kl}(q)\, p_l$$

and

$$\begin{aligned} dF(q,\,p) = \sum_{i,k,l} \left(\frac{\partial^2 V}{\partial q_i \partial q_k}(q) \tilde{a}_{kl}(q)\, p_l + \frac{\partial V}{\partial q_k}(q) \frac{\partial \tilde{a}_{kl}}{\partial q_i}(q)\, p_l \right) dq_i + \\ + \sum_{k,l} \left(\frac{\partial V}{\partial q_k}(q) \tilde{a}_{kl}(q) \right) dp_l. \end{aligned}$$

We would like to see that $(-\theta + \varepsilon\, dF) X_H < 0$ on S. Then we will conclude that $\beta = -\theta + \varepsilon\, dF$ must be a contact form on S (trivially $d\beta = \omega \mid S$). We compute

$$\begin{aligned} dF(q,0) X_H(q,0) &= -\sum_{j,k} \frac{\partial V}{\partial q_k}(q) \frac{\partial V}{\partial q_j}(q) \tilde{a}_{jk}(q) \\ &= -\langle A^{-1}(q) \frac{\partial V}{\partial q}(q),\, \frac{\partial V}{\partial q}(q) \rangle \end{aligned}$$

with $\frac{\partial V}{\partial q}(q) := (\frac{\partial V}{\partial q_1}(q), \ldots, \frac{\partial V}{\partial q_n}(q))^T$. We also see that $\frac{\partial H}{\partial p_k}(q,0) = 0$ for all k.

Because S is assumed to be regular, i.e. $dH(x) \neq 0$ for all $x \in S$, we conclude that $\frac{\partial H}{\partial q_k}(q,0) = \frac{\partial V}{\partial q_k}(q,0) \neq 0$ for some k. Now $A^{-1}(q)$ is positive definite , so we get

$$dF(q,0) X_H(q,0) < 0 \text{ if } (q,0) \in S.$$

On the other hand we have

$$\begin{aligned}\theta(q,p)X_H(q,p) &= \left(\sum_k p_k dq_k\right) X_H(q,p)\\ &= \frac{1}{2}\sum_{i,k}(\tilde{a}_{ik}(q)+\tilde{a}_{ki}(q))p_i p_k\\ &= g^*_{(q_1,\ldots,q_n)}((p_1,\ldots,p_n),(p_1,\ldots,p_n))\\ &\geq 0.\end{aligned}$$

So $\theta(q,p)X_H(q,p)=0$ if and only if $p=0$. Let S_0 be some neighborhood of the zero section such that $dF(X_H)<0$ on S_0. Remember that S is compact and take

$$M := \min_{(q,p)\in S\setminus S_0}\theta(q,p)X_H(q,p) > 0,$$

$$M' := \max_{(q,p)\in S\setminus S_0}\{0, dF(q,p)X_H(q,p)\}.$$

If $M'\neq 0$ then choose $\varepsilon < \frac{M}{M'}$. We see that

$$\beta(X_H) := -\theta(X_H) + \varepsilon\, dF(X_H) \;<\; 0 \text{ on } S_0$$

and on $S\setminus S_0$

$$\begin{aligned}\beta(X_H) &= -\theta(X_H)+\varepsilon\, dF(X_H)\\ &\leq -M+\varepsilon M'\\ &< 0.\end{aligned}$$

Summarizing the discussion above we found a 1-form β on T^*M such that $d\beta = \omega\,|_S$ and $\beta(X_H)<0$ on S. It remains to show that $\beta\wedge(d\beta)^{n-1}$ is a volume form on S.

Denote by $\mathcal{L}_S$ the characteristic line bundle over $S\subset T^*M$. We saw in the proof of Proposition 1.2.1 that $X_H\,|_S$ is a non vanishing section in $\mathcal{L}_S$, so $\beta(\xi)\neq 0$ for all nonzero $\xi\in\mathcal{L}_S$. Now $\ker\beta(x) := \{\eta\in T_xS \mid \beta_x(\eta)=0\}$ is a $(2n-2)$-dimensional subspace of T_xS, $\ker\beta(x)\oplus(\mathcal{L}_S)_x = T_xS$ and $(\omega\,|_S)_x$ is nondegenerate on $\ker\beta(x)$. Hence $d\beta(x)$ is nondegenerate on $\ker\beta(x)$ and because of $\beta_x\,|_{(\mathcal{L}_S)_x}\neq 0$ we conclude that $\beta_x\wedge(d\beta)_x^{n-1}$ is a volume form on T_xS. □

1.3 Relationship with Hydrodynamics

Reeb vector fields appear in hydrodynamics on Riemann manifolds of odd dimension [10, 37, 57]. The motion of a perfect incompressible fluid on an orientable Riemann manifold (M, g) with volume form μ is described by a time–dependent vector field X, the velocity of a fluid particle at a given point of M. The vector field X satisfies Euler's equation, which is

$$\begin{aligned} \frac{\partial X}{\partial t} + \nabla_X X &= -\nabla p, \\ L_X \mu &= 0, \\ X|_{t=0} &= X_0, \end{aligned} \tag{1.2}$$

for some time–dependent function p on M ('pressure'). The symbol ∇_X denotes covariant differentiation along X with respect to the Levi–Civita connection associated to the metric g and X_0 is a given initial vector field on M. If the manifold M has nonempty boundary, then we prescribe the boundary condition

$$g(X, n) \equiv 0$$

on $\partial M \times (0, \infty)$ and of course $g(X_0, n) = 0$, where n denotes the outward normal to ∂M. The condition $L_X \mu = 0$ means that the flow associated to X is volume preserving. It is an open question whether a solution X for any given smooth initial condition X_0 exists for all times. We will give an equivalent formulation for the Euler equation involving the curl $\nabla \times X$ of the vector field X which is defined by the equation

$$i_{\nabla \times X} \mu = d(i_X g). \tag{1.3}$$

Lemma 1.3.1 *The equation* $\frac{\partial X}{\partial t} + \nabla_X X = -\nabla p$ *is equivalent to the following equation:*

$$\frac{\partial (i_X g)}{\partial t} + i_X i_{\nabla \times X} \mu = -dP, \tag{1.4}$$

where $P := p + \frac{1}{2}\|X\|^2$.

Proof We calculate for an arbitrary vector field Z on M:

$$\begin{aligned} (i_X i_{\nabla \times X} \mu) Z &= (i_X d(i_X g)) Z \\ &= X((i_X g)(Z)) - Z((i_X g)(X)) - (i_X g)([X, Z]) \\ &= d(g(X, Z)) \cdot X - d(\|X\|^2) \cdot Z - g(X, [X, Z]) \end{aligned}$$

$$= g(\nabla_X X, Z) + g(X, \nabla_Z X) - d(\|X\|^2) \cdot Z$$
$$= g(\nabla_X X, Z) - \frac{1}{2} d(\|X\|^2) \cdot Z.$$

Since

$$-dP \cdot Z = g(-\nabla p, Z) - \frac{1}{2} d(\|X\|^2) \cdot Z$$

and

$$\frac{\partial i_X g}{\partial t} Z = g(\frac{\partial X}{\partial t}, Z),$$

the proof is complete. □

We will consider steady fluid flows in dimension three, i.e. solutions X of Eq. 1.2 which do not depend on time and where M is three dimensional. We are interested in a special class of steady solutions of the Euler equations, the so–called Beltrami fields:

Definition 1.3.2 A vector field X on a three dimensional Riemann manifold M is called a *Beltrami field* if it is parallel to its curl, i.e. $\nabla \times X = fX$ for some function f on M. A Beltrami field X is called *rotational* if f does not vanish anywhere, i.e. the curl of X is nonsingular.

It is easy to see that Beltrami fields are indeed solutions of the Euler equation. Use rather Eq. (1.4) with $p = -\frac{1}{2}\|X\|^2$. There is the following relation between nonsingular rotational Beltrami fields and Reeb vector fields [37]:

Theorem 1.3.3 *Let (M, g) be a three dimensional Riemann manifold with volume form μ. If X is a nonsingular rotational Beltrami field on M then there is a contact form λ on M so that its Reeb vector field X_λ equals $X/\|X\|^2$. Conversely, if λ is a contact form on M with Reeb vector field X_λ, then there exist a Riemann metric g' and a volume form μ' on M, so that hX_λ is a nonsingular rotational Beltrami field with respect to g' and μ', where h is an arbitrary non vanishing function on M.*

Proof We reproduce the proof of [37]. We define a 1–form on M by $\lambda := i_X g$ and we claim that this is the desired contact form. The rotational Beltrami condition $\nabla \times X = fX$ implies with (1.3) that

$$d\lambda = d(i_X g) = f\, i_X \mu,$$

which implies

$$i_X d\lambda = 0.$$

In order to verify that λ is a contact form, we consider a local orthonormal frame (e_1, e_2, e_3) for TM over a chart of M so that $e_1 = X/\|X\|$. If (e_1^*, e_2^*, e_3^*) is the dual frame then

$$\lambda = i_X g = \|X\| e_1^*.$$

Write $\mu = h e_1^* \wedge e_2^* \wedge e_3^*$ where h is a nowhere vanishing function. Then

$$\lambda \wedge d\lambda = f(i_X g \wedge i_X \mu) = fh\|X\|^2 e_1^* \wedge e_2^* \wedge e_3^* \neq 0,$$

hence λ is a contact form. In view of $i_X \lambda = \|X\|^2$, we have $X_\lambda = X/\|X\|^2$. This implies in particular that the flow lines of X and X_λ coincide up to parametrisation.

Assume now that λ is a contact form on M and choose a bundle metric γ on $\xi = \ker \lambda$. Define a metric g' on M by

$$g'_p(v, w) := \frac{1}{h}\lambda(p)v \cdot \lambda(p)w + \gamma_p(\pi_\lambda v, \pi_\lambda w), \ \ p \in M,$$

where $\pi_\lambda : TM \to \xi$ denotes the projection along X_λ and let $\mu' := h^{-1}\lambda \wedge d\lambda$. We claim that hX_λ is volume preserving with respect to μ' and equal to its curl (with respect to μ' and g'). We compute

$$\begin{aligned} L_{hX_\lambda}\mu' &= d(i_{hX_\lambda}\mu') \\ &= d(h^{-1}(i_{hX_\lambda}\lambda)d\lambda) \\ &= dd\lambda \\ &= 0 \end{aligned}$$

and

$$\begin{aligned} i_{hX_\lambda}\mu' &= d\lambda \\ &= d(i_{hX_\lambda} g'). \end{aligned}$$

Hence $\nabla \times hX_\lambda = hX_\lambda$. □

Because of Theorem 1.3.3 results about Reeb vector fields have hydrodynamic counterparts: For example, Martinet's theorem implies that every closed orientable three manifold has a nonsingular rotational Beltrami field for some Riemann metric and volume form. Theorems about the dynamics of Reeb vector fields imply corresponding results for Beltrami fields.

1.4 Pseudoholomorphic Curves and Contact Geometry

In this section we will state the results proved in this book and give the ideas of the proofs. Assume M is a manifold with contact form λ. Let

$$TM \;=\; (\ell, X_\lambda) \;\oplus\; (\xi, \omega)$$

be the associated splitting. We denote the projection onto ξ along ℓ by π_λ. Pick a complex structure $J : \xi \;\to\; \xi$ such that

$$g_J := d\lambda \;\circ\; (Id \times\; J)$$

defines a metric for ξ. We call such complex structures *compatible* with $d\lambda$.

Lemma 1.4.1 *The set $\mathcal{J}$ of smooth complex structures for ξ as specified above is a nonempty contractible space for the C^∞-topology.*

Proof We can associate an element $J \in \mathcal{J}$ to any bundle metric $\langle .\ ,\ . \rangle$ on $\pi : \xi \to M$ as follows: Write $\omega = \langle .\ , A\ . \rangle$ where A is a section in the bundle End(ξ). Define A^T by $\langle A^T .\ ,\ . \rangle = \langle .\ , A\ . \rangle$. Because ω is nondegenerate and skew–symmetric we have $A = -A^T$ and $\langle x, A^T A x \rangle > 0 \Longleftrightarrow x \neq 0$. One can define $\sqrt{A^T A} = \sqrt{-A^2}$ and $\sqrt{-A^2}$ is again symmetric and positive definite. Moreover $\sqrt{-A^2}$ commutes with A and A^{-1}. This can be seen by choosing a basis of the fiber ξ_x such that A consists of blocks of the form

$$\begin{pmatrix} 0 & -\lambda_1 \\ \lambda_1 & 0 \end{pmatrix}, \ldots, \begin{pmatrix} 0 & -\lambda_n \\ \lambda_n & 0 \end{pmatrix}.$$

Now define $J := A^{-1}\sqrt{-A^2}$ and we see that $J \in \mathcal{J}$. Pick $J_0 \in \mathcal{J}$ and write $g_{J_0} := \omega \circ (Id \times J_0)$. For each $J \in \mathcal{J}$

$$g_t := t g_{J_0} + (1-t) g_J := t g_{J_0} + (1-t)(\omega \circ (Id \times J))$$

is again a metric, i.e. $\omega = g_t(.\ , A_t\ .)$ with $A_t = (t J_0 + (1-t) J)^{-1}$. Now we define

$$H : [0,1] \times \mathcal{J} \longmapsto \mathcal{J}$$

$$H(t, J) \;:=\; A_t^{-1}\sqrt{-A_t^2}$$

and see that

$$H(0, .) \;=\; Id_{\mathcal{J}}$$

$$H(1, .) \;\equiv\; J_0.$$

Obviously H is continuous in the C^∞-topology. □

We will call $(\mathbf{R}\times M, d(e^t\lambda))$ the *symplectisation* of M because the 2-form $d(e^t\lambda)$ is a symplectic form on $\mathbf{R}\times M$ (t denotes the coordinate on the factor $\mathbf{R}$). We define an almost complex structure $\tilde{J}$ on $\mathbf{R}\times M$ as follows:

$$\tilde{J}(a,u)(h,k) := (-\lambda(u)k, J(u)\pi_\lambda k + h\cdot X_\lambda(u)).$$

Here $(a,u)\in\mathbf{R}\times M$, $(h,k)\in T_{(a,u)}(\mathbf{R}\times M)$ and J is a complex structure on ξ compatible with $d\lambda$. The geometric meaning of this definition is the following: $\tilde{J}$ restricted to the contact structure ξ is just J. Then we map $\partial/\partial t$ onto the Reeb vector field and the Reeb vector field onto $-\partial/\partial t$.

Now assume that (S,i) is a Riemann surface, where i denotes the complex structure on S. We will study maps $\tilde{u}=(a,u): S\to\mathbf{R}\times M$ which satisfy the partial differential equation

$$T\tilde{u}\circ i = \tilde{J}(\tilde{u})\circ T\tilde{u}. \tag{1.5}$$

We will call Eq. (1.5) the *nonlinear Cauchy Riemann equation* and $\tilde{u}$ as above a *pseudoholomorphic* or $\tilde{J}$*-holomorphic curve*. Let us discuss Eq. (1.5) in more detail. If $z=s+it$ are conformal coordinates on S then we may write the nonlinear Cauchy Riemann equation as

$$\partial_s\tilde{u} + \tilde{J}(\tilde{u})\partial_t\tilde{u} = 0$$

which is equivalent to

$$\begin{aligned} \pi_\lambda\partial_s u + J(u)\pi_\lambda\partial_t u &= 0\\ \partial_s a - \lambda(u)\partial_t u &= 0\\ \partial_t a + \lambda(u)\partial_s u &= 0. \end{aligned} \tag{1.6}$$

Let us now point out the relation between pseudoholomorphic curves in the symplectisation $\mathbf{R}\times M$ of a contact manifold M and the dynamics of the Reeb vector field X_λ. For this we define the notion of *energy* of a pseudoholomorphic curve $\tilde{u}: S\to\mathbf{R}\times M$. Denote by Σ the set of all smooth functions $\varphi:\mathbf{R}\to[0,1]$ with nonnegative derivative. Then we define the energy $E(\tilde{u})$ of $\tilde{u}$ by

$$E(\tilde{u}) := \sup_{\varphi\in\Sigma}\int_S \tilde{u}^* d(\varphi\lambda).$$

We claim that the energy of a pseudoholomorphic curve $\tilde{u}$ is always nonnegative and zero if and only if $\tilde{u}$ is constant. Using the nonlinear Cauchy Riemann equation,

we compute in conformal coordinates $s + it$ on S

$$\begin{aligned}
\tilde{u}^* d(\varphi\lambda) &= \tilde{u}^*(d\varphi \wedge \lambda + \varphi d\lambda) \\
&= \varphi'(a) da \wedge (u^*\lambda) + \varphi(a) u^* d\lambda \\
&= \Big(\varphi'(a)[\partial_s a \lambda(u)\partial_t u - \partial_t a \lambda(u)\partial_s u] + \\
&\quad + \frac{1}{2}\varphi(a)[\mid \pi_\lambda \partial_s u \mid_J^2 + \mid \pi_\lambda \partial_t u \mid_J^2] \Big) ds \wedge dt \\
&= \Big(\frac{1}{2}\varphi'(a)[(\partial_s a)^2 + (\partial_t a)^2 + (\lambda(u)\partial_s u)^2 + (\lambda(u)\partial_t u)^2] + \\
&\quad + \frac{1}{2}\varphi(a)[\mid \pi_\lambda \partial_s u \mid_J^2 + \mid \pi_\lambda \partial_t u \mid_J^2] \Big) ds \wedge dt,
\end{aligned}$$

proving the claim. If we chose S to be a closed Riemann surface then every $\tilde{J}$–holomorphic curve would be constant by Stokes' theorem:

$$E(\tilde{u}) = \sup_{\varphi \in \Sigma} \int_{\partial S = \emptyset} \tilde{u}^*(\varphi\lambda) = 0.$$

Hence we have to consider pseudoholomorphic curves on non compact domains. The simplest case would be $S = \mathbf{C}$, i.e. S is the two–sphere with one point removed. On the other hand, there are many uninteresting nontrivial pseudoholomorphic curves. If $x(t)$ is any trajectory of the Reeb vector field, then the map

$$\tilde{u} : \mathbf{C} \longrightarrow \mathbf{R} \times M$$

$$\tilde{u}(s,t) := (s, x(t))$$

is pseudoholomorphic, but $E(\tilde{u}) = \sup_{\varphi \in \Sigma} \int_{\mathbf{C}} \varphi'(s)\, ds\, dt = +\infty$. It turns out that finiteness and non triviality of the energy singles out interesting solutions: We will prove the following theorem in Chap. 4:

Theorem 1.4.2 *Let M be a contact manifold of arbitrary dimension with contact form λ and let J be a compatible complex structure on the contact structure* $\ker\lambda$. *Assume $\tilde{u} : \mathbf{C} \to \mathbf{R} \times M$ is a solution of the nonlinear Cauchy Riemann equation* (1.5) *so that $u(\mathbf{C})$ is contained in a compact subset of M and $0 < E(\tilde{u}) < +\infty$. Then*

$$T := \int_{\mathbf{C}} u^* d\lambda > 0$$

and every sequence $R_k' \to \infty$ *has a subsequence* R_k *such that*

$$u(R_k e^{2\pi i \frac{t}{T}}) \longrightarrow x(t)$$

for $k \to \infty$ *in* C^∞ *where* x *is a* T*-periodic solution of*

$$\dot{x}(t) = X_\lambda(x(t)).$$

So the existence of a nontrivial pseudoholomorphic plane in $\mathbf{R} \times M$ with finite energy and $u(\mathbf{C})$ bounded implies existence of a contractible periodic orbit of the Reeb vector field X_λ.

Later, in Chap. 9, we will prove an existence result for periodic Reeb orbits in the following cases:

(1) M is any closed three dimensional manifold with overtwisted contact form λ.
(2) M is a closed three dimensional manifold with tight contact form λ and nontrivial second homotopy group.
(3) M is the three sphere S^3 and λ is any contact form.

Here is the place where the theory of pseudoholomorphic curves connects with the so-called *Weinstein conjecture*, see [114].

Weinstein Conjecture *Let* λ *be a contact form on a compact manifold* M *without boundary. Then the associated Reeb vector field has a periodic solution.*

In the cases (1) and (2) above we will establish the existence of a nontrivial pseudoholomorphic plane with finite energy (*nontrivial finite energy plane*) by using so-called 'filling methods'. Case (3) is based on a deep theorem of Y. Eliashberg about classification of contact structures on the three sphere. Before we explain the idea of filling, let us compare case (3) above with the so-called Seifert conjecture.

In [101] it was conjectured that every smooth vector field on the three-sphere possesses a periodic orbit. It was shown by Schweitzer in [100] that the Seifert conjecture is at least false if we relax the smoothness assumption. Schweitzer constructed a C^1-counterexample. In 1994 K. Kuperberg, [78], succeeded to find even a smooth counterexample to the Seifert conjecture. Case (3) above shows that the Seifert conjecture is true for the restricted class of Reeb vector field. G. Kuperberg found a C^1-vector field on the three sphere with volume preserving flow that has no periodic orbits, [79]. It is still an open problem to find the most general class of vector fields on the three sphere so that the Seifert conjecture is true.

The Weinstein conjecture in dimension three has been completely resolved by C.H. Taubes in 2007 using Seiberg-Witten equations [108].

We will now summarize briefly how to prove the existence of a nontrivial finite energy plane in the cases (1) and (2) above. We will actually fill certain surfaces in M with pseudoholomorphic disks.

Let M be a three dimensional manifold with contact form λ and let F be a two dimensional oriented submanifold of M. If p is a point on the surface F then either $T_pF = \ker\lambda(p)$ or $T_pF + \ker\lambda(p) = T_pM$. In the first case we will call p a *singular*

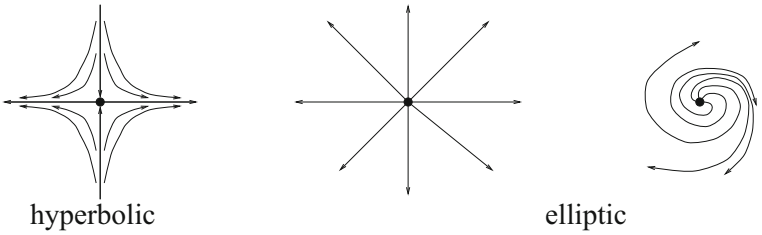


Fig. 1.3 Singular foliation near elliptic and hyperbolic points

point. In the second case, the intersection of T_pF and $\ker\lambda(p)$ is one dimensional. In Chap. 3 we will consider the *oriented singular foliation* on the surface F, the class of all vector fields V on F with $V(p) \in \ker\lambda(p)$ and $V(p) = 0$ if and only if p is a singular point. The oriented singular foliation on a surface in a three dimensional contact manifold is a central object in three dimensional contact geometry and was first studied by D. Bennequin, Y. Eliashberg and E. Giroux.

There are different kinds of singular points. Note that F was assumed to be oriented and $\ker\lambda(p)$ is canonically oriented by $d\lambda(p)$. If p is a singular point, we will call p *positive* if the orientations of T_pF and $\ker\lambda(p)$ coincide (otherwise p is called *negative*). Depending on the eigenvalues of the linearization of V in the singular point p we also distinguish *elliptic* and *hyperbolic* singular points. Figure 1.3 shows the oriented singular foliation near elliptic and hyperbolic points. In Chap. 3 we will show that a surface F can be C^0-small deformed near an elliptic singular point p without creating any new singular points and keeping the point p fixed, so that we have a certain normal form for F near the point p. Using this normal form we will establish in Chap. 7 the existence of a smooth 1-parameter family of pseudoholomorphic disks

$$\tilde{u}_\tau = (a_\tau, u_\tau) : D \longrightarrow \mathbf{R} \times M \ , \ \tau \in (0,1)$$

which satisfy the boundary condition

$$\tilde{u}_\tau(\partial D) \subset \{0\} \times (F \backslash \{p\}) \tag{1.7}$$

so that the disjoint union

$$\{p\} \cup \left(\bigcup_{0<\tau<1} u_\tau(\partial D) \right)$$

is an open neighborhood of the singular point p in F. Moreover the images of two such disks $\tilde{u}_\tau(D)$ and $\tilde{u}_\rho(D)$ are disjoint whenever $\rho \neq \tau$. We will call this family of disks a *Bishop family* because the first result of this kind was proved by E. Bishop (Fig. 1.4). We want to use Bishop families to prove existence of a nontrivial finite

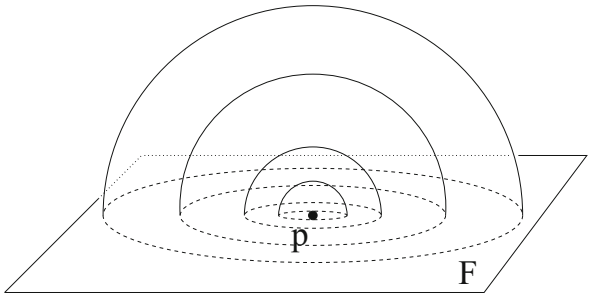


Fig. 1.4 Bishop family near an elliptic singular point p

energy plane which will imply existence of a periodic orbit of the Reeb vector field. We know that a Bishop family is coming out from an elliptic singular point but we also have to understand the global behavior of this family: How far can it be continued? Is it possible that the disk family suddenly stops or might it happen that a given Bishop family is splitting off into several disk families? We also have to understand what happens if two Bishop families intersect, do they fit together smoothly?

We will study all these questions in Chap. 7. The most important ingredient is a Fredholm theory for the nonlinear Cauchy Riemann operator on the disk with boundary condition (1.7) in a suitable functional analytic setting. The details of this analysis are provided in Appendix A.4. We summarize briefly the main result of Chap. 7:

- Given an embedded pseudoholomorphic disk $\tilde{u}_0$ in the symplectisation $\mathbf{R} \times M$ with Maslov index 2 that has its boundary in $\{0\} \times F^*$, where F^* is $F \backslash \{\text{singular points}\}$, then there exists an embedding

$$\tilde{u} : (-1, +1) \times D \to \mathbf{R} \times M$$

  so that $\tilde{u}(0, \,.\,) = \tilde{u}_0$ and each $\tilde{u}(\tau, \,.\,)$ is an embedded pseudoholomorphic disk with boundary in $\{0\} \times F^*$.
- Let $\tilde{v}$ be an embedded pseudoholomorphic disk and $\tilde{u}(\tau, \,.\,)$ a family of embedded pseudoholomorphic disks as above, both with boundaries in $\{0\} \times F^*$ and Maslov index 2. Assume that $\tilde{v}(D)$ does not intersect any of the disks $\tilde{u}(\tau, D)$ for $-1 < \tau < 0$, but $\tilde{v} \cap \tilde{u}(0, D) \neq \emptyset$. Then $\tilde{v}$ and $\tilde{u}(0, \,.\,)$ coincide up to parametrisation.

In Chap. 8 we will study some analytical properties of families of pseudoholomorphic disks. Let $\tilde{u} : [-1, 0) \times D \to \mathbf{R} \times M$ be an embedding so that each $\tilde{u}(\tau, \,.\,)$ is an embedded pseudoholomorphic disk with boundary in $\{0\} \times F^*$ and Maslov index 2. Then we either have a C^0-gradient bound on all the disks $\tilde{u}(\tau, \,.\,)$ uniformly in $\tau \in [-1, 0)$ or the C^0-norms of the gradients go to infinity as τ approaches zero. We will show the following. In the case of a uniform gradient bound the maps $\tilde{u}(\tau, \,.\,)$ converge in $C^\infty(D)$ to an embedded pseudoholomorphic disk $\tilde{u}_0$ with the right boundary condition and Maslov index. On the other hand, if the gradient blows up, a nontrivial finite energy plane will bubble off, i.e. after a suitable rescaling we

will extract from the disk family a sequence of pseudoholomorphic curves defined on larger and larger balls in the complex plane which will finally converge in C^∞_{loc} to a nontrivial finite energy plane.

Hence in order to prove existence of periodic orbits of the Reeb vector field we have to start with some surface F in M having an elliptic singular point, so that we get a local Bishop family near the singularity. Then we have to make sure that there is no uniform gradient bound for this Bishop family. In contrast to pseudoholomorphic curve techniques in symplectic geometry or Floer homology where C^0-gradient bounds are needed to prove compactness of certain moduli spaces we rather profit from the non compactness in our setup.

Let us come back to the settings where we want to prove existence of a periodic orbit of the Reeb vector field and let us explain how to obtain a Bishop family without gradient bound in these situations. Consider first the case of a closed three dimensional manifold M with tight contact form λ and nontrivial second homotopy group. Using a smooth version of the classical sphere theorem we find an embedded non contractible 2-sphere F in M. In Sect. 3.4 we will give the proof of the so-called elimination lemma due to E. Giroux after some preliminary work of Y. Eliashberg. In this book we will use a formulation due to Y. Eliashberg and D. Fuchs which roughly states the following. Let e be an elliptic and h a hyperbolic singular point of the singular characteristic foliation on F induced by the contact structure $\ker\lambda$. Assume that both singular points have the same sign and that there is a leaf of the characteristic foliation connecting e and h. Then there is a C^0-small isotopy deforming F into a surface F' eliminating the two singular points e and h, i.e. if $\mathcal{S}_F$ and $\mathcal{S}_{F'}$ denote the sets of singular points of F and F' respectively then $\mathcal{S}_{F'} = \mathcal{S}_F \backslash \{e, h\}$.

In the same chapter we will also give a proof of an inequality due to D. Bennequin and Y. Eliashberg which relates the numbers of elliptic and hyperbolic singular points of a surface F to topological data of F.

Combining the elimination lemma with Bennequin's inequality we are able to deform our embedded 2-sphere F in such a way that all singular points disappear except two elliptic points e^+ and e^- which have opposite signs. Figure 1.5 shows the result of this deformation process. Tightness of the contact form λ is needed for the deformation of F: Otherwise limit cycles for the singular characteristic foliation could pop up, but they would bound overtwisted disks and this is impossible if λ is tight.

Now we have two Bishop families coming out from the two elliptic singular points (see Fig. 1.6).

Let us assume that there is always a C^0-bound on the gradient since otherwise we are done. Then the two Bishop families could be continued further until they meet and they also have to fit together smoothly. But this means that we have filled the sphere F by embedded disjoint disks which is not possible since F was non contractible. This contradiction shows that our assumption of the existence of gradient bounds was false, but an explosion of the gradient leads to bubbling off of a nontrivial finite energy plane which implies existence of a periodic orbit of the Reeb vector field.

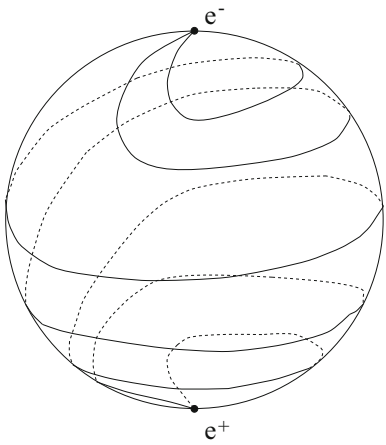


Fig. 1.5 A two-sphere in a tight contact manifold with optimal configuration of singular points

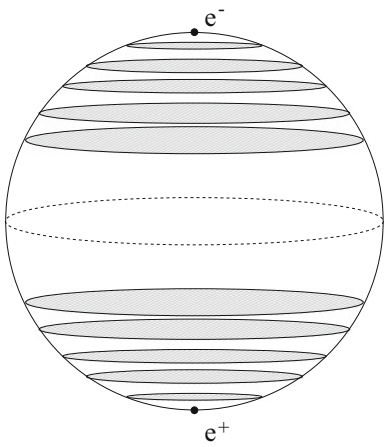


Fig. 1.6 S^2 with two elliptic singularities and local Bishop families

Consider now the case where M is a closed three dimensional manifold with overtwisted contact form λ. Then there exists an overtwisted disk $\mathcal{D}$ in M, i.e. the boundary $\partial\mathcal{D}$ is a leaf of the singular characteristic foliation on $\mathcal{D}$ induced by the contact structure λ and there are several singular points inside $\mathcal{D}$. Using again the elimination lemma together with Bennequin's inequality we can produce an overtwisted disk that looks as in Fig. 1.7. Now there is a Bishop family coming out from the elliptic singularity. We will show also that the boundary loops of the Bishop disks always have to be transverse to the leaves of the singular characteristic foliation on F (using a maximum principle), hence our Bishop family can never touch the boundary of the overtwisted disk. If we assume that there is a uniform C^0-gradient bound for all disks in the Bishop family then we could continue the family further and it would indeed hit the boundary of the overtwisted disk – contradiction. This shows again that bubbling off must occur and we are done.

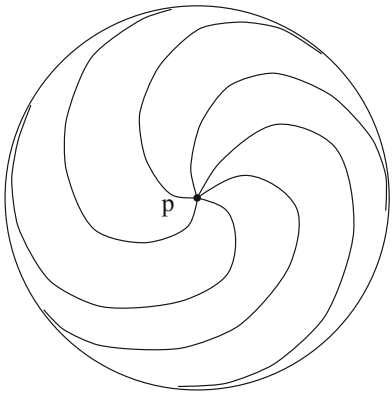


Fig. 1.7 An overtwisted disk with only one elliptic singular point p

We will use a similar argument to give a different proof of Bennequin's theorem that the standard contact structure on $\mathbf{R}^3$ is tight (Theorem 1.1.4) based on pseudoholomorphic curves. Assume that the standard contact form on $\mathbf{R}^3$ is overtwisted. Then there exists an overtwisted disk with singular characteristic foliation as in Fig. 1.7. By the same argument as above we derive existence of a nontrivial finite energy plane and therefore a periodic orbit of the Reeb vector field. But the Reeb vector field of the standard contact form on $\mathbf{R}^3$ is the constant vector field $(0, 0, 1)$ parallel to the z-axis which has no periodic orbits. So our assumption of existence of an overtwisted disk must have been false which proves tightness. In this proof we only have to handle the non compactness of $\mathbf{R}^3$, i.e. we have to derive uniform C^0-bounds for the Bishop disks which is of course not necessary in the case of a compact contact manifold M.

Chapter 2
Basic Results



In this chapter we will present several classical results. We will prove Darboux's theorem which states that all contact manifolds of the same dimension are locally diffeomorphic, hence the only local invariant of a contact manifold is its dimension. We will also prove the Legendre neighborhood theorem and its symplectic counterpart, the Lagrange neighborhood theorem. These results provide normal forms for neighborhoods of Legendrian submanifolds or Lagrangian submanifolds. We follow the presentation in [2] for these topics. Moreover we show that contact structures are stable on a compact manifold without boundary (which is called Gray's theorem), i.e. smoothly homotopic contact structures are diffeomorphic. Further, we will give a characterization of those vector fields whose flow preserves a contact structure ('contact vector fields'). Reeb vector fields are a special class of contact vector fields. Given a contact structure ξ, we will find a necessary and sufficient condition for a contact vector field to be the Reeb vector field of some contact form λ with $\ker \lambda = \xi$.

2.1 Darboux's Theorem

In this section we study local properties of contact forms.

Theorem 2.1.1 *Let (M, λ) and (N, τ) be contact manifolds of dimension $2n + 1$. Suppose $p \in M$ and $q \in N$ are given. Then there exists a local diffeomorphism $\psi : U \to V$ between open neighborhoods U of p and V of q such that*

$$\psi(p) = q, \text{ and } \psi^* \tau = \lambda.$$

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Obviously we may confine ourselves to the case where two contact forms λ and τ are given on a neighborhood of zero in $\mathbf{R}^{2n+1}$. Of course we may in addition assume that λ is the standard form on $\mathbf{R}^{2n+1}$.

Proof Let τ be a contact form in a neighborhood of the origin in $\mathbf{R}^{2n+1}$ with Reeb vector field X_τ and let λ be the standard form with Reeb vector field $X_\lambda = \frac{\partial}{\partial z}$. We have

$$\mathbf{R}^{2n+1} = \mathbf{R} \cdot X_\tau(0) \oplus \ker \tau(0)$$

and $d\tau(0)\ |_{\ker(\tau(0))\oplus\ker(\tau(0))}$ is a symplectic product. Then we can find a basis $e_1, \ldots, e_n, f_1, \ldots, f_n$ of $\ker(\tau(0))$ such that

$$d\tau(0)(e_i, e_j) = d\tau(0)(f_i, f_j) = 0,$$

$$d\tau(0)(e_i, f_j) = \delta_{ij} \ \text{ for all } \ 1 \leq i, j \leq n.$$

Define a linear isomorphism $A : \mathbf{R}^{2n+1} \longrightarrow \mathbf{R}^{2n+1}$ by

$$\frac{\partial}{\partial z} \longmapsto X_\tau(0)\ ;\ \frac{\partial}{\partial x_i} \longmapsto e_i\ ;\ \frac{\partial}{\partial y_i} \longmapsto f_i.$$

Then A satisfies

$$(A^*\tau)(0) = \lambda(0) = dz\ ;\ (A^*d\tau)(0) = d\lambda(0) = \sum_{j=1}^{n} dx_j \wedge dy_j$$

and

$$A \cdot X_\lambda(0) = X_\tau(0).$$

If we define $\tau' := A^*\tau$ then the Reeb vector fields of τ' and λ coincide at the origin.

Now we want to find a local diffeomorphism u on some neighborhood of zero in $\mathbf{R}^{2n+1}$ with $u(0) = 0$ such that $u^*\tau'$ and λ have the same Reeb vector field. Let ϕ^t be the flow of $X_{\tau'}$ and define

$$u(z, x_1, \ldots, x_n, y_1, \ldots, y_n) \ :=\ \phi^z(0, x_1, \ldots, x_n, y_1, \ldots, y_n).$$

Then

$$\begin{aligned} &(\phi^t \circ u)(z, x_1, \ldots, x_n, y_1, \ldots, y_n) \\ &= \phi^{t+z}(0, x_1, \ldots, x_n, y_1, \ldots, y_n) \\ &= (u \circ \psi^t)(z, x_1, \ldots, x_n, y_1, \ldots, y_n), \end{aligned}$$

where $\psi^t(z, x_1, \ldots, x_n, y_1, \ldots, y_n) := (z + t, x_1, \ldots, x_n, y_1, \ldots, y_n)$ is the flow of $X_\lambda = \frac{\partial}{\partial z}$. Hence

$$X_{\tau'}(u(x)) \;=\; du(x)X_\lambda(x)\;;\; x \in \mathbf{R}^{2n+1}.$$

We observe that

$$du(0) \;=\; \begin{pmatrix} X^1_{\tau'}(0) & 0\,0\cdots 0\,0 \\ X^2_{\tau'}(0) & 1\,0\cdots 0\,0 \\ X^3_{\tau'}(0) & 0\,1\cdots 0\,0 \\ \vdots & \vdots\,\vdots\,\ddots\,\vdots\,\vdots \\ X^{2n}_{\tau'}(0) & 0\,0\cdots 1\,0 \\ X^{2n+1}_{\tau'}(0) & 0\,0\cdots 0\,1 \end{pmatrix} \;=\; \mathrm{Id}_{R^{2n+1}}$$

because $X_{\tau'}(0) = X_\lambda(0) = \frac{\partial}{\partial z} \cong (1, 0, \ldots, 0)$. In particular, u is a local diffeomorphism. Moreover

$$\begin{aligned} i_{X_\lambda} d(u^*\tau') &= 0, \\ i_{X_\lambda}(u^*\tau') &= 1. \end{aligned}$$

So $\tau'' := u^*\tau'$ and λ are contact forms which have the same Reeb vector field. We note that τ'' and λ resp. $d\tau''$ and $d\lambda$ coincide in the origin because of $du(0) = \mathrm{Id}_{R^{2n+1}}$.

Define $\tau_t := t\tau'' + (1-t)\lambda$. Then

$$\begin{aligned} (\tau_t \wedge (d\tau_t)^n)(0) &= \lambda(0) \wedge (d\lambda(0))^n \\ &= \pm dz \wedge dx_1 \wedge \ldots \wedge dx_n \wedge dy_1 \wedge \ldots \wedge dy_n. \end{aligned}$$

Because of this we can find a small neighborhood of zero such that the forms $\tau_t \wedge (d\tau_t)^n$ are volume forms for all $t \in [0, 1]$, i.e. τ_t is a family of contact forms.

We calculate

$$i_{X_\lambda} d\tau_t \;=\; t\, i_{X_\lambda} d\tau'' + (1-t)\, i_{X_\lambda} d\lambda \;=\; 0,$$

$$i_{X_\lambda} \tau_t \;=\; t\, i_{X_\lambda} \tau'' + (1-t)\, i_{X_\lambda} \lambda \;=\; 1$$

and conclude that the Reeb vector field of each τ_t equals X_λ. Now define a time dependent vector field Y_t by

$$i_{Y_t} d\tau_t \;=\; -\frac{d}{dt}\tau_t \;=\; \lambda - \tau'',$$

$$i_{Y_t} \tau_t \;=\; 0.$$

We remark that Y_t would also be well–defined if we did not align the Reeb vector fields of λ and τ''on a neighborhood of 0. But then we would have the identity $i_{Y_t} d\tau_t = \lambda - \tau''$ only on $\ker \tau_t$. Plugging in $X_\lambda = X_{\tau_t} = X_{\tau''}$ shows that the 1–forms $i_{Y_t} d\tau_t$ and $\lambda - \tau''$ agree on all of $\mathbf{R}^{2n+1}$. The one parameter family (ϕ_t) generated by Y_t satisfies

$$\begin{aligned} \frac{d}{dt}\phi_t^*\tau_t &= \phi_t^*\left(\frac{d}{dt}\tau_t + L_{Y_t}\tau_t\right) \\ &= \phi_t^*\left(\frac{d}{dt}\tau_t + i_{Y_t} d\tau_t\right) \\ &= 0. \end{aligned}$$

Moreover, it exists until time 1 since $Y_t(0) = 0$ for all t. So $\phi_1^*\tau'' = \phi_0^*\lambda = \lambda$ and therefore $\phi^*\tau = \lambda$ with $\phi = A \circ u \circ \phi_1$. This completes the proof of Theorem 2.1.1. □

We note the following corollary:

Corollary 2.1.2 *Let M be a $2n+1$-dimensional manifold equipped with a contact form λ. Given a point $m_0 \in M$ there exists a chart ϕ defined on a neighborhood V of m_0 with image on some neighborhood of 0 in $\mathbf{R}^{2n+1}$ such that*

$$\phi^*\left(\sum_{i=1}^{n} x_i dy_i + dz\right) = \lambda. \qquad \square$$

Let us make a remark about the proof of Darboux's theorem. Everything until the construction of τ'' should be considered as preliminary work. We then have two contact forms, τ'' and λ; the aim is to find a diffeomorphism ϕ so that $\phi^*\tau'' = \lambda$. We would like to realize ϕ as the time–1–map of a time dependent vector field Y_t, i.e. we are looking for a differential equation whose time–1–map has the desired property. This technique was introduced by J. Moser in [90] and is often referred to as 'Moser technique'. It is very widely used in symplectic and contact geometry.

In the following example (provided by Herman Gluck) Moser's technique is used to find an explicit diffeomorphism of $\mathbf{R}^3$, transforming the standard contact form $\lambda = dz + xdy$ into the contact form $\lambda' = dz + xdy - ydx$. The contact structure $\ker \lambda'$ looks similar to the overtwisted one in Fig. 1.2, but the contact planes do not make a full turn, they actually never become vertical.

The required diffeomorphism is

$$h : \mathbf{R}^3 \longrightarrow \mathbf{R}^3$$

$$h(x, y, z) := (x, \frac{y}{2}, z + \frac{xy}{2})$$

since $h^*\lambda' = \lambda$. Here is how to find this diffeomorphism: We note first that both λ and λ' have the same Reeb vector field $\frac{\partial}{\partial z}$. Trying to look for a diffeomorphism of the form

$$(x, y, z) \longmapsto (X(x, y, z), Y(x, y, z), z)$$

does not work because $\lambda|_{\mathbf{R}^2}$ vanishes on the whole y–axis while $\lambda'|_{\mathbf{R}^2}$ vanishes only at the origin. Instead, we will use Moser's technique explicitly in order to find the diffeomorphism h. Define the following family of one–forms

$$\lambda_t := dz + x\, dy - ty\, dx\ ,\ 0 \leq t \leq 1.$$

These forms are all contact forms since $\lambda_t \wedge d\lambda_t = (1 + t)dx \wedge dy \wedge dz$ and they all have the same Reeb vector field $\frac{\partial}{\partial z}$. Following the proof of Darboux's theorem we define a time–dependent vector field W_t by requiring

$$i_{W_t} d\lambda_t = -\frac{d}{dt}\lambda_t = y\, dx$$

and

$$i_{W_t}\lambda_t = 0.$$

The one parameter family of diffeomorphisms ϕ_t generated by W_t satisfies

$$\begin{aligned}\frac{d}{dt}\phi_t^*\lambda_t &= \phi_t^*[\frac{d}{dt}\lambda_t + d(i_{W_t}\lambda_t) + i_{W_t}d\lambda_t] \\ &= \phi_t^*[\frac{d}{dt}\lambda_t + i_{W_t}d\lambda_t] \\ &= 0,\end{aligned}$$

hence $\phi_t^*\lambda_t = \phi_0^*\lambda_0 = \lambda_0$. Defining $h := \phi_1$, we get as desired $h^*\lambda' = \lambda$ (if ϕ_t exists until $t = 1$).

Let us write

$$W_t = u(x, y, z, t)\frac{\partial}{\partial x} + v(x, y, z, t)\frac{\partial}{\partial y} + w(x, y, z, t)\frac{\partial}{\partial z},$$

so that

$$i_{W_t}d\lambda_t = (1 + t)u\, dy - (1 + t)v\, dx = y\, dx,$$

which implies $u = 0$ and $v = -\frac{y}{1+t}$. Next,

$$i_{W_t}\lambda_t = -tyu + xv + w = 0,$$

hence $w = -xv = \frac{xy}{1+t}$. Thus

$$W_t = -\frac{y}{1+t}\frac{\partial}{\partial y} + \frac{xy}{1+t}\frac{\partial}{\partial z}.$$

Find now $\phi_t : \mathbf{R}^3 \to \mathbf{R}^3$ so that

$$\frac{d}{dt}\phi_t(x, y, z) = W_t(\phi_t(x, y, z)).$$

If we write $\phi_t(x, y, z) = (X(x, y, z, t), Y(x, y, z, t), Z(x, y, z, t))$ then we have to solve the following system of differential equations:

$$\begin{aligned}
\frac{\partial X}{\partial t} &= 0 \\
\frac{\partial Y}{\partial t} &= -\frac{Y}{1+t} \\
\frac{\partial Z}{\partial t} &= \frac{XY}{1+t},
\end{aligned}$$

subject to the initial conditions

$$X(x, y, z, 0) = x \ , \ Y(x, y, z, 0) = y \ , \ Z(x, y, z, 0) = z.$$

We obtain

$$X(x, y, z, t) = x \ , \ Y(x, y, z, t) = \frac{y}{1+t}.$$

Then

$$\frac{\partial Z}{\partial t} = \frac{XY}{1+t} = \frac{xy}{(1+t)^2}$$

yields

$$Z(x, y, z, t) = (z + xy) - \frac{xy}{1+t} = z + \frac{xyt}{1+t}.$$

Putting $t = 1$ we get the diffeomorphism h as claimed.

2.2 Lagrange and Legendre Neighborhood Theorems

In this section we will derive a normal form for a symplectic structure near a Lagrangian submanifold.

Lemma 2.2.1 *Let M be a $2n$-dimensional manifold and let $Q \subset M$ be a compact submanifold. Moreover let ω_0, ω_1 be closed non-degenerate 2-forms on M with $\omega_0(p) = \omega_1(p)$ for all $p \in Q$. Then there are neighborhoods U_0, U_1 of Q and a diffeomorphism*

$$\psi : U_0 \longrightarrow U_1$$

so that $\psi|_Q = id$ and $\psi^*\omega_1 = \omega_0$.

Proof

Step 1 First find a 1-form σ on U_0 with

$$\sigma|_{TM|_Q} \equiv 0 \text{ and } d\sigma = \omega_1 - \omega_0.$$

Choose any Riemann metric on M and let $(TQ)^\perp$ be the normal bundle with respect to this metric. We will write

$$\exp : (TQ)^\perp \longrightarrow M$$

for the restriction of the exponential map (with respect to the Levi-Civita connection). Now choose $\varepsilon > 0$ so small that the restriction of exp to

$$U_\varepsilon := \{v \in (TQ)^\perp \mid |v| < \varepsilon\}$$

is a diffeomorphism onto its image U_0. Then, for $0 < t \leq 1$, the maps defined by

$$\phi_t : U_0 \longrightarrow U_0$$

$$\phi_t(\exp(\eta)) = \exp(t\eta)$$

are diffeomorphisms which satisfy

$$\phi_1 = \mathrm{id}_{U_0},$$

$$\phi_0(U_0) \subset Q,$$

$$\phi_t|_Q = \mathrm{id}_Q.$$

Hence

$$\phi_0^*\tau = 0 \text{ and } \phi_1^*\tau = \tau,$$

where $\tau = \omega_1 - \omega_0$. Let us define the vector field

$$X_t(x) := \frac{d}{dt'}\phi_{t'}(\phi_t^{-1}(x))\Big|_{t'=t}$$

on U_0. Then

$$\frac{d}{dt}\phi_t(x) = X_t(\phi_t(x))$$

and

$$\begin{aligned}\frac{d}{dt}\phi_t^*\tau &= \phi_t^*(L_{X_t}\tau)\\ &= d\,(\phi_t^* i_{X_t}\tau)\\ &= d\sigma_t\end{aligned}$$

where the 1-form σ_t on U_0 is defined by

$$\sigma_t(p)(v) := \tau(\phi_t(p))\left(\frac{d}{dt}\phi_t(p), d\phi_t(p)v\right).$$

We note that $\sigma_t(p) = 0$ for $p \in Q$ since then $\frac{d}{dt}\phi_t(p) = 0$. Moreover

$$\begin{aligned}\tau &= \phi_1^*\tau - \phi_0^*\tau\\ &= \int_0^1 \frac{d}{dt}\phi_t^*\tau\, dt\\ &= \int_0^1 d\sigma_t \cdot dt.\end{aligned}$$

Then

$$\sigma := \int_0^1 \sigma_t \cdot dt$$

is the desired 1–form.

Step 2 ('Moser's argument') Consider the following family of 2-forms on U_0:

$$\omega_t := \omega_0 + t(\omega_1 - \omega_0) = \omega_0 + t\, d\sigma$$

where $0 \leq t \leq 1$. Shrinking U_0, if necessary, we may assume that $\omega_t|_{U_0}$ is nondegenerate for all $0 \leq t \leq 1$. Then we find a vector field X_t on U_0 so that

$$\sigma + i_{X_t}\omega_t = 0.$$

We note that $X_t|_Q \equiv 0$. If U_0 is chosen small enough we may assume that the time-t-map ψ_t of X_t exists for all $0 \leq t \leq 1$. We compute

$$\begin{aligned}\frac{d}{dt}\psi_t^*\omega_t &= \psi_t^*\left(\frac{d}{dt}\omega_t + di_{X_t}\omega_t + i_{X_t}d\omega_t\right)\\ &= \psi_t^*(d\sigma - d\sigma)\\ &= 0.\end{aligned}$$

Hence $\psi_1^*\omega_1 = \omega_0$ and we are done. □

Now we are ready to prove the Lagrange neighborhood theorem:

Theorem 2.2.2 *Let (M, ω) be a compact symplectic manifold and let $L \subset M$ be a Lagrangian submanifold. Then there are neighborhoods $U \subset T^*L$ of the zero section, $V \subset M$ of L and a diffeomorphism*

$$\phi : U \longrightarrow V$$

so that

$$\phi^*\omega = -d\theta$$

and

$$\phi|_L = id$$

where $-d\theta$ is the canonical symplectic structure on T^*L (see Sect. 1.2).

Proof Choose an almost complex structure J on TM which is compatible with ω, i.e.

$$J : TM \longrightarrow TM$$

$$J^2 = -\mathrm{Id}$$

and $g_J := \omega \circ (\mathrm{Id} \times J)$ is a Riemannian metric. Then $J(TL) = (TL)^{\perp}$ with respect to g_J and we continue each $v_p^* \in T_p^*L$, $p \in L$, onto T_pM by demanding $v_p^*|_{J_pT_pL} \equiv 0$.

Since ω is nondegenerate there is a uniquely determined tangent vector $\Phi(v_p^*) \in T_pM$ for each $v_p^* \in T_p^*L$ so that

$$\omega_p(\,.\,, \Phi(v_p^*)) = v_p^*.$$

Hence we have a uniquely determined embedding

$$\Phi : T^*L \longrightarrow TM|_L$$

with

- $\Phi(v_p^*) \in J_pT_pL$
- $\omega_p(v, \Phi(v_p^*)) = v_p^*(v)\ \forall\, v \in T_pM$

Now we choose a neighborhood $U \subset T^*L$ of the zero section so that $\exp|_{\Phi(U)}$ is a diffeomorphism onto its image V. Define

$$\phi := \exp \circ \Phi : T^*L \supset U \longrightarrow V \subset M.$$

We can write for $p \in L$ and $v \in T_{(p,0)}T^*L$

$$v = (v_0, v_1^*) \in T_pL \oplus T_p^*L = T_{(p,0)}T^*L.$$

Then

$$\begin{aligned} D\phi(p,0)(v) &= D\exp(p,0) \circ D\Phi(p,0)v \\ &= D\exp(p,0)\,(v_0 + \Phi(v_1^*)) \\ &= v_0 + \Phi(v_1^*) \end{aligned}$$

Hence

$$\begin{aligned} (\phi^*\omega)_{(p,0)}(v,w) &= \omega_p(D\phi(p,0)v, D\phi(p,0)w) \\ &= \omega_p(v_0 + \Phi(v_1^*), w_0 + \Phi(w_1^*)) \\ &= \omega_p(v_0, \Phi(w_1^*)) - \omega_p(w_0, \Phi(v_1^*)) \\ &= w_1^*(v_0) - v_1^*(w_0) \\ &= (-d\theta)_{(p,0)}(v,w), \end{aligned}$$

where θ is the canonical 1-form and $-d\theta$ is the canonical symplectic structure on T^*L. Summarizing we found a diffeomorphism

$$\phi : T^*L \supset U \longrightarrow V \subset M$$

between neighborhoods of the zero section and $L \subset M$ respectively so that $\phi^*\omega$ and $-d\theta$ coincide along the zero section in T^*L. By the previous lemma we find a diffeomorphism ψ between neighborhoods of the zero section in T^*L with $\psi^*(\phi^*\omega) = -d\theta$ and $\psi = \mathrm{Id}$ on the zero section. This completes the proof. □

We have a similar version of the Lagrange neighborhood theorem for Legendrian submanifolds of a contact manifold.

Definition 2.2.3 A submanifold L of a $(2n+1)$–dimensional contact manifold (M, λ) is called a *Legendrian submanifold* if L is n–dimensional and everywhere tangent to the contact hyperplanes, i.e. $\lambda|_{TL} \equiv 0$.

Theorem 2.2.4 *Let (M, λ) be a $2n+1$–dimensional contact manifold and let L be a Legendrian submanifold, i.e. $\dim L = n$ and $\lambda|_{TL} \equiv 0$. Then there is a neighborhood U of L in M and a diffeomorphism*

$$\Psi : \mathbf{R} \times T^*L \xrightarrow{\sim} U$$

so that $\Psi^*\lambda = \lambda_0 := dt + \theta$, where t is the coordinate in $\mathbf{R}$ and θ is the canonical 1–form on T^*L.

Proof The proof of the theorem consists of three steps:

Step 1 Find a diffeomorphism $\phi : \mathbf{R} \times T^*L \xrightarrow{\sim} U$ so that

$$(\phi^*\lambda)(0, 0_p) = (dt + \theta)(0, 0_p),$$

$\phi(0, 0_p) = p$ and $(\phi^* d\lambda)(0, 0_p) = d\theta(0, 0_p)$ for all $p \in L$ where $0_p \in T_p^*L$ is the zero section.

Step 2 Further modify the contact form in a neighborhood of L so that its Reeb vector field equals $\frac{\partial}{\partial t}$.

Step 3 Use the Moser technique in order to find the desired diffeomorphism Ψ.

Let us start with Step 1: We have a splitting $TM = \mathbf{R} \cdot X_\lambda \oplus \xi$, $\xi = \ker \lambda$, and $d\lambda$ is nondegenerate on ξ. Pick $J : \xi \to \xi$ so that $J^2 = -\mathrm{Id}$ and $g_J = d\lambda \circ (\mathrm{Id} \times J)$ is a bundle metric on ξ. Since $d\lambda(v, w) = v(\lambda(w)) - w(\lambda(v)) - \lambda([v, w])$ we see that for all $p \in L$, the vector space $J(p)T_pL$ is the orthogonal complement of T_pL in ξ with respect to g_J. We write $JTL = TL^\perp$.

We continue each $v_p^* \in T_p^*L$ onto all of ξ_p by requiring $v_p^*|_{J_pT_pL} \equiv 0$ and view v_p^* as an element in the dual space of ξ_p. Since $d\lambda(p)$ is nondegenerate on ξ_p, there is exactly one element $\Phi(v_p^*) \in \xi_p$ so that

$$d\lambda(p)(\Phi(v_p^*), \,.\,) = v_p^*.$$

We obtain an embedding

$$\Phi : T^*L \longrightarrow \xi|_L$$

with

$$\Phi(v_p^*) \in J(p)T_pL.$$

Define now the embedding

$$\tilde{\Phi} : \mathbf{R} \times T^*L \longrightarrow TM|_L$$

$$\tilde{\Phi}(a, v_p^*) := aX_\lambda(p) + \Phi(v_p^*)$$

and consider the following metric on M:

$$\langle v, w\rangle := \lambda(p)v \cdot \lambda(p)w + g_J(p)(\pi_\lambda v, \pi_\lambda w),$$

where $p \in M$ and $v, w \in T_pM$. The orthogonal complement $TL^\perp$ of TL with respect to this metric is $\mathbf{R} \cdot X_\lambda \oplus JTL$. We use $\tilde{\Phi}$ above to construct a tubular neighborhood of $L \subset M$ as follows:

$$\phi := \exp \circ \tilde{\Phi} : \mathbf{R} \times T^*L \supset I \times U \longrightarrow V \subset M,$$

where I is an open interval centered at 0 and U is an open neighborhood of the zero–section in T^*L. We see immediately that

$$\phi(0, 0_p) = p \quad \text{where } 0_p \in T_p^*L \text{ is the zero vector.}$$

Let us compute the derivative of ϕ at $(0, 0_p)$ for $p \in L$. We can decompose each $v \in T_{0_p}T^*L$ into its horizontal and vertical components

$$v = (v_0, v_1^*) \in T_pL \oplus T_p^*L \approx T_{0_p}T^*L.$$

Let γ be a curve with $\gamma(0) = 0_p$ and $\dot{\gamma}(0) = v = (v_0, v_1^*)$. Then with $\pi : T^*L \to L$ being the projection we obtain

$$\begin{aligned} D\tilde{\Phi}(0, 0_p) \cdot (a, v) &= \frac{d}{dt}\tilde{\Phi}(at, \gamma(t))\big|_{t=0} \\ &= \frac{d}{dt}\big(atX_\lambda(\pi\gamma(t)) + \Phi(\gamma(t))\big)\big|_{t=0} \\ &= aX_\lambda(p) + \frac{d}{dt}\Phi(\gamma(t))\big|_{t=0}. \end{aligned}$$

In order to calculate the last term we note that $\Phi : T^*L \to \xi|_L$ restricted to each fiber is linear. In a local trivialization of T^*L near p we may write $\gamma(t) = (\gamma_0(t), \gamma_1(t))$, where γ_0 is a curve in L with $\gamma_0(0) = p$, $\dot{\gamma}_0(0) = v_0$ and γ_1 is a

curve in T^*_pL with $\gamma_1(0) = 0$ and $\dot\gamma_1(0) = v_1^*$ (we assume that we have chosen γ_1 in such a way that $\gamma_1(t) = tv_1^*$). Then

$$\frac{d}{dt}\Phi(\gamma(t))\big|_{t=0} = v_0 + \Phi(v_1^*).$$

Hence

$$D\tilde\Phi(0, 0_p)\cdot(a, v) = aX_\lambda(p) + v_0 + \Phi(v_1^*)$$

and therefore

$$\begin{aligned} D\phi(0, 0_p)(a, v) &= D\exp(0_p)D\tilde\Phi(0, 0_p)(a, v)\\ &= aX_\lambda(p) + v_0 + \Phi(v_1^*). \end{aligned}$$

We compute

$$\begin{aligned} (\phi^*\lambda)(0, 0_p)\cdot(a, v) &= \lambda(p)(aX_\lambda(p) + v_0 + \Phi(v_1^*))\\ &= a\\ &= (dt + \theta)(0, 0_p)\cdot(a, v) \end{aligned}$$

and

$$\begin{aligned} (\phi^*d\lambda)(0, 0_p)\cdot\big((a, v), (b, w)\big) &= d\lambda(p)(aX_\lambda(p) + v_0 + \Phi(v_1^*), bX_\lambda(p) + w_0 + \Phi(w_1^*))\\ &= d\lambda(p)(v_0, \Phi(w_1^*)) + d\lambda(p)(\Phi(v_1^*), w_0)\\ &= v_1^*(w_0) - w_1^*(v_0)\\ &= d\theta(0_p)(v, w). \end{aligned}$$

This completes step 1 of the proof.

We are now in the following situation: Our contact manifold is $I\times U \subset \mathbf{R}\times T^*L$ with contact form $\tilde\lambda$ so that

$$\tilde\lambda(0, 0_p) = dt + \theta(0_p) = dt$$

and

$$d\tilde\lambda(0, 0_p) = d\theta(0_p),$$

in particular

$$X_{\tilde\lambda}(0, 0_p) = \frac{\partial}{\partial t}.$$

Step 2 Find a diffeomorphism ϕ defined near $\{0\} \times L \subset \mathbf{R} \times T^*L$ so that

$$\phi(0, 0_p) = (0, 0_p),$$

$$d\phi(0, 0_p) = \mathrm{Id}$$

for all $p \in L$ and

$$d\phi(a, v) \cdot \frac{\partial}{\partial t} = X_{\tilde{\lambda}}(a, v).$$

We consider the 'horizontal' ribbon

$$H := \{(0, v) \in \{0\} \times T^*L \,|\, |v| < \varepsilon\}$$

for sufficiently small $\varepsilon > 0$. Then the Reeb vector field $X_{\tilde{\lambda}}$ is transverse to H along L. Choosing ε small enough, we may assume that $X_{\tilde{\lambda}}$ is transverse to H everywhere. For $(t, v) \in \mathbf{R} \times T^*L$ with $(0, v) \in H$ and $|t|$ small, we may define a diffeomorphism ϕ near L by requiring:

$$\phi(t, v) = \psi_t(0, v),$$

where ψ_t denotes the flow of the Reeb vector field $X_{\tilde{\lambda}}$. The map ϕ has the desired properties, which completes step 2.

Step 3 Use Moser's method to complete the proof.

Define a family of 1–forms on M by

$$\lambda_t = t\tilde{\lambda} + (1 - t)\lambda_0 \ , \ t \in [0, 1].$$

These are all contact forms and they coincide on L. Moreover their Reeb–vector fields coincide near L by step 2 above. Define a time dependent vector field Y_t by

$$i_{Y_t}\lambda_t = 0 \ \text{ and } \ i_{Y_t}d\lambda_t = -\frac{d\lambda_t}{dt} = \lambda_0 - \tilde{\lambda}.$$

We note that $Y_t|_L \equiv 0$, hence the flow ϕ_t of Y_t exists until time 1 near L. Then

$$\begin{aligned} \frac{d}{dt}\phi_t^*\lambda_t &= \phi_t^*\left(\frac{d\lambda_t}{dt} + L_{Y_t}\lambda_t\right) \\ &= \phi_t^*\left(\frac{d\lambda_t}{dt} + i_{Y_t}d\lambda_t\right) \\ &= 0, \end{aligned}$$

hence $\phi_1^*\tilde{\lambda} = \lambda_0$ and also $\phi_1|_L = \mathrm{Id}_L$, which completes the proof of Theorem 2.2.4. □

2.3 Stability of Contact Structures

In this section we will prove Gray's theorem [59]:

Theorem 2.3.1 *Assume $(\lambda_t)_{t\in[0,1]}$ is a smooth family of contact forms on a closed manifold M and denote the associated contact structures by $(\xi_t)_{t\in[0,1]}$. Then there exists a smooth arc of diffeomorphisms $(\phi_t)_{t\in[0,1]}$ such that $\phi_0 = Id$ and*

$$T\phi_t(\xi_0) = \xi_t$$

for all $t \in [0, 1]$.

Darboux's theorem states that contact forms are locally diffeomorphic. On the other hand Gray's theorem says that contact structures on a closed manifold are the same globally up to diffeomorphism provided they can be connected by an arc of contact structures. Clearly if λ and τ are C^1-close contact forms then $t\lambda + (1-t)\tau$ are contact forms for every $t \in [0, 1]$. In particular, C^1-close contact forms have diffeomorphic contact structures. Gray's theorem does not allow any conclusion about the contact forms, i.e. we can not achieve that $\phi_t^*\lambda_t = \lambda_0$. in order to see this, we give an example. Define the 1-form λ on $\mathbf{R}^4$ by

$$\lambda = \frac{1}{2}\sum_{i=1}^{2}(x_i dy_i - y_i dx_i)$$

where the coordinates on $\mathbf{R}^4$ are given by x_1, x_2, y_1, y_2. Define

$$S^3 := \{(x_1, x_2, y_1, y_2) \in \mathbf{R}^4 \mid x_1^2 + x_2^2 + y_1^2 + y_2^2 = 1\}$$

and

$$E := \{(x_1, x_2, y_1, y_2) \in \mathbf{R}^4 \mid x_1^2 + r^2x_2^2 + y_1^2 + r^2y_2^2 = 1\}$$

where r^2 is an irrational number near 1. Then $\lambda\mid_E$ and $\lambda\mid_{S^3}$ are contact forms. The Reeb vector field of $\lambda\mid_{S^3}$ is given by

$$X_{S^3} = 2\left(-y_1\frac{\partial}{\partial x_1} - y_2\frac{\partial}{\partial x_2} + x_1\frac{\partial}{\partial y_1} + x_2\frac{\partial}{\partial y_2}\right).$$

A solution $\gamma = (a_1, b_1, a_2, b_2) \subset S^3$ of $\dot\gamma = X_{S^3}(\gamma)$ must satisfy

$$\begin{array}{ll}\dot a_1 = -2b_1 & \dot a_2 = -2b_2 \\ \dot b_1 = 2a_1 & \dot b_2 = 2a_2\end{array}.$$

So there are only π-periodic solutions. On the other hand the Reeb vector field of $\lambda\,|_E$ is given by

$$X_E \;=\; 2\left(-y_1\frac{\partial}{\partial x_1} - r^2 y_2\frac{\partial}{\partial x_2} + x_1\frac{\partial}{\partial y_1} + r^2 x_2\frac{\partial}{\partial y_2}\right).$$

A solution $\gamma = (a_1, b_1, a_2, b_2) \subset E$ of $\dot\gamma = X_E(\gamma)$ must satisfy

$$\begin{array}{ll} \dot a_1 = -2b_1 & \dot a_2 = -2r^2 b_2 \\ \dot b_1 = 2a_1 & \dot b_2 = 2r^2 a_2 \end{array}.$$

There are only two periodic solutions, namely one with initial condition $a_2(0) = b_2(0) = 0$ and period π and the other with initial condition $b_1(0) = a_1(0) = 0$ and period π/r^2. Now take the diffeomorphism

$$\psi : S^3 \longrightarrow E$$

$$z \longmapsto \frac{z}{(x_1^2 + y_1^2 + r^2(x_2^2 + y_2^2))^{-\frac{1}{2}}} \;=:\; \frac{z}{f(z)}.$$

Then $\hat\lambda := \psi^*\lambda\,|_E$ is again a contact form on S^3 and we calculate

$$\hat\lambda \;=\; \frac{1}{f^2}\cdot\lambda$$

i.e. λ and $\hat\lambda$ induce the same contact structure on S^3 but the dynamics of the corresponding Reeb vector fields are completely different.

Proof of Theorem 2.3.1 We have to find a smooth arc of diffeomorphisms $(\phi_t)_{t\in[0,1]}$ starting at Id and a smooth arc of functions $(f_t)_{t\in[0,1]}$ satisfying $f_t > 0$ such that

$$\phi_t^*\lambda_t \;=\; f_t\lambda_0.$$

We denote by $(X_t)_{t\in[0,1]}$ the arc of Reeb vector fields determined by the arc $(\lambda_t)_{t\in[0,1]}$. Define an arc of smooth functions by

$$r_t \;=\; i_{X_t}\frac{d}{dt}\lambda_t.$$

We see immediately that $\frac{d}{dt}\lambda_t \;-\; r_t\lambda_t$ vanishes on X_t. Hence we find a uniquely determined time-dependent vector field Y_t satisfying

$$\frac{d}{dt}\lambda_t \;=\; r_t\lambda_t \;-\; i_{Y_t}d\lambda_t$$

and

$$i_{Y_t}\lambda_t \;=\; 0.$$

Since $i_{Y_t}\lambda_t \;=\; 0$ and M is compact without boundary the one parameter family $(\phi_t)_{t\in[0,1]}$ generated by $(Y_t)_{t\in[0,1]}$ exists and satisfies

$$\begin{aligned}
&\frac{d}{dt}\phi_t^*\lambda_t \\
&= \phi_t^*\left(\frac{d}{dt}\lambda_t \;+\; L_{Y_t}\lambda_t\right) \\
&= \phi_t^*\left(\frac{d}{dt}\lambda_t \;+\; i_{Y_t}d\lambda_t\right) \\
&= \phi_t^*(r_t\lambda_t) \\
&= (r_t\circ\phi_t)\;(\phi_t^*\lambda_t).
\end{aligned}$$

This implies

$$\phi_t^*\lambda_t \;=\; \exp\left(\int_0^t r_s\circ\phi_s ds\right)\cdot\lambda_0$$

and the proof of theorem 2.3.1 is completed. □

2.4 Contact Hamiltonians

Assume (M,λ) is a manifold endowed with a contact form. We would like to characterize all vector fields on M whose flow preserves the contact structure $\xi \;=\; \ker\lambda$. Such vector fields are called *contact vector fields*. We first give a few equivalent conditions for a vector field being a contact vector field.

Lemma 2.4.1 *Let Y be a smooth vector field on a contact manifold (M,λ) with corresponding flow ϕ_t. Then the following are equivalent:*

(1) $L_Y\lambda = r\lambda$, *where* $r : M \to \mathbf{R}$ *is a smooth function.*
(2) $\phi_t^*\lambda = f_t\lambda$ *for all* t, *where* $f_t : M \to (0,\infty)$ *is smooth.*
(3) $(\phi_t)_*\xi = \xi$.

Proof Statement (3) trivially implies (2). We have

$$L_Y\lambda = \left.\frac{d}{dt}\phi_t^*\lambda\right|_{t=0} = \left.\frac{df_t}{dt}\right|_{t=0}\lambda,$$

hence (2) implies (1). On the other hand, statement (1) yields

$$(r \circ \phi_t)\phi_t^* \lambda = \phi_t^*(L_Y \lambda) = \frac{d}{dt}\phi_t^* \lambda$$

and

$$\phi_t^* \lambda = \left(e^{\int_0^t (r \circ \phi_\tau) d\tau}\right) \lambda,$$

which implies (2) and (3). □

Theorem 2.4.2 *For each smooth function $H : M \to \mathbf{R}$ there is a unique vector field $\hat{V}_H \in \ker \lambda$ so that*

$$V_H = H\, X_\lambda + \hat{V}_H$$

is a contact vector field.

The function H is called a *contact Hamiltonian*. We will sometimes call the vector field V_H the *contact Hamiltonian vector field associated to* H. The above theorem says also that for every contact vector field Y there is a unique smooth function H so that $Y = V_H$. This is true because Y can be decomposed in a unique way into a component along the Reeb vector field and another tangent to the contact structure.

Proof of Theorem 2.4.2 The condition that V_H above be a contact vector field is

$$\begin{aligned} L_{V_H} \lambda &= i_{V_H} d\lambda + d(i_{V_H} \lambda) \\ &= i_{\hat{V}_H} d\lambda + dH \\ &= r\, \lambda, \end{aligned}$$

that is

$$i_{\hat{V}_H} d\lambda + dH = r\lambda.$$

Evaluating both sides of this one–form equation on the Reeb vector field X_λ yields

$$r = dH(X_\lambda).$$

Hence the equation for the unknown vector field $\hat{V}_H$ tangent to $\ker \lambda$ becomes

$$i_{\hat{V}_H} d\lambda = dH(X_\lambda)\, \lambda - dH. \tag{2.1}$$

The one–form on the right hand side vanishes on X_λ, and hence, by the non degeneracy of $d\lambda$ on $\ker\lambda$, this equation uniquely determines the vector field $\hat{V}_H$ tangent to $\ker\lambda$. □

Similarly, if we have a one-parameter family of smooth functions $(H_t)_{t\in[0,1]}$ with $H_0 \equiv 0$, we obtain an arc of contact vector fields $(V_{H_t})_{t\in[0,1]}$ generating a smooth arc of contact diffeomorphisms starting at Id.

A trivial example is given by the following. Let $H = 1$. Then V_H is the Reeb vector field of λ.

Assume that $H : M \to \mathbf{R}$ is smooth and $F = H^{-1}(0)$ the zero-level surface for H. We compute for $x \in F$

$$(L_{V_H} H)(x) = dH(x)(\hat{V}_H(x)) = -d\lambda(x)(\hat{V}_H(x), \hat{V}_H(x)) = 0.$$

Hence the zero-level surface is invariant under the flow generated by the contact Hamiltonian vector field associated to H. So we have

$$V_H(x) = \hat{V}_H(x) \in T_x F \cap \xi_x \text{ for } x \in F.$$

We know that contact vector fields are in 1–1 correspondence with smooth functions H on M. We would like to answer the following question: Given a contact vector field V_H, when is it the Reeb vector field of some contact form inducing the same contact structure as λ? Here is the answer:

Proposition 2.4.3 *Let Z be a contact vector field for a contact structure ξ. Assume that Z is transverse to the contact hyperplanes. Then there exists a contact form λ inducing ξ such that $X_\lambda = Z$.*

Remark The above result illustrates clearly that the dynamics of the Reeb vector field can change completely if we pass from a contact form λ to $f\lambda$, where f is a nowhere vanishing function on M since every contact vector field transverse to ξ is a possible candidate for $X_{f\lambda}$.

Proof Pick any contact form τ inducing ξ. Since Z is transverse to ξ we have $H \in C^\infty(M, \mathbf{R}\backslash\{0\})$ where

$$H(m) := \tau(m)(Z(m)) \text{ for } m \in M.$$

Define

$$\lambda := \frac{1}{H} \cdot \tau.$$

Clearly λ is a contact form inducing ξ. Obviously

$$i_Z\lambda = \frac{1}{H} i_Z\tau = 1.$$

Moreover

$$\begin{aligned} dH &= di_Z\tau \\ &= L_Z\tau - i_Z d\tau \\ &= r\tau - i_Z d\tau \end{aligned}$$

and

$$i_Z dH \;=\; rH.$$

We compute by using the equalities above

$$\begin{aligned} i_Z d\lambda &= i_Z\left[-\frac{1}{H^2}dH\wedge\tau + \frac{1}{H}d\tau\right] \\ &= -\frac{1}{H^2}(i_Z dH)\tau + \frac{1}{H^2}(i_Z\tau)dH + \frac{1}{H}i_Z d\tau \\ &= -\frac{r}{H}\tau + \frac{dH}{H} - \frac{dH}{H} + \frac{r\tau}{H} \\ &= 0. \end{aligned}$$

Therefore

$$Z \;=\; X_\lambda \;\text{ with }\; \lambda \;=\; \frac{\tau}{i_Z\tau}. \qquad \square$$

In other words, Reeb vector fields are transverse contact vector fields. Assume next that Z is a contact vector field, which does not vanish anywhere. Let

$$N \;:=\; \{m \in M \;\mid\; Z(m) \in \xi_m\}$$

be the set of points of M, where Z is not transverse to the contact hyperplanes. In the following, we will investigate the structure of the set N.

Proposition 2.4.4 *If N is not empty then it is a submanifold of M of codimension one. If in addition M is three dimensional then each connected component of N is a torus.*

Proof Pick any contact form τ inducing ξ. Then define $H : M \longrightarrow \mathbf{R}$ by

$$H(m) \;=\; (i_Z\tau)(m),$$

so that

$$H^{-1}(0) \;=\; N.$$

We will show that $dH(m) \neq 0$ for every $m \in M$. We have

$$Z \;=\; H \cdot X_\tau \;+\; \hat{V}_H$$

with

$$i_{X_\tau}\tau = 1 \;\text{ and }\; i_{X_\tau} d\tau = 0$$

and

$$\begin{aligned} -dH + (i_{X_\tau} dH)\tau &= i_{\hat{V}_H} d\tau \\ i_{\hat{V}_H}\tau &= 0. \end{aligned}$$

We compute for $m \in M$ and $\eta \in \xi_m$

$$-dH(m)\eta \;=\; d\tau(m)(\hat{V}_H(m), \eta).$$

If this vanished for all $\eta \in \xi_m$ we would see that $\hat{V}_H(m) = 0$ since $i_{\hat{V}_H}\tau = 0$. But in this case

$$\begin{aligned} Z(m) &= H(m)X_\tau(m) + \hat{V}_H(m) \\ &= \hat{V}_H(m) \\ &= 0 \end{aligned}$$

contradicting the fact that Z does not vanish anywhere. Hence $dH(m)\eta \neq 0$ for a suitable $\eta \in \xi_m$ and the proof of the first part is finished.

Now let M be a compact oriented three dimensional manifold equipped with a contact structure ξ and a nowhere vanishing contact vector field Z such that

$$N \;:=\; \{m \in M \;\mid\; Z(m) \in \xi_m\} \;\neq\; \emptyset.$$

Clearly, the normal bundle $\nu := TM/TN$ is oriented by dH. Now we have a short exact sequence of bundle maps

$$0 \longrightarrow TN \hookrightarrow TM \longrightarrow \nu \longrightarrow 0.$$

Since M is oriented we see that N is oriented as well. Let us show that each component N' of N is diffeomorphic to the torus T^2. The Euler-characteristic of N' is given in terms of the genus g by

$$\chi(N') \;=\; 2 - 2g.$$

Since for $m \in M$

$$dH(m)Z(m) = r(m)H(m)$$
$$= 0,$$

we see that $Z|_N$ is a vector field on N. By assumption it does not vanish anywhere. Hence $\chi(N') = 0$ for every component and $g(N') = 1$, which means $N' \approx T^2$ by the classification of two dimensional compact oriented manifolds. □

Consider $\tilde{M} := M \backslash N$. Then $\tilde{M}$ decomposes into the pieces $H < 0$ and $H > 0$. On each such piece Z is a Reeb-vector field given by the one-form

$$\frac{\tau}{i_Z \tau} = \frac{\tau}{H}.$$

2.5 Martinet's Theorem

Let M^m be a $(2m + 1)$-dimensional closed, orientable manifold ($m \in \mathbf{N}$). The existence of a contact form on M^m allows a reduction of the structure group of the tangent bundle of M^m to $U(m) \times \mathrm{Id}$ because of the splitting $TM = \xi \oplus \mathbf{R} \cdot X_\lambda$. In dimensions greater than 5 this is a real obstruction; for existence theorems in special cases see [52, 53, 109] and [110].

J. Martinet gave the first affirmative answer for the case dim $M = 3$: Every closed orientable manifold of dimension three admits a contact form (see [83] and [81]). The proof of this theorem is the subject of this section. Martinet used a surgery description of the three-manifold M: Every closed orientable three-manifold M can be obtained by surgery on some link L in the three-sphere S^3 (see [80]). We will present here the proof of W.P. Thurston and H.E. Winkelnkemper [111] using Alexander's theorem [8]. We will state our main result now, which we will refer to as Martinet's theorem:

Theorem 2.5.1 *Every closed, orientable manifold M of dimension 3 admits a contact form λ.*

Our main tool is Alexander's theorem (Fig. 2.1):

Theorem 2.5.2 *Every closed, orientable manifold M of dimension 3 is diffeomorphic to*

$$W(h) \dot{\bigcup}_{Id} (\partial W \times D^2)$$

where D^2 is the unit disk in $\mathbf{R}^2$, W is an orientable 2-manifold with boundary and $h : W \to W$ is an orientation preserving diffeomorphism which restricts to the identity near ∂W. $W(h)$ denotes the manifold obtained from $W \times [0, 1]$ by identifying $(x, 0)$ with $(h(x), 1)$.

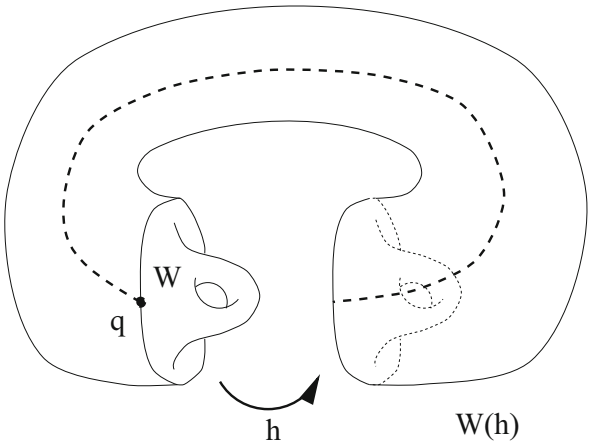


Fig. 2.1 Decomposition of M according to Alexander's theorem: The fat dashed circle is glued with $\{q\} \times \partial D^2 \subset \partial W \times D^2$

Proof of Theorem 2.5.1 We will first construct a contact form λ_1 on $W(h)$ that we control well near the boundary $\partial W(h) \approx \partial W \times S^1$. Then we construct a contact form λ_2 near $\partial W \times \{0\} \subset \partial W \times D^2$. Finally we will glue these forms together in a smooth way to obtain a contact form on $W(h) \dot{\bigcup}_{\mathrm{Id}} (\partial W \times D^2)$.

We can find a collar neighborhood C of ∂W so that $h(t, \theta) = (t, \theta)$ for all $(t, \theta) \in C$. Here we identify $(C, \partial W)$ with $([0, \varepsilon] \times (\dot{\bigcup}_n S^1), \{0\} \times (\dot{\bigcup}_n S^1))$ where we take an n–fold disjoint union of circles $S^1 \approx [0, 1]/\mathbf{Z}$ according to the number n of components of ∂W (Fig. 2.2). We claim that there is a volume form Ω on W that satisfies

- $\int_W \Omega = n$,
- $\Omega|_C = c_0\, dt \wedge d\theta$, where $c_0 > 0$ is a suitable constant.

Indeed, start with any volume form Ω' so that $\int_W \Omega' = n$. Then we have $\Omega'|_C = f'(t, \theta) dt \wedge d\theta$ with a positive smooth function f'. Pick now a new smooth positive function f which is constant if $t \leq \frac{1}{3}\varepsilon$ and agrees with f' if $t \geq \frac{2}{3}\varepsilon$ so that the resulting volume form Ω satisfies $\int_W \Omega = n$. Do one component of ∂W at a time. Rescaling the t–coordinate we may assume that $c_0 = 1$.

Let α_1 be any 1-form on W which equals $(1 + t)\, d\theta$ near ∂W. Then we obtain by Stokes' theorem:

$$\int_W (\Omega - d\alpha_1) = n - \int_{\partial W} \alpha_1 = n - \int_{\partial W} d\theta = 0.$$

The 2–form $\Omega - d\alpha_1$ on W is closed and vanishes near ∂W. By de Rham's theorem (see [113]) there exists a 1-form β on W with

$$d\beta = \Omega - d\alpha_1$$

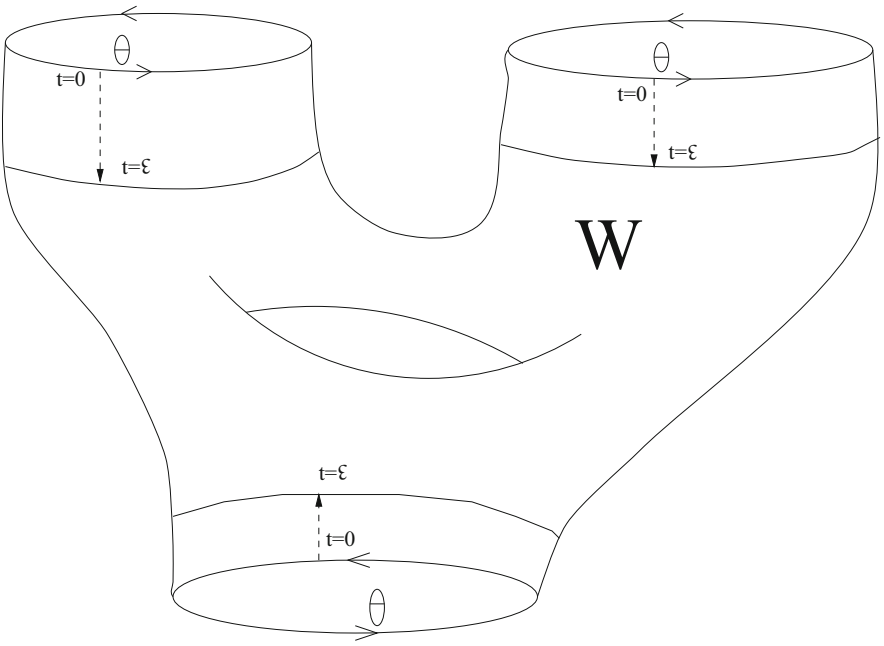


Fig. 2.2 The collar C

and $\beta \equiv 0$ near ∂W. Define now $\alpha_2 := \alpha_1 + \beta$. Then α_2 satisfies:

$$d\alpha_2 \text{ is a volume form on } W \text{ inducing the same orientation as } \Omega, \tag{2.2}$$

$$\alpha_2 = (1+t)\,d\theta \text{ near } \partial W. \tag{2.3}$$

The set of 1-forms on W satisfying (2.2) and (2.3) is therefore nonempty and also convex. We define the following 1-form on $W \times [0,1]$, where α is any 1–form on W satisfying (2.2) and (2.3).

$$\tilde{\alpha}(x,\tau) := \tau\alpha(x) + (1-\tau)(h^*\alpha)(x).$$

This 1-form descends to the quotient $W(h)$ and the restriction to each fiber of the fiber bundle $W(h) \xrightarrow{\pi} S^1$ satisfies condition (2.2). Moreover, since $h \equiv \mathrm{Id}$ near ∂W we have $\tilde{\alpha}(x,\tau) = (1+t)\,d\theta$ for all $(x,\tau) = ((t,\theta),\tau)$ near $\partial W(h) = \partial W \times S^1$.

Let $d\tau$ be a volume form on S^1. We claim that

$$\lambda_1 := \tilde{\alpha} + K\,\pi^* d\tau$$

is a contact form on $W(h)$ where $K > 0$ is some large constant (we will specify its size later). Pick $(x, \tau) \in W(h)$ and let $\{u, v, w\}$ be a basis of $T_{(x,\tau)}W(h)$ with $\pi_* u = \pi_* v = 0$. Then

$$\begin{aligned}
&(\lambda_1 \wedge d\lambda_1)(x, \tau)(u, v, w) \\
&= (\tilde{\alpha} \wedge d\tilde{\alpha})(x, \tau)(u, v, w) + K\,[d\tau(\pi_* w)\, d\tilde{\alpha}(x, \tau)(u, v)] \\
&\neq 0
\end{aligned}$$

for sufficiently large $K > 0$. Now we have to continue the contact form λ_1 beyond $\partial W(h) \approx \partial W \times S^1$ onto $\partial W \times D^2$. At this point it is convenient to change coordinates. We identify $C \times S^1$ with $\partial W \times (D^2_{1+\varepsilon} \backslash D^2_1)$, where D^2_ρ is the 2-disk of radius ρ. Using polar coordinates (r, ϕ) on $D^2_{1+\varepsilon}$ with $0 \leq \phi \leq 1$ and $0 < r \leq 1+\varepsilon$, our old coordinates are related to the new ones by

$$\partial W \times (D^2_{1+\varepsilon} \backslash D^2_1) \ni (\theta, r, \phi) \approx (\theta, 1 + t, \tau) \in C \times S^1$$

and λ_1 is given by

$$\lambda_1 = r\, d\theta + K\, d\phi$$

on $\partial W \times (D^2_{1+\varepsilon} \backslash D^2_1)$, with ε sufficiently small so that (2.3) holds.

We have to extend this now smoothly to a contact form on $\partial W \times D^2_{1+\varepsilon}$. Taking Cartesian coordinates $(x, y) = (r \cos 2\pi\phi, r \sin 2\pi\phi)$ on $D^2_{1+\varepsilon}$ we see that the 1-form

$$\lambda_2 := -d\theta + r^2 d\phi = -d\theta + \frac{1}{2\pi}(x\, dy - y\, dx)$$

is a contact form near $r = 0$. We want to find smooth functions

$$f_1, f_2 : [0, 1 + \varepsilon] \to \mathbf{R}$$

so that the 1-form

$$\lambda := f_1(r) d\theta + f_2(r) d\phi \tag{2.4}$$

is a contact form on $\partial W \times D^2_{1+\varepsilon}$ where $F = (f_1, f_2)$ satisfies

$$F(r) = \begin{cases} (-1, r^2) \text{ near} & r = 0 \\ (r, K) \quad \text{for} & 1 \leq r \leq 1 + \varepsilon \end{cases} \tag{2.5}$$

The 1-form λ is a contact form if and only if

$$\begin{aligned}\lambda \wedge d\lambda &= (f_1 d\theta + f_2 d\phi) \wedge (f_1' dr \wedge d\theta + f_2' dr \wedge d\phi) \\ &= f_1 f_2' d\theta \wedge dr \wedge d\phi + f_2 f_1' d\phi \wedge dr \wedge d\theta \\ &= (f_1 f_2' - f_2 f_1') d\theta \wedge dr \wedge d\phi \\ &\neq 0. \end{aligned} \tag{2.6}$$

Consider the curve $v(r) = (f_1(r), f_2(r))$ in $\mathbf{R}^2$. Then the above condition means that v and the 'velocity vector' v' have to be linear independent for all r (Fig. 2.3). Let $a : \mathbf{R} \to [-1, 0]$ be a smooth function satisfying

- $a \equiv -1$ on $(-\infty, \frac{1}{4}]$
- $a \equiv 0$ on $[\frac{3}{4}, +\infty)$
- $a' > 0$ on $(\frac{1}{4}, \frac{3}{4})$

and take

$$f_1(r) := a(r) + (1 + a(r))\, r$$

$$f_2(r) := -a(r)\, r^2 + (1 + a(r))\, K.$$

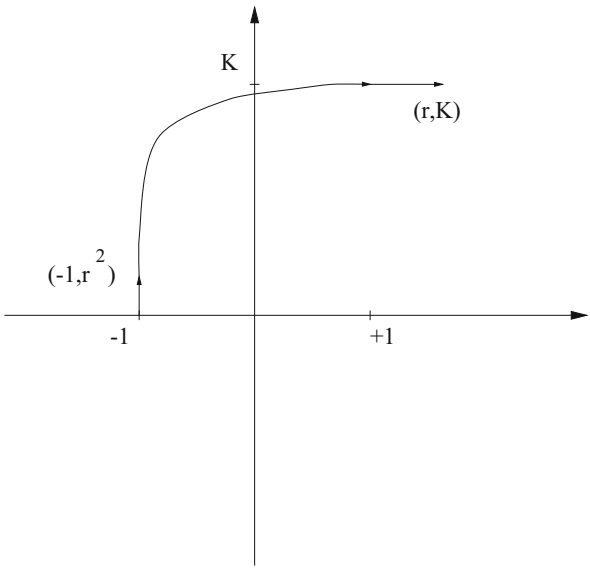


Fig. 2.3 Interpolating the two contact forms λ_1 and λ_2. The figure shows the image of the curve $v(r) = (f_1(r), f_2(r))$

We know already that (2.6) holds for $r \in (0, \frac{1}{4}] \cup [\frac{3}{4}, 1+\varepsilon)$. Then we compute for $r \in (\frac{1}{4}, \frac{3}{4})$:

$$\begin{aligned} f_1'(r) f_2(r) - f_1(r) f_2'(r) &= a'(r)\, r^3 + a(r)(1+a(r))\, r^2 + \\ &\quad + 2a^2(r)\, r + K\, a'(r)[r(1+a(r)) - a(r)] + \\ &\quad + K(1+a(r))^2 \\ &\geq -\frac{1}{4} r^2 + 2a^2(r)\, r + K(1+a(r))^2 \\ &\geq -\frac{9}{64} + 2a^2(r)\, r + K(1+a(r))^2. \end{aligned}$$

We choose $\delta > 0$ such that

$$2a^2(r)\, r > \frac{1}{4}$$

for all $r \in [\frac{1}{4}, \frac{1}{4} + \delta]$. Then we estimate

$$-\frac{9}{64} + 2a^2(r)\, r + K(1+a(r))^2 > 0$$

for $\frac{1}{4} \leq r \leq \frac{1}{4} + \delta$. On the other hand, if we choose the constant K so that

$$K > \frac{1}{\left(1 + a\left(\frac{1}{4} + \delta\right)\right)^2}$$

then we obtain for all $r \geq \frac{1}{4} + \delta$

$$\begin{aligned} -\frac{9}{64} + 2a^2(r)\, r + K(1+a(r))^2 &\geq -\frac{9}{64} + K(1+a(r))^2 \\ &\geq -\frac{9}{64} + K(1 + a(\frac{1}{4} + \delta))^2 \\ &> 0 \end{aligned}$$

because a is an increasing function. This completes the proof. □

Chapter 3
Surfaces in Three Dimensional Contact Manifolds



The aim of this chapter is to study surfaces in three dimensional contact manifolds. This will give us a so-called oriented singular foliation on the surface which is an important invariant of the contact structure. Roughly speaking, the construction is as follows. The contact planes and the tangent planes of a given surface are either transverse or they coincide at some point. In the latter case we will speak of a singular point. In the first case the contact planes define some direction field on the surface. An oriented singular foliation will be a class of vector fields on the surface which are pointing into this direction and which are zero exactly at the singular points. Surfaces in three dimensional contact manifolds will serve as boundary condition for pseudoholomorphic curves and it will be crucial to understand the oriented singular foliation on the surface. Again we assume that the contact structure on M is co-oriented, i.e. induced by a global 1-form λ.

3.1 Singular Points of a Surface

The first question we would like to study is quite local.

Definition 3.1.1 Let F be an orientable surface in (M, ξ). A point $x \in F$ is called *singular* if $T_x F = \xi_x$, otherwise we call it *regular*. If we fix an orientation on F and a contact form λ inducing ξ then we will call a singular point x *positive* if the given orientation on $T_x F$ and the orientation induced by $d\lambda$ coincide (otherwise x is called *negative*).

We denote the set of singular points on F by $\mathcal{S}_F$. It will be useful for the following to distinguish between different types of singularities.

Assume x_0 is a singular point. We write F near x_0 as $H^{-1}(0)$, where $dH(x_0) \neq 0$. By the discussion in Sect. 2.4 the contact Hamiltonian vector field $V_H|_F$ restricted

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to F is tangent to the surface F. Moreover $V_H(x_0) = \hat{V}_H(x_0) = 0$. We have the following simple fact:

Lemma 3.1.2 *A point $x \in F$ is singular if and only if $V_H(x) = 0$.*

Proof Assume first that x is a singular point, i.e. $T_x F = \xi_x$. Then we have for every $h \in T_x F = \ker dH(x)$

$$d\lambda(x)(V_H(x), h) = (i_{X_\lambda} dH)_x \, \lambda(x)h - dH(x)h = 0$$

and non degeneracy of $d\lambda(x)$ on $\xi_x = T_x F$ implies $V_H(x) = 0$. Conversely, if $V_H(x) = 0$ for some $x \in F$ then

$$dH(x) = (i_{X_\lambda} dH)_x \, \lambda(x)$$

which implies that ξ_x is contained in $T_x F$, but both spaces are two–dimensional, so they have to agree and x is a singular point. □

Definition 3.1.3 We call the singular point x_0 *non degenerate* if the spectrum of the linearization $\hat{V}'_H(x_0) : T_{x_0}F \longrightarrow T_{x_0}F$ of $\hat{V}_H$ is disjoint from the imaginary axis.

The definition of "non-degenerate" is independent of the choice of H as long as $dH(x_0) \neq 0$. This can be seen as follows: We remember that the linearization is given by

$$\hat{V}'_H(x_0) \cdot v := \frac{d}{dt} T\phi_t^H(x_0) \cdot v \mid_{t=0}$$

where ϕ_t^H is the flow of $\hat{V}_H$ (in local coordinates (x_1, x_2) this is just the matrix $((\partial \hat{V}_H^i(x_0)/\partial x_j)_{ij})$). Take a smooth function K with $F = K^{-1}(0)$ and $dK(x_0) \neq 0$. Then $dK(x) \neq 0$ for all $x \in F$ near x_0. Hence $\ker dH(x) = \ker dK(x)$ and therefore $dK(x) = \rho(x)\, dH(x)$ with some smooth non vanishing function ρ defined on a neighborhood of x_0 in F. From the definitions of $\hat{V}_H$ and $\hat{V}_K$ we see that

$$\hat{V}_K = \rho \hat{V}_H.$$

If $t : \mathbf{R} \to \mathbf{R}$; $t \longmapsto t(s)$ satisfies the differential equation

$$\begin{aligned} \frac{dt}{ds} &= \rho(\phi_s^K(x)) \ ; \ x \in F \text{ near } x_0 \text{ fixed} \\ t(0) &= 0 \end{aligned}$$

then the flows ϕ_t^H, ϕ_s^K of $\hat{V}_H$ and $\hat{V}_K$ are related by

$$\phi_t^H(x) = \phi_s^K(x).$$

This implies

$$\begin{aligned}\hat{V}'_K(x_0)\,v &= \frac{d}{ds}T\phi_s^K(x_0)\,v\mid_{s=0}\\ &= \frac{d}{ds}T\phi_t^H(x_0)\,v\mid_{s=0}\\ &= \frac{dt}{ds}\mid_{s=0}\ \frac{d}{dt}T\phi_t^H(x_0)\,v\mid_{t=0}\\ &= \rho(x_0)\hat{V}'_H(x_0)\,v\end{aligned}$$

so the linearization changes by multiplication with a nonzero real constant.

Let λ_1 and λ_2 be the eigenvalues of $\hat{V}'_H(x_0)$. Clearly, the complex conjugates $\overline{\lambda_1}$ and $\overline{\lambda_2}$ are also eigenvalues. Assume, $x_0 \in F$ is a non-degenerate singular point. Consider the cases:

1. $\lambda_1\lambda_2 > 0$, this means either $\lambda_1, \lambda_2 \in \mathbf{R}\backslash\{0\}$ having the same sign or $\lambda_2 = \overline{\lambda}_1 \in \mathbf{C}$ with nonzero real and imaginary part.
2. $\lambda_1\lambda_2 < 0$, which implies $\lambda_1, \lambda_2 \in \mathbf{R}\backslash\{0\}$ having opposite signs.

Definition 3.1.4 In the first case the singularity is called *elliptic* and in the second case *hyperbolic*. Moreover we call an elliptic singularity *nicely elliptic* if the numbers λ_1 and λ_2 are real.

Again, these definitions do not depend on the choice of H. We remark that our definition of a nondegenerate singular point corresponds to the notion of a hyperbolic singular point in the theory of dynamical systems (as in [94] for example). This should not be confused with our definition of hyperbolic singular points. Figure 3.1 shows the flow lines of the linear vector field $\hat{V}'_H$: Away from a singularity the tangent spaces of F and the contact structure intersect transversally. Therefore, we obtain a one-dimensional foliation on $F' = F \setminus \mathcal{S}_F$, which we will discuss in the next section. It will turn out that V_H generates this foliation (whenever F is given by $H^{-1}(0)$). By the Grobman-Hartman theorem (see [94]) Fig. 3.1 reflects the shape of this one-dimensional foliation near singular points.

We begin by investigating the neighborhood of a singularity. The question being local we may, in view of Darboux's theorem, assume that $M = \mathbf{R}^3$, the contact form is $dz + xdy$, F is a graph over a neighborhood of the origin in the x-y-plane and $x_0 = 0 \in F$. If the origin 0 in $\mathbf{R}^3$ is singular then T_0F is the x-y-plane. We write F as

$$F = \{(x, y, f(x, y)) \mid (x, y) \in U\}$$

where

$$f(0) = 0 \text{ and } df(0) = 0$$

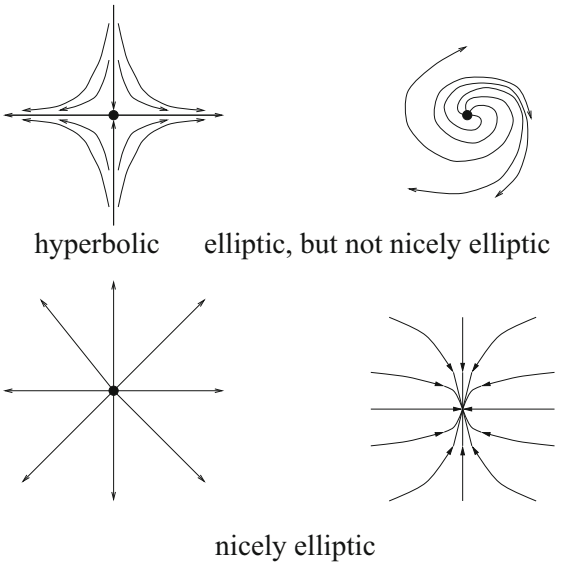


Fig. 3.1 The oriented singular foliation on F near non degenerate singular points

Let $f = q +$ "*higher order terms*", where q is a quadratic form. We have $F = H^{-1}(0)$, with

$$H(x, y, z) = z - f(x, y).$$

The contact Hamiltonian associated to H is given by

$$V_H(x, y, z) = H(x, y, z)(0, 0, 1) + \left(x + \frac{\partial f}{\partial y}(x, y), -\frac{\partial f}{\partial x}(x, y), x\frac{\partial f}{\partial x}(x, y)\right)$$

The vector field V_H with this choice of H represents the characteristic foliation if 0 is a positive singularity. If not then choose $-H$ instead of H. Let us calculate the linearization $\hat{V}'_H(p_0) : T_{p_0}F \to T_{p_0}F$ near a singular point $p_0 = (x_0, y_0, f(x_0, y_0))$. We must have $\hat{V}_H(p_0) = 0$, i.e. $f_x(x_0, y_0) = 0$ and $x_0 = -f_y(x_0, y_0)$. A basis of $T_{p_0}F$ is given by

$$\left\{\frac{\partial}{\partial x} + f_x(x_0, y_0)\frac{\partial}{\partial z}\,,\ \frac{\partial}{\partial y} + f_y(x_0, y_0)\frac{\partial}{\partial z}\right\} = \left\{\frac{\partial}{\partial x}\,,\ \frac{\partial}{\partial y} - x_0\frac{\partial}{\partial z}\right\}. \tag{3.1}$$

Now

$$\hat{V}'_H(p_0) = \begin{pmatrix} 1 + f_{xy}(x_0, y_0) & f_{yy}(x_0, y_0) & 0 \\ -f_{xx}(x_0, y_0) & -f_{xy}(x_0, y_0) & 0 \\ x_0 f_{xx}(x_0, y_0) & x_0 f_{xy}(x_0, y_0) & 0 \end{pmatrix}.$$

Hence

$$\frac{\partial}{\partial x} \longmapsto (1 + f_{xy}(x_0, y_0))\frac{\partial}{\partial x} - f_{xx}(x_0, y_0)\frac{\partial}{\partial y} + x_0 f_{xx}(x_0, y_0)\frac{\partial}{\partial z}$$

and

$$\frac{\partial}{\partial y} - x_0\frac{\partial}{\partial z} \longmapsto f_{yy}(x_0, y_0)\frac{\partial}{\partial x} - f_{xy}(x_0, y_0)\frac{\partial}{\partial y} + x_0 f_{xy}(x_0, y_0)\frac{\partial}{\partial z}.$$

So we can express $\hat{V}'_H(p_0) : T_{p_0}F \to T_{p_0}F$ with respect to the basis (3.1) of $T_{p_0}F$ by the matrix

$$\begin{pmatrix} 1 + f_{xy}(x_0, y_0) & f_{yy}(x_0, y_0) \\ -f_{xx}(x_0, y_0) & -f_{xy}(x_0, y_0) \end{pmatrix}.$$

Let us write q as

$$q(x, y) = \frac{1}{2}ax^2 + \frac{1}{2}by^2 + cxy.$$

Then we find

$$\hat{V}'_H(0) = \begin{pmatrix} 1 + c & b \\ -a & -c \end{pmatrix}.$$

Hence 0 is elliptic if and only if $-(1 + c)c + ab > 0$, hyperbolic if and only if $-(1 + c)c + ab < 0$ and nicely elliptic if and only if $0 < -(1 + c)c + ab \leq \frac{1}{4}$. Moreover 0 is degenerate if and only if $-(1 + c)c + ab = 0$ (then one eigenvalue is zero. The case of nonzero purely imaginary eigenvalues cannot occur).

Let us summarize our local investigation. If a surface F in $\mathbf{R}^3$ endowed with the standard contact structure is given by the graph of a function $f(x, y)$ so that the origin is a singular point then the type of the singularity $(0, 0, 0)$ depends only on the quadratic part of the function f. The purpose of this and the next section is the following. We would like to replace a given surface F in a three dimensional contact manifold by a 'better' one $\tilde{F}$ so that F and $\tilde{F}$ are isotopic via a smooth C^0–small isotopy and $|\mathrm{vol}_\lambda(F) - \mathrm{vol}_\lambda(\tilde{F})|$ is as small as we wish. What do we mean by a 'better' surface? Among other things, the surface $\tilde{F}$ should only have finitely many singular points and it should be given by the graph of a certain quadratic form near the (elliptic) singular points.

The following result is a first step into this direction. It shows that near a nondegenerate isolated singular point we can perturb the surface F to a new one $\tilde{F}$ having the same singular points, but being in normal form.

Theorem 3.1.5 *Let f be a smooth function defined on $\mathbf{R}^2$ with $f(0) = 0$ and $df(0) = 0$. Denote the graph of f by F. Assume 0 is a nondegenerate singularity*

on F for the standard contact structure on $\mathbf{R}^3$, and that there is no other singularity with (x, y)-coordinates being in some disk $B_{\varepsilon_0}(0)$. Then there exists for given ε with $\varepsilon_0 > \varepsilon > 0$ and $\delta > 0$ a smooth map $\tilde{f}$ such that the associated surface $\tilde{F}$ has the same singularities as F. Moreover $f = \tilde{f}$ outside the ε-ball around 0, and $| f - \tilde{f} | < \delta$ with

$$\tilde{f}(x, y) = \pm\frac{1}{2}xy$$

near to 0 with negative sign in the case of an elliptic singular point and positive sign in the case of a hyperbolic singular point. Moreover the new surface is obtained from the old one by an isotopy supported near the singular point.

The proof is based on modifying the surface F successively.

Lemma 3.1.6 *Consider the situation as described in Theorem 3.1.5. Then there exists a C^0-small isotopy supported near the singular point such that the new surface is again a graph. Moreover it has the same singular points, 0 is a hyperbolic or a nicely elliptic singularity and the surface near 0 is given by the graph of a quadratic form.*

Proof We write

$$f = q + h = \frac{1}{2}ax^2 + \frac{1}{2}by^2 + cxy + h$$

with h of order ≥ 3 in (x, y).

We will modify F near 0 such that it becomes the graph of q near 0. Take $\beta : [0, +\infty) \to [0, 1]$ vanishing on $[0, 1]$ and being identically 1 on $[2, +\infty)$. Moreover let $\beta'(s)$ be strictly positive on $(1, 2)$. For $\varepsilon > 0$ define

$$f_\varepsilon(x, y) := q(x, y) + \beta\left(\frac{x^2 + y^2}{\varepsilon^2}\right) h(x, y).$$

The graph of f_ε defines a new surface and we claim that for $\varepsilon > 0$ small enough, we do not create any additional singular points and $F_{f_\varepsilon} = F_q$ near 0.

To see this we first note that $f_\varepsilon = q$ for $x^2 + y^2 \leq \varepsilon^2$ and $f_\varepsilon = f$ for $x^2 + y^2 \geq 2\varepsilon^2$. If we choose

$$\varepsilon^2 \leq \frac{\varepsilon_0^2}{8}$$

then $f_\varepsilon \neq f$ only in the $\frac{\varepsilon_0}{2}$-ball around zero. Let us assume that for any sequence $\varepsilon_n \to 0$ with

$$\varepsilon_n^2 \leq \frac{\varepsilon_0^2}{8}$$

we create a new singular point $(x_n, y_n, f_{\varepsilon_n}(x_n, y_n))$ with $\varepsilon_n^2 \le x_n^2 + y_n^2 \le 2\varepsilon_n^2$ and $(x_n, y_n) \neq 0$ on the surface given by the graph of the function f_{ε_n}. Then by the definition of a singular point:

$$\begin{aligned}
0 &= x_n + \frac{\partial f_{\varepsilon_n}}{\partial y}(x_n, y_n) \\
&= (c+1)x_n + by_n + \frac{2y_n}{\varepsilon_n^2}\beta'\left(\frac{x_n^2+y_n^2}{\varepsilon_n^2}\right)h(x_n, y_n) \\
&\quad + \beta\left(\frac{x_n^2+y_n^2}{\varepsilon_n^2}\right)\frac{\partial h}{\partial y}(x_n, y_n), \\
0 &= -\frac{\partial f_{\varepsilon_n}}{\partial x}(x_n, y_n) \\
&= -cy_n - ax_n - \frac{2x_n}{\varepsilon_n^2}\beta'\left(\frac{x_n^2+y_n^2}{\varepsilon_n^2}\right)h(x_n, y_n) \\
&\quad - \beta\left(\frac{x_n^2+y_n^2}{\varepsilon_n^2}\right)\frac{\partial h}{\partial x}(x_n, y_n).
\end{aligned}$$

We omit the arguments and write shortly

$$0 \;=\; (c+1)x_n + by_n + \frac{2y_n}{\varepsilon_n^2}\beta' h + \beta h_y, \tag{3.2}$$

$$0 \;=\; -cy_n - ax_n - \frac{2x_n}{\varepsilon_n^2}\beta' h - \beta h_x. \tag{3.3}$$

or even

$$H \;=\; A\,\zeta_n$$

where $\zeta = (x_n, y_n)$,

$$A = \begin{pmatrix} c+1 & b \\ a & c \end{pmatrix} \text{ and } H = -\left(\frac{2y_n}{\varepsilon_n^2}\beta' h + \beta h_x, \frac{2x_n}{\varepsilon_n^2}\beta' h + \beta h_x\right).$$

We note that h is at least of order 3, so we have

$$\frac{2}{\varepsilon_n^2}\beta' h = O(\frac{1}{\varepsilon_n^2})O(\varepsilon_n^3) = O(\varepsilon_n)$$

because of $x_n^2 + y_n^2 \leq 2\varepsilon_n^2$. Then

$$\frac{H(x_n, y_n)}{\varepsilon_n} \longrightarrow 0$$

and we arrive at the contradiction

$$1 \leq \frac{1}{\varepsilon_n}\|\zeta\| = \frac{1}{\varepsilon_n}\|A^{-1}H\|.$$

Summarizing, we have shown that for sufficiently small $\varepsilon > 0$, the surface F_{f_ε} given by the graph of f_ε has the same singularities as F. Outside of a small neighborhood of the origin the surfaces F_{f_ε} and F coincide while F_{f_ε} is the graph of a quadratic form near the origin. We note that we can obtain the perturbed surface through a C^0-isotopy supported in a small neighborhood of the origin. □

In view of Lemma 3.1.6 we may therefore assume for the proof of Theorem 3.1.5 that F is given near 0 by the graph of a quadratic form

$$q = \frac{1}{2}ax^2 + \frac{1}{2}by^2 + cxy$$

with

$$0 < -c(1+c) + ab.$$

We define the following sets of matrices:

$$\mathcal{S}_\pm := \left\{ A = \begin{pmatrix} a & c \\ c & b \end{pmatrix} \middle| \; a, b, c \in \mathbf{R} \text{ so that } \pm\Delta(A) > 0 \right\},$$

where

$$\Delta(A) := -c(c+1) + ab.$$

Then graphs of quadratic forms q with matrices in $\mathcal{S}_+$ have an elliptic singular point at the origin, while matrices in $\mathcal{S}_-$ correspond to a hyperbolic singular point.

Lemma 3.1.7 *The sets $\mathcal{S}_\pm$ are path–connected.*

Proof We do the proof 'by hand'. Assume that

$$A = \begin{pmatrix} a & c \\ c & b \end{pmatrix} \in \mathcal{S}_+$$

We distinguish the cases

1. $c \in (-1, 0)$
2. $c \notin (-1, 0)$.

We begin with the first case, where $-c(c+1) > 0$. We define a path

$$\begin{pmatrix} (1-t)a & c \\ c & (1-t)b \end{pmatrix}, \ 0 \le t \le 1,$$

which lies entirely in $\mathcal{S}_+$ and connects A with $\begin{pmatrix} 0 & c \\ c & 0 \end{pmatrix}$ which can then be connected with $\begin{pmatrix} 0 & -1/2 \\ -1/2 & 0 \end{pmatrix}$ via

$$\begin{pmatrix} 0 & (1-t)c - \frac{t}{2} \\ (1-t)c - \frac{t}{2} & 0 \end{pmatrix}, \ 0 \le t \le 1. \tag{3.4}$$

The whole path lies in $\mathcal{S}_+$ since $c \in (-1, 0)$. In the second case we must have $ab > 0$, so we define the following path in $\mathcal{S}_+$:

$$\begin{pmatrix} a & (1-t)c - \frac{t}{2} \\ (1-t)c - \frac{t}{2} & b \end{pmatrix}.$$

which reduces case 2 to case 1. If $A \in \mathcal{S}_-$ we consider first the case where $ab \ge 0$ so that necessarily $-c(c+1) < 0$. We introduce

$$\begin{pmatrix} (1-t)a & c \\ c & (1-t)b \end{pmatrix}, \ 0 \le t \le 1,$$

which is a path in $\mathcal{S}_-$ and eliminates the entries a and b. If $c > 0$ then we scale it to $+1/2$, similarly as in (3.4). Otherwise we introduce another homotopy which creates entries a, b so that $ab < -\frac{1}{4}$ and we are able to homotop c to the value $+1/2$. After this is done, we can eliminate a and b. In the case $ab < 0$ we arrange first, by changing a or b, that $ab < -\frac{1}{4}$. Then we change c to $+1/2$ and finally we get rid of a and b. □

Proposition 3.1.8 *Assume that a surface $F \subset (\mathbf{R}^3, \lambda = dz + xdy)$ is given near 0 by the graph of a quadratic form as follows:*

$$F = \{(x, y, q(x, y)) \in \mathbf{R}^3 \mid (x, y) \in B_\rho(0)\},$$

where

$$q(x, y) = \frac{1}{2}\left\langle \begin{pmatrix} x \\ y \end{pmatrix}, A \cdot \begin{pmatrix} x \\ y \end{pmatrix} \right\rangle$$

with $A \in \mathcal{S}_{\pm}$. *Let* $\tilde{A} \in \mathcal{S}_{\pm}$ *be an arbitrary matrix in the same set as* A. *Then for every* $0 < \varepsilon < 2\varepsilon < \rho$ *there exists a surface* $\tilde{F}$ *given by the graph of a function* f *over* $B_\rho(0)$ *so that*

$$f \equiv q \text{ on } B_\rho(0) \backslash B_{2\varepsilon}(0)$$

and

$$f \equiv \tilde{q} := \frac{1}{2}\langle \,.\,,\ \tilde{A}\,.\,\rangle \text{ on } B_\varepsilon(0).$$

Moreover, the surface $\tilde{F}$ *has the same singularities as* F.

Proof Let

$$\gamma : [0, 1] \to \mathcal{S}_{\pm}$$

$$\gamma(t) = \begin{pmatrix} a(t) & c(t) \\ c(t) & b(t) \end{pmatrix}$$

be a smooth path connecting $\tilde{A}$ with A, i.e. $\gamma(0) = A$ and $\gamma(1) = \tilde{A}$. We have

$$\delta := \inf_{0 \le t \le 1} \Delta(\gamma(t)) > 0$$

and we introduce

$$|\gamma| := \max\{1,\ \sup_{0 \le t \le 1} (|a(t)| + |b(t)| + |c(t)|)\}.$$

We may assume without loss of generality that $\left|\frac{d\gamma}{dt}\right|$ is as small as we wish. Otherwise, we partition the path γ into finitely many small pieces and apply the result to each of these pieces separately.

We start by picking a smooth function $\beta : [0, \infty) \to [0, 1]$ with the following properties: $\beta \equiv 1$ on $[0, 1]$, $\beta \equiv 0$ on $[2, \infty)$ and $|\beta'| \le 2$ everywhere. We define now with $\beta(x, y) := \beta\left(\frac{x^2+y^2}{\varepsilon^2}\right)$

$$f(x, y) := \frac{1}{2}a(\beta(x, y))\, x^2 + \frac{1}{2}b(\beta(x, y))\, y^2 + c(\beta(x, y))\, xy$$

and

$$\tilde{F} := \{(x, y, f(x, y)) \in \mathbf{R}^3 \mid (x, y) \in B_\rho(0)\}.$$

We have $f \equiv q$ on $B_\varepsilon(0)$ and $f \equiv \tilde{q}$ on $B_\rho(0)\backslash B_{2\varepsilon}(0)$. The point $(0, 0, 0)$ is a singular point on $\tilde{F}$, obviously of the same type as on the original surface F. There are no singular points $(x, y, f(x, y))$ on $\tilde{F}$ if $(x, y) \in B_\rho(0)\backslash B_{\sqrt{2}\varepsilon}(0)$ and if $(x, y) \in B_\varepsilon(0)\backslash\{0\}$ in view of $A, \tilde{A} \in \mathcal{S}_\pm$. We have to verify also that there are no singular points either if $(x, y) \in B_{\sqrt{2}\varepsilon}(0)\backslash B_\varepsilon(0)$.

Assuming in the contrary that there is such a singular point, we obtain

$$0 = \frac{\partial f}{\partial x}(x, y)$$

and

$$0 = x + \frac{\partial f}{\partial y}(x, y)$$

in that point, which is

$$\begin{aligned} 0 \;=\; & a(\beta(x, y))\, x + c(\beta(x, y))\, y + \frac{x^3}{\varepsilon^2}\beta'\left(\frac{x^2 + y^2}{\varepsilon^2}\right) a'(\beta(x, y)) + \\ & + \frac{xy^2}{\varepsilon^2}\beta'\left(\frac{x^2 + y^2}{\varepsilon^2}\right) b'(\beta(x, y)) + \frac{2x^2 y}{\varepsilon^2}\beta'\left(\frac{x^2 + y^2}{\varepsilon^2}\right) c'(\beta(x, y)) \\ =: \; & ax + cy + \frac{x^3}{\varepsilon^2}\beta' a' + \frac{xy^2}{\varepsilon^2}\beta' b' + \frac{2x^2 y}{\varepsilon^2}\beta' c' \end{aligned}$$

and

$$0 = x + by + cx + \frac{x^2 y}{\varepsilon^2}\beta' a' + \frac{y^3}{\varepsilon^2}\beta' b' + \frac{2xy^2}{\varepsilon^2}\beta' c',$$

omitting the arguments of the functions involved in order to shorten the notation. We write these two equations as

$$\begin{aligned} 0 &= \begin{pmatrix} a + \frac{x^2}{\varepsilon^2}\beta' a' + \frac{y^2}{\varepsilon^2}\beta' b' & c + \frac{2x^2}{\varepsilon^2}\beta' c' \\ c + 1 + \frac{2y^2}{\varepsilon^2}\beta' c' & b + \frac{x^2}{\varepsilon^2}\beta' a' + \frac{y^2}{\varepsilon^2}\beta' b' \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \\ &= \begin{pmatrix} a & c \\ c + 1 & b \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + R \\ &= A \begin{pmatrix} x \\ y \end{pmatrix} + R. \end{aligned}$$

Because of $|\beta'| \leq 2$, $|x|, |y| \leq \sqrt{2}\,\varepsilon$ we have

$$|R| \leq 16\sqrt{2}\,\varepsilon \left|\frac{d\gamma}{dt}\right|.$$

Then

$$\varepsilon \leq \sqrt{x^2 + y^2} \leq 16\sqrt{2}\,\varepsilon \left|\frac{d\gamma}{dt}\right| |A^{-1}|$$

which is impossible had we chosen $|\gamma'| < (16\sqrt{2}\,|A^{-1}|)^{-1}$. □

Combining our local result in $\mathbf{R}^3$ with Darboux's theorem we obtain the following corollary:

Corollary 3.1.9 *Let F be any compact surface in a three-dimensional manifold equipped with a contact form λ and assume F has only non degenerate singularities.*

Then there exists a surface $\tilde{F}$ isotopic to F via a smooth C^0–small isotopy having the same singularities as F with the following property:

For each singular point on $\tilde{F}$ there exists a local Darboux-chart mapping the singular point to $0 \in (\mathbf{R}^3, dz + x\,dy)$, so that in local coordinates the surface $\tilde{F}$ is given by the graph of the following quadratic form:

$$(x, y) \longmapsto \pm\frac{1}{2}xy,$$

where the negative sign corresponds to an elliptic singular point and the positive sign corresponds to a hyperbolic singular point. In particular, every elliptic singular point becomes nicely elliptic. □

We remark that we have actually proved more: In fact, any matrix in $\mathcal{S}_-$ and $\mathcal{S}_+$ can serve as a local model near a hyperbolic and elliptic singular point respectively.

If we assume that F is a closed orientable surface in (M, λ) let us recall the definition of the number $\mathrm{vol}_\lambda(F)$. Pick a volume form σ on F and write $d\lambda$ as $d\lambda = f\sigma$ for some smooth function f on F. Define

$$\mathrm{vol}_\lambda(F) = \frac{1}{2}\int_F |f|\,\sigma.$$

Observe that this definition does not depend on the choices involved.

Also observe that the modifications which we carried out do not change the value of $\mathrm{vol}_\lambda(F)$ since near the singularity the surface may be viewed as a graph over the $x - y-$plane and the contribution to the λ-volume is precisely the area with respect to $dx \wedge dy$ of the projection of the surface onto the $x - y-$plane. This projection does not change under our local modification. Hence we have

Corollary 3.1.10 *Assume that F is a compact surface in (M, λ) and let the singularities be nondegenerate. If the boundary is nonempty we assume that $T\partial F \subset \xi|_F$ and that there is no singularity on ∂F.*

Then there exists a perturbation $\tilde{F}$ as described in Corollary 3.1.9 such that in addition the λ-volumes coincide. □

Our perturbation result has an obvious drawback: It only works under the assumption that all singular points on the given surface F are nondegenerate. This implies in particular that there are only finitely many of them. In order to get a satisfying perturbation result, we have to show that a given surface F can be slightly perturbed so that all singular points become nondegenerate. This is the aim of the next section.

3.2 The Characteristic Foliation on a Compact Oriented Surface

Let F be a compact oriented surface. An *oriented singular foliation* $[Z]$ on F consists of a set

$$[Z] = \{ fZ \mid f \in C^\infty(F, (0, \infty)) \}$$

where Z is a vector field on F. In other words an oriented singular foliation consists of an equivalence class of vector fields, where two vector fields are identified if one can be obtained from the other by multiplication with a smooth, strictly positive function. Note that the orbits obtained on F do not depend (modulo parametrisation) on the choice of $Y \in [Z]$. We denote the set of oriented singular foliations on F by $\mathcal{OSF}(F)$.

Assume that $F \subset M$, where M is an orientable three-manifold. We denote by $\mathrm{Distr}(M)$ the space of smooth co-oriented plane-field distributions σ on M, i.e. the line bundle TM/σ is oriented. Then this bundle must be trivial, hence there exist a non vanishing vector field X_σ transverse to σ (see [62]). There exists a natural map

$$\Delta : \mathrm{Distr}(M) \to \mathcal{OSF}(F) \ : \sigma \to [Z]$$

defined as follows: Pick a volume form Ω for F inducing the orientation. Let $j : F \to M$ be the inclusion and define a vector field Z on F by

$$i_Z \Omega = j^* \tau$$

where τ is a nowhere vanishing one-form with $\ker \tau(x) = \sigma(x)$ for all $x \in M$ so that τ is compatible with the co-orientation, i.e. $i_{X_\sigma} \tau > 0$. We see easily that $[Z]$ is well defined, i.e. it does not depend on the choices of Ω and τ. If we denote the set

of nowhere vanishing 1–forms on M by $\mathcal{E}^1_0(M)$ and if we fix an orientation form Ω on F, then we obtain a map

$$\Delta_\Omega : \mathcal{E}^1_0(M) \longrightarrow \mathrm{Vec}(F)\ ,\ Z = \Delta_\Omega \tau$$

defined by $i_Z\Omega = j^*\tau$.

If $[Z]$ is induced by a contact structure ξ on M, i.e. $[Z] = \Delta(\xi)$, we will call it the *characteristic foliation* on F induced by ξ.

We will topologize the spaces $\mathrm{Distr}(M)$ and $\mathrm{Vec}(F)$ by using the C^∞–Whitney topology (or 'fine' topology) which can be described as follows:

Let $f_0 : M_1 \to M_2$ be a C^r–map between C^r–differentiable manifolds M_1 and M_2. Let $\{K_i\}_{i\in\mathbf{N}}$ be a locally finite cover of M_1 by compact sets so that there are coordinate charts (U_i, ϕ_i) for M_1 and (V_i, ψ_i) for M_2 satisfying $K_i \subset U_i$ and $f_0(K_i) \subset V_i$. Moreover let $\delta = \{\delta_i\}_{i\in\mathbf{N}}$ be a sequence of positive numbers. Then define

$$\begin{aligned}\mathcal{U}^r(f_0, \delta) := \{f \in C^r(M_1, M_2) \mid f(K_i) \subset V_i\ ;\\ \|\psi_i \circ f|_{K_i} \circ \phi_i^{-1} - \psi_i \circ f_0|_{K_i} \circ \phi_i^{-1}\|_{C^r(\phi_i(K_i))} < \delta_i\}.\end{aligned}$$

The family

$$\{\mathcal{U}^r(f_0, \delta) \mid \delta = \{\delta_i\}_{i\in\mathbf{N}}\ ,\ \delta_i > 0\}$$

is a base for the neighborhood system of f_0 for a unique topology on $C^r(M_1, M_2)$ which is called the C^r–Whitney topology (or 'fine' topology). This topology does not depend on the choices of (U_i, ϕ_i), (V_i, ψ_i) and $\{K_i\}$ as long as $\{K_i\}$ satisfies the conditions specified above (see [93]). The C^∞–Whitney topology is defined as the union of all the C^r–Whitney topologies for $r \geq 0$.

We equip $\mathrm{Vec}(F) \subset C^\infty(F, TF)$ with the C^∞–Whitney topology. If $p : \mathrm{Vec}(F) \to \mathcal{OSF}(F)$ is the canonical projection then $\mathcal{OSF}(F)$ carries the quotient topology.

Lemma 3.2.1 *$p : Vec(F) \to \mathcal{OSF}(F)$ is open.*

Proof Pick $X_0 \in \mathrm{Vec}(F)$ and $\delta = \{\delta_i\}_{i\in\mathbf{N}}$ with $\delta_i > 0$. Take a cover of F by compact sets $\{K_i\}_{i\in\mathbf{N}}$ so that each K_i lies in a coordinate chart (U_i, ϕ_i) and $X_0(K_i)$ lies in a coordinate chart $(TF|_{U_i}, \psi_i)$ with $\psi_i(TF|_{U_i}) = \phi_i(U_i) \times \mathbf{R}^2$. It suffices to show that $p(\mathcal{U}^r(X_0, \delta))$ is open (where $r \geq 0$), i.e. the set

$$p^{-1}p(\mathcal{U}^r(X_0, \delta)) = \{fX \in \mathrm{Vec}(F) \mid f \in C^r(F, (0, \infty))\ ;\ X \in \mathcal{U}^r(X_0, \delta)\}$$

should be open in the C^r–Whitney topology. Pick $fX \in p^{-1}p(\mathcal{U}^r(X_0, \delta))$. Define the sequences

$$\tilde{\delta}_i := \|\psi_i \circ X|_{K_i} \circ \phi_i^{-1} - \psi_i \circ X_0|_{K_i} \circ \phi_i^{-1}\|_{C^r(\phi_i(K_i))} < \delta_i,$$

$$C_i := \left\| \frac{1}{f \circ \phi_i^{-1}} \right\|_{C^r(\phi_i(K_i))}$$

and choose $\hat{\delta}_i < C_i^{-1}(\delta_i - \tilde{\delta}_i)$ so that for all $Y \in \mathcal{U}^r(fX, \hat{\delta})$

$$\begin{aligned}
&\|\psi_i \circ X_0|_{K_i} \circ \phi_i^{-1} - \psi_i \circ (f^{-1}Y|_{K_i}) \circ \phi_i^{-1}\| \\
&\leq \|\psi_i \circ X|_{K_i} \circ \phi_i^{-1} - \psi_i \circ X_0|_{K_i} \circ \phi_i^{-1}\| + \\
&\quad + \|\psi_i \circ X|_{K_i} \circ \phi_i^{-1} - \psi_i \circ (f^{-1}Y|_{K_i}) \circ \phi_i^{-1}\| \\
&\leq \tilde{\delta}_i + C_i\hat{\delta}_i \\
&< \delta_i,
\end{aligned}$$

i.e. $f^{-1}Y \in \mathcal{U}^r(X_0, \delta)$ and therefore $\mathcal{U}^r(fX, \hat{\delta}) \subset p^{-1}p(\mathcal{U}^r(X_0, \delta))$ which proves the lemma. □

We will describe now the space $\mathrm{Distr}(M)$: We want to define a fiber bundle $G_2 \to M$ over M by

$$G_2 := \bigcup_{x \in M} G_2(T_xM)$$

where $G_2(T_xM)$ is the set of all two dimensional linear subspaces of T_xM.

The Grassmannian $G_2(\mathbf{R}^3)$ is a smooth manifold and we may construct charts as follows:

Take $E_0 \in G_2(\mathbf{R}^3)$ and pick $A_0 \in \mathrm{GL}(\mathbf{R}^3)$ such that $A_0E_0 = \mathbf{R}^2 \times \{0\}$. The following map Φ_{E_0} defines a chart for $G_2(\mathbf{R}^3)$ near E_0 with $\Phi_{E_0}(0, 0) = E_0$

$$\Phi_{E_0} : \mathbf{R}^2 \longrightarrow G_2(\mathbf{R}^3)$$

$$\Phi_{E_0}(\alpha, \beta) := A_0^{-1}(\{(x, y, \alpha x + \beta y) \in \mathbf{R}^3 \mid x, y \in \mathbf{R}\}).$$

If we pick a chart $\phi : M \supset U \widetilde{\to} \mathbf{R}^3$ so that $TM|_U$ is trivial then the trivialization $\hat{\Psi}_U : TM|_U \widetilde{\to} U \times \mathbf{R}^3$ defines a bijective map

$$\Psi_U : G_2|_U \widetilde{\longrightarrow} U \times G_2(\mathbf{R}^3)$$

so that $G_2 \to M$ becomes a smooth fiber bundle. We choose an atlas on G_2 consisting of coordinate charts (V, ψ) so that ψ is of the form $(\phi \times \Phi_{E_0}^{-1}) \circ \Psi_U$.

Then Distr(M) consists of those smooth sections σ in $G_2 \to M$ for which there exists a 1–form τ_σ on M such that

$$\ker \tau_\sigma(x) = \sigma(x)$$

for all $x \in M$. We equip Distr(M) with the topology induced from the C^∞–Whitney topology on $C^\infty(M, G_2)$. We want to prove the following:

Proposition 3.2.2 *The maps*

$$\Delta : Distr(M) \longrightarrow \mathcal{OSF}(F)$$

and

$$\Delta_\Omega : \mathcal{E}_0^1(M) \longrightarrow Vec(F)$$

are open.

Proof Let $\mathcal{E}_0^1(M)$ be the set of nowhere vanishing 1–forms on M equipped with the C^∞–Whitney topology. We call two such forms τ_1 and τ_2 equivalent if $\tau_2 = h \cdot \tau_1$ where $h : M \to (0, +\infty)$ is a smooth function. Let $[\mathcal{E}_0^1(M)]$ be the set of these equivalence classes equipped with the quotient topology. Note that we have the following commutative diagram

$$\begin{array}{ccccc} & & \mathcal{E}_0^1(M) & \xrightarrow{\Delta_\Omega} & \mathrm{Vec}(F) \\ & \pi & \downarrow & & \downarrow \quad p \\ \Delta : \mathrm{Distr}(M) & \xrightarrow{G} & [\mathcal{E}_0^1(M)] & \xrightarrow{\Delta_1} & \mathcal{OSF}(F), \end{array}$$

where $Z = \Delta_\Omega(\tau)$ is defined by $i_Z \Omega = j^*\tau$, π is the canonical projection and G assigns to $\sigma \in \mathrm{Distr}(M)$ the equivalence class $[\tau]$ of 1–forms τ on M whose kernel is given by σ so that $i_{X_\sigma}\tau > 0$. We will prove that G and Δ_0 are open. Then Δ_1 is also open because π is continuous and Δ_0, p are open (see Lemma 3.2.1).

Consider the map Δ_0 first: Let $\{K_i\}_{i \in \mathbf{N}}$ be a locally finite cover of M by compact sets so that the sets K_i are contained in coordinate charts (U_i, ϕ_i) of M. After maybe passing to a finer cover of M we may assume that $TM|_{U_i}$ is trivial and $\phi_i(U_i \cap F) = \mathbf{R}^2 \times \{0\} \approx \mathbf{R}^2$. If τ is a 1–form on M then

$$(\phi_i^{-1})^* \tau|_{U_i} = a_i dx + b_i dy + c_i dz$$

and

$$\left[(\phi_i|_{U_i\cap F})_*\Delta_0(\tau)|_{U_i\cap F}\right](q) = \frac{b_i(q,0)}{f_i(q)}\frac{\partial}{\partial x} - \frac{a_i(q,0)}{f_i(q)}\frac{\partial}{\partial y} =: \left(\frac{b_i(q,0)}{f_i(q)}, -\frac{a_i(q,0)}{f_i(q)}\right)$$

where we wrote $f_i dx \wedge dy$ for the orientation form $\Omega|_{U_i\cap F}$ in local coordinates. Let $\delta = \{\delta_i\}_{i\in\mathbf{N}}$ be a sequence of positive (small) numbers and consider $\mathcal{U}^r(\tau,\delta) \subset \mathcal{E}^1_0(M)$. We have to show that $\Delta_0(\mathcal{U}^r(\tau,\delta)) \subset \mathrm{Vec}(F)$ is open. Pick $\tilde{Z} \in \Delta_0(\mathcal{U}^r(\tau,\delta))$. Then $\tilde{Z}_{U_i\cap F}$ is represented in local coordinates by

$$\left(\frac{\tilde{b}_i(q,0)}{f_i(q)}, -\frac{\tilde{a}_i(q,0)}{f_i(q)}\right)$$

so that there is a 1–form $\tilde{\tau}$ on M represented locally by $\tilde{a}_i dx + \tilde{b}_i dy + \tilde{c}_i dz$ with

$$\|a_i - \tilde{a}_i\|_{C^r(\phi_i(K_i))}, \|b_i - \tilde{b}_i\|_{C^r(\phi_i(K_i))}, \|c_i - \tilde{c}_i\|_{C^r(\phi_i(K_i))} < \delta_i.$$

Take now $\hat{Z} \in \mathcal{U}^r(\tilde{Z},\hat{\delta}) \subset \mathrm{Vec}(F)$ and show that $\hat{Z} \in \Delta_0(\mathcal{U}^r(\tau,\delta))$ if the numbers $\hat{\delta}_i \in \hat{\delta}$ are chosen small enough:

In local coordinates $\hat{Z}$ is represented by couples $(\hat{b}_i, -\hat{a}_i)$ so that

$$\left\|\hat{a}_i - \frac{\tilde{a}_i}{f_i}\right\|_{C^r(\phi_i(K_i\cap F))}, \left\|\hat{b}_i - \frac{\tilde{b}_i}{f_i}\right\|_{C^r(\phi_i(K_i\cap F))} < \hat{\delta}_i.$$

Now $i_{\hat{Z}}\Omega$ is a 1–form on F represented locally by $f_i(\hat{a}_i dx + \hat{b}_i dy)$. We have

$$\|f_i\hat{a}_i - \tilde{a}_i\|_{C^r(\phi_i(K_i\cap F))} \le C\left\|\hat{a}_i - \frac{\tilde{a}_i}{f_i}\right\|_{C^r(\phi_i(K_i\cap F))} \le C\cdot\hat{\delta}_i$$

and

$$\|f_i\hat{b}_i - \tilde{b}_i\|_{C^r(\phi_i(K_i\cap F))} \le C\left\|\hat{b}_i - \frac{\tilde{b}_i}{f_i}\right\|_{C^r(\phi_i(K_i\cap F))} \le C\cdot\hat{\delta}_i$$

where $C > 0$ is a constant depending on r, Ω and the charts (U_i,ϕ_i). Now we can continue the 1–form $i_{\hat{Z}}\Omega$ to some non vanishing 1–form $\hat{\tau}$ on M, represented locally by $f_i(\hat{a}_i dx + \hat{b}_i dy + \hat{c}_i dz)$ (the coefficients defined on U_i), so that $\hat{\tau}$ coincides with

$\tilde{\tau}$ outside W and moreover

$$\|f_i\hat{a}_i - \tilde{a}_i\|_{C^r(\phi_i(K_i))}, \|f_i\hat{b}_i - \tilde{b}_i\|_{C^r(\phi_i(K_i))}, \|f_i\hat{c}_i - \tilde{c}_i\|_{C^r(\phi_i(K_i))} \leq C'\hat{\delta}_i$$

with some constant $C' > 0$ depending also on W. But this implies

$$\|f_i\hat{a}_i - a_i\|_{C^r(\phi_i(K_i))} \leq C'\hat{\delta}_i + \|\tilde{a}_i - a_i\|_{C^r(\phi_i(K_i))} < \delta_i,$$

$$\|f_i\hat{b}_i - b_i\|_{C^r(\phi_i(K_i))} \leq C'\hat{\delta}_i + \|\tilde{b}_i - b_i\|_{C^r(\phi_i(K_i))} < \delta_i,$$

and

$$\|f_i\hat{c}_i - c_i\|_{C^r(\phi_i(K_i))} \leq C'\hat{\delta}_i + \|\tilde{c}_i - c_i\|_{C^r(\phi_i(K_i))} < \delta_i,$$

provided $\hat{\delta}_i$ is chosen sufficiently small which proves that the map Δ_0 is open. Let us explain why it is possible to continue the 1–form $i_{\hat{Z}}\Omega$ as required above:

Since both F and M are orientable, the surface F has a trivial normal bundle because of the splitting short exact sequence of bundles (see [62])

$$0 \longrightarrow TF \hookrightarrow TM \longrightarrow TM/TF \longrightarrow 0.$$

Hence F has a tubular neighborhood which looks like $\mathbf{R} \times F$. Let τ_0 be a non vanishing 1–form on $\mathbf{R} \times F$ and τ_F be a 1–form on F. Then take a smooth function $\chi : \mathbf{R} \to [0, 1]$ with $\chi|_{[-1,1]} \equiv 1$ and $\chi(t) = 0$ for $|t| \geq 2$. If we continue τ_F onto $\mathbf{R} \times F$ by

$$\tau(t, x) \cdot (a, v) := \tau_0(t, x) \cdot (a, v) + \chi(t)[\tau_F(x)v - \tau_0(0, x) \cdot (0, v)],$$

where $(t, x) \in \mathbf{R} \times F$ and $(a, v) \in \mathbf{R} \times T_xF$. Then the 'size' of $\tau - \tau_0$ depends only on χ and $\tau_F - j^*\tau_0$ where $j : F \approx \{0\} \times F \hookrightarrow \mathbf{R} \times F$ is the inclusion and τ does not vanish if $|\tau_F(x) - (j^*\tau_0)(x)|$ is sufficiently small.

Let us show now that G is open. In order to examine the map G we take a locally finite cover of M by compact sets $\{K_i\}_{i\in\mathbf{N}}$ so that each K_i is contained in a coordinate chart (U_i, ϕ_i). Pick $\sigma \in \mathrm{Distr}(M)$. Then we assume in addition that $TM|_{U_i}$ is trivial and that each $\sigma(K_i)$ is contained in a coordinate chart (V_i, ψ_i) of G_2 so that $\sigma|_{U_i}$ is represented in local coordinates by

$$\mathbf{R}^3 \supset \phi_i(U_i) \longrightarrow \mathbf{R}^2$$

$$p \longmapsto (\alpha_i(p), \beta_i(p))$$

with suitable smooth functions α_i, β_i. If $\tau_\sigma \in \mathcal{E}^1_0(M)$ is given in such a way that $\ker \tau_\sigma(x) = \sigma(x)$ for $x \in M$ then $\tau_\sigma|_{U_i}$ is represented locally by

$$h_i(\alpha_i dx + \beta_i dy - dz)$$

where $h_i : \phi_i(U_i) \to \mathbf{R}\backslash\{0\}$ are suitable smooth functions depending on the choice of τ_σ. The co–orientation of σ fixes the sign of $h_i|_{\phi_i(U_i)}$. Let $\delta = \{\delta_i\}_{i\in\mathbf{N}}$ be a sequence of positive numbers. We want to show that the set $G(\mathcal{U}^r(\sigma, \delta)) \subset [\mathcal{E}^1_0(M)]$ is open. This is the case if and only if $\pi^{-1}G(\mathcal{U}^r(\sigma, \delta)) \subset \mathcal{E}^1_0(M)$ is open. Pick some $\tilde{\tau} \in \pi^{-1}G(\mathcal{U}^r(\sigma, \delta))$, i.e. there exist $\tilde{\sigma} \in \mathrm{Distr}(M)$ represented locally by $(\tilde{\alpha}_i, \tilde{\beta}_i)$ so that

$$\|\tilde{\alpha}_i - \alpha_i\|_{C^r(\phi_i(K_i))}\ ,\ \|\tilde{\beta}_i - \beta_i\|_{C^r(\phi_i(K_i))}\ < \delta_i$$

and $\tilde{\tau}|_{U_i}$ is represented locally by

$$\tilde{h}_i(\tilde{\alpha}_i dx + \tilde{\beta}_i dy - dz)$$

where $\tilde{h}_i : \phi_i(U_i) \to \mathbf{R}\backslash\{0\}$ are suitable smooth functions (again the sign of each $\tilde{h}_i$ is determined by the co–orientation of $\tilde{\sigma}$). Take now $\hat{\tau} \in \mathcal{U}^r(\tilde{\tau}, \hat{\delta}) \subset \mathcal{E}^1_0(M)$ where $\hat{\delta} = \{\hat{\delta}_i\}_{i\in\mathbf{N}}$ is some sequence of positive numbers, and show that

$$\hat{\tau} \in \pi^{-1}G(\mathcal{U}^r(\sigma, \delta))$$

if the numbers $\hat{\delta}_i$ are chosen small enough. In local coordinates $\hat{\tau}|_{U_i}$ looks like

$$\hat{\alpha}_i dx + \hat{\beta}_i dy + \hat{c}_i dz$$

with

$$\|\tilde{h}_i\tilde{\alpha}_i - \hat{\alpha}_i\|_{C^r(\phi_i(K_i))}\ ,\ \|\tilde{h}_i\tilde{\beta}_i - \hat{\beta}_i\|_{C^r(\phi_i(K_i))}\ ,\ \|\tilde{h}_i + \hat{c}_i\|_{C^r(\phi_i(K_i))}\ < \hat{\delta}_i.$$

Note that we have $\tilde{h}_i(K_i) \subset (-\infty, -2\varepsilon_i] \cup [2\varepsilon_i, \infty)$ for a suitable $\varepsilon_i > 0$. If $\hat{\delta}_i$ is chosen small enough then we also have $\hat{c}_i(K_i) \subset (-\infty, -\varepsilon_i] \cup [\varepsilon_i, \infty)$. Then $\hat{\sigma} := \ker \hat{\tau} \in \mathrm{Distr}(M)$ is represented locally by

$$\left(-\frac{\hat{\alpha}_i}{\hat{c}_i}, -\frac{\hat{\beta}_i}{\hat{c}_i}\right).$$

Now

$$\begin{aligned}\left\|\alpha_i+\frac{\hat\alpha_i}{\hat c_i}\right\|_{C^r(\phi_i(K_i))} &\le \|\alpha_i-\tilde\alpha_i\|_{C^r(\phi_i(K_i))}+\left\|\tilde\alpha_i-\frac{\hat\alpha_i}{\tilde h_i}\right\|_{C^r(\phi_i(K_i))}+\\ &\quad+\left\|\frac{\hat\alpha_i(\hat c_i+\tilde h_i)}{\hat c_i\tilde h_i}\right\|_{C^r(\phi_i(K_i))}\\ &=: T_1+T_2+T_3.\end{aligned}$$

We have $T_2 \le C_2\hat\delta_i$ with a positive constant C_2 depending on $\tilde\tau, r$ and the charts (U_i,ϕ_i). Since $\|\hat c_i\|_{C^r(\phi_i(K_i))}\ge\varepsilon_i$, we can estimate

$$\begin{aligned}T_3 &\le C_3\|\frac{\hat\alpha_i}{\tilde h_i}(\hat c_i+\tilde h_i)\|_{C^r(\phi_i(K_i))}\\ &\le C_3(\|\tilde\alpha_i(\hat c_i+\tilde h_i)\|_{C^r(\phi_i(K_i))}+\|(\frac{\hat\alpha_i}{\tilde h_i}-\tilde\alpha_i)(\hat c_i+\tilde h_i)\|_{C^r(\phi_i(K_i))})\\ &\le C_4\cdot\hat\delta_i\end{aligned}$$

where the constant C_4 depends on $\tilde\tau, r$ and ε_i. This shows that

$$\left\|\alpha_i+\frac{\hat\alpha_i}{\hat c_i}\right\|_{C^r(\phi_i(K_i))}<\delta_i$$

if $\hat\delta_i$ is chosen sufficiently small. We obtain similarly

$$\left\|\beta_i+\frac{\hat\beta_i}{\hat c_i}\right\|_{C^r(\phi_i(K_i))}<\delta_i$$

and therefore $\hat\sigma\in\mathcal{U}^r(\sigma,\delta)$, i.e. $[\hat\tau]\in G(\mathcal{U}^r(\sigma,\delta))$ which proves that G is an open map. □

Let us consider a simple example: The local "normal form" of Corollary 3.1.9. Take $(M,\lambda)=(\mathbf{R}^3, dz+xdy)$ and $H(x,y,z):=z+\frac12 xy$. define F by $H^{-1}(0)=\{(x,y,-\frac12 xy)\mid x,y\in\mathbf{R}\}$. Then

$$dH(x,y,z)=\frac12 y\,dx+\frac12 x\,dy+dz$$

and $(0,0,0)$ is a singular point, i.e.

$$T_0F=\ker dH(0)=\operatorname{span}\left\{\frac{\partial}{\partial x},\ \frac{\partial}{\partial y}\right\}=\ker\lambda(0).$$

Take the global chart

$$\phi : \mathbf{R}^2 \to F \; ; \; (\alpha, \beta) \longmapsto (\alpha, \beta, -\frac{1}{2}\alpha\beta),$$

the orientation $\sigma := (\phi^{-1})^*(d\alpha \wedge d\beta)$ on F and let us calculate the characteristic foliation $[Z]$ on F which is induced by λ.

A representative Z of $[Z]$ must satisfy

$$\begin{aligned}(j \circ \phi)^*\lambda &= \phi^*(i_Z((\phi^{-1})^* d\alpha \wedge d\beta)) \\ &= (d\alpha \wedge d\beta)(\phi_*^{-1} Z, \, .\,).\end{aligned}$$

Now

$$\begin{aligned}(j \circ \phi)^*(dz + x\,dy) &= d(-\frac{1}{2}\alpha\beta) + \alpha\, d\beta \\ &= \frac{1}{2}\alpha\, d\beta - \frac{1}{2}\beta\, d\alpha.\end{aligned}$$

Therefore we get

$$\phi_*^{-1} Z \; = \; \frac{1}{2}\alpha\frac{\partial}{\partial\alpha} + \frac{1}{2}\beta\frac{\partial}{\partial\beta}$$

and

$$Z \; = \; \frac{1}{2}x\frac{\partial}{\partial x} + \frac{1}{2}y\frac{\partial}{\partial y} + z\frac{\partial}{\partial z}.$$

Figure 3.2 shows the flow lines of $\phi_*^{-1} Z$. Using the definition of the contact Hamiltonian we get

$$\hat{V}_H = \frac{1}{2}x\frac{\partial}{\partial x} + \frac{1}{2}y\frac{\partial}{\partial y} - \frac{1}{2}xy\frac{\partial}{\partial z}$$

and

$$V_H = \frac{1}{2}x\frac{\partial}{\partial x} + \frac{1}{2}y\frac{\partial}{\partial y} + z\frac{\partial}{\partial z} = Z$$

(the proof of Lemma 3.3.3 in the next section will imply that we always have $Z = V_H$ near singularities). We invite the reader to carry out the above calculations also in the case of a hyperbolic singular point in 'normal form', i.e. $H(x, y, z) = z - \frac{1}{2}xy$. We then obtain the 'hyperbolic picture' as in Fig. 3.1.

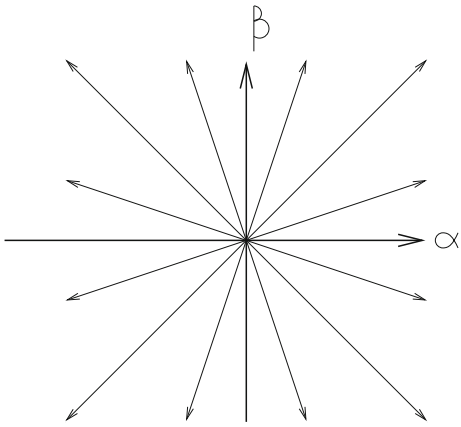


Fig. 3.2 The characteristic foliation near a positive nicely elliptic point in normal form

Another key result is the following:

Theorem 3.2.3 *Let F be a compact oriented surface without boundary in a three dimensional manifold M, which is equipped with a contact form λ and the associated contact structure $\xi = \ker \lambda$. Then there exists an arbitrarily small C^∞-perturbation $\tilde{F}$ of F such that the characteristic foliation induced by ξ on $\tilde{F}$ has the following properties:*

- *There are only non-degenerate singular points*
- *The α- and ω-limit set of an orbit is either a singular point or a limit cycle*
- *The limit cycles are non-degenerate; i.e. the linearization of the Poincaré-map has no eigenvalue of modulus one*
- *There are no connecting orbits between hyperbolic singular points*

The theorem says that the characteristic foliation becomes *Morse-Smale* (see [5, 94]) under a small perturbation of F. We reduce the proof to the following result due to Peixoto (see chapter 4 of [94]). We give an appropriate modification.

Theorem 3.2.4 *Let F be a compact, orientable surface without boundary. Then the smooth vector fields which are Morse-Smale are C^∞-dense among all vector fields.* □

Remark Theorem 3.2.4 holds also for F compact, orientable with **nonempty** boundary if we consider only the class of smooth vector fields on F so that the components of ∂F are closed trajectories of the vector field.

Proof of Theorem 3.2.3 Let $[Z]$ be the oriented singular foliation induced on F by ξ, i.e. $\Delta(\xi) = [Z]$. Fix an orientation form Ω on F inducing the given orientation and let $Y := \Delta_\Omega(\lambda)$. Take a small convex neighborhood U of λ in $\mathcal{E}^1_0(M)$ with respect to the C^∞–Whitney topology so that all forms in U are contact forms. Then $\Delta_\Omega(U)$ is an open neighborhood of Y in $\mathrm{Vec}(F)$ and contains therefore a Morse–Smale vector field $\tilde{Y}$. Pick now $\tilde{\lambda} \in U$ so that $\Delta_\Omega(\tilde{\lambda}) = \tilde{Y}$ and $\tilde{\lambda}$ coincides with λ

outside some compact neighborhood K of F. Then the 1–forms $\lambda_t := t\tilde{\lambda} + (1 - t)\lambda,\ 0 \leq t \leq 1$ are all contact forms and Gray's theorem (or rather its proof) provides the existence of a smooth family of diffeomorphisms (ϕ_t) so that $\phi_0 = \mathrm{Id}$ and $T\phi_t(\ker\lambda) = \ker\lambda_t$. Then $\tilde{F} = \phi_1^{-1}(F)$ is a C^∞–small perturbation of F and the characteristic foliation on $\tilde{F}$ induced by $\ker\lambda$ has the desired property. □

We can even say more

Theorem 3.2.5 *Let F be an oriented closed surface in a three dimensional contact manifold (M, λ). Assume $\varepsilon > 0$ is given. Then there exists a C^0-small isotopy $\tilde{F}$ of F such that*

$$\mid vol_\lambda(F) - vol_\lambda(\tilde{F}) \mid < \varepsilon.$$

Moreover the characteristic foliation on $\tilde{F}$ is Morse-Smale. Further all elliptic singularities of $\tilde{F}$ are nicely elliptic and there exist local Darboux-charts mapping the singular points to $0 \in \mathbf{R}^3$, so that in local coordinates the surface $\tilde{F}$ is given near every singularity by the graph of the quadratic form

$$(x, y) \longmapsto \pm\frac{1}{2}xy$$

with negative sign in the case of an elliptic singularity and positive sign in the case of a hyperbolic singularity.

Proof First one carries out the C^∞-small isotopy bringing F in the form described by Theorem 3.2.3. Then one carries out a C^0-small perturbation to achieve the normal forms guaranteed by Theorem 3.1.5 near the meanwhile finitely many singularities. The new characteristic foliation is again Morse–Smale. □

3.3 Characterization of Singular Foliations

Assume $F \subset M$ is an embedded compact oriented surface in an oriented three dimensional manifold M equipped with an oriented singular foliation $[Z]$. The surface F then has a tubular neighborhood U which is diffeomorphic to a product $\mathbf{R} \times F$ and $\{0\} \times F \approx F \subset U$.

Via $(t, x) \longmapsto t$ we get a smooth function $H : U \longrightarrow \mathbf{R}$ with $H^{-1}(0) = F$ and $dH \neq 0$ on F. Since the map Δ of Proposition 3.2.2 is surjective, there is a co–oriented 2–plane–distribution σ on M which induces the oriented singular foliation $[Z]$ on F. We will investigate the following problem:

Under which conditions does there exist a germ of a contact form on $\mathbf{R} \times F$ near $F = \{0\} \times F$ inducing $[Z]$ and what does it look like?

We will give, following E. Giroux [58], a necessary and sufficient condition on $[Z]$. Moreover we will see by a version of Gray's theorem that the associated germ

of contact structure is unique modulo diffeomorphism fixed on F if it exists. So the characteristic foliation on a surface F induced by λ determines the contact structure already in a neighborhood of F up to diffeomorphism.

Definition 3.3.1 Let F be a compact orientable surface without boundary, Y a vector field on F and σ some orientation form on F. Then the *divergence* of Y is defined by

$$\mathrm{div}_\sigma(Y) \cdot \sigma = L_Y \sigma.$$

We call a zero x of a vector field Y *divergence free* if the divergence of Y vanishes at x.

Remark Let Y be a vector field on F with $Y(x_0) = 0$ for some $x_0 \in F$, where F is a surface with orientation form σ. Moreover let g be a strictly positive function on F and let f be a nowhere vanishing function on F. Then

$$\begin{aligned}
\mathrm{div}_{f\sigma}(gY)_{x_0} \cdot (f\sigma)(x_0) &= L_{gY}(f\sigma)_{x_0} \\
&= d(i_{gY}(f\sigma))_{x_0} \\
&= f(x_0)g(x_0)d(i_Y\sigma)_{x_0} \\
&= g(x_0)\mathrm{div}_\sigma(Y)_{x_0} \cdot (f\sigma)(x_0),
\end{aligned}$$

hence

$$\mathrm{div}_{f\sigma}(gY)_{x_0} = g(x_0)\mathrm{div}_\sigma(Y)_{x_0}.$$

So the notion of a *divergence free singularity* and the *sign of the divergence at a singularity* can also be defined for oriented singular foliations on F and they do not depend on the orientation form σ on F and the vector field representing the oriented singular foliation.

We have the following interesting observation due to E. Giroux [58]:

Proposition 3.3.2 *Let F be an oriented compact surface. Assume $\mathcal{F}$ is an oriented singular foliation on F without divergence-free singularities. View $F = \{0\} \times F$ as a subset of $\mathbf{R} \times F$.*

- *Then there exists a germ of a contact structure on $\mathbf{R} \times F$ near F which induces $\mathcal{F}$, i.e. $\mathcal{F}$ is the characteristic foliation of some contact structure defined near F.*
- *If F has no boundary then any two germs of contact structures near F inducing the same oriented singular foliation on F are equivalent by a diffeomorphism ϕ defined on a neighborhood of F. Moreover ϕ is the identity on F and is isotopic to the identity by an isotopy fixed on F.*

We want to see first that oriented singular foliations induced by contact structures can never have singularities with vanishing divergence.

So let F be an oriented surface in (M, λ) and let $[Z]$ be the characteristic foliation on F induced by the contact structure $\xi = \ker \lambda$. Assume that x_0 is a singular point on F. Then, with $\omega = \lambda \mid_F$, we see that $d\omega$ defines a volume form on F near x_0. The aim is to show that necessarily $\mathrm{div}_{d\omega}(Z)_{x_0} \neq 0$ and we will do this by comparing Z with a contact Hamiltonian vector field where we know its divergence. If x_0 is a positive singular point then the given orientation on F and the one given by $d\omega$ near x_0 agree, otherwise if x_0 is a negative singular point then the given orientation on the surface and the one given by $-d\omega$ agree. Let H be a smooth function on a neighborhood U of x_0 in M so that $H^{-1}(0) = F \cap U$ and $dH(x) \neq 0$ for all $x \in F \cap U$. Then we compute with V_H being the contact Hamiltonian vector field induced by H:

$$\begin{aligned} L_{V_H} d\lambda &= (i_{V_H} d + d\, i_{V_H}) d\lambda \\ &= d\, i_{V_H} d\lambda \\ &= d((i_{X_\lambda} dH)\lambda - dH) \\ &= d((i_{X_\lambda} dH)\lambda) \\ &= d(i_{X_\lambda} dH) \wedge \lambda + (i_{X_\lambda} dH) d\lambda. \end{aligned}$$

Evaluating this in x_0 and restricting to $T_{x_0} F = \ker \lambda(x_0)$ we obtain

$$(L_{V_H} d\omega)_{x_0} = (i_{X_\lambda} dH)_{x_0} d\omega_{x_0}$$

and therefore

$$\mathrm{div}_{d\omega}(V_H)_{x_0} = (i_{X_\lambda} dH)_{x_0}.$$

On the other hand, we compute for $x \in F$ near x_0 using the definition of V_H

$$(i_{V_H} d\omega)_x = (i_{X_\lambda} dH)_x\, \omega,$$

hence if Z is a representative of the oriented singular foliation $[Z]$ satisfying $\pm i_Z d\omega = \omega$ (positive sign for calculations near a positive singularity) we get $V_H = \pm(i_{X_\lambda} dH) Z$ by non degeneracy of $d\omega$ near x_0 and

$$\begin{aligned} (i_{X_\lambda} dH)_{x_0} &= \mathrm{div}_{d\omega}(V_H)_{x_0} \\ &= \mathrm{div}_{d\omega}(\pm(i_{X_\lambda} dH)\, Z) \\ &= \pm(i_{X_\lambda} dH)_{x_0}\, \mathrm{div}_{d\omega}(Z)_{x_0}. \end{aligned}$$

In view of $(i_{X_\lambda} dH)_{x_0} \neq 0$ we conclude

$$\mathrm{div}_{d\omega}(Z)_{x_0} = \pm 1.$$

We sum up our discussion as follows:

Lemma 3.3.3 *Let F be an oriented surface in (M, λ) and let $[Z]$ be the characteristic foliation on F induced by the contact structure $\xi = \ker \lambda$.*

- *Then a singular point x_0 is positive (negative) if and only if its divergence $div_{d\lambda|F}(Z)(x_0)$ is positive (negative). Moreover, positive elliptic singularities are sources and negative elliptic singularities are sinks.*
- *If F is given near a singular point x_0 as the regular zero set of a smooth function H so that $sign(x_0) = sign(i_{X_\lambda} dH)_{x_0}$ then the contact Hamiltonian V_H is a representative of the oriented singular foliation $[Z]$ near x_0.* □

Remark We said earlier that the divergence of a vector field on F at a singularity does not depend on the orientation of F. In the situation of the above lemma, changing the orientation of F turns positive singularities into negative ones and vice versa, but this also changes Z to $-Z$.

Lemma 3.3.3 shows that the condition in Proposition 3.3.2 on the singularities having nonzero divergence is necessary. Now we prove that this is also sufficient.

Proof of Proposition 3.3.2 Consider $\mathbf{R} \times F$. We have to construct a contact form near $F := F \times \{0\}$ inducing some given oriented singular foliation $\mathcal{F}$ on F where $\mathcal{F}$ has no divergence free singularities. We may view $\mathbf{R}$ as a parameter. Consider a family $(\beta_t)_{t\in\mathbf{R}}$ of one forms on F and similarly a family of smooth functions on F denoted by $(u_t)_{t\in\mathbf{R}}$. We define a one-form α on $\mathbf{R} \times F$ by

$$\alpha \;=\; \beta_t + u_t dt$$

which is a contact form if and only if $\Omega := \alpha \wedge d\alpha$ is a volume form on $\mathbf{R} \times F$. We compute:

$$\begin{aligned}\Omega &= (\beta_t + u_t dt) \wedge (d\beta_t - \frac{\partial}{\partial t}\beta_t \wedge dt + du_t \wedge dt)\\ &= \beta_t \wedge d\beta_t - \beta_t \wedge \frac{\partial}{\partial t}\beta_t \wedge dt + \beta_t \wedge du_t \wedge dt + u_t dt \wedge d\beta_t\\ &= (-\beta_t \wedge \frac{\partial}{\partial t}\beta_t + \beta_t \wedge du_t + u_t d\beta_t) \wedge dt.\end{aligned}$$

This means precisely that

$$\beta_t \wedge (du_t - \frac{\partial}{\partial t}\beta_t) + u_t\, d\beta_t$$

is a volume form on F for every t. In particular, α defines a contact form near F if and only if the pair $(u_0, (\frac{\partial}{\partial t}\beta_t)\,|_{t=0})$ satisfies

$$u_0\, d\beta_0 + \beta_0 \wedge \left(du_0 - \frac{\partial}{\partial t}\beta_t\Big|_{t=0}\right) \neq\; 0$$

everywhere on F. We try to construct a form α in such a way that u is not depending on t. Let σ be a volume form on F inducing the given orientation. We define β_0 by

$$\beta_0 \;=\; i_Z\sigma,$$

where σ is a 2–form on F inducing the given orientation and Z represents $\mathcal{F}$. Further, we define the function u_0 by

$$d\beta_0 \;=\; u_0\sigma.$$

Denote by $\mathcal{S}_F$ the set of singular points on F, i.e. the zero set of Z. We compute

$$d\beta_0 \;=\; d\, i_Z\sigma \;=\; L_Z\sigma.$$

By assumption the singularities are not divergence-free, so we deduce that $d\beta_0 \neq 0$ near $\mathcal{S}_F$. Therefore $u_0 \neq 0$ on an open neighborhood U of $\mathcal{S}_F$. Now pick a 1-form γ on F such that $\gamma(Z) \leq 0$ and $\gamma(Z) < 0$ on $F\backslash U$, for example take a metric $(.\,,\,.)$ on F and choose $\gamma := -(Z,\,.)$. We calculate

$$\beta_0 \wedge \gamma = -\gamma(Z)\,\sigma$$

because of $\sigma \wedge \gamma \equiv 0$. Define now β_t by

$$\beta_t \;=\; \beta_0 + t(du_0 - \gamma)$$

and let α be the associated one-form on $\mathbf{R} \times F$. We compute

$$\begin{aligned} u_0\, d\beta_0 + \beta_0 \wedge \left(du_0 - \frac{\partial}{\partial t}\beta_t\bigg|_{t=0} \right) &= u_0\, d\beta_0 + \beta_0 \wedge (du_0 - (du_0 - \gamma)) \\ &= u_0^2\,\sigma + \beta_0 \wedge \gamma \\ &= (u_0^2 - \gamma(Z))\,\sigma. \end{aligned}$$

By construction this is a volume-form implying that α defines a contact form near F in $\mathbf{R} \times F$. The above shows that $[Z]$ is induced by a contact form.

Next we want to show that the germ of contact structure is unique up to diffeomorphism fixed on F. So assume that α_0 and α_1 are germs of contact forms defined near F on $\mathbf{R} \times F$ inducing the same characteristic foliation $[Z]$ on F. We also assume that α_0 and α_1 induce the same orientation on the part of $\mathbf{R} \times F$ where they are defined. If σ is an orientation form on F then

$$j^*\alpha_0 = i_{Z_0}\sigma \text{ and } j^*\alpha_1 = i_{fZ_0}\sigma,$$

where j denotes the injection $\{0\} \times F \hookrightarrow \mathbf{R} \times F$, σ is an orientation form on F, Z_0 is a vector field on F representing the characteristic foliation and f is a suitable nowhere vanishing function on F. Since we are only interested in the contact structures, not the contact forms, we may multiply α_1 with f^{-1} and assume that $f \equiv 1$. Writing

$$\alpha_0 = \beta_t + u_t\, dt \text{ and } \alpha_1 = \tilde{\beta}_t + \tilde{u}_t\, dt,$$

where t denotes the coordinate on $\mathbf{R}$, u_t is a function and β_t is a 1–form on F, we obtain

$$\beta_0 = j^*\alpha_0 = j^*\alpha_1 = \tilde{\beta}_0.$$

The condition of α_0, α_1 being contact forms yields

$$\sigma_t := u_t d\beta_t + \beta_t \wedge \left(du_t - \frac{\partial \beta_t}{\partial t} \right) \neq 0$$

and

$$\tilde{\sigma}_t := \tilde{u}_t d\tilde{\beta}_t + \tilde{\beta}_t \wedge \left(d\tilde{u}_t - \frac{\partial \tilde{\beta}_t}{\partial t} \right) \neq 0$$

on F, i.e. both 2–forms above are orientation forms on F for all t. Since α_0 and α_1 induce the same orientation on $\mathbf{R} \times F$, both 2–forms above induce the same orientation on F for all t. We define now the following family of 1–forms for $0 \leq s \leq 1$:

$$\begin{aligned} \alpha_s &:= (1-s)\alpha_0 + s\,\alpha_1 \\ &= \beta_t + ((1-s)u_t + s\tilde{u}_t)\, dt + s(\tilde{\beta}_t - \beta_t). \end{aligned}$$

We claim that all the forms α_s are contact forms on a sufficiently small neighborhood of $\{0\} \times F$ in $\mathbf{R} \times F$. It suffices to verify that $\alpha_s \wedge d\alpha_s|_{t=0} \neq 0$ for all $0 \leq s \leq 1$. We compute

$$\begin{aligned} \alpha_s \wedge d\alpha_s = (1-s)^2 \alpha_0 \wedge d\alpha_0 + s^2 \alpha_1 \wedge d\alpha_1 + \\ + s(1-s)(\alpha_0 \wedge d\alpha_1 + \alpha_1 \wedge d\alpha_0), \end{aligned}$$

and

$$(\alpha_0 \wedge d\alpha_1 + \alpha_1 \wedge d\alpha_0)|_{t=0} = [\sigma_0 + \tilde{\sigma}_0] \wedge dt,$$

so that

$$\alpha_s \wedge d\alpha_s|_{t=0} = [(1-s)\sigma_0 + s\tilde{\sigma}_0] \wedge dt$$

which does not vanish anywhere.

We define a time-dependent vector field Y_s near F by

$$i_{Y_s}\alpha_s \;=\; 0$$

and

$$i_{Y_s}d\alpha_s \;=\; \left(i_{X_s}\frac{d\alpha_s}{ds}\right)\alpha_s - \frac{d\alpha_s}{ds}$$

where X_s is the Reeb vector field of α_s. The time-dependent vector field defines a 1-parameter family of local diffeomorphisms $(\phi_s)_{s\in[0,1]}$ near F having the property that $\phi_0 = id$, $\phi_s(x) = x$ for $x \in F$ and $(T\phi_1)\xi_0 = \xi_1$ (see proof of Gray's result, Theorem 2.3.1). This completes the proof of Proposition 3.3.2. □

3.4 The Elimination Theorem

Let (M, λ) be a three-manifold equipped with a contact form λ. Denote by ξ the associated contact structure. The fibers of ξ are oriented by $d\lambda$. We assume that M carries the orientation given by the volume form $\lambda \wedge d\lambda$. Let F be an embedded compact oriented surface. We assume that all singular points are nondegenerate. Hence the number of singular points is finite.

We will prove an important result due to Eliashberg, Fuchs and Giroux which will permit us to perturb a surface in a three dimensional contact manifold so that the perturbed surface has less singularities than the original one. If λ is a tight contact form and F is diffeomorphic to S^2 we will even show that there is a C^0 small isotopy to a surface F' so that F' has only two nicely elliptic singular points and no limit cycles.

We will also show that, in the overtwisted case, there is an overtwisted disk in M so that the characteristic foliation contains exactly one elliptic singularity in the interior. We will sketch a proof of Bennequin's inequality, which relates the number and type of singularities on a surface to its Euler characteristic [13, 33].

In this section we prove an important result due to Giroux [58]. It was motivated by an earlier result of Eliashberg [32]. The general form of the result as we present it here is due to Fuchs and Eliashberg [34].

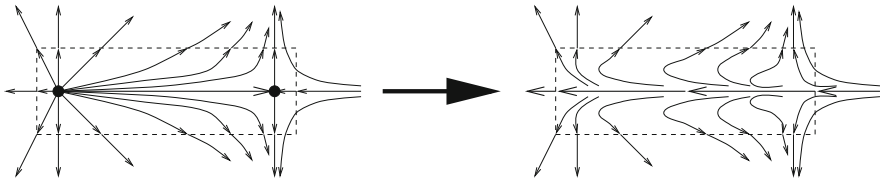


Fig. 3.3 Eliminating a pair of singularities

Theorem 3.4.1 (Fig. 3.3) *Let F be a compact oriented surface in a three dimensional contact manifold (M, λ). Assume the following:*

- *the characteristic foliation on F is Morse-Smale,*
- *There exists a trajectory of the characteristic foliation γ connecting an elliptic singularity q and a hyperbolic singularity p of the same sign.*

Then there exists a C^0-small isotopy supported in an arbitrary small neighborhood U of $\gamma \cup \{q, p\}$, keeping γ fixed, such that the new surface $\tilde{F}$ has two singularities less than the original one.

Moreover, by an additional C^∞-perturbation, which is fixed in a small neighborhood of the singular points the latter surface can be perturbed again such that its singular foliation is Morse-Smale.

Given any $\varepsilon > 0$ the perturbations can be carried out in such a way that the new surface $\tilde{F}$ satisfies

$$| \, vol_\lambda(\tilde{F}) - vol_\lambda(F) \, | < \varepsilon.$$

Proof of Theorem 3.4.1 First, we apply Theorem 3.2.5 and the remark after Theorem 3.2.4 in the case $\partial F \neq \emptyset$. Hence we may assume that all the elliptic singularities are in fact nicely elliptic and that the surface F is in normal form near each singularity. We would like to find an embedded curve $\tilde{\gamma} : (-1-\varepsilon, 1+\varepsilon) \to F$ so that $\tilde{\gamma}(-1) = q$, $\tilde{\gamma}(+1) = p$, $\tilde{\gamma}([-1, +1]) = \gamma$ and the whole curve $\tilde{\gamma}$ is tangent to the characteristic foliation on F. We may assume without loss of generality that the elliptic singularity q is a source. If we parameterize a compact piece $\hat{\gamma}$ of the stable manifold of p so that $\hat{\gamma}(t)$ (where $0 \leq t \leq 1 + 2\varepsilon$) satisfies $\hat{\gamma}(+1) = p$ and $\hat{\gamma}(0)$ lies in a neighborhood of q described by Corollary 3.1.9 then $\hat{\gamma}$ is an embedded curve. In Chap. 3, Sect. 3.2 we calculated the characteristic foliation near a nicely elliptic and a hyperbolic singularity. From this we see directly that $\hat{\gamma}$ can be continued smoothly to a curve $\tilde{\gamma}$ as shown in Fig. 3.4, i.e. we continue the curve $\hat{\gamma}$ beyond the point q. Because the curve $\tilde{\gamma}$ is an embedded Legendrian curve, we can apply Theorem 2.2.4 (Legendrian neighborhood theorem). We can then find a diffeomorphism Ψ defined on some open neighborhood U of $\gamma([-1, +1])$ to V, a small neighborhood of $\{(\theta, 0, 0) \in \mathbf{R}^3 \mid -1 \leq \theta \leq +1\}$, such that

$$\Psi(\tilde{\gamma}) = \{(\theta, 0, 0) \in \mathbf{R}^3 \mid |\theta| \leq 1 + \hat{\delta}\}$$

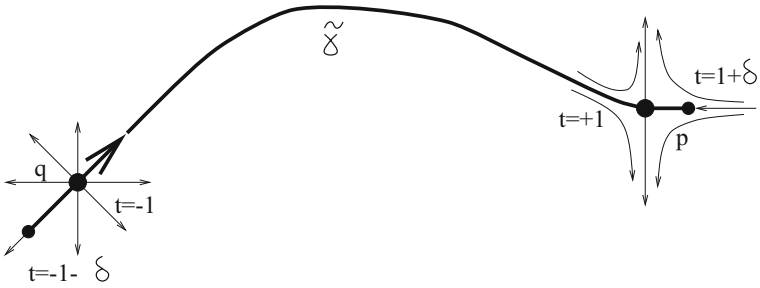


Fig. 3.4 Extending the connecting orbit beyond the singular points

with a suitable $\hat{\delta} > 0$ and

$$\begin{aligned}\Psi(\gamma) &= \{(\theta, 0, 0) \in \mathbf{R}^3 \mid -1 \le \theta \le +1\},\\ \lambda|_U &= \Psi^*(dy + x\,d\theta),\\ \Psi(q) &= (-1, 0, 0),\\ \Psi(p) &= (+1, 0, 0).\end{aligned}$$

By shrinking U if necessary, we may assume in addition that $U \cap F$ only contains the singular points p and q. Summarizing, we can map an 'extension' $\tilde{\gamma}$ of the connecting orbit γ diffeomorphically onto an interval $(-1 - \hat{\delta}, +1 + \hat{\delta}) \times \{0\} \times \{0\} \subset \mathbf{R}^3$ so that the contact form λ corresponds to the standard form on $\mathbf{R}^3$. In the coordinates (θ, x, y) defined near the connecting orbit γ, the piece $\mathcal{D} := \Psi(U \cap F)$ of our surface can be parameterized as follows:

$$\mathcal{D} = \{(\theta, x(\theta, r), y(\theta, r)) \in \mathbf{R}^3 \mid |\theta| \le 1 + \hat{\delta}\ ,\ r \in I\},$$

where $I \in \mathbf{R}$ is a suitable interval and x, y are smooth functions with

$$x(\theta, 0) \equiv y(\theta, 0) \equiv 0.$$

The interval I is of the form $[0, 1)$ if the connecting orbit γ lies on ∂F, otherwise I is of the form $(-1, 1)$. Moreover, we assume that $\mathcal{D}$ is oriented in such a way that the above parameterization $([0, 1] \times [0, 1], d\theta \wedge dr) \to \mathcal{D}$ is orientation preserving. We orient $\mathcal{L}$ by $v = \frac{d}{d\theta}$, so that $(v, \nu = \frac{\partial}{\partial r})$ is positively oriented. A point $(\theta_0, 0, 0)$ is a singular point if and only if $\partial_r y(\theta_0, 0) = 0$. Since also $\partial_\theta y(\theta_0, 0) = \partial_\theta x(\theta_0, 0) = 0$ and $\mathcal{D}$ is embedded, we conclude that $\partial_r x(\theta_0, 0) \neq 0$. The tangent space $T_{(\theta_0,0,0)}\mathcal{D}$ is oriented by the basis $\left(\frac{\partial}{\partial\theta}, \partial_r x(\theta_0, 0)\frac{\partial}{\partial x}\right)$. On the other hand, the contact structure $\ker\lambda(\theta_0, 0, 0)$ is oriented by $\left(\frac{\partial}{\partial\theta}, -\frac{\partial}{\partial x}\right)$. The singular point $(\theta_0, 0, 0)$ is positive if these two orientations coincide, which is the case for $\partial_r x(\theta_0, 0) < 0$, otherwise $(\theta_0, 0, 0)$ is negative.

Let us first point out how to recognize the type of a singularity $(\theta_0, 0, 0)$ in the above parametrization. Since the Jacobian of the map $M(\theta, r) = (\theta, x(\theta, r))$ at the point $(\theta_0, 0)$ has rank 2, there is a local inverse and we parameterize $\mathcal{D}$ by

$$\mathcal{D} = \{(\theta, x, (y \circ M^{-1})(\theta, x))\},$$

where (θ, x) is sufficiently near to $(\theta_0, 0)$. We write

$$f(\theta, x) := (y \circ M^{-1})(\theta, x)$$

and note that

- $f(\theta, 0) \equiv 0$,
- $\partial_x f(\theta_0, 0) = 0$ since $(\theta_0, 0, 0)$ is a singular point.

We write near the point $(\theta_0, 0, 0)$

$$f(\theta, x) = \frac{a}{2}(\theta - \theta_0)^2 + \frac{b}{2}x^2 + c(\theta - \theta_0)x + h(\theta, x)$$

with $a = \partial_{\theta\theta} f(\theta_0, 0)$, $b = \partial_{xx} f(\theta_0, 0)$, $c = \partial_{x\theta} f(\theta_0, 0)$, and h of order at least 3 in $(\theta - \theta_0, x)$. Note that $a = 0$ and also $h(\theta, 0) = 0$, hence

$$f(\theta, x) = \frac{b}{2}x^2 + c(\theta - \theta_0)x + h(\theta, x).$$

Investigate now the admissible values for the constants b and c. Near a singular point $(\theta_0, 0, 0)$, the surface $\mathcal{D}$ is given by $H^{-1}(0)$, where

$$H(\theta, x, y) := y - f(\theta, x).$$

Then the vector field $\hat{V}_H$, which is defined by $i_{\hat{V}_H} d\lambda = (i_{X_\lambda} dH)\lambda - dH$ and $i_{\hat{V}_H}\lambda = 0$, is given by

$$\hat{V}_H(\theta, x, y) = -\partial_x f(\theta, x)\frac{\partial}{\partial\theta} + (x + \partial_\theta f(\theta, x))\frac{\partial}{\partial x} + x\partial_x f(\theta, x)\frac{\partial}{\partial y}.$$

Its linearization $\hat{V}'_H(\theta, x, y)$ is given by

$$\begin{pmatrix} -\partial_{x\theta} f(\theta, x) & -\partial_{xx} f(\theta, x) & 0 \\ \partial_{\theta\theta} f(\theta, x) & 1 + \partial_{x\theta} f(\theta, x) & 0 \\ x\partial_{x\theta} f(\theta, x) & \partial_x f(\theta, x) + x\partial_{xx} f(\theta, x) & 0 \end{pmatrix}.$$

The contact structure $\ker\lambda(\theta_0, 0, 0)$ is generated by the vectors $(1, 0, 0)$ and $(0, 1, 0)$. We represent $\hat{V}'_H(\theta_0, 0, 0)$ by the matrix

$$\begin{pmatrix} -\partial_{x\theta} f(\theta_0, 0) & -\partial_{xx} f(\theta_0, 0) \\ 0 & 1 + \partial_{x\theta} f(\theta_0, 0) \end{pmatrix} = \begin{pmatrix} -c & -b \\ 0 & 1 + c \end{pmatrix}.$$

The singular point $(\theta_0, 0, 0)$ is then by definition hyperbolic if $c(c + 1) > 0$ and elliptic if $c(c + 1) < 0$. Let us translate this into our original parameterization $(\theta, x(\theta, r), y(\theta, r))$ of $\mathcal{D}$. Using $f(\theta, x(\theta, r)) = y(\theta, r)$, we compute

$$\partial_{\theta r} y(\theta_0, 0) = c\, \partial_r x(\theta_0, 0),$$

so that we are in the following situation: The point $(\theta_0, 0, 0)$ is

1. positive if $\partial_r x(\theta_0, 0) < 0$,
2. negative if $\partial_r x(\theta_0, 0) > 0$,
3. elliptic if $\frac{b'(\theta_0)}{a(\theta_0)} \in (-1, 0)$ and
4. hyperbolic if $\frac{b'(\theta_0)}{a(\theta_0)} \in (-\infty, -1) \cup (0, +\infty)$.

Invoking Theorem 3.1.5, we may assume that after a small perturbation near the singular points $(\pm 1, 0, 0)$, we have the following relation between the functions $x(\theta, r)$ and $y(\theta, r)$:

$$y(\theta, r) = c\, x(\theta, r)(\theta - \theta_0), \tag{3.5}$$

where $\theta_0 \in \{\pm 1\}$ and $|r|$, $||\theta| - 1|$ are small. In the case of an elliptic singularity, i.e. $\theta_0 = -1$, we may assume that $c = -\frac{1}{2}$. Parameterizing $\mathcal{D}$ by

$$\mathcal{D} = \{(\theta, x(\theta, r), y(\theta, r)) \in \mathbf{R}^3 \mid |\theta| \leq 1 + \hat{\delta}\,,\ r \in I\}$$

we expand x, y as follows:

$$x(\theta, r) = a(\theta)\, r + h(\theta, r)$$
$$y(\theta, r) = b(\theta)\, r + k(\theta, r),$$

where h, k are of order at least 2 in r.

Step 1 Get rid of the higher order terms.

We want to perturb $\mathcal{D}$ near the set $\Psi(\gamma) = [-1, +1] \times \{0\} \times \{0\}$, leaving $\Psi(\gamma)$ fixed, so that the higher order terms h and k disappear. Pick a number $0 < \tau < \hat{\delta}$ and a smooth function $\beta_\tau : [0, \infty) \to [0, 1]$ with $\beta_\tau \equiv 0$ on $[0, \frac{1}{1+\tau}]$, $\beta_\tau \equiv 1$ on

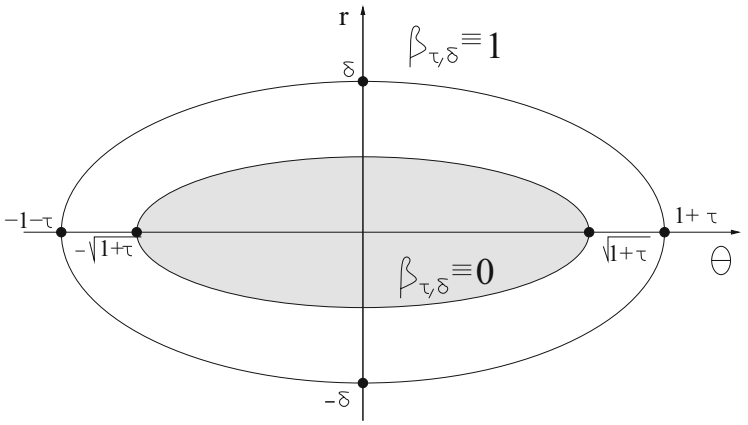


Fig. 3.5 Perturbing near the connecting orbit γ

$[1, \infty)$ and $0 \le \beta_\tau'(s) \le 2 + 2\tau^{-1}$ for all $s \ge 0$. We define for $0 < \delta \le \tau$

$$\beta_{\tau,\delta}(\theta, r) := \beta_\tau \left(\frac{\theta^2}{(1+\tau)^2} + \frac{r^2}{\delta^2} \right).$$

Figure 3.5 shows the support of the function $\beta_{\tau,\delta}$. Moreover, we define

$$x_\delta(\theta, r) := a(\theta)r + \beta_{\tau,\delta}(\theta, r)h(\theta, r),$$
$$y_\delta(\theta, r) := b(\theta)r + \beta_{\tau,\delta}(\theta, r)k(\theta, r)$$

and

$$\mathcal{D}_\delta = \{(\theta, x_\delta(\theta, r), y_\delta(\theta, r)) \in \mathbf{R}^3 \mid |\theta| \le 1 + \hat{\delta}\,,\ r \in I\}.$$

For $\frac{\theta^2}{(1+\tau)^2} + \frac{r^2}{\delta^2} \ge 1$ the perturbed surface $\mathcal{D}_\delta$ coincides with $\mathcal{D}$ and we also have $\Psi(\gamma) \subset \mathcal{D}_\delta$. We will show that $\mathcal{D}_\delta$ has the same singularities as $\mathcal{D}$, provided $\delta > 0$ was chosen sufficiently small. Moreover, the type of the singularities has to be the same as well which is left as an exercise to the reader.

Proceeding indirectly, we assume that there are sequences $\delta_n \searrow 0$, (θ_n, r_n) with

$$\frac{\theta_n^2}{(1+\tau)^2} + \frac{r_n^2}{\delta_n^2} \le 1$$

so that $(\theta_n, x_{\delta_n}(\theta_n, r_n), y_{\delta_n}(\theta_n, r_n))$ is a singular point for the characteristic foliation on $\mathcal{D}_{\delta_n}$ different from $(\pm 1, 0, 0)$. This means that we have

$$\begin{aligned} 0 &= \partial_\theta y_{\delta_n}(\theta_n, r_n) + x_{\delta_n}(\theta_n, r_n) \\ &= b'(\theta_n) r_n + \beta_{\tau,\delta_n}(\theta_n, r_n)\partial_\theta k(\theta_n, r_n) + \\ &\quad + \frac{2\theta_n}{(1+\tau)^2}\beta'_\tau \left(\frac{\theta_n^2}{(1+\tau)^2} + \frac{r_n^2}{\delta_n^2} \right) k(\theta_n, r_n) + a(\theta_n) r_n + \\ &\quad + \beta_{\tau,\delta_n}(\theta_n, r_n) h(\theta_n, r_n) \end{aligned} \tag{3.6}$$

and

$$\begin{aligned} 0 &= b(\theta_n) + \beta_{\tau,\delta_n}(\theta_n, r_n)\partial_r k(\theta_n, r_n) + \\ &\quad + \frac{2r_n}{\delta_n^2}\beta'_\tau \left(\frac{\theta_n^2}{(1+\tau)^2} + \frac{r_n^2}{\delta_n^2} \right) k(\theta_n, r_n). \end{aligned} \tag{3.7}$$

After passing to a subsequence of $(\delta_n)_{n\in\mathbf{N}}$, we may assume that the sequence $(\theta_n)_{n\in\mathbf{N}}$ converges to some θ_0. We know that $r_n^2 \le \delta_n^2 \to 0$ and we claim that also $\theta_0 \in \{\pm 1\}$. Passing to the limit in Eq. (3.7) and using the fact that k is of order at least 2 in r, we obtain $b(\theta_0) = 0$. Because the only singular points on $\Psi(\tilde{\gamma}) \cap \mathcal{D}$ are the points $(\pm 1, 0, 0)$ and

$$(\theta_n, x_{\delta_n}(\theta_n, r_n), y_{\delta_n}(\theta_n, r_n)) \to (\theta_0, 0, 0)$$

we have indeed $\theta_0 \in \{\pm 1\}$. This implies that we may use the normal form (3.5) for large n and insert it into Eq. (3.6). Doing so, we obtain from (3.5)

$$\partial_{\theta r} y(\theta, r) = c\partial_{\theta r} x(\theta, r)(\theta - \theta_0) + c\partial_r x(\theta, r)$$

and setting $r = 0$

$$b'(\theta) = c\, a(\theta) + c\, a'(\theta)(\theta - \theta_0).$$

Because $r_n = 0$ would lead to $\theta_n = \pm 1$ in contradiction to our assumption, Eq. (3.6) reduces to

$$\begin{aligned} 0 &= (c+1)a(\theta_n) + c\, a'(\theta_n)(\theta_n - \theta_0) + \\ &\quad + \beta_{\tau,\delta_n}(\theta_n, r_n)\frac{\partial_\theta k(\theta_n, r_n)}{r_n} + \beta_{\tau,\delta_n}(\theta_n, r_n)\frac{h(\theta_n, r_n)}{r_n} + \\ &\quad + \frac{2\theta_n}{(1+\tau)^2}\beta'_\tau \left(\frac{\theta_n^2}{(1+\tau)^2} + \frac{r_n^2}{\delta_n^2} \right) \frac{k(\theta_n, r_n)}{r_n} \\ &= (c+1)\, a(\theta_n) + \rho_n, \end{aligned}$$

where $\rho_n \to 0$ and $a(\theta_n) \neq 0$ for n large. This however contradicts the assumption that the singular points p and q are nondegenerate, i.e. $c \neq -1$. Summing up, we have shown that for sufficiently small $\delta > 0$, the surface $\mathcal{D}_\delta$ has the same number and type of singularities as $\mathcal{D}$. Hence we may assume that near $\Psi(\gamma)$ we have

$$\mathcal{D} = \{(\theta, a(\theta)r, b(\theta)r) \in \mathbf{R}^3 \mid |\theta| \leq 1 + \hat{\delta}\,,\ r \in I\},$$

where $a(\theta) := \partial_r x(\theta, 0)$ and $b(\theta) := \partial_r y(\theta, 0)$ having the following properties:

- $b(-1) = b(+1) = 0$, but b is nonzero otherwise,
- $\text{sign}\, a(-1) = \text{sign}\, a(+1) \neq 0$,
- $$-1 < \frac{b'(-1)}{a(-1)} < 0,$$
- $$\frac{b'(+1)}{a(+1)} < -1 \text{ or } \frac{b'(+1)}{a(+1)} > 0.$$

We will confine ourselves for the remainder of the proof to the case, where $(\pm 1, 0, 0)$ are both negative singular points, i.e. $a(\pm 1) > 0$. The case of two positive singular points is left as an easy exercise to the reader.

Step 2 Remove zeros of the function a.

We use the same function $\beta_{\tau,\delta}$ as in step 1 to change the function $a(\theta)$ as follows: We pick a smooth function $\hat{a}(\theta)$ so that $\hat{a}(\theta) \geq C > 0$ for all $\mathbf{R}$, where C is a constant larger than $\max\{|a|_{C^0}, 3|b'|_{C^0}\}$. We define a perturbed surface $\mathcal{D}_\delta$ by

$$\mathcal{D}_\delta := \{(\theta, a(\theta, r), b(\theta)r) \in \mathbf{R}^3 \mid |\theta| \leq 1 + \hat{\delta}\,,\ r \in I\},$$

where

$$a(\theta, r) := (1 - \beta_{\tau,\delta}(\theta, r))\hat{a}(\theta)r + \beta_{\tau,\delta}(\theta, r)a(\theta)r,$$

and $\delta, \tau > 0$ are sufficiently small. As before, this surface agrees with the original one outside a small neighborhood of $[-1, +1] \times \{0\} \times \{0\} \subset \mathbf{R}^3$. If this perturbation does not create any new singularities then we may assume that our surface is given by

$$\mathcal{D}_\delta := \{(\theta, \hat{a}(\theta)r, b(\theta)r) \in \mathbf{R}^3 \mid |\theta| \leq 1 + \hat{\delta}\,,\ r \in I\}.$$

Arguing indirectly, we assume that there are $\delta > 0$, $|r| \leq \delta$ and $|\theta| \leq 1 + \tau$, so that the point

$$(\theta, a(\theta, r), b(\theta)r) \neq (\pm 1, 0, 0)$$

is singular on $\mathcal{D}_\delta$. This leads to the conditions

$$b(\theta) = 0 \text{ and } \theta = \pm 1 \tag{3.8}$$

and

$$0 = b'(\theta)r + \hat{a}(\theta)r + \beta_{\tau,\delta}(\theta, r)r(a(\theta) - \hat{a}(\theta)). \tag{3.9}$$

We recall that the singular points $(\pm 1, 0, 0)$ are both negative, which implies $a(\pm 1) > 0$. Moreover,

$$-a(-1) < b'(-1) < 0$$

since $(-1, 0, 0)$ is elliptic and

$$b'(+1) > 0$$

since $(+1, 0, 0)$ is hyperbolic, the alternative $\frac{b'(+1)}{a(+1)} < -1$ is impossible since b only vanishes at the points ± 1. Condition (3.8) implies $\theta \in \{\pm 1\}$, since these are the only zeros of the function b, and $r \neq 0$. Condition (3.9) then becomes

$$0 = b'(\pm 1) + \hat{a}(\pm 1) + \mu(a(\pm 1) - \hat{a}(\pm 1))$$

with some number $0 \leq \mu \leq 1$. This leads to a contradiction because the above expression depends linearly on μ, but it is strictly positive for $\mu = 1$ and for $\mu = 0$ as well. This completes the discussion of step 2.

Step 3 Remove the singular points $(\pm 1, 0, 0)$.

By step 2 we may assume that

$$\mathcal{D} = \{(\theta, a(\theta)r, b(\theta)r) \in \mathbf{R}^3 \mid |\theta| \leq 1 + \hat{\delta}\,,\ r \in I\},$$

where the function a never vanishes and satisfies in addition

$$a(\theta) > 3|b'|_{C^0}.$$

We pick now a smooth function $\hat{b}$, so that $\hat{b}(\theta) = b(\theta)$ for $|\theta| \geq 1 + \tau$, where $\tau > 0$ is a sufficiently small number, $\hat{b} > 0$ everywhere and $|\hat{b}'|_{C^0} \leq |b'|_{C^0}$ (Fig. 3.6). Let

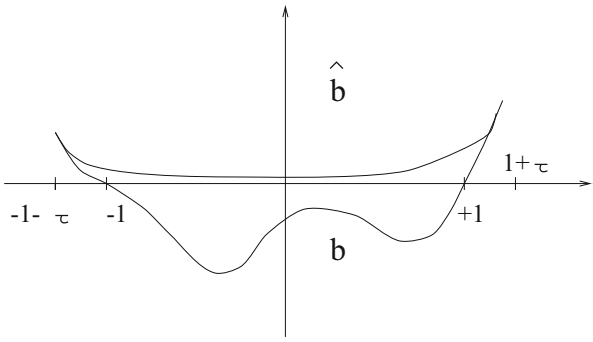


Fig. 3.6 The final step of the perturbation process

$\beta : \mathbf{R} \to [0, 1]$ be a smooth function which vanishes on $[-1, +1]$ and equals 1 on $(-\infty, -2] \cup [2, \infty)$. Moreover, let $\delta > 0$ be some small number. Then we define

$$b(\theta, r) := \hat{b}(\theta)r + \beta\left(\frac{r}{\delta}\right) r(b(\theta) - \hat{b}(\theta)).$$

The surface $\mathcal{D}'$ given by the points $(\theta, a(\theta)r, b(\theta, r))$ agrees with the original surface if $r > 2\delta$ or $|\theta| \geq 1 + \tau$. Let us look for singular points in the range $r \leq 2\delta$ and $|\theta| \leq 1 + \tau$. The conditions for $(\theta, a(\theta)r, b(\theta, r))$ to be a singular point are

$$0 = \hat{b}(\theta) + \left[\beta\left(\frac{r}{\delta}\right) + \frac{r}{\delta}\beta'\left(\frac{r}{\delta}\right)\right](b(\theta) - \hat{b}(\theta)) \tag{3.10}$$

and

$$\begin{aligned} 0 &= \partial_\theta b(\theta, r) + a(\theta)r \\ &= \hat{b}'(\theta)r + \beta\left(\frac{r}{\delta}\right) r(b'(\theta) - \hat{b}'(\theta)) + a(\theta)r. \end{aligned} \tag{3.11}$$

Assuming $r \neq 0$, we obtain from (3.11) the following contradiction

$$\begin{aligned} 0 &= \hat{b}'(\theta) + \beta\left(\frac{r}{\delta}\right)(b'(\theta) - \hat{b}'(\theta)) + a(\theta) \\ &\geq a(\theta) - 3|b'|_{C^0} \\ &> 0. \end{aligned}$$

On the other hand, assuming $r = 0$, Eq. (3.10) yields $\hat{b}(\theta) = 0$, which is also impossible since $\hat{b}$ does not vanish anywhere.

Summarizing, our final surface $\mathcal{D}'$ has the same singular points as $\mathcal{D}$ outside a small neighborhood U of $[-1, +1] \times \{0\} \times \{0\}$, but there are no singular points inside U anymore. This completes the proof of the theorem. □

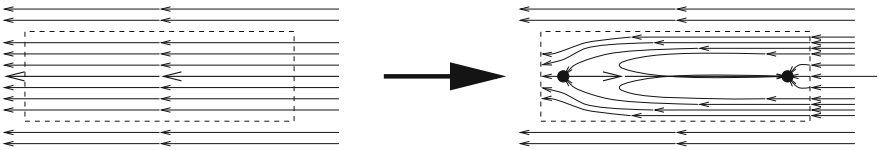


Fig. 3.7 Creating a pair of singular points

In the same way how we deleted a pair of singular points, it is also possible to create one (Fig. 3.7):

Theorem 3.4.2 *Let F be a compact oriented surface in (M, λ). Assume that γ is a compact piece of a leaf of the characteristic foliation not containing any singular points.*

Then there exist a C^0-small isotopy supported in an arbitrary small neighborhood U of γ keeping γ fixed, such that the new surface $\tilde{F}$ has two singularities more than F, namely an elliptic and a hyperbolic singularity of the same sign.

Every leaf γ' of the oriented singular foliation on $\tilde{F}$, which is sufficiently near to γ, is caught by one of the newly created singularities.

Proof Starting in the same way as in the proof of the elimination lemma, Theorem 3.4.1, we may assume that the curve γ is actually an interval $[-1-\hat{\delta}, 1+\hat{\delta}] \times \{0\} \times \{0\} \subset \mathbf{R}^3$ and the contact form is given by $\lambda = dy + x d\theta$. Moreover, the piece of the surface containing γ is given by

$$\mathcal{D} = \{(\theta, x(\theta, r), y(\theta, r)) \in \mathbf{R}^3 \mid |\theta| \leq 1 + \hat{\delta}\,,\ r \in I\},$$

using the same notation as in the proof of Theorem 3.4.1. We have $x(\theta, 0) \equiv y(\theta, 0) \equiv 0$ and $\partial_r y(\theta, 0)$ never vanishes since there are no singular points. We perform Step 1 as in the proof of Theorem 3.4.1 and we obtain already a contradiction after Eq. (3.7) if we pass to the limit $n \to \infty$. Then we may assume that $\mathcal{D}$ is parameterized by $(\theta, a(\theta)r, b(\theta)r)$ with suitable functions a, b, where b has the property that it never vanishes. We now pick a function $\hat{b}$ which agrees with b for $|\theta| \geq 1+\tau$, where $\tau > 0$ is sufficiently small, so that in addition $\hat{b}(\pm 1) = 0$ and $\hat{b} \neq 0$ anywhere else. We perform now Step 2 of the proof of Theorem 3.4.1, where we ask for $|\hat{a}|_{C^0} > |b'|_{C^0} + |\hat{b}'|_{C^0}$. Choosing $\hat{a}$ positive will create a pair of negative singularities, and negative $\hat{a}$ yields a pair of positive singularities. In contrast to the elimination process, we can do in Step 2 what we wish since condition (3.8) is never satisfied. Finally, we perform now Step 3 as in the proof of Theorem 3.4.1. The roles of b and $\hat{b}$ are reversed now. We have arranged a in Step 2 in such a way that Step 3 does not produce any new singular points apart from $(\pm 1, 0, 0)$. Moreover, the point $(-1, 0, 0)$ is elliptic and $(+1, 0, 0)$ is hyperbolic. □

3.5 Bennequin's Inequality in Tight Contact Manifolds

In this section we consider an oriented surface F in a three dimensional contact manifold (M, λ). We assume that the characteristic foliation on F induced by $\xi = \ker \lambda$ has only finitely many singular points. We first discuss Bennequin's inequality for the case where F is a closed surface, which relates the number of singular points to the Euler characteristic of F. We start with the following definition.

Definition 3.5.1 We denote by $e_\pm$ and $h_\pm$ the number of positive (negative) elliptic and hyperbolic points, respectively. Define the algebraic number of positive (negative) points $d_\pm$ by

$$d_\pm := e_\pm - h_\pm.$$

Let us denote by $[F] \in H_2(F, \mathbf{Z})$ the fundamental class of F. The vectorbundle $(\xi, \omega = d\lambda)$ is a symplectic vector bundle and has therefore a distinguished class of compatible complex multiplications. Since the space $\mathcal{J}$ of such complex multiplications is contractible (see Lemma 1.4.1), the symplectic vector bundle (ξ, ω) has a well-defined first Chern class. Consider $c_1(\xi \mid F) \in H^2(F, \mathbf{Z})$ and the associated first Chern number $c_1(F) \in \mathbf{Z}$ defined by

$$c_1(F) := (c_1(\xi \mid F))([F])$$

Denote the *Euler-characteristic* of F by $\chi(F)$. The following proposition is well-known:

Proposition 3.5.2 *Let E be a complex vector bundle over an oriented surface F and Z a section of E with a finite number of zeros, which are non-degenerate, i.e. the linearization $Z'(s) : T_sF \to E_s$ is an isomorphism for every zero s of Z. Then the following formula holds*

$$(c_1(E))([F]) = \#\{s \mid Z'(s) \text{ is o. p.}\} - \#\{s \mid Z'(s) \text{ is o. r.}\} \tag{3.12}$$

Here o. p. (o. r.) stands for orientation-preserving (-reversing). □

Formula (3.12) above is also a possible definition of the first Chern number, which is the same as the Euler number. Note that if E is the tangent bundle of the surface F we observe that the left-hand side in the formula above is the Euler-characteristic (which is Hopf's theorem, see [62]). From this we obtain easily the following result:

Proposition 3.5.3 *Let F be a compact oriented surface without boundary in (M, λ). Then the following formulas hold:*

$$\begin{aligned} \chi(F) &= d_+ + d_- \\ c_1(F) &= d_+ - d_-. \end{aligned} \tag{3.13}$$

Proof The normal bundle of F is trivial since it is an orientable line bundle. Take any smooth function $H : M \to \mathbf{R}$ such that 0 is a regular value and $F = H^{-1}(0)$. Consider the associated contact Hamiltonian V_H. It defines, restricted to F, the vector field $\hat{V}_H$ which is tangent to F and lies also in the contact structure $\xi|_F$. The zeros of the vector field $\hat{V}_H$ correspond to the singular points of the characteristic foliation. The type of a singular point x_0 depends on the sign of the determinant of the linearization $\hat{V}'_H(x_0)$. If $\det \hat{V}'_H(x_0) > 0$ then x_0 is an elliptic singular point, if $\det \hat{V}'_H(x_0) < 0$ then x_0 is hyperbolic. As a map from $T_{x_0}F$ onto itself (endowed with the orientation of F), $\hat{V}'_H(x_0)$ is orientation preserving if x_0 is elliptic and orientation reversing if x_0 is hyperbolic. We apply Proposition 3.5.2 and find

$$\chi(F) = e_+ + e_- - h_+ - h_- = d_+ + d_-,$$

which is the first equation. If x_0 is a singular point then $T_{x_0}F = \xi_{x_0}$. We view now $\hat{V}'_H(x_0)$ as a map from $T_{x_0}F$ to ξ_{x_0}, the first space endowed with the orientation from F, the second endowed with the orientation from $d\lambda$. These two orientations are different if x_0 is a negative singularity. In order to compute $c_1(F)$, the negative singularities now contribute with the opposite sign as before, hence

$$c_1(F) = e_+ - e_- - h_+ + h_- = d_+ - d_-.$$

□

Daniel Bennequin obtained his inequality in the case of the three–sphere with standard contact structure. We will sketch the proof of a generalization due to Y. Eliashberg which holds for general tight contact manifolds.

Theorem 3.5.4 *Let (M, λ) be a contact manifold with a tight contact form λ. Assume that $F \subset M$ is an embedded oriented closed surface. If F is diffeomorphic to the 2-sphere then $c_1(F) = 0$. Otherwise we have the inequality*

$$|c_1(F)| \leq -\chi(F).$$

Observe that Proposition 3.5.3 gives us the following equalities for the case $F = S^2$

$$2 = d_+ + d_-, \tag{3.14}$$

$$c_1(F) = d_+ - d_-. \tag{3.15}$$

Hence Bennequin's inequality tells us that the tightness of the contact structure implies

$$d_+ = 1,$$

$$d_- = 1.$$

We need the following proposition:

Proposition 3.5.5 *Let (M, λ) be a three-manifold together with a tight contact form. Assume F is oriented and diffeomorphic to the 2-sphere. Given any $\varepsilon > 0$ there exists a C^0-small isotopy F' of F having precisely two singular points, both being nicely elliptic (one of them positive, the other one negative). Moreover for suitable local coordinates the singularities of F look like in Theorem 3.1.5. Further the oriented singular foliation is Morse-Smale and the following estimate holds*

$$| \, vol_\lambda(F') \, - \, vol_\lambda(F) \, | \leq \varepsilon.$$

We note that the perturbation method exhibited in the elimination theorem can be carried out in such a way that vol_λ changes as little as we wish. Moreover a small C^∞-perturbation also gives a small change of vol_λ.

Proof of Proposition 3.5.5 Via a C^0-small perturbation we may assume that the oriented singular foliation on F is Morse-Smale and that all the elliptic points are nicely elliptic. We may assume also that F is oriented in such a way that positive elliptic singularities are sources for the characteristic foliation and negative elliptic singularities are sinks. Let N be the number of singular points. Applying the elimination theorem successively (at most $[\frac{N}{2}]$-times) we may assume that the new surface F' has an oriented singular foliation of Morse-Smale type and that vol_λ differs by at most $\varepsilon/2$.

Moreover there are no connecting orbits between an hyperbolic and an elliptic singularity of the same sign. Since ξ is tight there cannot be any closed leaf P of the characteristic foliation. Indeed a component $\mathcal{D}$ bounded by P would be an overtwisted disk. Since

$$2 \, = \, d_+ + d_-$$

we must have at least two elliptic points on F'. We would like to see that there are exactly two elliptic points. Assume that there is a positive hyperbolic point h^+ and look at some orbit arriving at h^+. Because the foliation is Morse-Smale and there are no limit cycles, the α-limit set of this orbit must be a singular point x. There are no connecting orbits between hyperbolic points, so x must be a positive elliptic singularity. But this is not possible since there are no connecting orbits between hyperbolic and elliptic points of the same sign. In the same way we exclude the existence of a negative hyperbolic singularity h^- by considering an orbit starting from h^-. So there are only two elliptic points. Because there are no limit cycles they must have different signs. This completes the proof of Proposition 3.5.5. □

Now we are able to address Theorem 3.5.4. For more details we refer the reader to [33].

Let us first consider the case, where F is diffeomorphic to the 2–sphere. In this case we invoke Proposition 3.5.5 and we may find another surface F', isotopic to F, so that the characteristic foliation on F' has only two elliptic singularities of

opposite sign. Since $d_+ = d_- = 1$ we deduce that $c_1(F') = 0$. Clearly $c_1(F) = c_1(F')$, since F and F' are isotopic.

By a C^∞–small perturbation of the surface F, we may assume that the characteristic foliation on F is Morse–Smale. This of course does not change $c_1(F)$ and $\chi(F)$. We obtain the following identities from Proposition 3.5.3:

$$d_\pm = \frac{1}{2}(\chi(F) \pm c_1(F)).$$

There are surfaces F' and F'' isotopic to F, so that F' has no positive elliptic singular points and F'' has no negative elliptic singular points. Then Bennequin's inequality follows:

$$\begin{aligned}\frac{1}{2}(\chi(F) + c_1(F)) &= \frac{1}{2}(\chi(F') + c_1(F')) \\ &= d_+ \\ &= -h_+ \\ &\leq 0\end{aligned}$$

and

$$\begin{aligned}\frac{1}{2}(\chi(F) - c_1(F)) &= \frac{1}{2}(\chi(F'') - c_1(F'')) \\ &= d_- \\ &= -h_- \\ &\leq 0.\end{aligned}$$

Let us end this section with the following results which we will use later.

Theorem 3.5.6 *Let M be a closed three dimensional manifold with non-vanishing second homotopy group. Assume that λ is a tight contact form on M. Then there exists a non-contractible embedded surface F in M, diffeomorphic to S^2, such that the characteristic foliation is Morse-Smale.*

It has precisely two singularities, one of them positively nicely elliptic, the other negatively nicely elliptic. Each non-constant leaf of the characteristic foliation connects the two elliptic points.

Proof Because of the smooth sphere theorem there is an embedded oriented surface which is diffeomorphic to the 2-sphere. Now apply Theorem 3.5.5. A possible reference for the smooth version of the sphere theorem is [88], although they prove much more than what we need. □

Theorem 3.5.7 *Let M be a three dimensional manifold with overtwisted contact form λ. Then there exists an overtwisted disk $\mathcal{D}$ so that the characteristic foliation*

has exactly one singular point, an elliptic singularity e in the interior, and all leaves of the characteristic foliation connect e with $\partial\mathcal{D}$.

Proof This result can also be obtained by successively simplifying the characteristic foliation as follows: If $\mathcal{D}$ is just any overtwisted disk with singular points $d_\pm = e_\pm - h_\pm$, we double it and obtain a 2-sphere. Then

$$2 = \chi(S^2) = 2d_+ + 2d_-$$

so that

$$d_+ + d_- = 1.$$

We eliminate hyperbolic-elliptic pairs of singular points of the same sign. If a limit cycle appears we proceed with the smaller disk which has this limit cycle as its boundary. Iterating this procedure we obtain an overtwisted disk without any hyperbolic-elliptic pairs of singular points of the same sign. Then either $e_+ = 1$ or $e_- = 1$. □

Chapter 4
Finite Energy Planes and Periodic Orbits



4.1 Introduction

In this chapter we will prove the main result on finite energy planes due to H. Hofer [64] (see also [2]). Namely, given any manifold M equipped with a contact form λ, denote by $\xi \to M$ the associated contact structure and by X_λ the associated Reeb vector field. Let us fix any complex structure $J : \xi \to \xi$ compatible with $d\lambda|(\xi \oplus \xi)$ in the sense that

$$g_J(x)(h,k) \ := \ d\lambda(h, J(x)k)$$

defines a metric on the bundle $\xi \to M$ (we will write $|h|_J := g_J(h,h)^{1/2}$).

Let $\tilde{J}$ be the associated almost complex structure on $\mathbf{R} \times M$ defined by

$$\tilde{J}(a,u)(h,k) \ := \ (\,-\lambda(u)k\,,\ J(u)\pi_\lambda k + hX_\lambda(u)\,)$$

Define a Riemannian metric on $\mathbf{R} \times M$ by

$$g_{\tilde{J}}(\,(h_1,k_1),(h_2,k_2)\,) \ := \ h_1h_2 + \lambda(k_1)\lambda(k_2) + g_J(\pi_\lambda k_1, \pi_\lambda k_2),$$

where $\pi_\lambda : TM \longrightarrow \xi$ is the projection along the Reeb vector field X_λ.

We study solutions of

$$\begin{aligned} \tilde{u} = (a,u) : \mathbf{C} &\longrightarrow \mathbf{R} \times M \\ \tilde{u}_s + \tilde{J}(\tilde{u})\tilde{u}_t \ &= \ 0 \end{aligned} \tag{4.1}$$

having finite, but nontrivial, energy

$$0 < E(\tilde{u}) < +\infty \tag{4.2}$$

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with

$$E(\tilde{u}) = \sup_{\varphi \in \Sigma} \int_{\mathbf{C}} \tilde{u}^* d(\varphi\lambda) \tag{4.3}$$

and $\Sigma = \{\varphi \in C^\infty(\mathbf{R}, [0,1]) | \varphi' \geq 0\}$. We remark that Eq. (4.1) is equivalent to the one that we saw already in Chap. 1.4:

$$\pi_\lambda \partial_s u + J(u)\pi_\lambda \partial_t u = 0$$
$$(u^*\lambda) \circ i = da,$$

where $i\frac{\partial}{\partial s} = \frac{\partial}{\partial t}$ on $T_z\mathbf{C} = \mathbf{C}$. Moreover, the integrand in Eq. (4.3) is nonnegative:

$$\begin{aligned}
\tilde{u}^* d(\varphi\lambda) &= \tilde{u}^*(d\varphi \wedge \lambda + \varphi d\lambda) \\
&= \varphi'(a) da \wedge (u^*\lambda) + \varphi(a) u^* d\lambda \\
&= \Big(\varphi'(a)[a_s \lambda(u) u_t - a_t \lambda(u) u_s] + \\
&\quad + \frac{1}{2}\varphi(a)[\, |\pi_\lambda u_s|_J^2 + |\pi_\lambda u_t|_J^2 \,] \Big) ds \wedge dt \\
&= \Big(\frac{1}{2}\varphi'(a)[a_s^2 + a_t^2 + [\lambda(u)u_s]^2 + [\lambda(u)u_t]^2] + \\
&\quad + \frac{1}{2}\varphi(a)[\, |\pi_\lambda u_s|_J^2 + |\pi_\lambda u_t|_J^2 \,] \Big) ds \wedge dt.
\end{aligned}$$

The following theorem by H. Hofer [64] is the main result of this chapter.

Theorem 4.1.1 *Assume there exists a compact subset $K \subset M$ and a solution $\tilde{u} = (a, u)$ of (4.1), (4.2) with $u(\mathbf{C}) \subset K$. Then*

$$T := \int_{\mathbf{C}} u^* d\lambda > 0$$

and for every sequence $0 < R_k' \to +\infty$ there exists a subsequence $(R_k)_{k \in \mathbf{N}}$ such that

$$u(R_k e^{2\pi i \frac{t}{T}}) \longrightarrow x(t)$$

in C^∞ for $k \to \infty$, where x is a T-periodic solution of

$$\dot{x}(t) = X_\lambda(x(t)).$$

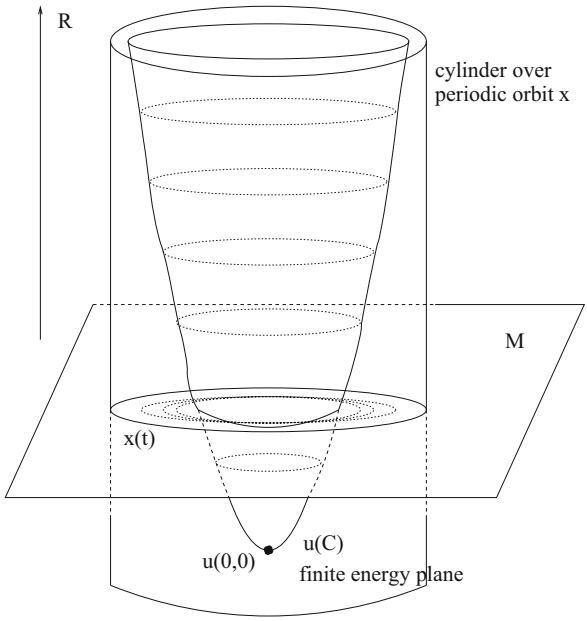


Fig. 4.1 A finite energy plane approaching asymptotically a cylinder over an isolated periodic orbit x of X_λ

We begin by deriving estimates for the linear Cauchy Riemann operator. Using this, we will prove under quite general assumptions that a uniform gradient bound for a family of J-holomorphic maps implies uniform C^∞-bounds.

Figure 4.1 should visualize the map $\tilde{u}$ of Theorem 4.1.1 which we will call a *finite energy plane*. The finite energy plane approaches asymptotically the cylinder $(s,t) \longmapsto (s, x(t)) \in \mathbf{R} \times M$ over a periodic orbit x. The reader easily verifies that the map $\tilde{v} : (s,t) \longmapsto (s, x(t))$ is a pseudoholomorphic plane with infinite energy $E(\tilde{v})$, but '$d\lambda$–energy' $\int_{\mathbf{C}} v^* d\lambda = 0$. The figure shows the idealistic case where there is convergence for all sequences $R_k \to +\infty$, i.e. there is only one asymptotic limit $\tilde{v}$. This actually happens if the periodic orbit x is isolated among all T–periodic orbits of X_λ.

4.2 L^2–Estimates for the Linear Cauchy Riemann Operator

Let $z = (z_1, \ldots, z_n)$ be the usual coordinates on $\mathbf{C}^n$. Write $z_j = q_j + ip_j$. The standard symplectic form ω on $\mathbf{C}^n$ is by definition

$$\omega = \sum_{k=1}^{n} dq_k \wedge dp_k.$$

If $(\eta, \tau) := \sum_{j=1}^{n} \eta_j \bar{\tau}_j$ is the Hermitian inner product on $\mathbf{C}^n$ we denote by $\langle \cdot, \cdot \rangle := \mathrm{Re}\,(\cdot, \cdot)$ the real inner product. We have the following formula

$$\omega(\eta, \tau) = \langle i\eta, \tau \rangle$$

Assume $\phi\colon \mathbf{C} \to \mathbf{C}^n$ is a compactly supported smooth map. Since $\omega = d\lambda$ for a suitable 1-form λ we see that by Stokes' Theorem applied on a large ball B_R containing $\mathrm{supp}(\phi)$

$$\int_{\mathbf{C}} \phi^*\omega = \int_{B_R} \phi^*\omega = \int_{\partial B_R} \phi^*\lambda = 0.$$

Observe that

$$\phi^*\omega = \omega(\phi_s, \phi_t) ds \wedge dt = \langle i\phi_s, \phi_t \rangle ds \wedge dt.$$

Then we compute with $\bar{\partial} = \frac{\partial}{\partial s} + i\frac{\partial}{\partial t}$

$$\begin{aligned} \left|\bar{\partial}\phi\right|^2 &= |\phi_s + i\phi_t|^2 \\ &= |\phi_s|^2 + |\phi_t|^2 + 2\mathrm{Re}(\phi_s, i\phi_t) \\ &= |\phi_s|^2 + |\phi_t|^2 + 2\langle \phi_s, i\phi_t \rangle \\ &= |\phi_s|^2 + |\phi_t|^2 - 2\langle i\phi_s, \phi_t \rangle, \end{aligned}$$

which implies that

$$\left|\bar{\partial}\phi\right|^2 ds \wedge dt = (|\phi_s|^2 + |\phi_t|^2) ds \wedge dt - 2\phi^*\omega.$$

Integrating this equality over $\mathbf{C}$ yields

$$\|\bar{\partial}\phi\|^2_{L^2(\mathbf{C})} = \|\phi_s\|^2_{L^2(\mathbf{C})} + \|\phi_t\|^2_{L^2(\mathbf{C})}. \tag{4.4}$$

Assume again that the support of ϕ is inside some ball $B_R = B_R(0)$ of radius R. Then

$$\phi(s,t) = \phi(-R,t) + \int_{-R}^{s} \phi_s(\sigma, t) d\sigma = \int_{-R}^{s} \phi_s(\sigma, t) d\sigma$$

and using Hölder's inequality

$$\begin{aligned}|\phi(s,t)|^2 &= \left|\int_{-R}^{s} \phi_s(\sigma,t)d\sigma\right|^2 \\ &= \left(\int_{-R}^{s} 1\cdot|\phi_s(\sigma,t)|\,d\sigma\right)^2 \\ &\le |s+R|\int_{-R}^{s}|\phi_s(\sigma,t)|^2 d\sigma \\ &\le |s+R|\int_{-R}^{R}|\phi_s(\sigma,t)|^2\,d\sigma.\end{aligned}$$

Integrating over the ball B_R we obtain

$$\begin{aligned}\|\phi\|^2_{L^2(\mathbf{C})} &= \|\phi\|^2_{L^2(B_R)} \\ &\le \int_{B_R}\left[|s+R|\cdot\int_{-R}^{R}|\phi_s(\sigma,t)|^2 d\sigma\right] ds\,dt \\ &\le \int_{-R}^{R}|s+R|ds\cdot\int_{B_R}|\phi_s(\sigma,t)|^2 d\sigma\,dt \\ &\le 4R^2\,\|\phi_s\|^2_{L^2(B_R)} \qquad (4.5)\\ &= 4R^2\,\|\phi_s\|^2_{L^2(\mathbf{C})}.\end{aligned}$$

Combining (4.4) and (4.5) yields the following estimate for $\phi\in C_0^\infty(B_R,\mathbf{C}^n)$, i.e. a smooth function $\phi\colon B_R\to\mathbf{C}^n$ with compact support in $\overset{\circ}{B}_R$ (0):

$$\|\phi_s\|^2_{L^2(\mathbf{C})}+\|\phi_t\|^2_{L^2(\mathbf{C})}+\|\phi\|^2_{L^2(\mathbf{C})}\le(4R^2+1)\,\|\bar\partial\phi\|^2_{L^2(\mathbf{C})}.$$

For $1\le p<\infty$ and Ω a domain in $\mathbf{C}$ we consider the Sobolev norm

$$\|\phi\|_{W^{1,p}(\Omega)} := \|\phi\|_{1,p,\Omega} := \left[\|\phi_s\|^p_{L^p(\Omega)}+\|\phi_t\|^p_{L^p(\Omega)}+\|\phi\|^p_{L^p(\Omega)}\right]^{1/p},$$

We summarize our discussion as follows:

Theorem 4.2.1 *For every $\phi\in C_0^\infty(B_R,\mathbf{C}^n)$ we have the estimate*

$$\|\phi\|^2_{1,2,\mathbf{C}}\le(4R^2+1)\,\|\bar\partial\phi\|^2_{0,2,\mathbf{C}}$$

where $\|\ \|_{0,2,\mathbf{C}}$ is the usual L^2-norm on $\mathbf{C}$. □

The same estimates remain true if '2' is replaced by $1 < p < \infty$ and $4R^2 + 1$ by a different constant.

Theorem 4.2.2 *Given $1 < p < \infty$, there exists a constant $c = c(p, R) > 1$ such that for every $\phi \in C_0^\infty(B_R, \mathbf{C}^n)$ we have the estimate*

$$\|\phi\|_{1,p,\mathbf{C}} \leq c\|\bar{\partial}\phi\|_{0,p,\mathbf{C}} \qquad \square$$

This result is much harder to prove and we refer the reader to [106] and [99].

Remark If we consider derivatives of ϕ one gets easily

$$\|\phi\|_{k,p,\mathbf{C}} \leq c\,\|\bar{\partial}\phi\|_{k-1,p,\mathbf{C}}$$

for any integer k.

In the following section we will use Theorem 4.2.2 to derive estimates for the nonlinear Cauchy Riemann operator.

4.3 Gradient Bounds Imply C^∞-Bounds

Let W be any $2n$-dimensional manifold equipped with an almost complex structure J. Denote by $K \subset W$ a compact subset. We will be interested mainly in $W = \mathbf{R} \times M$, where M is either a closed contact manifold or $\mathbf{R}^3$, hence we may assume by Whitney's embedding theorem that $W \subset \mathbf{R}^N$ for some sufficiently large N. So any map $u\colon \mathbf{C} \to W$ may be viewed as a map into $\mathbf{R}^N$. We will take advantage of that because we will consider Sobolev–spaces $W^{l,p}(\Omega, \mathbf{R}^N)$ and use the linear structure of $\mathbf{R}^N$. Let us introduce the following notation for the derivatives of u: Write $D^\alpha u$, where

$$D^\alpha = \left(\frac{\partial}{\partial s}\right)^{\alpha_1}\left(\frac{\partial}{\partial t}\right)^{\alpha_2}, \quad \alpha = (\alpha_1, \alpha_2) \in \mathbf{N}^2$$

Let $\Gamma = \Gamma(c, K)$ be a set of smooth solutions of

$$\begin{aligned} u &: D \longrightarrow W \\ u_s + J(u)\,u_t &= 0 \quad \text{on } D, \end{aligned}$$

where $D \subset \mathbf{C}$ is the unit disk, such that

$$\begin{aligned} \sup_{z \in D} |\nabla u(z)| &\leq c \\ u(D) &\subset K. \end{aligned}$$

The crucial estimate is the following:

Proposition 4.3.1 *Let $2 < p < \infty$. Given any $\delta \in (0,1)$ and $l \in \mathbf{N}$, then there exists a constant $d > 0$ depending on l, p, c, δ and K, such that*

$$\sup_{u \in \Gamma(c,K)} \|u\|_{l,p,B_\delta(0)} \leq d.$$

Remark The consequence of this result is the following. If there is a uniform C^1–bound on a set of solutions u of the nonlinear Cauchy Riemann equation then this implies a uniform C^∞_{loc} bound for all u in this set, i.e. for each integer $k \geq 1$ and each compact set K there is a u–uniform bound in $C^k(K)$. If a set of solutions of the nonlinear Cauchy Riemann equation fails to be compact in the C^∞_{loc}–topology then the reason for that must be the lack of a C^1–bound; existence of a uniform C^0–bound would then reduce the phenomenon of non compactness to the explosion of the gradient in the C^0–norm ('bubbling off'). The reason why we are using Sobolev–norms here while actually dealing with smooth functions is that there are regularity estimates for the Cauchy Riemann operator in those norms in contrast to the C^k–norms.

Proof For $l = 0$ and $l = 1$ the result is true by definition of the set $\Gamma(c, K)$. So assume it has been proved for $l \geq 1$. By induction we will show that it is true for $l + 1$. In order to prove the proposition it is actually sufficient to show that every sequence (u_k) in $\Gamma(c, K)$ has a subsequence which is bounded in the $W^{l+1,p}$–norm. Fix some number $\delta' \in (\delta, 1)$ and let $(u_k)_{k \in \mathbf{N}}$ be any sequence in $\Gamma(c, K)$.

Since $\|u_k\|_{1,p,B_{\delta'}(0)} \leq c_{\delta'}$ for some constant $c_{\delta'} > 0$ not depending on k, we may assume by the compact embedding

$$W^{1,p}(B_{\delta'}(0), \mathbf{R}^N) \hookrightarrow C^0(\overline{B_{\delta'}(0)}, \mathbf{R}^N)$$

that

$$u_k \longrightarrow u \text{ in } C^0(\overline{B_{\delta'}(0)}, \mathbf{R}^N)$$

after taking a subsequence. This implies in particular that the map u has image in W as well. For every $x_0 \in \overline{B_\delta(0)}$ we find $0 < \varepsilon_0 < \delta' - \delta$, an open neighborhood $U \subset W$ of $u(x_0)$ and a coordinate chart

$$\varphi : U \xrightarrow{\sim} \mathbf{R}^{2n} \approx \mathbf{C}^n$$

with $\varphi(u(x_0)) = 0$ such that

$$u_k(\overline{B_{\varepsilon_0}(x_0)}) \subset U$$

for all k large enough. Define now an almost complex structure $\hat{J}$ on $\varphi(U) \subset \mathbf{C}^n$ by

$$\hat{J}(z) := T\varphi(\varphi^{-1}(z)) \circ J(\varphi^{-1}(z)) \circ T\varphi^{-1}(z)$$

Composing φ with a real linear isomorphism we may even assume that $\hat{J}(0) = i$. Now $v_k := \varphi \circ u_k$ satisfies

$$\frac{\partial v_k}{\partial s} + \hat{J}(v_k)\frac{\partial v_k}{\partial t} = 0$$

on $B_{\varepsilon_0}(x_0)$. It is sufficient to prove that the sequence $(v_k)_{k\in\mathbf{N}}$ is bounded in $W^{l+1,p}(B_\varepsilon(x_0), \mathbf{R}^{2n})$ for some $\varepsilon \in (0, \varepsilon_0]$. Indeed, composing v_k with the smooth map φ^{-1}, we would obtain that the sequence $(u_k)_{k\in\mathbf{N}}$ is bounded in $W^{l+1,p}(B_\varepsilon(x_0), \mathbf{R}^N)$ as well. Since we can cover $B_\delta(0)$ by finitely many of these balls $B_\varepsilon(x_0)$ we will obtain the desired result. The reasons why we had to squeeze all the sets $u_k(\overline{B_\varepsilon(x_0)})$ into a single coordinate chart U of W are the following: First, we will compare the Sobolev norm of $J(u_k)$ with the one of u_k, i.e. we will apply Proposition 4.3.2 below,[1] and we will use that in local coordinates J may be viewed as a smooth map $\varphi(U) \to \mathrm{GL}(\mathbf{R}^{2n})$. Second, we have to compare the nonlinear Cauchy Riemann operator with the standard linear one, for which we have estimates. We will need the fact that $J(u_k(x))$ does not differ much from the constant $J(u(x_0)) \approx i$ if x is close to x_0.

Summing up, we have arrived at the following situation

$$v_k : \overline{B_{\varepsilon_0}(x_0)} \longrightarrow \mathbf{C}^n,$$

$$\partial_s v_k + \hat{J}(v_k)\partial_t v_k = 0,$$

$$v_k(x_0) \longrightarrow 0 \text{ since } u_k(x_0) \to u(x_0),$$

the sequence $(v_k)_{k\in\mathbf{N}}$ converges in $C^0(\overline{B_{\varepsilon_0}(x_0)}, \mathbf{C}^n)$ to some v and

$$\|D^\alpha v_k\|_{L^p(B_{\varepsilon_0}(x_0))} \le c_l \text{ for } 1 \le |\alpha| \le l$$

[1] **Propositions 4.3.2** Let $\Omega \subset \mathbf{R}^n$ be a bounded domain with Lipschitz boundary and $u \in W^{m,p}(\Omega, \mathbf{R}^l)$ with $mp > n$. Moreover let $f \in C^\infty(\mathbf{R}^l, \mathbf{R}^k)$ with bounded derivatives up to order m. Then $f \circ u \in W^{m,p}(\Omega, \mathbf{R}^k)$ and there is a constant $c > 0$ so that

$$\|f \circ u - f(0)\|_{m,p,\Omega} \le c\,\|u\|_{m,p,\Omega}$$

for all $u \in W^{m,p}(\Omega)$.

□

for suitable positive constants c_l. Take a smooth function $\beta\colon [0,+\infty) \to [0,1]$ such that

$$\begin{aligned} \beta(s) &= 1 \text{ for } 0 \le s \le \tfrac{1}{2} \\ \beta(s) &= 0 \text{ for } \quad s \ge 1 \end{aligned}$$

and $\beta'(s) < 0$ if $\frac{1}{2} < s < 1$. For $\varepsilon \in (0, \varepsilon_0/2]$ define

$$\beta_\varepsilon(z) := \beta\left(\frac{|z - x_0|}{2\varepsilon}\right)$$

so that $\operatorname{supp}(\beta_\varepsilon) \subset B_{2\varepsilon}(x_0) \subset B_{\varepsilon_0}(x_0)$ and $\beta_\varepsilon \equiv 1$ on $B_\varepsilon(x_0)$. We compute with $\|.\|_{l,p} := \|.\|_{l,p,B_\delta(0)}$ and c being the constant from Theorem 4.2.2.

$$\begin{aligned} c^{-1}\|v_k\|_{l+1,p,B_\varepsilon(x_0)} &\le c^{-1}\|\beta_\varepsilon v_k\|_{l+1,p} \\ &\le \|\bar\partial(\beta_\varepsilon v_k)\|_{l,p} \\ &\le \|(\beta_\varepsilon v_k)_s + \hat J(v_k)(\beta_\varepsilon v_k)_t\|_{l,p} + \\ &\quad + \|(i - \hat J(v_k))(\beta_\varepsilon v_k)_t\|_{l,p} \qquad (4.6) \\ &\le \|\beta_\varepsilon[(v_k)_s + \hat J(v_k)(v_k)_t]\|_{l,p} + \\ &\quad + \|(\beta_\varepsilon)_s v_k + (\beta_\varepsilon)_t \hat J(v_k) v_k\|_{l,p} \\ &\quad + \|(i - \hat J(v_k))(\beta_\varepsilon v_k)_t\|_{l,p} \\ &= \|(\beta_\varepsilon)_s v_k + (\beta_\varepsilon)_t \hat J(v_k) v_k\|_{l,p} + \\ &\quad + \|(i - \hat J(v_k))(\beta_\varepsilon v_k)_t\|_{l,p}. \end{aligned}$$

The first expression

$$\|(\beta_\varepsilon)_s v_k + (\beta_\varepsilon)_t \hat J(v_k) v_k\|_{l,p}$$

is bounded by a constant $c(\varepsilon)$ depending on ε because of the induction hypothesis. We have used that $W^{k,p}$ has the Banach algebra property for $k \ge 1$, $p > 2$ and the fact that $\|\hat J(v_k)\|_{l,p} \le c\|v_k\|_{l,p} + \text{const}$, which is Proposition 4.3.2. The only parts of the second expression

$$\|(i - \hat J(v_k))(\beta_\varepsilon v_k)_t\|_{l,p}$$

which are not bounded by the induction hypothesis are terms of the form

$$(i - \hat J(v_k)) D^\alpha (\beta_\varepsilon v_k)_t \qquad (4.7)$$

with $|\alpha| = l$ which also contain derivatives of v_k of degree $l+1$. The others are of the form

$$D^{\beta}(i - \hat{J}(v_k))D^{\alpha-\beta}(\beta_\varepsilon v_k)_t \tag{4.8}$$

with $0 < |\beta| \leq |\alpha| = l$. Hence all the expressions as in (4.8) can be bounded in L^p by the induction hypothesis. For expressions of the form (4.7) we obtain

$$\|(i - \hat{J}(v_k))D^{\alpha}(\beta_\varepsilon v_k)_t\|_{L^p(B_\delta(0))} \leq \|i - \hat{J}(v_k)\|_{L^\infty(B_{2\varepsilon}(x_0))}\|\beta_\varepsilon v_k\|_{l+1,p}. \tag{4.9}$$

Combining (4.6) and (4.9) we conclude

$$c^{-1}\|\beta_\varepsilon v_k\|_{l+1,p} \leq c(\varepsilon) + \|i - \hat{J}(v_k)\|_{L^\infty(B_{2\varepsilon}(x_0))}\|\beta_\varepsilon v_k\|_{l+1,p}.$$

We note that for $k \to \infty$

$$\|i - \hat{J}(v_k)\|_{L^\infty(B_{2\varepsilon}(x_0))} \longrightarrow \|i - \hat{J}(v)\|_{L^\infty(B_{2\varepsilon}(x_0))}.$$

Moreover, $\|i - \hat{J}(v)\|_{L^\infty(B_{2\varepsilon}(x_0))} \to 0$ for $\varepsilon \to 0$. So we end up with the estimate

$$(c^{-1} - \delta(\varepsilon) - \tau(k))\,\|\beta_\varepsilon v_k\|_{l+1,p} \leq c(\varepsilon),$$

where $\delta(\varepsilon) \to 0$ as $\varepsilon \to 0$ and $\tau(k) \to 0$ as $k \to \infty$. For $\varepsilon > 0$ small and k large enough we can arrange that

$$c^{-1} - \delta(\varepsilon) - \tau(k) > \frac{1}{2c} > 0.$$

Then

$$\begin{aligned} \|v_k\|_{l+1,p,B_\varepsilon(x_0)} &\leq \|\beta_\varepsilon v_k\|_{l+1,p} \\ &\leq \frac{c(\varepsilon)}{c - \delta(\varepsilon) - \tau(k)} \\ &\leq 2c\,c(\varepsilon) \end{aligned}$$

and we obtain the desired bound. □

By the Sobolev embedding theorem we have a (compact) embedding

$$W^{k+1,p}(\Omega) \hookrightarrow C^k(\bar{\Omega})$$

where Ω is any bounded 2-dimensional domain with Lipschitz boundary and $2 < p < \infty$. Hence the previous result gives us bounds for the C^k-Norms:

Theorem 4.3.3 *Given any $\delta \in (0,1)$ and $l \in \mathbf{N}$, there exists a constant d' depending on l, c, δ and K such that*

$$\|u\|_{C^l(\overline{B_\delta(0)})} \leq d'$$

for every $u \in \Gamma(c, K)$. □

We would like to apply these results to the case $W = \mathbf{R} \times M$ where M is a closed $(2n+1)$-dimensional manifold with contact form λ (or $\mathbf{R}^{2n+1}$). Let $\xi \to M$ be the associated contact structure and $J : \xi \to \xi$ a complex multiplication compatible with $d\lambda$. As before denote by $\tilde{J}$ the associated almost complex structure on $\mathbf{R} \times M$.

For $K \subset M$ being a compact subset and $c > 0$ we are interested in sets of the form

$$\Gamma'(c, K) = \left\{\tilde{u} = (a, u) \colon D \to \mathbf{R} \times M \mid \quad \tilde{u} \text{ satisfies (4.10) below}\right\},$$

where

$$\begin{aligned} \tilde{u}_s + \tilde{J}(\tilde{u})\, \tilde{u}_t &= 0 \qquad \text{on } D \\ \left|\frac{\partial \tilde{u}}{\partial s}\right|^2 + \left|\frac{\partial \tilde{u}}{\partial t}\right|^2 &\leq c^2 \quad \text{on } D \\ u(D) &\subset K. \end{aligned} \tag{4.10}$$

We assume again that $\mathbf{R} \times M \subset \mathbf{R}^N$ for some sufficiently large N. Observe that if $\tilde{u} = (a, u) \in \Gamma'(c, K)$ then also $\tilde{u}_\tau = (a + \tau, u) \in \Gamma'(c, K)$, where $\tau \in \mathbf{R}$. For this reason there can not be any uniform C^0-bound for $\tilde{u}$ if $\Gamma'(c, K) \neq \emptyset$. Nevertheless, the following is true.

Theorem 4.3.4 *Given $\delta \in (0,1)$ and $\alpha \in \mathbf{N}^2$ with $|\alpha| \geq 1$ then there exits a constant $d = d(c, K, \alpha, \delta) > 0$ such that*

$$\|D^\alpha \tilde{u}\|_{C^0(\overline{B_\delta(0)})} \leq d$$

for all $\tilde{u} \in \Gamma'(c, K)$.

Proof For every $|\alpha| \geq 1$ we have

$$\|D^\alpha \tilde{u}_\tau\|_{C^0} = \|D^\alpha \tilde{u}\|_{C^0}.$$

So we only need to consider the subset $\Gamma^*(c, K)$ of $\Gamma'(c, K)$ consisting of all $\tilde{u} = (a, u)$ with

$$a(0) = 0.$$

Since we have the uniform gradient bound (4.10), we obtain a C^0-bound for $\Gamma^*(c, K)$. Now apply Theorem 4.3.3 to the set $\Gamma^*(c, K)$. □

4.4 The Main Result for Finite Energy Planes

Take M, λ, ξ as before and let $\pi_\lambda : TM \to \xi$ be the projection along the Reeb vector field X_λ. Again we fix any compatible complex structure $J : \xi \to \xi$ for the contact structure $\xi \to M$ and we study the solutions of

$$\begin{aligned} \pi_\lambda u_s + J(u)\pi_\lambda u_t &= 0 \\ (u^*\lambda) \circ i &= da, \end{aligned} \tag{4.11}$$

which is equivalent to

$$\tilde{u}_s + \tilde{J}(\tilde{u})\tilde{u}_t = 0,$$

where $\tilde{u} = (a, u) : \mathbf{C} \to \mathbf{R} \times M$. We assume that there exists a compact subset $K \subset M$ such that

$$u(\mathbf{C}) \subset K. \tag{4.12}$$

With

$$\Sigma = \left\{\varphi \in C^\infty(\mathbf{R}, [0, 1]) | \varphi' \geq 0\right\}$$

let

$$E(\tilde{u}) = \sup_{\varphi \in \Sigma} \int_{\mathbf{C}} \tilde{u}^* d(\varphi\lambda).$$

We recall that $\tilde{u}^* d(\varphi\lambda)$ is a nonnegative integrand if $\tilde{u}$ solves (4.11). We will prove now the main result of this chapter:

Theorem 4.4.1 *Let $\tilde{u} : \mathbf{C} \to \mathbf{R} \times M$ be a solution of (4.11) satisfying also (4.12). If moreover*

$$0 < E(\tilde{u}) < +\infty$$

then $T := \int_{\mathbf{C}} u^* d\lambda > 0$ *and for every sequence* $0 < R'_k \to \infty$ *there exists a subsequence* $(R_k)_{k\in\mathbf{N}}$ *such that the* C^∞*–limit*

$$x(t) := \lim_{k\to\infty} u(R_k e^{2\pi i \frac{t}{T}})$$

exists and defines a T*-periodic solution of*

$$\dot{x}(t) = X_\lambda(x(t)).$$

We split the theorem into several parts.

Proposition 4.4.2 *Let* $\tilde{u} : \mathbf{C} \to \mathbf{R} \times M$ *solve* (4.11)*. Assume moreover that*

$$\int_{\mathbf{C}} u^* d\lambda = 0$$

and

$$E(\tilde{u}) < +\infty.$$

Then $\tilde{u}$ *is constant.*

Before we start with the proof we note the following corollary

Corollary 4.4.3 *Let* $\tilde{u} : \mathbf{C} \to \mathbf{R} \times M$ *be a non constant solution of* (4.11) *with finite energy* $E(\tilde{u})$*. Then*

$$\int_{\mathbf{C}} u^* d\lambda > 0.$$

The pseudoholomorphic plane $(s,t) \longmapsto (s, x(t))$, with x being any trajectory of X_λ, shows that the assumption of finite energy in Proposition 4.4.2 cannot be dropped.

Proof of Proposition 4.4.2 We know that

$$0 = \int_{\mathbf{C}} u^* d\lambda = \frac{1}{2} \int_{\mathbf{C}} (|\pi_\lambda u_s|_J^2 + |\pi_\lambda u_t|_J^2) ds \wedge dt.$$

This means precisely that $|\pi_\lambda u_s|_J^2 = |\pi_\lambda u_t|_J^2 = 0$ or $u^* d\lambda = 0$. Hence

$$\pi_\lambda \circ Du(z) : \mathbf{C} \to \xi_{u(z)}$$

is the zero map for every $z \in \mathbf{C}$ which means that $\partial_s u$ and $\partial_t u$ are both parallel to $X_\lambda(u)$:

$$\partial_s u = (\lambda(u)\partial_s u)\, X_\lambda(u)\ ,\ \partial_t u = (\lambda(u)\partial_t u)\, X_\lambda(u).$$

The 1–form $u^*\lambda$ on $\mathbf{C}$ is closed and therefore exact, hence there is a function f on $\mathbf{C}$ so that $df = u^*\lambda$. We recall that also $u^*\lambda = -da \circ i$ which is part of the nonlinear Cauchy Riemann equation. Now $df = -da \circ i$ is equivalent to the function $\Phi = a + if : \mathbf{C} \to \mathbf{C}$ being holomorphic. In particular, the function a is harmonic.

Consider the 2-form τ on $\mathbf{C}$ defined by

$$\tau := d(\phi(s)\, dt)$$

where $z = s + it$ and $\phi \in \Sigma$. We compute

$$\begin{aligned}\int_{\mathbf{C}} \tilde{u}^* d(\phi\lambda) &= \int_{\mathbf{C}} d(\phi(a)\, df) \\ &= \int_{\mathbf{C}} \Phi^* d(\phi(s) dt) \\ &= \int_{\mathbf{C}} \Phi^*\tau \le c < +\infty\end{aligned}$$

for all $\phi \in \Sigma$. Let us summarize what has happened so far. The derivative f_s is just the component of $\partial_s u$ along the Reeb vector field, similarly for the t–derivative. If we view f as a map of $\mathbf{C}$ into the 1–dimensional contact manifold $\mathbf{R}$ with contact form dt (t being the coordinate on $\mathbf{R}$) and Reeb vector field $\frac{d}{dt}$ then Φ can be viewed as a 'pseudoholomorphic plane' in the symplectisation $(\mathbf{R} \times \mathbf{R} \approx \mathbf{C}, \tau)$ of $(\mathbf{R}, dt)$. So we have replaced the pseudoholomorphic plane $\tilde{u}$ in $\mathbf{R} \times M$ by the genuinely holomorphic plane Φ in the symplectisation $\mathbf{C}$ of a 1–dimensional contact manifold with the feature that both planes have the same energy and one is constant if and only if the other is constant. The condition $\int u^* d\lambda = 0$ says that we may discard all but two dimensions of the symplectisation.

If Φ is constant then a is constant. Then we conclude immediately from $(u^*\lambda) \circ i = da = 0$ that $Du(z) : \mathbf{C} \to T_{u(z)}M$ has image in $\xi_{u(z)} = \ker \lambda(u(z))$. On the other hand we know that $\pi_\lambda \circ Du(z)$ is the zero map, so u must be constant too.

Now we have to show that our assumptions will always yield a constant Φ. Assume first that Φ is non constant and has bounded derivatives. Then Φ is an affine map by Liouville's theorem, i.e.

$$\Phi(z) = \alpha z + \beta$$

with $\alpha \neq 0$. Hence Φ is biholomorphic and we obtain

$$\text{const} \cdot \int_{\mathbf{C}} \Phi^*\tau = \int_{\mathbf{C}} \tau = \int_{\mathbf{C}} \phi'(s) ds \wedge dt,$$

but this integral is $+\infty$ for non constant $\phi \in \Sigma$ which contradicts the assumption $E(\tilde{u}) < +\infty$. Hence we are left with the proof that Φ can not have unbounded derivatives either. Arguing indirectly, we pick sequences $0 < (\varepsilon_k)_{k\in\mathbf{N}} \subset \mathbf{R}$ and

$(z_k)_{k\in\mathbf{N}} \subset \mathbf{C}$ such that

$$R_k := |\nabla\Phi(z_k)| \to +\infty \tag{4.13}$$

and

$$\begin{aligned} R_k\varepsilon_k &\longrightarrow +\infty \\ \varepsilon_k &\longrightarrow 0. \end{aligned} \tag{4.14}$$

We will now use a so–called 'bubbling–off' argument in order to show that explosion of the gradient of Φ contradicts the assumption that the energy is finite. This argument works for all maps satisfying an elliptic partial differential equation invariant under conformal transformations of the domain, like the (non)linear Cauchy Riemann equation, the Yang–Mills equation, the harmonic map equation or the Yamabe–equation. The phenomenon of 'bubbling' was first discovered by Sacks and Uhlenbeck for harmonic maps [98]. Bubbling is intimately connected with the compactness of the spaces of solutions of the above differential equations. However, we are considering a very simple case at the moment because our map Φ is holomorphic.

We need the following lemma due to H. Hofer [63]:

Lemma 4.4.4 *Let (X, d) be a complete metric space. Then for every continuous map $f\colon X \to [0, +\infty)$ the following holds: Given $\varepsilon_0 > 0$, $x_0 \in X$ there exist $\varepsilon \in (0, \varepsilon_0]$, $x \in X$ with*

(a) $\varepsilon f(x) \geq \varepsilon_0 f(x_0)$,
(b) $d(x, x_0) \leq 2\varepsilon$,
(c) $f(y) \leq 2f(x)$ *for all y with* $d(y, x) \leq \varepsilon$.

Remark The converse of the lemma is also true, i.e. it provides a characterization of complete metric spaces.

Proof Define $\varepsilon := \varepsilon_0$ and $x := x_0$ if condition (c) holds for $\varepsilon = \varepsilon_0$. Otherwise put $\varepsilon_1 := \frac{\varepsilon_0}{2}$ and pick x_1 with $d(x_1, x_0) \leq \varepsilon_0$ and $f(x_1) > 2f(x_0)$. Set $\varepsilon := \varepsilon_1, x := x_1$ if (c) holds for $\varepsilon = \varepsilon_1$. Otherwise define $\varepsilon_2 := \frac{\varepsilon_1}{2}$ and pick x_2 with $d(x_2, x_1) \leq \varepsilon_1$ and $f(x_2) > 2f(x_1)$. If this procedure stops after finitely many iterations then we are done, otherwise we will get infinite sequences $(x_k)_{k\in\mathbf{N}}$ and $(\varepsilon_k)_{k\in\mathbf{N}}$. In this case, we estimate for $l \geq 1$

$$d(x_{k+l}, x_k) \leq \sum_{j=k}^{k+l-1} d(x_{j+1}, x_j) \leq \sum_{j=k}^{\infty} \varepsilon_j = \sum_{j=k}^{\infty} \frac{\varepsilon_0}{2^j} = \frac{\varepsilon_0}{2^{k-1}}.$$

Hence $(x_k)_{k\in\mathbf{N}}$ is a Cauchy sequence and has a limit by completeness of X. Let $x_\infty := \lim_{k\to\infty} x_k$. Since $f(x_1) > 0$ we deduce

$$f(x_{k+1}) > 2^k f(x_1)$$

implying that $\lim_{k\to\infty} f(x_k) = +\infty$. On the other hand $\lim_{k\to\infty} f(x_k) = f(x_\infty) \in \mathbf{R}^+$ by continuity. This contradiction shows that the iteration above stops at some k_0 such that

$$f(y) \le 2f(x_{k_0}) \quad \text{for all } y \text{ with } d(y, x_{k_0}) \le \varepsilon_{k_0}.$$

By construction

$$\varepsilon_{k_0} f(x_{k_0}) = \frac{\varepsilon_0}{2^{k_0}} f(x_{k_0}) \ge \frac{\varepsilon_0}{2^{k_0}} 2^{k_0} f(x_0) = \varepsilon_0 f(x_0)$$

and

$$d(x_{k_0}, x_0) \le \sum_{j=0}^{k_0-1} \varepsilon_j \le \sum_{j=0}^{\infty} \frac{\varepsilon_0}{2^j} \le 2\varepsilon_0$$

Put $x := x_{k_0}$, $\varepsilon := \varepsilon_{k_0}$ and the proof of the lemma is finished. □

Returning to the proof of Proposition 4.4.2, we recall that

$$\begin{aligned} R_k := |\nabla\Phi(z_k)| &\longrightarrow +\infty \\ R_k\varepsilon_k &\longrightarrow +\infty \\ 0 < \varepsilon_k &\longrightarrow 0 \end{aligned} \tag{4.15}$$

By applying the lemma to the continuous function $f(z) := |\nabla\Phi(z)|$, we may choose new sequences $(\varepsilon'_k)_{k\in\mathbf{N}}$ and $(z'_k)_{k\in\mathbf{N}}$ so that (4.15) still holds and in addition there is an estimate of $|\nabla\Phi|$ on the ε'_k-neighborhood of z'_k. Therefore, we may assume without loss of generality that

$$|\nabla\Phi(z)| \le 2\,|\nabla\Phi(z_k)| \quad \text{if } |z - z_k| \le \varepsilon_k. \tag{4.16}$$

Define now maps $\Phi_k : \mathbf{C} \to \mathbf{C}$ by conformal rescaling

$$\Phi_k(z) = \Phi(z_k + \frac{z}{R_k}) - \Phi(z_k).$$

Then the maps Φ_k are entire holomorphic maps with

$$|\nabla\Phi_k(0)| = 1\,,\ \Phi_k(0) = 0 \tag{4.17}$$

and

$$|\nabla \Phi_k(z)| \leq 2 \quad \text{on } B_{\varepsilon_k R_k}(0). \tag{4.18}$$

We see that for every compact subset $K \subset \mathbf{C}$ the sequence $(\Phi_k)_{k\in\mathbf{N}}$ is uniformly bounded in $C^\infty(K)$: Use the gradient bound and $\Phi_k(0) = 0$ to obtain uniform C^0–bounds and Cauchy's integral formula for the higher derivatives. After passing to a subsequence we may assume by the Arzela-Ascoli theorem that the sequence $(\Phi_k)_{k\in\mathbf{N}}$ converges with all derivatives on compact sets, i.e. $\Phi_k \to \Psi$ in $C^\infty_{loc}(\mathbf{C}, \mathbf{C})$ where $\Psi : \mathbf{C} \to \mathbf{C}$ is holomorphic and satisfies

$$|\nabla \Psi(0)| = 1,$$
$$|\nabla \Psi(z)| \leq 2 \text{ on every ball } B_R(0).$$

Hence Ψ is an affine non constant map. Now we take a non constant function $\phi \in \Sigma$ and define $\phi_k(s) := \phi(s - \mathrm{Re}(\Phi(z_k)))$. We calculate with $\tau_\phi := d\,(\phi(s)\,dt)$ using $\Phi^*\tau_{\phi_k} = \tilde{u}^* d(\phi_k \lambda)$:

$$\begin{aligned}
\int_{B_{\varepsilon_k R_k}(0)} \Phi_k^* \tau_\phi &= \int_{B_{\varepsilon_k}(z_k)} \Phi^* \tau_{\phi_k} \\
&\leq \int_{\mathbf{C}} \Phi^* \tau_{\phi_k} \\
&= \int_{\mathbf{C}} \tilde{u}^* d(\phi_k \lambda) \\
&\leq E(\tilde{u}) \\
&< +\infty.
\end{aligned}$$

Fix $R > 0$. If k is sufficiently large then $R_k \varepsilon_k \geq R$. We have

$$\int_{B_R(0)} \Phi_k^* \tau_\phi \longrightarrow \int_{B_R(0)} \Psi^* \tau_\phi$$

for $k \to \infty$. This implies that for every $R > 0$

$$E(\tilde{u}) \geq \int_{B_R(0)} \Psi^* \tau_\phi$$

and therefore

$$E(\tilde{u}) \geq \int_{\mathbf{C}} \Psi^* \tau_\phi.$$

Since Ψ is affine and non constant we see that the right hand side equals $+\infty$ because ϕ is not constant. This is a contradiction, hence we have finished the proof of Proposition 4.4.2.

We want to remark that we did not need to demand for $\varepsilon_k \searrow 0$ at the beginning of the proof, $\varepsilon_k \equiv 1$ would have been fine. Our choice simply emphasizes that Lemma 4.4.4 adds property (4.16) to properties (4.15). □

Let us make some remarks about gradients of pseudoholomorphic maps before we continue. Lemma 4.4.4 and reparameterisation of Φ gave us a sequence of holomorphic maps $\Phi_k : \mathbf{C} \to \mathbf{C}$ with uniform gradient bound

$$|\nabla \Phi_k(z)| \leq 2 \text{ on } B_{\varepsilon_k R_k}(0)$$

and $\varepsilon_k R_k \to +\infty$. Since also $\Phi_k(0) = 0$ we got k–uniform bounds in C^0 on each compact set as well. Using Cauchy's integral formula we could bound the sequence Φ_k uniformly in C^∞ on compact subsets of $\mathbf{C}$. Then we obtained convergence in C^∞_{loc} to some map Ψ. Finally we obtained a contradiction by considering the energy: Ψ does not exist.

Later, we will consider gradients of pseudoholomorphic maps and we will proceed in the same way. We will only need a different argument to go from gradient bounds to C^∞ bounds, replacing Cauchy's integral formula. This is where the results from last section come into play.

Next, we will use Proposition 4.4.2 to derive gradient bounds for finite energy planes. Let $\phi\colon \mathbf{R} \times S^1 \to \mathbf{C} \setminus \{0\}$ be the biholomorphic map defined by

$$\phi(s,t) = e^{2\pi(s+it)}$$

where $S^1 = \mathbf{R}/\mathbf{Z}$. Let $\tilde{u} = (a,u)$ be a finite energy plane with nontrivial energy (as in Theorem 4.4.1). It will be more convenient to consider a $\tilde{J}$-holomorphic cylinder instead of a plane, so we define

$$\tilde{v}\colon \mathbf{R} \times S^1 \to \mathbf{R} \times M$$

by

$$\tilde{v} = \tilde{u} \circ \phi.$$

Then

$$\begin{aligned} &\tilde{v}_s + \tilde{J}(\tilde{v})\tilde{v}_t = 0 \quad \text{on } \mathbf{R} \times S^1 \\ &\int_{\mathbf{R}\times S^1} v^* d\lambda = \int_{\mathbf{C}} u^* d\lambda > 0 \\ &0 < E(\tilde{v}) = E(\tilde{u}) < \infty \\ &v(\mathbf{R} \times S^1) \subset K, \end{aligned} \tag{4.19}$$

where $K \subset M$ is a compact subset.

Proposition 4.4.5 *Let $\tilde{v}$ be a solution of* (4.19). *Then*

$$|\nabla \tilde{v}(s,t)| \leq c$$

for all $(s,t) \in \mathbf{R} \times S^1$ *for a suitable constant* $c > 0$.

Proof The argument is quite similar to the one used in the proof of Proposition 4.4.2, where we assumed that the gradient of Φ is not bounded. Clearly if $(s_k, t_k) \in \mathbf{R} \times S^1$ with

$$R_k := |\nabla \tilde{v}(s_k, t_k)| \to +\infty$$

we must have $s_k \to +\infty$ since the gradient is bounded on $\mathbf{R}^- \times S^1$ (note that ϕ maps $\mathbf{R}^- \times S^1$ onto the unit disk without the origin $D\backslash\{0\} \subset \mathbf{C}$ but $\sup_D |\nabla \tilde{u}| < \infty$). View $\tilde{v} : \mathbf{R}^2 \approx \mathbf{C} \to \mathbf{R} \times M$ as being 1-periodic in the second argument. Applying Lemma 4.4.4 to R_k and a sequence $(\varepsilon_k)_{k \in \mathbf{N}}$ with

$$\begin{aligned} \varepsilon_k &\longrightarrow 0, \quad \varepsilon_k > 0 \\ \varepsilon_k R_k &\longrightarrow +\infty \end{aligned}$$

we may also assume that

$$|\nabla \tilde{v}(s,t)| \leq 2\,|\nabla \tilde{v}(s_k, t_k)| \quad \text{for } (s - s_k)^2 + (t - t_k)^2 \leq \varepsilon_k^2.$$

Define with $z_k := (s_k, t_k)$, $s_k \to +\infty$, $t_k \in [0,1]$

$$\begin{aligned} \tilde{v}_k(z) &:= (b_k(z), v_k(z)) \\ &:= \left(b(z_k + \frac{z}{R_k}) - b(z_k), v(z_k + \frac{z}{R_k}) \right), \end{aligned}$$

where we wrote $\tilde{v} = (b, v) : \mathbf{C} \to \mathbf{R} \times M$. Then

$$\begin{aligned} |\nabla \tilde{v}_k(0)| &= 1 \\ |\nabla \tilde{v}_k(z)| &\leq 2 \quad \text{for } z \in B_{\varepsilon_k R_k}(0) \\ \frac{\partial \tilde{v}_k}{\partial s} + \tilde{J}(\tilde{v}_k) \frac{\partial \tilde{v}_k}{\partial t} &= 0 \quad \text{on } \mathbf{C}. \end{aligned} \tag{4.20}$$

Moreover with $\varphi_k \in \Sigma$ given by $\varphi_k(s) = \varphi(s - b(z_k))$, where $\varphi \in \Sigma$, we find for large k

$$\begin{aligned} \int_{B_{R_k\varepsilon_k}(0)} \tilde{v}_k^* d(\varphi\lambda) &= \int_{B_{\varepsilon_k}(z_k)} \tilde{v}^* d(\varphi_k\lambda) \\ &\leq \int_{\mathbf{R}\times[0,1]} \tilde{v}^* d(\varphi_k\lambda) \\ &\leq E(\tilde{v}) \\ &< +\infty. \end{aligned} \tag{4.21}$$

For every $R > 0$ we have

$$\int_{B_R(0)} v_k^* d\lambda = \int_{B_{R/R_k}(z_k)} v^* d\lambda \overset{k\to\infty}{\longrightarrow} 0 \tag{4.22}$$

since

$$\int_{\mathbf{R}\times[0,1]} v^* d\lambda = \int_{\mathbf{R}\times[0,1]} \tilde{v}^* d(\phi_0\lambda) \leq E(\tilde{v}) < +\infty,$$

where $\phi_0 \equiv 1$ and $R/R_k \to 0$. Because of $v_k(\mathbf{C}) \subset K$ and $b_k(0) = 0$, we deduce from (4.20), Theorem 4.3.4 and the Ascoli–Arzela theorem that a subsequence of $(\tilde{v}_k)$ converges in C^∞_{loc} to some

$$\tilde{w} = (\beta, w) \colon \mathbf{C} \to \mathbf{R} \times M$$

satisfying

$$\begin{aligned} \tilde{w}_s + \tilde{J}(\tilde{w})\tilde{w}_t &= 0 \quad \text{on } \mathbf{C} \\ |\nabla\tilde{w}(0)| &= 1 \\ |\nabla\tilde{w}(z)| &\leq 2 \quad \text{for } z \in \mathbf{C}. \end{aligned} \tag{4.23}$$

Employing (4.21) gives in addition

$$E(\tilde{w}) \leq E(\tilde{u}) < +\infty \tag{4.24}$$

and we have because of (4.22)

$$\int_{\mathbf{C}} w^* d\lambda = 0. \tag{4.25}$$

By Proposition 4.4.2 $\tilde{w}$ must be constant contradicting however $|\nabla\tilde{w}(0)| = 1$. This proves Proposition 4.4.5. □

Now we are ready to prove Theorem 4.4.1.

Proof of Theorem 4.4.1 By the previous discussions the proof of Theorem 4.4.1 is reduced to the following problem. Let $K \subset M$ be a compact subset and

$$\tilde{v} = (b, v) \colon \mathbf{R} \times S^1 \longrightarrow \mathbf{R} \times M$$

a map satisfying

$$\begin{aligned} \tilde{v}_s + \tilde{J}(\tilde{v})\tilde{v}_t &= 0 \quad \text{on } \mathbf{R} \times S^1 \\ \int_{\mathbf{R}\times S^1} v^* d\lambda &> 0 \\ 0 < E(\tilde{v}) &< +\infty \\ v(\mathbf{R} \times S^1) &\subset K \\ v(s,t) &\to \{\text{point}\} \qquad \text{for } s \to -\infty. \end{aligned}$$

By Proposition 4.4.5 we know that the gradient has to be bounded:

$$|\nabla \tilde{v}(s,t)| \leq c \quad \text{for } (s,t) \in \mathbf{R} \times S^1.$$

Define maps $\tilde{v}_k \colon \mathbf{R} \times S^1 \to \mathbf{R} \times M$ for some sequence $s_k \to +\infty$ by

$$\tilde{v}_k(s,t) = (b(s + s_k, t) - b(s_k, 0), v(s_k + s, t)).$$

Then we have with $\tilde{v}_k = (b_k, v_k)$

$$b_k(0,0) = 0$$

and moreover by Theorem 4.3.4

$$\left| D^\alpha \tilde{v}_k(s,t) \right| \leq c_{\alpha, K_0} \quad \text{on compact subsets } K_0 \subset \mathbf{R} \times S^1$$

for $|\alpha| \geq 0$, here we assumed that $M \subset \mathbf{R}^N$. Eventually taking a subsequence we have by the Ascoli–Arzela theorem

$$\tilde{v}_k \to \tilde{w} \quad \text{on } C^\infty_{loc}(\mathbf{R} \times S^1, \mathbf{R} \times M)$$

$$|\nabla \tilde{w}(s,t)| \leq c \ \text{ for } \ (s,t) \in \mathbf{R} \times S^1$$

and we get

$$\tilde{w}_s + \tilde{J}(\tilde{w})\tilde{w}_t = 0 \quad \text{on } \mathbf{R} \times S^1 \tag{4.26}$$

$$E(\tilde{w}) = \sup_{\phi \in \Sigma} \int_{\mathbf{R}\times S^1} \tilde{w}^* d(\phi\lambda) = E(\tilde{u}) < +\infty$$

because of

$$\sup_{\phi\in\Sigma}\int_{\mathbf{R}\times S^1}\tilde{v}_k^*d(\phi\lambda)=\sup_{\phi\in\Sigma}\int_{\mathbf{R}\times S^1}\tilde{v}^*d(\phi\lambda).$$

Fix any $s_0\in\mathbf{R}$. Then

$$\int_{\{s_0\}\times S^1}v_k^*\lambda=\int_{(-\infty,s_0+s_k]\times S^1}v^*d\lambda$$

converges to

$$\int_{\mathbf{R}\times S^1}v^*d\lambda=\int_{\mathbf{C}}u^*d\lambda=:T\ >\ 0$$

as $k\to\infty$ and for every $R>0$ we have

$$\int_{[-R,R]\times S^1}v_k^*d\lambda=\int_{[-R+s_k,R+s_k]\times S^1}v^*d\lambda\to 0.$$

Passing to the limit $k\to\infty$, we obtain

$$\int_{\{s_0\}\times S^1}w^*\lambda=\int_{\mathbf{C}}u^*d\lambda=T>0 \tag{4.27}$$

and

$$\int_{\mathbf{R}\times S^1}w^*d\lambda=0. \tag{4.28}$$

Remark We remark that (4.28) and $0<E(\tilde{w})<\infty$ is no contradiction to Proposition 4.4.2. The proposition does not apply. Using a biholomorphic transformation ϕ^{-1} between the cylinder and the punctured complex plane we see that $\tilde{w}\circ\phi^{-1}$ cannot be continued over $\{0\}$. On the other hand, viewing $\tilde{w}$ as a map on $\mathbf{C}$ which is periodic in the t–variable, we would get $\int_{\mathbf{C}}\tilde{w}^*d(\phi\lambda)=+\infty$ for non constant $\phi\in\Sigma$.

Let us view $\tilde{w}=(\beta,w)\colon\mathbf{C}\to\mathbf{R}\times M$ as being 1-periodic in the second argument. We infer from (4.28) that $\pi_\lambda\circ Tw(z):\mathbf{C}\to\xi_{w(z)}$ is the zero map for all $z\in\mathbf{C}$. As in the proof of Proposition 4.4.2 we construct a smooth map $f:\mathbf{C}\to\mathbf{R}$ so that

$$\Phi:=\beta+if$$

is holomorphic. Note that the holomorphic map $\Phi = \beta + if$ is non-constant and has bounded gradient. Indeed, if Φ were to be constant then we would have

$$0 = d\beta = (w^*\lambda) \circ i$$

i.e. the components of w_s and w_t along the Reeb vector field would vanish. But we also have $\pi_\lambda w_s = \pi_\lambda w_t = 0$ so w would be constant contradicting

$$\int_{\{s_0\}\times S^1} w^*\lambda = T > 0.$$

The gradient of Φ is bounded because of

$$\sup_{\mathbf{C}} |\nabla \Phi|^2 = 2 \sup_{\mathbf{C}} |\nabla \beta|^2 \leq 2 \sup_{\mathbf{C}} |\nabla \tilde{w}|^2 < +\infty.$$

Hence by Liouville's theorem the function Φ must be an affine non constant function, i.e.

$$\begin{aligned} \Phi(z) &= az + b \\ &= \beta(z) + if(z) \end{aligned}$$

where $a, b \in \mathbf{C}$, $a = a_1 + ia_2 \neq 0$. So

$$\beta(z) = \beta(s,t) = a_1 s - a_2 t + b_1 = a_1 s + b_1$$

since β is 1-periodic in t. Now

$$\begin{aligned} w_s &= \pi_\lambda w_s + \lambda(w_s) X_\lambda(w) \\ &= \lambda(w_s) X_\lambda(w) \\ &= -\beta_t X_\lambda(w) \\ &= 0 \end{aligned}$$

and

$$\begin{aligned} w_t &= \pi_\lambda w_t + \lambda(w_t) X_\lambda(w) \\ &= \lambda(w_t) X_\lambda(w) \\ &= \beta_s X_\lambda(w) \\ &= a_1 \cdot X_\lambda(w). \end{aligned}$$

Hence $x(t) := w(s, a_1^{-1}t)$ is an orbit of the Reeb vector field (actually not depending on s). We compute

$$T = \int_{\{s_0\}\times S^1} w^*\lambda = \int_0^1 \lambda(w_t)\,dt = \int_0^1 a_1\,dt = a_1$$

and

$$\tilde{w}(s,t) = (Ts + b_1, x(Tt)).$$

By construction

$$v(s_k, t) \to x(tT) \quad \text{in } C^\infty$$

as $k \to \infty$ or equivalently

$$u(e^{2\pi(s_k+it)}) \to x(tT)$$

in C^∞. Replacing tT by t we find

$$u\left(e^{2\pi(s_k+i\frac{t}{T})}\right) \to x(t)$$

in C^∞, where $t \longmapsto x(t)$ is a T-periodic solution of the Reeb vector field. □

Chapter 5
Properties of Pseudoholomorphic Curves



5.1 Local Behavior

Let (W, J) be an *almost complex manifold*, i.e. J is a section in $\mathrm{End}(TW)$ with $J^2 = -\mathrm{Id}$. Let $D \subset \mathbf{C}$ be the closed unit disk. We would like to study pseudoholomorphic curves on D, i.e. maps satisfying

$$\begin{aligned} & u : D \longrightarrow W \\ & \bar{\partial}_J u(z) \ := \ u_s(z) + J(u(z))\, u_t(z) = 0 \ \text{ on } \ \mathring{D}, \end{aligned} \tag{5.1}$$

from the local point of view. Moreover, we will assume that the following boundary condition holds:

$$u(\partial D) \subset F,$$

where $F \subset W$ is a *totally real submanifold*, i.e.

$$T_x F \oplus J(x) T_x F \ = \ T_x W$$

for all $x \in F$. We will show that pseudoholomorphic curves share certain properties with ordinary holomorphic curves. Since we are only concerned with local studies near some point $z_0 \in D$, we will pass to local coordinates. If $z_0 \in \mathring{D}$ then our local model is a map $v : D \to \mathbf{C}^n$ satisfying $\partial_s v + \bar{J}(v)\partial_t v = 0$ for a suitable almost complex structure $\bar{J}$ on $\mathbf{C}^n$ and z_0 corresponds to the origin in D. If $z_0 \in \partial D$ then we choose a chart $\sigma : U \to V \subset \mathbf{C}^n$, where U is a neighborhood of $u(z_0)$ in W so that

$$\sigma(F \cap V) \ = \ \mathbf{R}^n \cap U$$

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and $\sigma(u(z_0)) = 0$. Define an almost complex structure $\bar{J}$ on U by

$$\bar{J}(w) := T\sigma(\sigma^{-1}(w)) \circ J(\sigma^{-1}(w)) \circ T\sigma^{-1}(w)$$

then $\mathbf{R}^n \cap U$ is totally real for $\bar{J}$. The biholomorphic map

$$\psi_\varepsilon : H^+ := \{z \in \mathbf{C} \mid \operatorname{Im} z \geq 0\} \longrightarrow D\backslash\{-z_0\}$$

$$\psi_\varepsilon(z) := z_0 \frac{i - \varepsilon z}{i + \varepsilon z} \;\; ; \;\; \varepsilon > 0$$

maps the upper half disk D^+ onto a neighborhood of z_0. Choose $\varepsilon > 0$ sufficiently small so that $v := \sigma \circ u \circ \psi_\varepsilon : D^+ \to U \subset \mathbf{C}^n$ is well-defined. Composing σ with a suitable real linear map A we may assume in addition that $\bar{J}(0) = i$. Since $\mathbf{R}^n$ is totally real in both $(\mathbf{C}^n, i)$ and $(\mathbf{C}^n, \bar{J}(0))$, we can find A with $A\bar{J}(0)A^{-1} = i$ and $A \cdot \mathbf{R}^n = \mathbf{R}^n$. Summarizing, we have

$$\begin{aligned} v_s(z) + \bar{J}(v(z))\, v_t(z) &= 0 \quad \text{on } D^+, \\ v((-1,+1)) &\subset \mathbf{R}^n \\ v(0) &= 0, \\ \bar{J}(0) &= i. \end{aligned} \tag{5.2}$$

If we want furthermore $\bar{J} \equiv i$ on $\mathbf{R}^n \cap U$ then we may arrange this at the expense of having an additional zero order term of the form $C(z) \cdot v(z)$ in the equation for v in (5.2).

Theorem 5.1.1 *Let $u, v : D \longrightarrow W$ be pseudoholomorphic disks with*

$$u(\partial D), v(\partial D) \subset F,$$

where $F \subset W$ is a totally real submanifold. Then the set

$$\Omega := \{z \in \overline{D} \mid D^k u(z) = D^k v(z) \text{ for all } k \geq 0\}$$

is either empty or equals D.

Proof Assume that the set Ω is not empty. Then we will show that it is open and closed in D and therefore equals D. Assume $z_0 \in \Omega \cap \partial D$. Passing to local coordinates as in (5.2), we compute with $w := u - v$

$$\begin{aligned} 0 &= \partial_s w + \bar{J}(u)\partial_t u - \bar{J}(v)\partial_t v \\ &= \partial_s w + \bar{J}(u)\partial_t w + (\bar{J}(u) - \bar{J}(v))\partial_t v \end{aligned}$$

$$= \partial_s w + \bar{J}(u)\partial_t w + \\ + \left(\int_0^1 D\bar{J}(v + t(u - v)) \cdot w \, dt \right) \cdot \partial_t v.$$

We define a smooth map $C : D^+ \longrightarrow \mathcal{L}_{\mathbf{R}}(\mathbf{C}^n)$ by

$$C(z)\zeta := \left(\int_0^1 D\bar{J}(v + t(u - v)) \cdot \zeta \, dt \right) \cdot \partial_t v,$$

so that

$$0 = \partial_s w + \bar{J}(u)\partial_t w + C \cdot w.$$

If $\{e_1, \ldots, e_n\}$ is a basis of $\mathbf{R}^n \subset \mathbf{C}^n$ then

$$\{e_1, \ldots, e_n, ie_1, \ldots, ie_n\}$$

and

$$\{e_1, \ldots, e_n, \bar{J}(u(z))e_1, \ldots, \bar{J}(u(z))e_n\}$$

are both real bases of $\mathbf{C}^n$ since $\mathbf{R}^n$ is totally real for both i and $\bar{J}(u(z))$. We define a smooth map

$$A : D^+ \longrightarrow \mathrm{GL}_{\mathbf{R}}(\mathbf{C}^n)$$

by requiring $A(z)e_k = e_k$ and $A(z)\bar{J}(u(z))e_k = ie_k$, where $\mathrm{GL}_{\mathbf{R}}(\mathbf{C}^n)$ denotes the set of invertible real linear maps $\mathbf{C}^n \to \mathbf{C}^n$, so that

$$A(z)\,\mathbf{R}^n = \mathbf{R}^n \text{ and } i\,A(z) = A(z)\bar{J}(u(z)).$$

We have

$$\hat{C} := ACA^{-1} - (\partial_s A + i\,\partial_t A)A^{-1} \in L^\infty(D^+, \mathbf{C}^n)$$

and

$$\partial_s(Aw) + i\,\partial_t(Aw) = A(\partial_s w + A^{-1} i A \partial_t w) + (\partial_s A + i\,\partial_t A)\,w \\ = -\hat{C}\,Aw.$$

The similarity principle in Appendix A.5 states that for $z \in D_\varepsilon^+$

$$A(z)w(z) = \Phi(z)h(z),$$

where $D_\varepsilon^+ := \{z \in \mathbf{C} \,|\, \mathrm{Im}(z) \geq 0\,,\ |z| \leq \varepsilon\}$ for a suitable $\varepsilon > 0$, $\Phi : D_\varepsilon^+ \to \mathrm{GL}_{\mathbf{C}}(\mathbf{C}^n)$ is a smooth map and $h : D_\varepsilon^+ \to \mathbf{C}^n$ is holomorphic. Since all derivatives of w vanish in 0, this is also the case for Aw and hence also for the holomorphic map h. But this implies that $h \equiv 0$ and therefore $u \equiv v$ in a neighborhood of the point $z_0 \in \Omega$. If z_0 is an interior point of the disk, we argue similarly (replace the domain D^+ above by D). This shows that the set Ω is open. Since it is obviously closed we obtain $\Omega = D$. □

Proposition 5.1.2 *Let $u : \Omega \to \mathbf{C}^n$ be a pseudoholomorphic curve, where Ω is either the unit disk in $\mathbf{C}$ or the upper half of the unit disk, where we assume in addition that $u((-1,+1)) \subset \mathbf{R}^n$. Assume that $Du(0) \neq 0$. Then there exists a diffeomorphism $\varphi : U \to \mathbf{C}^n$ where U is an open neighborhood of $u(0)$ so that*

$$(\varphi \circ u)(z) \;=\; (z, 0, \ldots, 0)$$

and

$$T\varphi(u(z)) \circ \bar{J}(u(z)) \;=\; i \circ T\varphi(u(z))$$

for z in a sufficiently small neighborhood of 0.

Proof If we restrict u to a sufficiently small neighborhood of zero it is an embedding and we may find a chart $\tilde{\varphi}$ so that

$$(\tilde{\varphi} \circ u)(z) \;=\; (z, 0, \ldots, 0) \in \mathbf{C}^n.$$

We define near $0 = \tilde{\varphi}(u(0))$ the almost complex structure $\tilde{J}$ by

$$D\tilde{\varphi}(w) \circ \bar{J}(w) \;=\; \tilde{J}(\tilde{\varphi}(w)) \circ D\tilde{\varphi}(w).$$

Now we have to modify $\tilde{J}$ so that $\tilde{J}(\tilde{\varphi}(u(z))) = i$ for z near zero.

We note that $\tilde{J}(\tilde{\varphi}(u(z)))$ is multiplication by i on $\mathbf{C} \times \{0\} \subset \mathbf{C} \times \mathbf{C}^{n-1}$ because of

$$(\tilde{\varphi} \circ u)_s + \tilde{J}(\tilde{\varphi} \circ u)\,(\tilde{\varphi} \circ u)_t = 0.$$

First we would like to find a real linear invertible map A on $\mathbf{C}^n$ so that

$$A\,\tilde{J}(0) \;=\; i\,A$$

and

$$A(h, 0, \ldots, 0) = (h, 0, \ldots, 0)$$

for $(h, 0, \ldots, 0) \in \mathbf{C} \times \{0\} \subset \mathbf{C} \times \mathbf{C}^{n-1}$. We pick any $C \in \mathrm{GL}_{\mathbf{R}}(\mathbf{C}^n)$ with

$$C\,\tilde{J}(0) \;=\; i\,C.$$

Then we have

$$\begin{aligned} C(ih, 0, \ldots, 0) &= C\,\tilde{J}(0)(h, 0, \ldots, 0) \\ &= i\,C(h, 0, \ldots, 0). \end{aligned}$$

Take now $B \in \mathrm{GL}(\mathbf{C}^n)$ so that $B(C(1, 0, \ldots, 0)) = (1, 0, \ldots, 0)$ and put $A := BC$. Indeed, A does the job since

$$\begin{aligned} A(h, 0, \ldots, 0) &= B(C(h, 0, \ldots, 0)) \\ &= B(C(\mathrm{Re}(h) + i\,\mathrm{Im}(h), 0, \ldots, 0)) \\ &= (\mathrm{Re}(h), 0, \ldots, 0) \;+\; i\,B(C(\mathrm{Im}(h), 0, \ldots, 0)) \\ &= (h, 0, \ldots, 0) \end{aligned}$$

and $A\tilde{J}(0) = BC\tilde{J}(0) = BiC = iBC = iA$. Now define $\hat{v} := A \circ \tilde{\varphi} \circ u$ and $\hat{J}$ by $A\tilde{J}(w) = \hat{J}(Aw)A$. Then $\hat{v}$ satisfies

$$\hat{v}_s(z) + \hat{J}(\hat{v}(z))\hat{v}_t(z) = 0$$

$$\hat{v}(z) = (z, 0, \ldots, 0)$$

and we have $\hat{J}(0) = i$. If $|w|$ is small we infer $\mathbf{R}^n \oplus \hat{J}(w)\mathbf{R}^n = \mathbf{C}^n$. Let $\{e_1, \ldots, e_n\}$ be the standard basis of $\mathbf{R}^n$ and let $\delta > 0$ be a sufficiently small number; then we define $\Phi \in C^\infty(B_\delta(0), \mathrm{GL}_{\mathbf{R}}(\mathbf{C}^n))$ by

$$\Phi(z)\,e_k \;:=\; e_k \;\;,\;\; 1 \le k \le n$$

$$\Phi(z)\hat{J}(\hat{v}(z))\,e_k \;:=\; i\,e_k.$$

Then $\Phi(z)\hat{J}(\hat{v}(z)) = i\,\Phi(z)$ and $\Phi(z)(h, 0, \ldots, 0) = (h, 0, \ldots, 0)$. Let pr: $\mathbf{C}^n \to \mathbf{C}$ be the projection onto the first factor and

$$\tau : B_\delta(0) \times \mathbf{C}^{n-1} \longrightarrow \mathbf{C}^n$$

$$\tau(u) := \Phi(\mathrm{pr}(u)) \cdot u.$$

We calculate for $u_0 = (z, 0, \ldots, 0) \in B_\delta(0) \times \{0\}$ and $\xi \in \mathbf{C}^n$:

$$\tau(u_0) \;=\; u_0$$

and

$$D\tau(u_0)\,\xi \;=\; \Phi(z)\,\xi.$$

There exists a neighborhood V of $B_\delta(0)\times\{0\}$ in $\mathbf{C}^n$ so that $\tau\,|_V$ is a diffeomorphism onto its image (see Lemma 7.4.2). Now put $v := \tau\circ\hat{v}$, define J' by

$$J'(v(z)) := D\tau(\hat{v}(z))\circ\hat{J}(\hat{v}(z))\circ D\tau^{-1}(v(z))$$

and calculate

$$\begin{aligned} J'(v(z)) &= \Phi(z)\circ\hat{J}(\hat{v}(z))\circ\Phi(z)^{-1} \\ &= i. \end{aligned}$$

Moreover, $v(z) = \tau(\hat{v}(z)) = \tau(z,0,\ldots,0) = (z,0,\ldots,0)$ and we obtain the desired diffeomorphism φ if we take $\varphi := \tau\circ A\circ\tilde{\varphi}$. □

The following theorem shows that intersection points of distinct pseudoholomorphic disks can only accumulate in points which are critical on both disks.

Theorem 5.1.3 *Let $u, v \in C^\infty(D, W)$ satisfy*

$$\bar{\partial}_J u(z) \;=\; \bar{\partial}_J v(z) \;=\; 0 \;\; \textit{on} \;\; \overset{\circ}{D}$$

and

$$u(\partial D)\;,\;\; v(\partial D)\subset F$$

where F is a totally real submanifold of W. Moreover assume that the images of u and v are not identical and let $p \in u(D)\cap v(D)$ be an accumulation point of intersections, i.e. there are sequences $(z_k)\,,\;(z_k')\subset D$ with

$$z_k \longrightarrow \zeta \in D \;\;;\;\; z_k \neq \zeta$$

$$z_k' \longrightarrow \zeta' \in D \;\;;\;\; z_k' \neq \zeta'$$

so that $u(z_k) = v(z_k')$ and $u(\zeta) = p = v(\zeta')$. Then $Du(\zeta) = 0 = Dv(\zeta')$.

Since the points ζ and ζ' may be interior or boundary points, we will examine this problem locally distinguishing between the possible cases.

Proposition 5.1.4 *Let $u, v : D \to W$ be pseudoholomorphic disks with*

$$u(0) = v(0) \;\;;\;\; Du(0) \neq 0$$

Assume further that there are sequences (z_k), $(z'_k) \subset D$ *with* z_k, $z'_k \to 0$ *and* $z_k \neq 0 \neq z'_k$ *so that* $u(z_k) = v(z'_k)$. *Then there exists a holomorphic function* $\phi : B_{\varepsilon'}(0) \to D$ *for some* $\varepsilon' > 0$ *so that* $\phi(0) = 0$ *and*

$$v = u \circ \phi.$$

Proof By Proposition 5.1.2 we may assume that we are in the following situation (after restricting u and v to a small neighborhood of zero)

$$u, v : B_{\varepsilon'}(0) \longrightarrow \mathbf{C}^n,$$

$$u(z) = (z, 0 \ldots, 0),$$

$$J(z, 0, \ldots, 0) = i.$$

Let us write $v(z) = (v_1(z), \tilde{v}(z))$ where $v_1 : B_{\varepsilon'}(0) \to \mathbf{C}$ and $\tilde{v} : B_{\varepsilon'}(0) \to \mathbf{C}^{n-1}$. Then

$$\begin{aligned} 0 &= (\partial_s v_1(z), \partial_s \tilde{v}(z)) + J(v_1(z), \tilde{v}(z))\big(\partial_t v_1(z), \partial_t \tilde{v}(z)\big) \\ &= (\partial_s v_1(z), \partial_s \tilde{v}(z)) + \\ &\quad + \left(J(v_1(z), 0) + \int_0^1 D_2 J(v_1(z), \tau \tilde{v}(z)) \cdot \tilde{v}(z)\, d\tau\right)\big(\partial_t v_1(z), \partial_t \tilde{v}(z)\big). \end{aligned} \tag{5.3}$$

Define now $A \in L^\infty(B_{\varepsilon'}(0), \mathcal{L}_{\mathbf{R}}(\mathbf{C}^n))$ by

$$A(z)\,\xi := \left(\int_0^1 D_2 J(v_1(z), \tau \tilde{v}(z)) \cdot \xi\, d\tau\right)\big(\partial_t v_1(z), \partial_t \tilde{v}(z)\big)$$

We decompose this into components in $\mathbf{C}$ and $\mathbf{C}^{n-1}$

$$A(z)\xi = (\alpha(z)\xi, \beta(z)\xi) \in \mathbf{C} \times \mathbf{C}^{n-1}.$$

Then Eq. (5.3) becomes

$$0 = (\partial_s v_1(z) + i\,\partial_t v_1(z) + \alpha(z)\tilde{v}(z)\,,\ \partial_s \tilde{v}(z) + i\,\partial_t \tilde{v}(z) + \beta(z)\tilde{v}(z))$$

and we note that $\tilde{v}(z'_k) = 0$. Using the Similarity principle A.5.1 we may write for $z \in B_\varepsilon(0)$ with some $\varepsilon \leq \varepsilon'$

$$\tilde{v}(z) = \Phi(z)\sigma(z),$$

where $\Phi(z) \in \mathrm{GL}(\mathbf{C}^n)$ and $\sigma : B_\varepsilon(0) \to \mathbf{C}^{n-1}$ is holomorphic. Hence 0 is an accumulation point of zeros for σ, so σ must vanish identically. Therefore $\tilde{v} \mid_{B_\varepsilon(0)} \equiv$

0 and $v_1 \mid_{B_\varepsilon(0)}$ is holomorphic. Because of Theorem 5.1.1 this is even true for $B_{\varepsilon'}(0)$ instead of $B_\varepsilon(0)$. Choosing $\phi := v_1$ completes the proof. □

There is also a boundary version of Proposition 5.1.4

Proposition 5.1.5 *Let $u : D^+ \to W$, $v : \Omega \to W$ be pseudoholomorphic maps, where Ω is either the unit disk D or the upper half disk D^+. We assume that $u(0) = v(0)$, $Du(0) \neq 0$ and $u((-1,+1)) \subset F$ where $F \subset W$ is a totally real submanifold. If $\Omega = D^+$, then we also assume that $v((-1,+1)) \subset F$. Assume further that there are sequences $(z_k) \subset D^+$, $(z_k') \in \Omega$ with $z_k, z_k' \to 0$ and $z_k \neq 0 \neq z_k'$ so that $u(z_k) = v(z_k')$. In the case $\Omega = D^+$ there exists a holomorphic function $\phi : D^+_{\varepsilon'} \to D^+$ for some $\varepsilon' > 0$ so that*

$$\phi(0) = 0 \ ; \ \ \phi((-\varepsilon', +\varepsilon')) \subset (-1,+1)$$

and

$$v = u \circ \phi.$$

In the case $\Omega = D$ the map u may be extended from D^+_ε onto the ε–disk D_ε and the conclusion of Proposition 5.1.4 holds.

Proof If we consider a small neighborhood of zero in D^+ we may assume by Proposition 5.1.2 that our situation looks as follows:

$$u : D^+_\varepsilon \longrightarrow \mathbf{C}^n,$$

$$u(z) = (z, 0, \ldots, 0),$$

$$u((-\varepsilon, +\varepsilon)) \subset \mathbf{R}^n.$$

If $\Omega = D^+$ then we argue similarly as in the proof of Proposition 5.1.4 invoking the boundary version of the similarity principle A.5.2 instead. In the case $\Omega = D$ we extend the map u above to the ε–disk D_ε and we are in the setup of Proposition 5.1.4. □

Proof of Theorem 5.1.3 Arguing indirectly we assume that $Du(\zeta) \neq 0$. We define a subset $\Omega \subset D$ as follows: The set Ω should consist of all points z_0 so that $u(z_0) = v(z_0')$ for some $z_0' \in D$ and $Du(z_0) \neq 0$. Moreover, we require that there are sequences $z_k \to z_0$ and $z_k' \to z_0'$, such that $u(z_k) = v(z_k')$ and $z_k \neq z_0$, $z_k' \neq z_0'$. Our assumption implies that $\Omega \neq \emptyset$. Clearly, the set Ω is closed, but Propositions 5.1.4 and 5.1.5 imply that Ω is also open. Hence $\Omega = D$ since it is not empty by assumption contradicting $u(D) \neq v(D)$. □

Theorem 5.1.6 *Let $u \in C^\infty(D, W)$ be a non constant solution of $\bar\partial_J u(z) = 0$ on $\overset{\circ}{D}$ and $u(\partial D) \subset F$ where F is a totally real submanifold of W. Then the critical points of u are isolated.*

Proof Take $z_0 \in D$ with $Du(z_0) = 0$ and let us consider a small neighborhood of z_0. If $z_0 \in \overset{\circ}{D}$ then we may assume that our problem looks like (5.2), i.e.

$$u : D \longrightarrow \mathbf{C}^n$$
$$\bar{\partial}_J u(z) = 0 \text{ on } \overset{\circ}{D}$$
$$u(0) = 0 \ ; \ \ J(0) = i \ ; \ \ Du(0) = 0.$$

In the case $z_0 \in \partial D$ we have

$$u : D^+ \longrightarrow \mathbf{C}^n$$
$$\bar{\partial}_J u(z) = 0 \text{ on } \overset{\circ}{D}{}^+$$
$$u((-1, +1)) \subset \mathbf{R}^n$$
$$u(0) = 0 \ ; \ \ J(0) = i \ ; \ Du(0) = 0.$$

Let $v(z) := u_s(z)$. Then

$$\begin{aligned} 0 \ &= \ u_{ss}(z) + J(u(z))\, u_{st}(z) + (DJ(u(z)) \cdot u_s(z)) \cdot u_t(z) \\ &= \ v_s(z) + J(u(z))\, v_t(z) + (DJ(u(z)) \cdot v(z)) \cdot u_t(z) \\ &=: v_s(z) + \tilde{J}(z)\, v_t(z) + A(z) \cdot v(z). \end{aligned}$$

Composing v with a suitable matrix–valued function $B(z) \in \mathrm{GL}_{\mathbf{R}}(\mathbf{C}^n)$ we may assume that $\tilde{J}(z) = i$ and the zero order term $A(z)$ changes. Then we apply the Similarity principle (Theorem A.5.1 or A.5.2) and we can write near $z_0 = 0$

$$v(z) \ = \ \Phi(z)\, f(z),$$

where $\Phi(z) \in \mathrm{GL}_{\mathbf{R}}(\mathbf{C}^n)$ and f is holomorphic. If the zero $v(0) = 0$ is not isolated we must have $f \equiv 0$ near $z_0 = 0$, i.e. the ∞-jet of v vanishes at $z_0 = 0$. But this would imply by Theorem 5.1.1 that $v \equiv 0$, i.e. u is constant which contradicts our assumption. □

5.2 Pseudoconvexity

In this section we consider an almost complex manifold (W, J) and an orientable codimension one submanifold M dividing W into two disjoint parts W^- and W^+, i.e. $W^+ \cap W^- = \emptyset$, $W^+ \cup W^- \cup M = W$ and $\overline{W^+} \cap \overline{W^-} = M$. We will define the notion of M being pseudoconvex with respect to W^- and we will see that J-holomorphic disks with boundary values in such a pseudoconvex M will stay on one side of M (namely W^-) at all interior points.

Let us start with the definition of pseudoconvexity: M is called *pseudoconvex* with respect to W^- if there exists a smooth function $H : W \to \mathbf{R}$ so that the following holds:

- $H^{-1}(0) = M$
- $dH(x) \neq 0$ for all $x \in M$
- $W^- = \{x \in W \mid H(x) < 0\}$
- $-d(dH \circ J)_x(v_x, J(x)v_x) > 0$ for all nonzero $v_x \in T_xW$, $x \in W$

Before we continue we will give a more geometric interpretation of pseudoconvexity. Each tangent space T_xM, $x \in M$, contains a unique complex vector space ξ_x of real dimension $\dim_{\mathbf{R}}(M) - 1$. i.e. a co-dimension one linear subspace that is invariant under $J(x) : T_xW \to T_xW$. We will call ξ_x the *complex tangency* to M at x. The complex tangency is canonically oriented by taking a complex basis $e_1, \ldots, e_n$ for ξ_x, where $2n + 1 = \dim(M)$, and taking the orientation associated to $(e_1, Je_1, \ldots, e_n, Je_n)$. Since M is orientable the line bundle TM/ξ is also orientable and therefore trivial. Hence we find a nonzero vector field Y so that

$$TM = \xi \oplus \mathbf{R} \cdot Y$$

where Y is determined up to multiplication by a nonzero function. Now the kernel of the one form $(dH \circ J)_x|_{T_xM}$, $x \in M$, is exactly the complex tangency ξ_x. Clearly $\xi_x \subset \ker (dH \circ J)_x|_{T_xM}$ since $J(x)\xi_x = \xi_x$ and $dH(x)|_{T_xM} \equiv 0$. On the other hand if we had $\mathrm{Ker}(dH \circ J)_x|_{T_xM} = T_xM$ then we would get $dH(x) \equiv 0$ which contradicts our assumptions on H. We compute with $x \in M$, $0 \neq e \in \xi_x$:

$$\begin{aligned} &(dH \circ J)_x \wedge d(dH \circ J)_x(e, J(x)e, Y(x)) \\ &= (dH \circ J)_x Y(x) \cdot d(dH \circ J)_x(e, J(x)e) \\ &\neq 0. \end{aligned}$$

Summarizing, we have proved that the complex tangencies on a pseudoconvex hypersurface define a contact structure which is induced by the one form $dH \circ J$.

If (M, λ) is a three dimensional contact manifold there exists an almost complex four manifold $(W, \tilde{J})$ so that M divides W into two parts W^- and W^+, and M is pseudoconvex with respect to W^-.

Let us choose a complex multiplication $J : \xi \to \xi$ on $\xi = \ker \lambda$ so that $d\lambda \circ (\mathrm{Id} \times J)$ is a bundle metric. Define $W := \mathbf{R} \times M$ and an almost complex structure $\tilde{J}$ on W as follows:

$$\tilde{J}(a, u)(h, k) = (-\lambda(u)k, J(u)\pi_\lambda k + h\, X_\lambda(u))$$

Moreover choose

$$H : \mathbf{R} \times M \longrightarrow \mathbf{R}$$

$$(t, x) \longmapsto e^t - 1$$

and

$$W^- := (-\infty, 0) \times M \ , \ \ W^+ := (0, +\infty) \times M.$$

We compute for $(t, x) \in \mathbf{R} \times M$ and $0 \neq (h, k) \in T_{(t,x)}(\mathbf{R} \times M)$:

$$\begin{aligned} & -d(dH \circ \tilde{J})_{(t,x)}((h, k), \tilde{J}(t, x)(h, k)) \\ &= -d(dH \circ \tilde{J})_{(t,x)}((h, k), (-\lambda(x)k, J(x)\pi_\lambda k + hX_\lambda(x))) \\ &= e^t \left(d\lambda(x)(\pi_\lambda k, J(x)\pi_\lambda k) + h^2 + (\lambda(x)k)^2 \right) \end{aligned}$$

since

$$(dH \circ \tilde{J})_{(t,x)} = -e^t \lambda(x)$$

and therefore

$$-d(dH \circ \tilde{J})_{(t,x)} = e^t \, d\lambda(x) + e^t \, dt \wedge \lambda(x)$$

Hence

$$-d(dH \circ \tilde{J})_{(t,x)}((h, k), \tilde{J}(t, x)(h, k)) > 0$$

for nontrivial $(h, k) \in T_{(t,x)}(\mathbf{R} \times M)$ and therefore $M \approx \{0\} \times M \subset (\mathbf{R} \times M, \tilde{J})$ is pseudoconvex with respect to W^-. Now let us study the behavior of pseudoholomorphic disks

$$u : D \longrightarrow W$$

$$u_s + J(u)u_t = 0 \ \ (\text{or } du + J(u)\, du \circ i = 0\,) \tag{5.4}$$

with boundary condition

$$u(\partial D) \subset M \tag{5.5}$$

where M is pseudoconvex with respect to W^-.

Theorem 5.2.1 *Let u be a solution of (5.4) and (5.5) so that all its derivatives extend continuously to the boundary. Then either*

$$u(D) \subset M$$

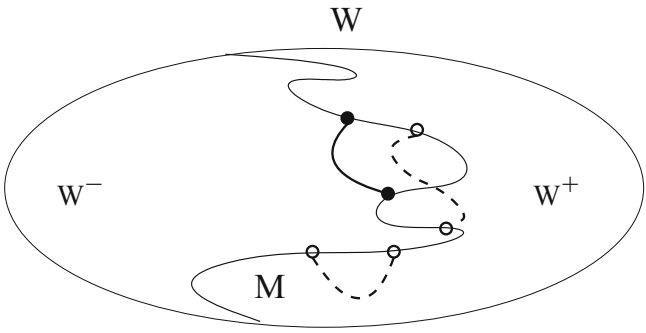


Fig. 5.1 Pseudoholomorphic disks with boundary in a pseudoconvex hypersurface M lie in $W^- \cup M$

or

$$u(\overset{\circ}{D}) \subset W^- \text{ and } u \text{ is transverse to } M \text{ at each point } z \in \partial D \text{ , i.e.}$$

$$du(z)T_zD + T_{u(z)}M = T_{u(z)}W$$

In Fig. 5.1, we display W as a two–dimensional set, the pseudoholomorphic disks and M as one dimensional. The dashed disks are impossible by Theorem 5.2.1.

Proof of Theorem 5.2.1 Define $\alpha := H \circ u : D \to \mathbf{R}$. Then

$$\begin{aligned}
(-\Delta\alpha)\, ds \wedge dt &= d(d\alpha \circ i) \\
&= d(dH \circ du \circ i) \\
&= d(dH \circ J \circ du) \\
&= d\, u^*(dH \circ J) \\
&= u^* d(dH \circ J) \\
&= d(dH \circ J)(u_s, u_t) ds \wedge dt \\
&= d(dH \circ J)(u_s, J(u)u_s) ds \wedge dt
\end{aligned}$$

Hence $\Delta\alpha \geq 0$ and $\alpha|_{\partial D} \equiv 0$. By the strong maximum principle (see [51] and [55], Theorems 2.1., 2.2.) we either have

$$\alpha \equiv 0 \text{ on } D$$

or

$$\alpha(z) < 0 \text{ on } \overset{\circ}{D}$$

and

$$\frac{\partial \alpha}{\partial \nu}(z) > 0 \text{ for all } z \in \partial D,$$

where $\frac{\partial \alpha}{\partial \nu}$ denotes the outer normal derivative. By the properties of H the first possibility is equivalent to

$$u(D) \subset M.$$

If the second possibility occurs then we must have

$$u(\overset{\circ}{D}) \subset W^-$$

We conclude for $z \in \partial D$ that

$$0 < \frac{d}{dt} H(u(tz))\Big|_{t=1},$$

hence $du(z)T_z D$ cannot be contained in $T_{u(z)}M$ which proves the theorem. □

Let us formulate Theorem 5.2.1 for $\tilde{J}$-holomorphic disks in the symplectisation $\mathbf{R} \times M$ with boundary values in $\{0\} \times F^* \approx F^*$ where F is a compact orientable surface in M and $F^* := F \backslash \Sigma_{F,\xi}$ is the surface without the singular points. Recall from Chap. 3 that a point $x \in F$ is called singular if $T_x F = \xi_x$.

$$\tilde{u} = (a, u) : D \longrightarrow \mathbf{R} \times M$$

$$\tilde{u}_s + \tilde{J}(\tilde{u})\tilde{u}_t = 0 \tag{5.6}$$

$$\tilde{u}(\partial D) \subset F^* \approx \{0\} \times F^* \subset \{0\} \times M.$$

We know that $\tilde{u}$ takes boundary values in a pseudoconvex hypersurface with respect to $W^- = (-\infty, 0] \times M$. Applying the previous theorem we get:

Corollary 5.2.2 *Let $\tilde{u}$ be a solution of (5.6). Then either*

$$\tilde{u} \equiv \textit{ constant } (0, p)$$

or

- $\int u^* d\lambda > 0$
- $a(\overset{\circ}{D}) < 0$
- $|\nabla a(z)| \neq 0$ *for all* $z \in \partial D$
- $u|_{\partial D} : \partial D \to F^*$ *is an immersion transverse to the one-dimensional distribution* $\xi|_{F^*} \cap TF^* \to F^*$.

Proof The alternative $\tilde{u}(D) = (a, u)(D) \subset M \approx \{0\} \times M$ of Theorem 5.2.1 would imply $a \equiv 0$ and therefore $u^*\lambda = -da \circ i \equiv 0$, i.e. $\lambda(u)u_s = \lambda(u)u_t \equiv 0$. Then we also have

$$0 \equiv u^*d\lambda = \frac{1}{2}(|\pi_\lambda u_s|_J^2 + |\pi_\lambda u_t|_J^2)ds \wedge dt.$$

Hence

$$\begin{aligned} u_s = \lambda(u)u_s \cdot X_\lambda(u) + \pi_\lambda u_s &= 0, \\ u_t = \lambda(u)u_t \cdot X_\lambda(u) + \pi_\lambda u_t &= 0 \end{aligned}$$

and $\tilde{u}$ is constant in this case. If $\tilde{u}$ is not constant then $\tilde{u}(\overset{\circ}{D}) \subset W^-$, i.e. $a(\overset{\circ}{D}) \subset (-\infty, 0)$, and $\tilde{u}$ is transverse to M at each boundary point $z \in \partial D$, which implies

$$\begin{aligned} & d\tilde{u}(z)T_zD + T_{\tilde{u}(z)}M \\ &= (da(z)T_zD, du(z)T_zD) + (0, T_{u(z)}M) \\ &= T_{\tilde{u}(z)}W \\ &= (\mathbf{R}, T_{u(z)}M) \end{aligned}$$

so that $|\nabla a(z)| \neq 0$ for all $z \in \partial D$, in particular, $\tilde{u}$ is not constant. If we had $\int_D u^*d\lambda = 0$ then we would conclude from

$$\begin{aligned} (-\Delta a)ds \wedge dt &= d(da \circ i) \\ &= d(-u^*\lambda) \\ &= -u^*d\lambda \\ &= -\frac{1}{2}(|\pi_\lambda u_s|_J^2 + |\pi_\lambda u_t|_J^2)ds \wedge dt \end{aligned}$$

that $\Delta a = 0$. The boundary condition $a|_{\partial D} \equiv 0$ then yields $a \equiv 0$ by the maximum principle which contradicts $|\nabla a(z)| \neq 0$ for $z \in \partial D$. Hence we have shown that

$$\int_D u^*d\lambda > 0$$

if $\tilde{u}$ is not constant. Let us show now that

$$d(u|_{\partial D})(z) \cdot T_z\partial D$$

is never contained in $\xi_{u(z)}$ for all $z \in \partial D$. Choose $z \in \partial D$ and $0 \neq v \in T_z\partial D$. We have to show that

$$(u^*\lambda)(z) \cdot v = -da(z)(iv)$$

is not zero. We know, since $\tilde{u}$ is transverse to M at each point $z \in \partial D$, that

$$da(z)T_zD = \mathbf{R}$$

We decompose T_zD for $z \in \partial D$ into its normal and tangent direction:

$$T_zD = \mathbf{R} \cdot v + \mathbf{R} \cdot iv.$$

Because of $a|_{\partial D} \equiv 0$ we must have $da(z)v = 0$ for all $v \in T_z\partial D$, hence $da(z)(iv) \neq 0$ for all $z \in D$ and nonzero $v \in T_z\partial D$. This shows that $u|_{\partial D} : \partial D \to F^*$ is an immersion transverse to the one-dimensional distribution $\xi|_{F^*} \cap TF^* \to F^*$. □

In the special cases of surfaces F that we are actually interested in, we can even conclude that $u|_{\partial D}$ is an embedding:

Proposition 5.2.3 *Assume that F is either an embedded S^2 with precisely two elliptic singularities and no limit cycles for the characteristic foliation or F is an embedded disk with ∂F being Legendrian and precisely one elliptic singularity in $F \setminus \partial F$ for the characteristic foliation.*

If $\tilde{u}$ solves (5.6), $\tilde{u}$ not constant, and the winding number around an elliptic singularity with respect to some orientation is ± 1, then $u|_{\partial D} : \partial D \to F^*$ is an embedding.

Proof We know from Corollary 5.2.2 that $u|_{\partial D} : \partial D \to F^*$ is an immersion. Moreover $u(\partial D)$ is transverse to the characteristic foliation $\xi|_{F^*} \cap TF^* \to F^*$ on F^*. This implies that $u|_{\partial D}$ is also injective since the winding number is ± 1 so that the curve $u(\partial D)$ hits each leaf of the characteristic foliation only once. □

5.3 Removal of Singularities

Let (M, ω) be a compact *symplectic manifold* with *compatible almost complex structure* J, i.e. ω is a nondegenerate closed two-form on M and $J : TM \to TM$ is chosen in such a way that $g_J := \omega \circ (\mathrm{Id} \times J)$ is a Riemannian metric. Moreover let $L \subset M$ be a *Lagrangian submanifold*, i.e. $\dim L = \frac{1}{2}\dim M$ and $\omega|_{TL} \equiv 0$.

Theorem 5.3.1 *Let $D_1^+ := \{z \in \mathbf{C} \mid |z| < 1 \text{ and } \mathrm{Im}\, z \geq 0\}$ and let $u : D_1^+ \setminus \{0\} \to M$ be a smooth map satisfying*

$$u_s + J(u)u_t = 0,$$

$$u((-1, 0) \cup (0, +1)) \subset L$$

and

$$\int_{D_1^+} u^*\omega < +\infty.$$

Then u can be extended smoothly over D_1^+.

Proof By Whitney's embedding result we can embed (M, g_J) into some $\mathbf{R}^N$. Because M is compact, u is bounded in $L^p(D_1^+, \mathbf{R}^N)$ for any $1 \leq p \leq +\infty$. On the other hand

$$\begin{aligned}\int_{D_1^+} u^*\omega &= \int_{D_1^+} \omega(u_s, u_t)ds \wedge dt \\ &= \int_{D_1^+} \omega(u_s, J(u)u_s)ds \wedge dt \\ &= \frac{1}{2}\int_{D_1^+} (|u_s|_J^2 + |u_t|_J^2)ds \wedge dt \\ &=: \frac{1}{2}\int_{D_1^+} |\nabla u(z)|^2 ds \wedge dt \\ &< +\infty.\end{aligned}$$

Hence we have a bound in $W^{1,2}(D_1^+, \mathbf{R}^N)$ which does not help us much since we cannot apply our regularity results from Appendix A.2 unless we also have proved that u can be extended continuously over D_1^+. The strategy is the following: First we will transform the problem from D_1^+ to the infinite strip $(-\infty, -1] \times [0, \frac{1}{2}] =: S^-$. Using a bubbling off argument and the L^2-bound on the gradient we will get a uniform gradient bound on S^- and finally bounds on all derivatives. Then $W^{1,2}$-convergence of arcs "going to $-\infty$" will provide C^0-convergence and finally this will lead to continuity of u on D_1^+.

Let us consider the biholomorphic map

$$\phi : S := (-\infty, 0) \times [0, \frac{1}{2}] \xrightarrow{\sim} D_1^+ \backslash \{0\}$$

$$\phi(s, t) := e^{2\pi(s+it)}.$$

Then $S^- \subset S$ and define $v := u \circ \phi : S \to M$. We have

- $v_s + J(v)v_t = 0$
- $v(\partial S) \subset L$ where $\partial S := ((-\infty, 0) \times \{0\}) \cup ((-\infty, 0) \times \{\frac{1}{2}\})$
- $\int_S v^*\omega = \int_{D_1^+ \backslash \{0\}} u^*\omega < +\infty$.

We claim that

$$\sup_{S^-} |\nabla v| < +\infty.$$

Arguing indirectly, we assume that there is a sequence $z'_k = (s'_k, t'_k) \in S^-$ so that $R'_k := |\nabla v(z'_k)| \to +\infty$. Note that we must have $s'_k \to -\infty$. Now choose a sequence $\varepsilon'_k \searrow 0$ so that still $R'_k \varepsilon'_k \to +\infty$. Applying Lemma 4.4.4 to the map

$$S^- \longrightarrow [0, +\infty)$$

$$z \longmapsto |\nabla v(z)|$$

we find sequences (ε_k), $(z_k) = ((s_k, t_k))$ so that

- $\varepsilon_k R_k := \varepsilon_k |\nabla v(z_k)| \geq \varepsilon'_k R'_k$
- $|z_k - z'_k| \leq 2\varepsilon_k$
- $|\nabla v(y)| \leq 2|\nabla v(z_k)|$ for all y with $|y - z_k| \leq \varepsilon_k$.

After choosing a subsequence we may assume that $t_k \to t_0 \in [0, \frac{1}{2}]$. Now define for $z \in \Omega_k := \{z = (s, t) \in \mathbf{C} |\ s \leq -R_k s_k\ ;\ -t_k R_k \leq t \leq R_k(\frac{1}{2} - t_k)\}$

$$v_k(z) := v\left(\frac{z}{R_k} + z_k\right).$$

Then

$$|\nabla v_k(0)| = 1$$

and

$$|\nabla v_k(z)| \leq 2 \ \text{ for } \ z \in B_{\varepsilon_k R_k}(0) \cap \Omega_k \ (\neq 0).$$

Now we have to consider the following cases:

1. $t_k R_k \to l < +\infty$ (then necessarily $R_k(\frac{1}{2} - t_k) \to +\infty$)
2. $t_k R_k \to +\infty$
   (a) $R_k(\frac{1}{2} - t_k) \to +\infty$
   (b) $R_k(\frac{1}{2} - t_k) \to \tilde{l} < +\infty$.

We claim that in case (1.) for every compact set

$$K \subset H_l := \{z = (s, t) \in \mathbf{C} \mid t > -l\}$$

there is a number $k_0 \in \mathbf{N}$ so that for each $k > k_0$

$$K \subset \Omega_k \cap B_{\varepsilon_k R_k}(0).$$

Indeed, we can find $k_0 \in \mathbf{N}$ so that for each $k > k_0$:

- $\sup\{s|(s,t) \in K\} \leq -R_k(s_k + 1) \to +\infty$
- $\sup\{t|(s,t) \in K\} \leq R_k(\frac{1}{2} - t_k) \to +\infty$
- $\inf\{t|(s,t) \in K\} \geq -t_k R_k$
- $\sup_{z \in K} |z| \leq \varepsilon_k R_k \to +\infty$

By the discussion in Chap. 4.3 and Appendix A.2 we have uniform C^∞-bounds for the sequence (v_k) on each compact set $K \subset H_l$ (provided $k > k_0$). Then by the Ascoli-Arzela theorem we conclude that the sequence (v_k) has a subsequence that converges in C^∞_{loc} to some

$$\tilde{v} : H_l \longrightarrow M$$

which satisfies

- $\tilde{v}_s + \tilde{J}(\tilde{v})\tilde{v}_t = 0$
- $|\nabla \tilde{v}(0)| = 1$
- $|\nabla \tilde{v}(z)| \leq 2$ for all $z \in H_l$.

Let $R > 0$ and

$$Q_R := \{(s,t) \in \mathbf{C} \mid -R \leq s \leq R\,;\ \frac{1}{R} - l \leq t \leq R\} \subset H_l.$$

If $k_0 \in \mathbf{N}$ is large enough then

$$Q_R \subset \Omega_k \cap B_{\varepsilon_k R_k}(0).$$

We compute

$$\int_{Q_R} \tilde{v}_k^* \omega = \int_{Q'_{R,k}} \tilde{v}^* \omega \longrightarrow 0 \ \text{ as } \ k \to +\infty$$

with

$$Q'_{R,k} := \{(s,t) \in \mathbf{C} \mid -\frac{R}{R_k} + s_k \leq s \leq \frac{R}{R_k} + s_k$$
$$\text{and } \ t_k + (R^{-1} - l)R_k^{-1} \leq t \leq \frac{R}{R_k} + t_k\},$$

because the volume of the cube $Q'_{R,k}$ tends to zero as $k \to +\infty$. By Fatou's lemma we have for all $R > 0$ large

$$\int_{Q_R} \tilde{v}^* \omega \leq \liminf_k \int_{Q_R} v_k^* \omega = 0.$$

Hence $\int_{H_l} \tilde{v}^*\omega = 0$ and therefore $\tilde{v}$ is constant in contradiction to $|\nabla\tilde{v}(0)| = 1$.

Now let us consider the case (2a.). For every compact set $K \subset \mathbf{C}$ there is a number $k_0 \in \mathbf{N}$ so that for each $k > k_0$ we have $K \subset \Omega_k \cap B_{\varepsilon_k R_k}(0)$. As before we get convergence in C^∞_{loc} to a pseudoholomorphic map

$$\tilde{v} : \mathbf{C} \longrightarrow M$$

satisfying $|\nabla\tilde{v}(0)| = 1$ and $|\nabla\tilde{v}(z)| \leq 2$ for all $z \in \mathbf{C}$. Looking at $v_k|_{Q_R}$ with $Q_R := \{(s,t) \in \mathbf{C} \mid -R \leq s,t \leq R\}$ we get again a contradiction.

The case (2b.) is similar to the first case, instead of H_l we have to use the half plane $\{z \in \mathbf{C} \mid \operatorname{Im} z \leq l\}$.

Hence we have proved that there is a uniform gradient bound

$$\sup_{z \in S^-} |\nabla v(z)| < +\infty.$$

Let us point out how to get uniform bounds for the higher derivatives of v on $S^-_\delta := (-\infty, -\delta] \times [0, 1/2]$ for each $\delta > 0$. We define

$$v_k : \tilde{S} := (-1, 0) \times [0, \frac{1}{2}] \longrightarrow M \quad ; k \in \mathbf{N} \cup \{0\}$$

$$v_k(s,t) := v(s-k, t).$$

Each map v_k also satisfies the Cauchy-Riemann equation

$$\frac{\partial}{\partial s} v_k + J(v_k)\frac{\partial}{\partial t} v_k = 0$$

and we have for all $k \in \mathbf{N} \cup \{0\}$

$$\sup_{z \in \tilde{S}} |\nabla v_k(z)| \leq \sup_{z \in S^-} |\nabla v(z)| < +\infty.$$

Since M is compact the results of Chap. 4.3 yield uniform bounds on the slightly smaller domains $\tilde{S}_\delta := (-1+\delta, -\delta) \times [0, \frac{1}{2}]$ for each $\delta > 0$:

$$\begin{aligned} &\sup \left\{ |D^\alpha v(z)| \mid -1 + \delta - k \leq s \leq -k - \delta \,,\ t \in [0, 1/2] \right\} \\ &= \sup_{z \in \tilde{S}_\delta} |D^\alpha v_k(z)| \\ &\leq c = c(\alpha, \delta) < +\infty \end{aligned}$$

which implies

$$\sup_{z \in S^-_\delta} |D^\alpha v(z)| \leq c.$$

Now we would like to show that there is a point $p \in L$ with $v(z_k) \to p$ for every sequence $(z_k) \subset S^-$ where $s_k \to -\infty$. Then u can be extended continuously over D_1^+. Define the function

$$\Phi : (-\infty, -1] \longrightarrow [0, +\infty)$$

by

$$\Phi(s) := \frac{1}{2} \int_{-\infty}^{s} \int_0^{1/2} (|v_s(\sigma, \tau)|^2 + |v_t(\sigma, \tau)|^2)\, d\tau\, d\sigma.$$

We observe that $\Phi(s)$ is always finite since it is bounded by $\int_{S^-} v^*\omega < +\infty$. Moreover, the function Φ is increasing and differentiable with

$$\frac{d}{ds}\Phi(s) = \int_0^{1/2} |v_s(s,t)|^2 dt = \int_0^{1/2} |v_t(s,t)|^2 dt$$

Let us write $v(s) := v(s, \,.\,)$, $v'(s) := \frac{\partial}{\partial s} v(s, \,.\,)$ and $\dot{v}(s) := \frac{\partial}{\partial t} v(s, \,.\,)$. We note that

$$\lim_{s \to -\infty} \Phi(s) = 0$$

by Lebesgue's theorem. Now

$$\begin{aligned} \frac{d^2\Phi}{ds^2}(s) &= \frac{d}{ds} \int_0^{1/2} \langle v_s(s,t), v_s(s,t) \rangle dt \\ &= 2 \int_0^{1/2} \langle v_{ss}(s,t), v_s(s,t) \rangle dt \\ &\leq c \end{aligned}$$

for all $s \leq -1$ since all the derivatives of v are bounded. Pick $\varepsilon > 0$ and choose $s_0 \leq -1$ so that for all $s \leq s_0$

$$\Phi(s) = \int_{-\infty}^{s} \left(\frac{d}{ds}\Phi(s) \right) ds \leq \frac{\varepsilon^2}{4c}.$$

Now take $s' \leq s$ with $|s' - s| \leq \frac{\varepsilon}{2c}$. Then

$$\begin{aligned} \left| \frac{d}{ds}\Phi(s)|_s - \frac{d}{ds}\Phi(s)|_{s'} \right| &\leq c\, |s' - s| \\ &\leq \frac{\varepsilon}{2}. \end{aligned}$$

Therefore

$$\frac{d}{ds}\Phi(s)|_{s'} \geq \frac{d}{ds}\Phi(s)|_s - \frac{\varepsilon}{2}$$

which implies

$$\frac{\varepsilon^2}{4c} \geq \int_{s-\frac{\varepsilon}{2c}}^{s} \frac{d}{ds}\Phi(s)|_{s'}\, ds' \geq \left(\frac{d}{ds}\Phi(s)|_s - \frac{\varepsilon}{2}\right)\frac{\varepsilon}{2c}$$

and

$$\frac{d}{ds}\Phi(s) \leq \varepsilon$$

for all $s \leq s_0$, hence $\lim_{s\to-\infty} \frac{d}{ds}\Phi(s) = 0$. The length of the arc $v(s) : [0, 1/2] \to M$ is given by

$$\begin{aligned}\int_0^{1/2} |\dot{v}(s)(t)|\, dt &\leq \frac{1}{\sqrt{2}}\|\dot{v}(s)\|_{L^2([0,1/2])} \\ &= \left(\frac{1}{2}\frac{d}{ds}\Phi(s)\right)^{1/2} \longrightarrow 0 \\ &\quad \text{as } s \to -\infty.\end{aligned}$$

In order to establish convergence of the arcs $v(s)$ in C^0 as $s \to -\infty$ it suffices by the Sobolev embedding theorem if we prove $W^{1,2}$-convergence. Since we have L^2-convergence of $\dot{v}(s)$ already, we are left with the L^2-convergence of $v(s)$:

We define the set of "small arcs"

$$\Lambda_L^\varepsilon := \{x \in C^\infty([0, 1/2], M) \mid x(\{0, 1/2\}) \subset L\ ;\ \|\dot{x}\|_{L^2([0,1/2])} < \varepsilon\}.$$

For any $\varepsilon > 0$ we can find some $s_0 = s_0(\varepsilon) \leq -1$ so that $v(s) \in \Lambda_L^\varepsilon$ for all $s \leq s_0$. By the Lagrange neighborhood theorem (see Theorem 2.2.2) there exist neighborhoods U of $L \subset M$; $V \subset T^*L$ of the zero section and a diffeomorphism

$$\phi : U \longrightarrow V$$

so that $\phi^*(-d\theta) = \omega$ where θ is the canonical one-form on T^*L and $\phi|_L = \mathrm{id}$. Since L is compact we can find some small $\varepsilon > 0$ so that all the arcs $x \in \Lambda_L^\varepsilon$ have image in some open neighborhood U' of L with $U' \subset \bar{U}' \subset U$. Write

$$\lambda := -\phi^*\theta.$$

Then $d\lambda = \omega|_U$ and λ vanishes on L. We define for some $\varepsilon > 0$ as above

$$a(x) := \int_0^{1/2} x^*\lambda \quad ; x \in \Lambda_L^\varepsilon. \tag{5.7}$$

Now we claim that

$$\frac{d}{ds} a(v(s)) = \frac{d}{ds}\Phi(s)$$

if $s \le s_0$ where s_0 is chosen so small that $v(s) \in \Lambda_L^\varepsilon$ for all $s \le s_0$. Indeed

$$\begin{aligned}
\frac{d}{ds} a(v(s)) &= \lim_{h\to 0} \frac{1}{h}\left[\int_0^{1/2} \lambda(\partial_t v(s+h))\,dt - \int_0^{1/2} \lambda(\partial_t v(s))\,dt\right] \\
&= \lim_{h\to 0} \frac{1}{h} \int_{\partial([s,s+h]\times[0,1/2])} v^*\lambda \\
&\quad \text{since } v(\partial S) \subset L \text{ and } \lambda \text{ vanishes on } L \\
&= \lim_{h\to 0} \frac{1}{h} \int_s^{s+h} \int_0^{1/2} d\lambda(v_s, v_t)\,dt\,ds \\
&= \int_0^{1/2} \omega(v_s(s,t), v_t(s,t))\,dt \\
&= \int_0^{1/2} |v_s(s,t)|^2\,dt \\
&= \frac{d}{ds}\Phi(s)
\end{aligned}$$

and $a(v(s)) - \Phi(s)$ is constant. Now

$$\begin{aligned}
a(v(s)) &= \int_0^{1/2} v(s)^*\lambda \\
&= \int_0^{1/2} \lambda(v(s,t)) \cdot v_t(s,t)\,dt \qquad (5.8) \\
&\le \text{ const } \cdot \int_0^{1/2} |v_s(s,t)|\,dt \\
&\longrightarrow 0 \quad \text{as } s \to -\infty.
\end{aligned}$$

Therefore $a(v(s)) = \Phi(s)$.

In order to prove the L^2-convergence for the arcs $v(s)$ as $s \to -\infty$ we will need an estimate of the form

$$a(x) \le \text{ const } \cdot \|\dot{x}\|^2_{L^2([0,1/2])}.$$

Note that (5.8) implies already $a(x) \leq \text{const} \cdot \|\dot{x}\|_{L^2([0,1/2])}$ but this estimate is not good enough as we will see. We will have to "localize" the situation:

Recall that M is embedded into some $\mathbf{R}^N$. We choose finitely many open balls $\{B(x_i)\}_{1\leq i\leq K}$ centered at points $x_i \in \bar{U}'$ so that

- $U \supset \bigcup_{1\leq i\leq K} B(x_i) \supset \bar{U}'$
- $\phi(\bar{U}' \cap B(x_i)) \subset V_i \subset T^*L$ where $\psi_i : V_i \xrightarrow{\sim} \mathbf{R}^{2n}$ is a coordinate chart of T^*L

If $\varepsilon > 0$ is chosen smaller than the Lebesgue number of the covering $\{B(x_i)\}_{1\leq i\leq K}$ then the image of every arc $x \in \Lambda_L^\varepsilon$ is entirely contained in one of the sets $U_i := \bar{U}' \cap B(x_i)$. Let us review the definition of Lebesgue number and the proof of this fact. Let (X, d) be a metric space. A number $a > 0$ is called the Lebesgue number of an open cover $\{G_i\}$ of X if each subset $Y \subset X$ with diameter less or equal a is contained in at least one of the sets G_i.

Lemma 5.3.2 *If the metric space X is sequentially compact then every open cover has a Lebesgue number.*

Proof Let us call a set $A \subset X$ large if it is not contained in any of the sets G_i. Define

$$a' := \inf\{\, \text{diam}(Y) \mid Y \text{ is large}\,\}.$$

We have to show that $a' > 0$. So assume that $a' = 0$. Let us pick a sequence of large sets Y_n with $0 < \text{diam}(Y) < \frac{1}{n}$ and choose $x_n \in Y_n$. Since X is sequentially compact, we may assume that $x_n \to x \in X$ after passing to a subsequence. We have $x \in G_{i_0}$ for some i_0. The sets G_i are all open, hence there exists a ball $B_r(x)$ which is contained in G_{i_0} and a number N so that $x_n \in B_r(x)$ whenever $n \geq N$. Choosing N so that $N > \frac{2}{r}$, we conclude

$$Y_n \subset B_r(x) \subset G_{i_0}$$

for $n \geq N$ large enough in contradiction to the assumption that Y_n is large. □

Hence if $\varepsilon > 0$ is small enough the definition (5.7) of $a(x)$ makes sense for all $x \in \Lambda_L^\varepsilon$ and since $\phi \circ x$ has image in a coordinate patch V_i of T^*L we can compute $a(x)$ in terms of local coordinates $(q_1, \ldots, q_n, p_1, \ldots, p_n)$ of T^*L.

$$\begin{aligned} a(x) &= \int_0^{1/2} x^*\lambda \\ &= -\int_0^{1/2} (\phi \circ x)^*\theta \\ &= -\int_0^{1/2} z^* \left(\sum_{j=1}^n p_j dq_j \right) \end{aligned}$$

where

$$\begin{aligned} z(t) &:= (\psi_i \circ \phi \circ x)(t) \\ &= (\alpha_1(t), \ldots, \alpha_n(t), \beta_1(t), \ldots, \beta_n(t)). \end{aligned}$$

Hence

$$\begin{aligned} a(x) &= -\sum_{j=1}^{n} \int_0^{1/2} \beta_j(t)\dot{\alpha}_j(t)\, dt \\ &= -\frac{1}{2}\sum_{j=1}^{n} \int_0^{1/2} \beta_j(t)\dot{\alpha}_j(t)\, dt + \frac{1}{2}\sum_{j=1}^{n} \int_0^{1/2} \dot{\beta}_j(t)\alpha_j(t)\, dt - \\ &\quad -\frac{1}{2}\sum_{j=1}^{n} \int_0^{1/2} \frac{d}{dt}(\beta_j(t)\alpha_j(t))\, dt \\ &= \frac{1}{2}\sum_{j=1}^{n} \int_0^{1/2} (\dot{\beta}_j(t)\alpha_j(t) - \beta_j(t)\dot{\alpha}_j(t))\, dt. \end{aligned}$$

Since $\beta_j(1/2) = \beta_j(0) = 0$ because of $x(1/2), x(0) \in L$. If $\langle\, .\,,\, .\,\rangle$ is the usual scalar product on $\mathbf{R}^{2n} = \mathbf{C}^n$ and i is the standard complex structure on $\mathbf{C}^n$ this becomes

$$a(x) = \frac{1}{2}\langle -i\dot{z}, z\rangle_{L^2([0,1/2])}$$

Now we define the piecewise smooth loop

$$\zeta : \mathbf{R}/\mathbf{Z} \approx S^1 \longrightarrow \mathbf{C}^n$$

$$\zeta(t) := \begin{cases} z(t) & \text{for } 0 \le t \le 1/2 \\ (\alpha_1(1-t), \ldots, \alpha_n(1-t), 0, \ldots, 0) & \text{for } 1/2 \le t \le 1 \end{cases}.$$

Then

$$a(x) = \frac{1}{2}\langle -i\dot{\zeta}, \zeta\rangle_{L^2([0,1])}$$

Let

$$\zeta(t) = \sum_{k\in\mathbf{Z}} a_k e^{2\pi i k t}$$

be the Fourier expansion of ζ. Then

$$-i\dot{\zeta}(t) = \sum_{k\in\mathbf{Z}} 2\pi k a_k e^{2\pi ikt}$$

and

$$\begin{aligned} a(x) &= \sum_{k\in\mathbf{Z}} \pi k|a_k|^2 \\ &\le \sum_{k\in\mathbf{Z}} 4\pi^2 k^2 |a_k|^2 \\ &= \| - i\dot{\zeta}\|^2_{L^2([0,1])} \\ &= \|\dot{\zeta}\|^2_{L^2([0,1])} \\ &\le \text{const} \cdot \|\dot{x}\|^2_{L^2([0,1/2])}. \end{aligned} \tag{5.9}$$

Moreover

$$\begin{aligned} \frac{d}{ds} a(v(s)) &= \|v_t(s)\|^2_{L^2([0,1/2])} \\ &= \|v_t(s)\|_{L^2([0,1/2])} \|v_s(s)\|_{L^2([0,1/2])} \\ &\ge c \cdot a(v(s))^{1/2} \|v_s(s)\|_{L^2([0,1/2])} \end{aligned} \tag{5.10}$$

We consider the following cases:

(1) There is some $s \le s_0$ with $a(v(s)) = 0$.

Since the function $s \longmapsto a(v(s)) = \Phi(s)$ is nonnegative and increasing this implies $a(v(s')) = 0$ for all $s' \le s$ and therefore

$$\frac{d}{ds} a(v(s))\bigg|_{s=s'} = 0$$

Because of

$$\begin{aligned} \frac{d}{ds} a(v(s)) &= \frac{d}{ds}\Phi(s) \\ &= \|v_t(s)\|^2_{L^2([0,1/2])} \\ &= \|v_s(s)\|^2_{L^2([0,1/2])} \end{aligned}$$

we must have $v(s') \equiv$ const for all $s' \le s$ and we are done.

(2) $a(v(s)) > 0$ for all $s \leq -1$
This implies for $s_1 \leq s_2 \leq s_0$

$$\begin{aligned} \|v(s_2) - v(s_1)\|_{L^2([0,1/2])} &\leq \int_{s_1}^{s_2} \|v_s(s)\|_{L^2([0,1/2])} ds \\ &\leq c \int_{s_1}^{s_2} \frac{d}{ds}(a(v(s)))^{1/2} ds \\ &= c\,(a(v(s_2))^{1/2} - a(v(s_1))^{1/2}) \\ &\longrightarrow 0 \quad \text{for } s_1, s_2 \to -\infty \end{aligned}$$

since $a(v(s)) \to 0$ for $s \to -\infty$. Summarizing we obtain convergence

$$v(s) \longrightarrow w \in C^0([0, 1/2], \mathbf{R}^N) \cap W^{1,2}([0, 1/2], \mathbf{R}^N)$$

for $s \to -\infty$ and since $\|\dot{v}(s)\|_{L^2([0,1/2])} \to 0$ the limit is constant $w(t) \equiv p \in L \subset M \subset \mathbf{R}^N$.

Going back to the half-disk D_1^+ we know now that

- $u \in C^0(D_1^+, \mathbf{R}^N) \cap W^{1,2}(D_1^+, \mathbf{R}^N)$
- $u((-1, +1)) \subset L$
- u is smooth on $D_1^+ \backslash \{0\}$ and satisfies $u_s + J(u)u_t = 0$

Considering u near the point $0 \in D_1^+$ and applying the regularity results of Appendix A.2 we obtain smoothness.

We mention the following version of Theorem 5.3.1 for interior singularities:

Theorem 5.3.3 *Let $D := \{z \in \mathbf{C} \mid |z| < 1\}$ and $u : D^* := D\backslash\{0\} \to M$ be a smooth map satisfying*

$$u_s + J(u)u_t = 0$$

and

$$\int_D u^*\omega < +\infty$$

Then u can be extended smoothly over D.

Proof The proof is similar to the one of Theorem 5.3.1, but simpler. We will just indicate the necessary modifications.

Instead of transforming to the infinite strip we have to consider here the infinite cylinder $(-\infty, 0) \times S^1$. If one carries out the bubbling off analysis things become simpler since it is not necessary any more to consider points approaching the boundary. Instead of Λ_L^ε we have to consider smooth loops x in M with

$$\|\dot{x}\|_{L^2(S^1)} < \varepsilon.$$

Now we cover the target manifold M by finitely many (symplectic) Darboux-charts (see Chap. 2). Choosing ε small enough, the image of each loop x with $\|\dot{x}\|_{L^2(S^1)} < \varepsilon$ will be contained entirely in one of the Darboux-charts on M so that we can define (as before) $a(x)$ with respect to one of these charts which contain $x(S^1)$. It can be shown that the definition of $a(x)$ does not depend on the choice of the chart. Now we just proceed as in Theorem 5.3.1 except that we need not replace z by ζ; we expand z in a Fourier series. □

Chapter 6
Intersection Theory for Pseudoholomorphic Disks



The chapter is devoted to the study of the intersection theory of holomorphic curves in (real) four-dimensional almost complex manifolds. Some of the ideas originate from McDuff's work in [84], see also [85], which in turn was influenced by Gromov's seminal paper [60]. Other important work in this direction is [89] and [104].

6.1 Generalized Tangent Space and Order in a Point

In this section we derive a number of useful quantities describing local properties of pseudoholomorphic curves.

We begin with the local model. Assume therefore we have a smooth map $u : D_\varepsilon \to \mathbf{C}^n$ solving

$$u_s + J(u)u_t = 0 \text{ and } u(0) = 0.$$

Here J is an almost complex structure defined on a neighborhood of 0. We may also assume that $J(0) = i$. By the similarity principle either $u = 0$ near 0 or there exists an integer $\ell \geq 1$ such that the following limit exists

$$\lim_{|z|\to\infty} \frac{1}{z^\ell} \cdot_{J(u(z))} u(z) = a_\ell \in \mathbf{C}^n \setminus \{0\} \tag{6.1}$$

We will use the following notation: If $a + ib \in \mathbf{C}$ and $\zeta \in \mathbf{C}^n$ then we define

$$(a + ib) \cdot_J \zeta := a\,\zeta + b\,J\,\zeta.$$

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Indeed, employing the similarity principle, we find that

$$\frac{1}{z^\ell} \cdot_{J(u(z))} u(z) = \Phi(z)(\frac{1}{z^\ell} \cdot_i \sigma(z)).$$

Since σ has the form $\sigma(z) = a_\ell z^\ell + h(z)$, where h is of higher order and $a_\ell \neq 0$, it follows that the limit in (6.1) is a_ℓ. We call ℓ the order of u at the point 0. We denote it by $o(u, 0)$. If $u = 0$ near 0 we define the order to be ∞.

In the general case assume that (W, J) is an almost complex manifold and S a Riemann surface without boundary. We denote its complex structure by i. Assume that $u : S \to W$ is a pseudoholomorphic curve, i.e.

$$Tu \circ i = J \circ Tu.$$

For $p_0 \in S$ we define the order $o(u, p_0) \in \mathbf{N}$ of u at p_0 as follows. The order is ∞ if u is constant near p_0. In the case that u is not constant near p_0, we take a holomorphic embedding $\phi : (D, 0) \to (S, p_0)$ and a chart Γ defined on a neighborhood of $u(p_0)$ mapping $u(p_0)$ to $0 \in \mathbf{C}^n$ such that

$$i \circ T\Gamma(u(p_0)) = T\Gamma(u(p_0)) \circ J(u(z_0)).$$

Perhaps, after restricting ϕ to a smaller disk, the map

$$w = \Gamma \circ u \circ \phi$$

satisfies $w(0) = 0$ and

$$w_s + j(w)w_t = 0$$

for an almost complex structure on $\mathbf{C}^n$ defined near 0 (conjugated to J via Γ, $j(0) = i$). The order of u at p_0 is now by definition the order of w at 0:

$$o(u, p_0) := o(w, 0).$$

We leave it as an exercise that the definition of order does not depend on the choices involved. It follows immediately from the definition and the similarity principle that the following corollary holds:

Corollary 6.1.1 *Let $u : S \to W$ as described above. Then $o(u, p_0) = 1$ if and only if u is an embedding near p_0. Moreover $o(u, p_0) = \infty$ if and only if u is constant near p_0.*

Moreover we have the following

Corollary 6.1.2 *Assume (S, i) is a connected Riemann surface without boundary and (W, J) an almost complex manifold. If $u : S \to W$ is a pseudoholomorphic curve and $p_0 \in S$ is a point with $o(u, p_0) = \infty$, then u is a constant.*

Proof Let Θ be the set of points in S having an open neighborhood, where u is constant. Clearly $p_0 \in \Theta$ by assumption. By definition Θ is open. Assume next that p is a point in the closure of Θ. Then on some open neighborhood of p the similarity principle is valid. We may identify p with 0 and the open neighborhood with the open unit disk in $\mathbf{C}$. Hence $u(z) = \Phi(z)\sigma(z)$ and $u(0) = 0$. Moreover, by assumption there exists a sequence $z_k \to 0$ such that $u(z_k) = u(z_j)$ for all k and j. This implies that $u(z_k) = u(0) = 0$. Hence $\sigma(z_k) = 0$ implying that $\sigma = 0$. This shows that Θ is closed. By assumption S is connected. Consequently, since $\Theta \neq \emptyset$ and Θ is closed and open we find that $\Theta = S$. This implies the desired result. □

Another immediate consequence of the similarity principle is the following:

Corollary 6.1.3 *Let $u : S \to W$ be a pseudoholomorphic curve in the almost complex manifold (W, J). Assume that u is not constant near a point $p_0 \in S$, but $Tu(p_0) = 0$. Then $Tu(p) \neq 0$ for $p \in U(p_0) \setminus \{p_0\}$, where $U(p_0)$ is a suitable neighborhood of p_0.*

Proof We may work in local coordinates. Hence

$$u : D \to \mathbf{C}^n \text{ and } u(0) = 0$$
$$u_s + J(u)u_t = 0$$
$$u_s(0) = 0$$

We find with $v = u_s$, differentiating the equation for u with respect to s:

$$\begin{aligned} 0 &= v_s + J(u)v_t + (DJ(u)u_s)u_t \\ &=: v_s + J(u(z))v_t + Av. \end{aligned}$$

Moreover $v(0) = 0$ by assumption. Observe that $z \mapsto A(z)$ is a smooth $\mathcal{L}_{\mathbf{R}}(\mathbf{C}^n)$-valued map. Pick a smooth map $z \mapsto \Gamma(z)$ with $\Gamma(z) \in \mathrm{GL}_{\mathbf{R}}(\mathbf{C}^n)$ and

$$J(u(z))\Gamma(z) = \Gamma(z)i.$$

Define w by

$$v(z) = \Gamma(z)w(z).$$

Then

$$\begin{aligned} 0 &= \Gamma w_s + \Gamma_s w + J(u)\Gamma_t w + J(u)\Gamma w_t + A\Gamma w \\ &= \Gamma[w_s + i w_t + (\Gamma^{-1}\Gamma_s + \Gamma^{-1}J(u)\Gamma_t + \Gamma^{-1}A\Gamma)w] \\ &=: \Gamma[w_s + i w_t + Bw] \end{aligned}$$

Here $z \mapsto B(z)$ is a smooth $\mathcal{L}_{\mathbf{R}}(\mathbf{C}^n)$-valued map. By the similarity principle

$$w(z) = \Psi(z)\sigma(z).$$

with $\sigma(0) = 0$ and $\Psi(z) \in \mathrm{GL}_{\mathbf{C}}(\mathbf{C}^n)$, $\Psi(0) = Id$. Summing up

$$v(z) = \Gamma(z)\Psi(z)\sigma(z) =: \Phi(z)\sigma(z).$$

By the properties of Ψ and Γ we find that Φ is of class $\cap_{2<p<\infty} W^{1,p}_{loc}$ and moreover $\Phi(z)$ is invertible and satisfies $\Phi(z)i = J(u(z))\Phi(z)$. In particular, either $v = 0$ which is not possible since u is not constant near 0, or $v(z) \neq 0$ for $z \neq 0$, but $|z|$ small. □

We use now the previous result to define the generalized tangent bundle associated to a non-constant pseudoholomorphic curve. Assume $u : S \to W$ is a non-constant pseudoholomorphic curve. Assume further that S is connected with smooth boundary (perhaps empty). If u is an immersion we define the generalized tangent space $\mathcal{T}_u$ to be the one-dimensional complex sub-bundle of u^*TW whose fibre at the point z is the image of $Tu(z)$. This clearly is a smooth line-bundle.

Assume next that u is allowed to have singularities, i.e. there are isolated points, where $Tu(z)$ vanishes. Let us assume that the singularities are not on ∂S. Denote the collection of all singular points of u by $\mathcal{S}$. Away from $\mathcal{S}$ we can define the tangent space as before. However we shall see that it can be extended over the singularities smoothly. So put $\dot{S} = S \setminus \mathcal{S}$ and consider for $w = u|\dot{S}$ the tangent bundle $\mathcal{T}_w$. Let p_0 be a singular point and denote by U a small neighborhood only containing the singular point p_0. For $p \in \dot{U} := U \setminus \{p_0\}$ the tangent bundle at p is the image of $Tu(p)$. Let ℓ be the order of u at the point p. By assumption $\ell \geq 2$. Take a holomorphic chart $\tau : (B, 0) \to (U, p_0)$ around p_0. Then the tangent bundle at p is the image of $Tu(p) \circ T\tau(\tau^{-1}(p))$ which has the same image as

$$\frac{1}{(\tau^{-1}(p))^{\ell-1}} \cdot_{J(u(p))} Tu(p) \circ T\tau(\tau^{-1}(p)).$$

As an immediate consequence of the similarity principle this map can however be extended as a nowhere vanishing section over p_0, so that the extension belongs to $W^{1,p}_{loc}$ for every $2 < p < \infty$. To see this and the fact that the section is actually smooth let us consider the local situation. Therefore we may assume that $u : D \to \mathbf{C}^n$ is a non-constant J-holomorphic disk satisfying

$$u(0) = 0 \text{ and } Tu(0) = 0.$$

Moreover we assume that $J(0) = i$. We denote by $\ell \geq 2$ the order of u at 0. From

$$u_s + J(u)u_t = 0$$

we infer that $v = u_s$ satisfies

$$v_s + J(u)v_t + (DJ(u)v)u_t = 0.$$

By the similarity principle

$$v(z) = \Phi(z)\sigma(z)$$

for some holomorphic map σ satisfying $\sigma(0) = 0$ and a matrix valued map Φ of class $W^{1,p}$ for every $2 < p < \infty$ satisfying

$$\Phi(z) \circ i = J(u(z)) \circ \Phi(z)$$

and

$$\Phi(0) = Id.$$

Applying the similarity principle to u yields

$$u(z) = \Psi(z)\beta(z),$$

where β and Ψ have similar properties. Denote the first non-vanishing coefficient in the power series of β by a_ℓ (recall that a_ℓ is the order). Denote the coefficients of σ by b_j. We compute

$$\begin{aligned} a_\ell &= \lim_{s\to 0} \frac{u(s,0)}{s^\ell} \\ &= \lim_{s\to 0} \frac{u_s(s,0)}{s^{\ell-1}} \\ &= b_{\ell-1}. \end{aligned}$$

This implies that the first non-vanishing coefficient of σ is $b_{\ell-1}$, which takes the value a_ℓ. The nowhere vanishing section of the tangent bundle we mentioned earlier is now defined by

$$\begin{aligned} \Phi(z)\alpha(z) &:= \frac{1}{z^{\ell-1}} \cdot_{J(u(z))} u_s(z) \\ &= \Phi(z)\left[a_\ell + \sum_{k=0}^{\infty} b_{\ell+k} z^{k+1}\right] \end{aligned}$$

where α is a holomorphic function which does not vanish near 0. The map w defined by

$$w(z) := \Phi(z)\alpha(z)$$

is of class $W^{1,p}$ as a product of a matrix valued function of class $W^{1,p}$ and a holomorphic map of class $W^{1,p}$ near 0. We will now show that this map is actually smooth. The map

$$z \longmapsto w_s(z) + J(u(z))w_t(z) + (DJ(u(z))w(z))u_t(z)$$

is of class L^p in a neighborhood of 0. We show that this map vanishes for $z \neq 0$ and z close to 0. This implies that w is a $W^{1,p}$-solution of the partial differential equation

$$w_s + J(u)w_t + (DJ(u)w)u_t = 0.$$

Interior elliptic regularity (see Appendix A.2) then implies that w is smooth in a neighborhood of 0. Note that w is already smooth away from 0. Indeed for $z \neq 0$ we have

$$w(z) = \frac{1}{z^{\ell-1}} \cdot_{J(u(z))} u_s(z) \neq 0.$$

Hence our proof will show that there is no singularity at 0. Denote by $a(z)$ the real part of $\frac{1}{z^{\ell-1}}$ and by $b(z)$ the imaginary part. For $z \neq 0$ we have

$$a_s - b_t = 0 \text{ and } a_t + b_s = 0,$$

since $\frac{1}{z^{\ell-1}}$ is holomorphic. We note that

$$w(z) = a(z)u_s(z) + b(z)u_t(z).$$

We compute

$$\begin{aligned} w_s + J(u)w_t + (DJ(u)w)u_t &= a_s u_s + a u_{ss} + b_s u_t + b u_{ts} \\ &\quad + J(u)[a_t u_s + a u_{st} + b_t u_t + b u_{tt}] \\ &\quad + a(DJ(u)u_s)u_t + b(DJ(u)u_t)u_t \\ &= a_s[u_s + J(u)u_t] + b_s[u_t - J(u)u_s] \\ &\quad + a[u_{ss} + J(u)u_{st} + (DJ(u)u_s)u_t] \\ &\quad + b[u_{ts} + J(u)u_{tt} + (DJ(u)u_t)u_t] \\ &= [a\frac{\partial}{\partial s} + b\frac{\partial}{\partial t}](u_s + J(u)u_t) \\ &= 0. \end{aligned}$$

This proves that w is a smooth nonvanishing map near 0. Let us note this result as follows.

Theorem 6.1.4 *Let $u : S \to W$ be a non-constant pseudoholomorphic curve defined on some connected Riemann surface without boundary. Then there exists a uniquely defined smooth complex line bundle $\mathcal{T}_u \to S$, which is a smooth subbundle of u^*TW such that for every p with $o(u, p) = 1$ the fiber is the image of $Tu(z)$. The line bundle $\mathcal{T}_u$ is a called the generalized tangent bundle of u.* □

6.2 Intersections with Different Generalized Tangent Spaces

In this section we study the intersections of two pseudoholomorphic curves in real four-dimensional manifolds (W, J) equipped with an almost complex structure J.

Assume that we have two nontrivial holomorphic curves u and v

$$\begin{aligned} u &: D \to \mathbf{C}^2,\ u(0) = 0 \\ v &: D \to \mathbf{C}^2,\ v(0) = 0 \\ u_s &+ J(u)u_t = 0 \\ v_s &+ J(v)v_t = 0. \end{aligned} \tag{6.2}$$

Again we may assume $J(0) = i$. Denote by Γ_u and Γ_v the generalized tangent spaces of u and v respectively at 0. They are complex lines in $(\mathbf{C}^2, i)$. We either have $\Gamma_u = \Gamma_v$ or $\Gamma_u \oplus \Gamma_v = \mathbf{C}^2$. We begin with the following lemma.

Lemma 6.2.1 *Let u and v be as described in (6.2) and assume that $J(0) = i$. If the generalized tangent spaces at 0 satisfy*

$$\Gamma_u \neq \Gamma_v$$

then 0 is an isolated intersection point.

Proof Arguing indirectly, we assume that

$$u(z_k) = v(z_k^{'})$$

for sequences $z_k \to 0$ and $z_k^{'} \to 0$. Observe that by the similarity principle we have $u(z) \neq 0$ for $0 < |z| \leq \delta$ and the same for v, provided δ is small enough. In particular for k large $u(z_k) \neq 0$. Let m be the order of u and let n the order of v at 0. Then

$$\frac{u(z_k)}{z_k^m} = \frac{(z_k^{'})^n}{z_k^m} \cdot_i \frac{v(z_k^{'})}{(z_k^{'})^n}.$$

Since the following limits exist

$$a_u := \lim_{k\to\infty} \frac{1}{z_k^m} \cdot_{J(u(z_k))} u(z_k) \overset{(J(0)=i)}{=} \lim_{k\to\infty} \frac{1}{z_k^m} \cdot_i u(z_k) \neq 0$$

$$a_v := \lim_{k\to\infty} \frac{v(z_k^{'})}{(z_k^{'})^n} \neq 0$$

and because they generate the generalized tangent spaces Γ_u and Γ_v respectively, we see that the limit

$$\Delta := \lim_{k\to\infty} \frac{(z_k^{'})^n}{z_k^m} \in \mathbf{C}$$

exists as well, and it satisfies

$$a_u = \Delta \cdot a_v.$$

This implies however $\Gamma_u = \Gamma_v$ contrary to our assumption that the two generalized tangent spaces are different. □

Summing up the whole discussion we deduce that for sufficiently small $\varepsilon > 0$ the map

$$\Theta : D_\varepsilon \times D_\varepsilon \to \mathbf{C}^2 : \Theta(z, z') = u(z) - u(z')$$

has only one zero, namely $(0, 0)$. The local intersection index

$$\mathrm{int}((u, 0), (v, 0))$$

is therefore well-defined. It is independent of ε provided ε is small enough. We can homotop Θ to a rather special map without introducing zeros at the boundary. Indeed, using the similarity principle, we write for $\varepsilon > 0$ small enough

$$\Theta(z, z') = \Phi(z)\sigma(z) - \Psi(z')\delta(z'),$$

with

$$\sigma(z) = a_m z^m + \text{‘higher order’ and } \delta(z) = b_n z^n + \text{ ‘higher order’}$$

which is homotopic to

$$\Theta^1(z, z') = a_m z^m - b_n (z')^n.$$

The vectors a_m and a_n are linearly independent over $\mathbf{C}$ by our assumption that Γ_u and Γ_v are different. The map Θ^1 is homotopic to Θ^2 defined by

$$\Theta^2(z, z') = (z^m, -(z')^n).$$

Here the local intersection index is obviously given by

$$\text{int}((u, 0), (v, 0)) = m \cdot n.$$

We have proved the following theorem:

Theorem 6.2.2 *If u and v are nontrivial pseudoholomorphic curves with $u(0) = v(0) = 0$, so that the generalized tangent spaces at 0 are different, then the local intersection index at 0 is the product of the local orders:*

$$int((u, 0), (v, 0)) = o(u, 0) \cdot o(v, 0).$$

□

In the following we study the analytically more interesting (and more difficult) case of two curves intersecting at 0, but having identical generalized tangent spaces. This study is relying on some sophisticated normal forms. In the following sections we will give a concrete description of a holomorphic curve in the neighborhood of a point as well as a relative normal form.

6.3 A Normal Form for Holomorphic Curves Near a Point

Assume we have a map $v : D \to \mathbf{C}^2$, $v(0) = 0$ satisfying $v_s + J(v)v_t = 0$. Assume that $J(0) = i$. We introduce polar coordinates

$$\sigma(s, t) = e^{-2\pi(s+it)},\ (s, t) \in \mathbf{R}^+ \times S^1,\ S^1 = \mathbf{R}/\mathbf{Z}.$$

Replacing v by $u = v \circ \sigma$ we see that u satisfies

$$u : [0, \infty) \times S^1 \to \mathbf{C}^2$$
$$u_s + J(u)u_t = 0$$
$$\lim_{s\to\infty} u(s, t) = 0 \text{ in } C^\infty \text{ uniformly in } t.$$

The following estimates are easily established.

Lemma 6.3.1 *For every multi-index $\beta \in \mathbf{N}^2$ there exists a constant c_β such that*

$$|(D^\beta u)(s, t)| \le c_\beta e^{-2\pi s} \text{ for } s \to \infty.$$

Proof By construction $u(s,t) = v(e^{-2\pi(s+it)})$ with $v(0) = 0$. By the similarity principle applied to v we find $v(z) = \Phi(z)h(z)$, where h is holomorphic and $h(0) = 0$. Moreover Φ has the usual regularity properties. If $\ell \geq 1$ is the order of v at 0 then

$$\begin{aligned} |u(s,t)| &= |v(\sigma(s,t))| \\ &= |\Phi(\sigma(s,t))h(\sigma(s,t))| \\ &\leq c\,\sigma(s,t)^{\ell} \\ &= c\ e^{-2\pi\ell s} \\ &\leq c\ e^{-2\pi s} \end{aligned}$$

for some generic constant $c > 0$. If $|\beta| \geq 1$ we obtain

$$(D^{\beta}u)(s,t) = e^{-2\pi(s+it)} f(s,t)$$

for some function $f = f_{\beta}$ which is bounded uniformly in s and t. Hence we arrive at the same conclusion as before. □

By the similarity principle $u(s,t) \neq 0$ for s large, unless of course, $v = 0$. Therefore we can define a function $\alpha : \mathbf{R}^{+} \to \mathbf{R}$ by

$$\alpha(s) = \frac{(u(s), \frac{\partial u}{\partial s}(s))}{(u(s), u(s))}.$$

Here $(.,.)$ denotes the real L^2-inner product where integration is over the variable $t \in S^1$, and $u(s)$ is the loop defined by $u(s)(t) = u(s,t)$, $t \in S^1$. Again as a consequence of the similarity principle it is possible to give a precise description of the behaviour of α. Clearly α is a smooth map. But moreover

Lemma 6.3.2 *Let ℓ be the order of u at 0. Then*

$$\lim_{s\to\infty} \alpha(s) = -2\pi\ell.$$

Proof We note that

$$\|u(s)\|^2 = \int_0^1 \left| \Phi(\sigma(s,t)) \left(\sum_{j=\ell}^{\infty} a_j \sigma(s,t)^j \right) \right|^2 dt.$$

This yields

$$e^{4\pi\ell s}\, \|u(s)\|^2 \longrightarrow |a_{\ell}|^2.$$

On the other hand, we compute with

$$u_s(s,t) = -\Phi(\sigma(s,t)) \sum_{j=\ell}^{\infty} 2\pi j a_j \, e^{-2\pi j(s+it)} + (\Phi \circ \sigma)_s \sum_{j=\ell}^{\infty} a_j \, e^{-2\pi j(s+it)}$$

and

$$(\Phi \circ \sigma)_s = -2\pi \, D\Phi(\sigma(s,t)) \, \sigma(s,t)$$

$$\begin{aligned} & e^{4\pi \ell s}(u(s), \frac{\partial u}{\partial s}(s)) \\ = & \int_0^1 \langle \Phi(\sigma(s,t))[a_\ell + \#], -\Phi(\sigma(s,t))[2\pi \ell a_\ell + \#] \rangle \, dt + \\ & + \int_0^1 \langle \Phi(\sigma(s,t))[a_\ell + \#], -2\pi \, D\Phi(\sigma(s,t))\sigma(s,t)[a_\ell + \#] \rangle \, dt \\ \longrightarrow & -2\pi \ell |a_\ell|^2, \end{aligned}$$

where # stands for higher order terms in the series expansion, and using that we can bound the second term by

$$\|(\Phi \circ \sigma)(s,*)[a_\ell + \#]\|_{L^q([0,1])} \, \|2\pi \, D\Phi(\sigma(s,*))\sigma(s,*)[a_\ell + \#]\|_{L^p([0,1])} \leq c \, e^{-2\pi s}$$

where $1/q + 1/p = 1$. Consequently

$$\alpha(s) = \frac{e^{4\pi \ell s} \, (u(s), \frac{\partial u}{\partial s}(s))}{e^{4\pi \ell s} \, \|u(s)\|^2} \longrightarrow -2\pi \ell.$$

□

Define

$$\xi(s,t) = \frac{u(s,t)}{\|u(s)\|}.$$

and recall that

$$\alpha(s) = \frac{(u(s), u_s(s))}{(u(s), u(s))}.$$

We know that $u(s) \neq 0$ for large s unless v vanishes in a neighborhood of 0. Before we are going to find a relative normal form we have to understand the behaviour of a pseudoholomorphic curve in a neighborhood of a point much better. For this we need the following important theorem.

Theorem 6.3.3 *Let u, α and ξ as described above and denote by $\ell < \infty$ the order of v at 0. Then we have uniform estimates for every partial derivative D^β and derivative D^j:*

$$\sup_{s,t} |D^\beta \xi(s,t)| < \infty \ , \ \sup_{s} |D^j \alpha(s)| < \infty. \tag{6.3}$$

Proof Recall the definition of α and ξ:

$$\alpha(s) = \frac{(u(s), u_s(s))}{(u(s), u(s))} \ , \ \xi(s,t) = \frac{u(s,t)}{\|u(s)\|}.$$

We see that ξ solves the following differential equation

$$\begin{aligned} \xi_s(s) &= \frac{u_s(s)}{\|u(s)\|} - \frac{(u(s), u_s(s))}{\|u(s)\|^2}\xi \\ &= \frac{-J(u(s))u_t(s)}{\|u(s)\|} - \alpha(s)\xi(s) \\ &= -J(u(s))\xi_t(s) - \alpha(s)\xi(s). \end{aligned}$$

Hence

$$\xi_s + J(u)\xi_t + \alpha\xi = 0.$$

We know that

$$\lim_{s\to\infty} \alpha(s) = -2\pi\ell,$$

where ℓ is the order of v at 0. We deduce from $J(0) = i$

$$\begin{aligned} 0 &= \xi_s + i\xi_t + \left(\int_0^1 DJ(\tau u)ud\tau\right)\xi_t + \alpha\xi \\ &= \xi_s + i\xi_t + \left(\int_0^1 DJ(\tau u)\xi d\tau\right)u_t + \alpha\xi \\ &=: \xi_s + i\xi_t + \Delta(u_t)\xi + \alpha\xi. \end{aligned} \tag{6.4}$$

As a consequence of the decay estimates for u we find that

$$|D^\beta(\Delta(u_t))(s,t)| \le c_\beta \cdot e^{-2\pi s}.$$

Now we are in the position to derive the desired estimates (6.3) from (6.4). Pick any $\delta_0 > 0$ and $s_0 > 0$ large, and define a monotone decreasing sequence $\delta_j \searrow \frac{1}{2}\delta_0$ by

$$\delta_j = \frac{1}{2}\delta_0(1 + 2^{-j}).$$

We chose a smooth bump function $\beta_j : \mathbf{R} \to [0,1]$ vanishing outside of $(s^* - \delta_{j-1}, s^* + \delta_{j-1})$ and equal to 1 on $[s^* - \delta_j, s^* + \delta_j]$ where $s^* \geq s_0$. We choose β_j such that the shape of its graph is independent of s^*, by picking such a function for some fixed s^* and then translating it over $\mathbf{R}$. Introducing the nested intervals $I_j = [s^* - \delta_j, s^* + \delta_j]$ and $Q_j = I_j \times S^1$ we claim that for every $N \geq 1$ and every $2 < p < \infty$ there exists a constant $C_{N,p} > 0$ such that

$$\|\xi\|_{W^{N,p}(Q_{N+1})} \leq C_{N,p} \ , \ \|\alpha\|_{W^{N,p}(I_{N+1})} \leq C_{N,p}, \tag{6.5}$$

where the constants are independent of s^*. Using the Sobolev embedding theorem, and the fact that the bounds in (6.5) are translation invariant, this would then conclude the proof of the theorem.

We proceed inductively making use of the standard L^p regularity estimate for the linear $\bar{\partial}$-operator. In particular

$$\|\xi\|_{W^{j,p}(Q_j)} \leq M_{p,j} \|\bar{\partial}(\beta_j \xi)\|_{W^{j-1,p}(Q_{j-1})}, \tag{6.6}$$

where $M_{p,j}$ only depends on p and j but not on s^*. Starting with $j = 1$ we first show that ξ is uniformly bounded. Using the differential equation for ξ in (6.4), taking $p = 2$ and using the fact that α is bounded, we find

$$\begin{aligned} \|\xi\|_{W^{1,2}(Q_1)} &\leq c' \|\bar{\partial}(\beta_1 \xi)\|_{L^2(Q_0)} \\ &\leq c' \|\bar{\partial}\beta_1 \cdot \xi\|_{L^2(Q_0)} + c' \|\beta_1(\Delta(u_t) + \alpha)\xi\|_{L^2(Q_0)} \\ &\leq c \, \|\xi\|_{L^2(Q_0)}. \end{aligned}$$

The constant c depends on the C^1-norm of β_1, $\sup|\Delta(u_t)|$ and $\sup|\alpha|$, but not on s^*. Since $\|\xi(s)\|_{L^2(S^1)} = 1$, we infer that $\|\xi\|_{L^2(Q_0)} = 2\delta_0$. This yields the bound

$$\|\xi\|_{W^{1,2}(Q_1)} \leq c_1$$

for some constant independent of s^*. By the Sobolev embedding theorem this implies that for every $1 < p < \infty$ there is a constant $c'_p > 0$ independent of s^* such that

$$\|\xi\|_{L^p(Q_1)} \leq c'_p,$$

Inequality (6.6) and the above L^p-estimate together with the Sobolev embedding theorem implies that

$$\|\xi\|_{C^0(Q_2)} \leq c'' \text{ and } \|\xi\|_{W^{1,p}(Q_2)} \leq c'''$$

where $c''' > 0$ does not depend on s^*, hence

$$\sup_{s,t} |\xi| < \infty. \tag{6.7}$$

We now derive a $W^{1,p}$–bound for α over I_2. Observe that

$$\alpha(s) = (-i\xi_t(s), \xi(s)) - (\Delta(u_t)\xi(s), \xi(s)). \tag{6.8}$$

We differentiate and use the fact that the operator

$$-i\frac{d}{dt} : L^2(S^1) \supset W^{1,2}(S^1) \longrightarrow L^2(S^1)$$

is self–adjoint and that $(\xi(s), \xi_s(s)) = 0$

$$\begin{aligned}
\alpha'(s) &= 2(-i\xi_t(s), \xi_s(s)) - ((\Delta(u_t))_s\xi(s), \xi(s)) \\
&\quad -(\Delta(u_t)\xi_s(s), \xi(s)) - (\Delta(u_t)\xi(s), \xi_s(s)) \\
&= 2(\xi_s(s) + \alpha(s)\xi(s) + \Delta(u_t)\xi(s), \xi_s(s)) \\
&\quad -((\Delta(u_t))_s\xi(s), \xi(s)) - (\Delta(u_t)\xi_s(s), \xi(s)) \\
&\quad -(\Delta(u_t)\xi(s), \xi_s(s)) \\
&= 2\|\xi_s(s)\|^2_{L^2(S^1)} + (\Delta(u_t)\xi(s), \xi_s(s)) \\
&\quad -(\Delta(u_t)\xi_s(s), \xi(s)) - ((\Delta(u_t))_s\xi(s), \xi(s)).
\end{aligned}$$

Consequently

$$|\alpha'(s)| \leq c_1\|\xi_s(s)\|^2_{L^2(S^1)} + c_2\|\xi_s(s)\|_{L^2(S^1)} + c_3$$

with $c_1, c_2, c_3 > 0$ independent of s. Taking the p-th power and integrating over I_2 we obtain

$$\begin{aligned}
\int_{I_2} |\alpha'(s)|^p\, ds &\leq c \int_{I_2} (\|\xi_s(s)\|^{2p}_{L^2(S^1)} + \|\xi_s(s)\|^p_{L^2(S^1)} + 1)\, ds \\
&= c \int_{I_2} \left(\left(\int_{S^1} |\xi_s(s,t)|^2 dt \right)^p + \left(\int_{S^1} |\xi_s(s,t)|^2 dt \right)^{p/2} + 1 \right) ds \\
&\leq c'(\|\xi_s\|^{2p}_{L^{2p}(Q_2)} + \|\xi_s\|^p_{L^{2p}(Q_2)} + 1).
\end{aligned}$$

Here c and c' are constants not depending on s^*. Hence (6.5) is proved for $N = 1$ and every $1 < p < \infty$. Now we proceed inductively by differentiating (6.8) and using (6.4), (6.6) in order to derive at the desired result. The details are left to the reader (see the proof of Lemma 3.10 in [1]). □

The theorem implies the following corollary.

Corollary 6.3.4 *Let α be as described above. Then*

$$\lim_{s\to\infty} \alpha(s) = -2\pi\ell$$

$$\lim_{s\to\infty} \frac{d^m\alpha}{ds^m}(s) = 0 \, for \, m \geq 1.$$

Proof Since we have uniform bounds on all the derivatives and

$$\lim_{s\to\infty} \alpha(s) = -2\pi\ell,$$

we can argue as follows. Arguing indirectly, assume there exists a positive integer j and a sequence $s_k \to \infty$ such that $|(D^j\alpha)(s_k)| \geq c$ for some positive constant c. Define

$$\alpha_k(s) = \alpha(s + s_k).$$

Using the C^∞-Ascoli-Arzela theorem we may assume after taking a subsequence that $\alpha_k \to \alpha^0$ in $C^\infty_{loc}(\mathbf{R})$. We know already that the sequence (α_k) has to converge in C^0_{loc} to the constant map $s \mapsto -2\pi\ell$. Consequently $\alpha^0(s) = -2\pi\ell$. Hence $(D^j\alpha_k)(0) \to 0$. We obtain the contradiction

$$c \leq |(D^j\alpha)(s_k)| = |(D^j\alpha_k)(0)| \to 0$$

completing the proof of Corollary 6.3.4. □

Take now any sequence $s_k \to \infty$ and consider

$$\alpha_k(s) = \alpha(s + s_k)$$
$$\xi_k(s,t) = \xi(s + s_k, t).$$

Then, after taking a subsequence, we may assume by the Ascoli–Arzela theorem that

$$\alpha_k \to \alpha^0 \text{ and } \xi_k \to \xi^0$$

in C^∞_{loc}, where $\alpha^0(s) = -2\pi\ell$. Passing to the limit in the differential equation (6.4) for ξ we obtain the equation

$$\xi^0_s + i\xi^0_t - (2\pi\ell)\xi^0 = 0 \text{ on } \mathbf{R}\times S^1 \tag{6.9}$$
$$\|\xi^0(s)\|_{L^2(S^1)} = 1 \text{ for } s \in \mathbf{R}.$$

We leave it as an easy exercise in Fourier series to show that the following holds:

For a solution of (6.9) necessarily

$$\xi^0(s,t) = a_0 e^{-2\pi i \ell t}$$

for some $a_0 \in S^3$.

Next we sharpen Theorem 6.3.3 considerably.

Theorem 6.3.5 *There exists a point $a_0 \in S^3$, so that for every multi-index β and every number $\sigma > 0$ there exists a constant c_β^σ such that*

$$| D^\beta [\xi(s,t) - a_0 e^{-2\pi i \ell t}] | \leq c_\beta^\sigma e^{-(2\pi - \sigma)s}.$$

Moreover

$$| D^j [\alpha(s) + 2\pi \ell] | \leq c_j e^{-2\pi s}.$$

Further, we have the representation formula

$$u(s,t) = c \cdot e^{-2\pi \ell s + \delta(s)} [a_0 e^{-2\pi i \ell t} + r(s,t)].$$

Here c is positive constant and $\delta(s)$ and r are smooth functions satisfying

$$|D^j \delta(s)| \leq c_j e^{-2\pi s}$$
$$|D^\beta r(s,t)| \leq c_\beta^\sigma e^{-(2\pi - \sigma)s}.$$

Proof The function ξ satisfies the differential equation

$$\xi_s + i\xi_t + \Delta\xi + \alpha\xi = 0. \tag{6.10}$$

Recall that α converges to $-2\pi\ell$ and that all its derivatives converge to 0 as $s \to \infty$. Moreover Δ and all its derivatives satisfy an exponential decay estimate

$$|D^\beta \Delta(s,t)| \leq c_\beta e^{-2\pi s}$$

(we are now using the notation $\Delta(s,t)$ instead of $\Delta(u_t)(s,t)$ used previously). We conclude that

$$0 = (D^\beta \xi)_s + i(D^\beta \xi)_t + \sum_{0 \leq \gamma \leq \beta} c_\gamma (D^\gamma \Delta) D^{\beta - \gamma} \xi$$
$$+ \sum_{0 \leq \gamma \leq \beta} (D^\gamma \alpha) D^{\beta - \gamma} \xi.$$

Denote by Q the orthogonal projection in L^2 onto the orthogonal complement of the kernel of the self-adjoint operator $-i\frac{d}{dt} + 2\pi\ell$. Then, with $v = Q(D^\beta \xi)$

$$0 = v_s + iv_t + \sum c_\gamma Q((D^\gamma \Delta)(D^{\beta-\gamma}\xi)) + \sum c_\gamma (D^\gamma \alpha)(QD^{\beta-\gamma}\xi) = 0.$$

Denote by W the column vector $(D^\beta \xi)_{|\beta|\le k}$ for some positive integer k. By $\mathcal{Q}$ we denote the diagonal operator associated to Q of the appropriate size. Then, observing that $Q(\xi_t) = (Q\xi)_t$ etc.

$$(\mathcal{Q}W)_s + i(\mathcal{Q}W)_t + \Gamma(s)\mathcal{Q}W + \Theta(s,t)W = 0.$$

Here $\Gamma(s)$ is a matrix converging for $s \to \infty$ to the matrix $-2\pi\ell Id$. Moreover its derivatives $\Gamma^{(j)}(s)$ converge to the zero matrix. Further $\Theta(s,t)$ satisfies together with its derivatives exponential decay estimates of the form

$$|D^\beta \Theta(s,t)| \le c_\beta e^{-2\pi s}.$$

Since all the derivatives of ξ are uniformly bounded, we have a uniform bound for W and its derivatives. Therefore we obtain that

$$(\mathcal{Q}W)_s + i(\mathcal{Q}Q)_t + \Gamma(s)\mathcal{Q}W = b, \tag{6.11}$$

where b and its derivatives satisfy bounds of the type

$$|(D^\beta b)(s,t)| \le ce^{-2\pi s}.$$

Differentiating (6.11) with respect to s we obtain

$$((\mathcal{Q}W)_s)_s + i((\mathcal{Q}Q)_s)_t + \Gamma_s(s)\mathcal{Q}W + \Gamma(s)(\mathcal{Q}W)_s = b_s. \tag{6.12}$$

Next consider the map a defined by

$$a(s) = \frac{1}{2}\|\mathcal{Q}W(s)\|^2,$$

where the norm is the L^2-norm. Differentiating two times with respect to s we obtain

$$a''(s) = ((\mathcal{Q}W)_s(s), (\mathcal{Q}W)_s(s)) + (\mathcal{Q}W(s), (\mathcal{Q}W)_{ss}(s)). \tag{6.13}$$

Inserting (6.12) into (6.13) gives

$$\begin{aligned}
a''(s) &= \|(\mathcal{Q}W)_s(s)\|^2 \\
&\quad +(\mathcal{Q}W(s), -i(\mathcal{Q}W)_{st}(s)) \\
&\quad +(\mathcal{Q}W(s), -\Gamma_s(s)\mathcal{Q}W(s) - \Gamma(s)(\mathcal{Q}W)_s(s) + b_s(s)) \\
&= \|(\mathcal{Q}W)_s(s)\|^2 + (-i(\mathcal{Q}W)_t(s), (\mathcal{Q}W)_s(s)) \\
&\quad -(\Gamma_s(s)\mathcal{Q}W(s), \mathcal{Q}W(s)) \\
&\quad -(\Gamma(s)(\mathcal{Q}W)_s(s), \mathcal{Q}W(s)) + (b_s(s), \mathcal{Q}W(s)) \\
&= \|(\mathcal{Q}W)_s(s)\|^2 + ((\mathcal{Q}W)_s(s) + \Gamma(s)\mathcal{Q}W(s), \mathcal{Q}W_s(s)) \\
&\quad -(b(s), (\mathcal{Q}W)_s(s)) \\
&\quad -(\Gamma_s(s)\mathcal{Q}W(s), \mathcal{Q}W(s)) - (\Gamma(s)(\mathcal{Q}W)_s(s), \mathcal{Q}W(s)) \\
&\quad +(b_s(s), \mathcal{Q}W(s)) \\
&= 2 \cdot \|(\mathcal{Q}W)_s(s)\|^2 + (\Gamma(s)\mathcal{Q}W(s), (\mathcal{Q}W)_s(s)) \\
&\quad -(\Gamma(s)(\mathcal{Q}W)_s(s), \mathcal{Q}W(s)) - (b(s), (\mathcal{Q}W)_s(s)) \\
&\quad -(\Gamma_s(s)\mathcal{Q}W(s), \mathcal{Q}W(s)) + (b_s(s), \mathcal{Q}W(s)).
\end{aligned}$$

First we discuss the terms occurring in the above equation. We note that

$$\begin{aligned}
&(\Gamma(s)\mathcal{Q}W(s), (\mathcal{Q}W)_s(s)) - (\Gamma(s)(\mathcal{Q}W)_s(s), \mathcal{Q}W(s)) \\
&\qquad \geq -\varepsilon(s)\|(\mathcal{Q}W)_s(s)\| \cdot \|\mathcal{Q}W(s)\|,
\end{aligned} \tag{6.14}$$

where $\varepsilon(s) \to 0$ as $s \to \infty$ and $\varepsilon(s) > 0$. Recall that

$$\mathcal{Q}W_s = -i\mathcal{Q}W_t - \Gamma(s)\mathcal{Q}W + b(s), \tag{6.15}$$

where

$$\Gamma(s) \to -2\pi\ell Id.$$

Hence

$$\begin{aligned}
&(2\pi - \delta(s)) \cdot \|\mathcal{Q}W(s)\| \\
&\leq \|\mathcal{Q}W_s(s)\| + ce^{-2\pi s}.
\end{aligned} \tag{6.16}$$

Here $\delta(s) > 0$ and $\delta(s) \to 0$ as $s \to \infty$. Combining (6.14) and (6.16) we obtain

$$\begin{aligned}
&(\Gamma(s)\mathcal{Q}W(s), (\mathcal{Q}W)_s(s)) - (\Gamma(s)(\mathcal{Q}W)_s(s), \mathcal{Q}W(s)) \\
&\qquad \geq -\varepsilon_1(s)\|\mathcal{Q}W_s(s)\|^2 - \varepsilon_2(s)e^{-2\pi s}\|\mathcal{Q}W_s(s)\|.
\end{aligned}$$

Similarly we obtain estimates of the form

$$\begin{aligned}
-(b(s), \mathcal{Q}W_s(s)) &\geq -ce^{-2\pi s}\|\mathcal{Q}W_s(s)\|.\\
-(\Gamma_s(s)\mathcal{Q}W(s), \mathcal{Q}W(s)) &\geq -\varepsilon(s)\|\mathcal{Q}W_s(s)\|^2\\
-\varepsilon(s)e^{-4\pi s} &- \varepsilon(s)e^{-2\pi s}\|\mathcal{Q}W_s(s)\|.\\
(b_s(s), \mathcal{Q}W(s)) &\geq -ce^{-2\pi s}\|\mathcal{Q}W_s(s)\| - ce^{-4\pi s}.
\end{aligned}$$

Here $c > 0$ is a positive constant and $\varepsilon(s)$ a positive function of s converging for $s \to \infty$ to 0. Using all these estimates we obtain an estimate of the form

$$\begin{aligned}
a''(s) \geq (2 - \delta_1(s))\|(\mathcal{Q}W)_s(s)\|^2 - ce^{-2\pi s}\|\mathcal{Q}W_s(s)\|\\
-ce^{-4\pi s}.
\end{aligned}$$

Here $\delta_1(s) \to 0$ for $s \to \infty$, where $\delta_1(s) > 0$. This of course leads to an estimate of the form

$$a''(s) \geq (2 - \tau)\|(\mathcal{Q}W)_s(s)\|^2 - c_\tau e^{-4\pi s}$$

for every $\tau > 0$ and $c_\tau > 0$ a suitable constant. Using (6.16) we replace $\|\mathcal{Q}W_s(s)\|$ by $\|\mathcal{Q}W(s)\|$. Further we use the definition of a and finally we obtain an estimate of the form

$$a''(s) \geq 8\pi^2(2 - \tau)a(s) - c_\tau e^{-2\pi s}a^{\frac{1}{2}}(s) - c_\tau e^{-4\pi s}$$

for a suitable constant $c_\tau > 0$. This estimate implies immediately that for every $\sigma > 0$ there exists a constant $d_\sigma > 0$ such that

$$a(s) \leq d_\sigma e^{-(4\pi - \sigma)s}.$$

Next, starting with Eq. (6.10), take at time s the inner product with $\xi(s)$. This gives

$$\begin{aligned}
\alpha(s) &= (-i\xi_t(s), \xi(s)) - (\Delta(s)\xi(s), \xi(s))\\
&= -2\pi\ell + (-i(Q\xi)_t(s) + 2\pi\ell Q\xi(s), Q\xi(s))\\
&\quad -(\Delta(s)\xi(s), \xi(s)).
\end{aligned}$$

Since Δ and $Q\xi$ satisfy including their derivatives exponential decay estimates and ξ and its derivatives are bounded as discussed earlier, we conclude that

$$|D^j[\alpha(s) + 2\pi\ell]| \leq c_j e^{-2\pi s} \text{ for } j = 0, 1, .. \tag{6.17}$$

Denote by W the column vector associated to partial derivatives up to order k. Then W satisfies an equation of the form

$$W_s + iW_t + \tilde{\Gamma}(s)W + \tilde{\Theta}(s,t)W = 0. \tag{6.18}$$

Here $\tilde{\Gamma}(s)$ as indicated is a matrix only depending on s, converging to $-2\pi\ell Id$, with the derivatives of α as coefficients, satisfying

$$D^j[\tilde{\Gamma}(s) + 2\pi\ell Id] \to 0$$

as $s \to \infty$. In view of (6.17) this implies

$$|D^j[\tilde{\Gamma}(s) + 2\pi\ell Id]| \leq c_j^{'} e^{-2\pi s}.$$

Moreover

$$|D^\beta \tilde{\Theta}(s,t)| \leq c_\beta e^{-2\pi s}.$$

Denote by P the orthogonal projection onto the kernel of $-i\frac{d}{dt} + 2\pi\ell$ and by $\mathcal{P}$ the associated diagonal operator. Applying $\mathcal{P}$ to (6.18) we find

$$(\mathcal{P}W)_s(s) + 2\pi\ell\mathcal{P}W(s) + \tilde{\Gamma}(s)\mathcal{P}W(s) + \mathcal{P}\tilde{\Theta}(s,t)W(s) = 0.$$

Using the fact that W is uniformly bounded and the exponential decay estimates we deduce

$$|\mathcal{P}W(s_1) - \mathcal{P}W(s_0)| \leq const \cdot \int_{s_0}^{s_1} e^{-2\pi\tau} d\tau.$$

Hence the limit of $\mathcal{P}W(s)$ exists for $s \to \infty$ and must component-wise belong to the kernel of $-i\frac{d}{dt} + 2\pi\ell$. Denote the zero order component by ξ^0. Clearly it must have the form

$$\xi^0(t) = a_0 e^{-2\pi\ell it},\ a_o \in S^3.$$

It is also clear that the higher order components must be zero. Summing up the whole discussion (and using the Sobolev embedding theorem) we conclude that for every $\sigma > 0$ and multi-index β there exists a constant c_β^σ such that

$$|D^\beta[\xi(s,t) - \xi^0(t)]| \leq c_\beta^\sigma e^{-(2\pi-\sigma)s}$$

as required.

Now we are in the position to complete the proof of Theorem 6.3.5. By the above theorem we have

$$\xi(s,t) = a_0 e^{-2\pi i \ell t} + r(s,t)$$

with

$$|(D^\beta r)(s,t)| \leq c_\beta^\sigma \cdot e^{-(2\pi-\sigma)s}.$$

In view of the definition of α we see that

$$\|u(s)\| = \|u(s_0)\| \cdot e^{\int_{s_0}^{s_1} \alpha(\tau)d\tau}.$$

Now using the definitions of ξ and α we conclude that

$$u(s,t) = \|u(s_0)\| e^{\int_{s_0}^{s_1} \alpha(\tau)d\tau}[a_0 e^{-2\pi i \ell t} + r(s,t)].$$

Invoking the exponential estimates we can write more precisely:

$$u(s,t) = c \cdot e^{-2\pi\ell s + \delta(s)}\,[a_0 e^{-2\pi \ell i t} + r(s,t)],$$

where $c > 0$ is a positive constant $a_0 \in S^3$ and δ and r satisfy the estimates

$$|D^j \delta(s)| \leq c_j e^{-2\pi s}$$
$$|D^\beta r(s,t)| \leq c_\beta^\sigma e^{-(2\pi-\sigma)s}.$$

This completes the proof of Theorem 6.3.5.

Before we prove a relative normal form for two holomorphic curves let us draw a consequence of the above normal form.

Denote by $(.,.)_m$ any smooth family of Hermitian metrics on $\mathbf{C}^n$, so that $(.,.)_0$ is the standard Hermitian inner product on $\mathbf{C}^n$. Consider the map

$$(s,t) \to |u_s(s,t)|^{-1}_{u(s,t)} u_s(s,t).$$

From the normal form we obtain immediately:

Lemma 6.3.6 *The following estimate holds for every multi-index β:*

$$|D^\beta[|u_s(s,t)|^{-1}_{u(s,t)} u_s(s,t) - a_0 e^{-2\pi i \ell t}]| \leq c_\beta^\sigma e^{-(2\pi-\sigma)s}.$$

We refer the reader to the papers by R. Siefring, [102, 103], for related results and strong refinements.

6.4 A Relative Normal Form

In this section we derive a relative normal form given two holomorphic curves, which intersect at a point and have a common generalized tangent plane. Let u be as described in the above section. Without loss of generality we may assume that $a_0 = (1, 0)$. Denote by $(.,.)$ the standard Hermitian inner product. The associated real inner product is $\langle .,. \rangle = Re(.,.)$ and the associated (standard) symplectic form is $\omega = -Im(.,.)$. Fix a smooth family of Hermitian inner products $(.,.)_m$ defined for m close to $0 \in \mathbf{C}^2$, where the complex multiplication on $\mathbf{C}^2$ for the inner product $(.,.)_m$ is $J(m)$. Denote by $\Pi_{(s,t)}$ the orthogonal projection for the metric $(.,.)_{u(s,t)}$ onto $\mathcal{T}^{\perp}_{u,(s,t)}$. Recall that $\mathcal{T}_{u,(s,t)}$ is the generalized tangent space to u at $u(s,t)$. Note that it is actually the real tangent space for s large, since there are no singularities. Hence

$$\Pi_{(s,t)} : (\mathbf{C}^2, J(u(s,t))) \to ([\mathbf{C}\cdot_{J(u(s,t))}]^{\perp}, J(u(s,t))),$$

which is $\mathbf{C}$-linear for the indicated complex multiplication. Hence the image of $\Pi_{(s,t)}$ is the complex normal bundle to the tangent space and we denote it for simplicity by $\mathcal{N}_{(s,t)}$. Consequently

$$(\mathbf{C}^2, J(u(s,t))) = \mathcal{T}_{u,(s,t)} \oplus \mathcal{N}_{(s,t)}.$$

We note that

$$\Pi_{(s,t)}(0,1) \neq 0 \text{ for } s \text{ large enough.}$$

Finally we define a smooth section of $\mathcal{N}$ of unit length by

$$\gamma(s,t) = \frac{\Pi_{(s,t)}(0,1)}{|\Pi_{(s,t)}(0,1)|_{u(s,t)}}.$$

For $s_0 > 0$ sufficiently large and $\varepsilon > 0$ sufficiently small we define a smooth map

$$\Phi : [s_0, \infty) \times S^1 \times ([-\varepsilon, \varepsilon] + i[-\varepsilon, \varepsilon]) \to \mathbf{C}^2$$

by

$$\Phi(s,t,z) = c \cdot e^{-2\pi\ell s + \delta(s)}[(1,0)e^{-2\pi\ell i t} + r(s,t) + z \cdot_{J(u(s,t))} \gamma(s,t)].$$

We will see that Φ gives us nice immersive coordinates around u which can be used to describe suitable nearby curves. From the definition it follows immediately that

$$\Phi(s,t,0) = u(s,t).$$

Moreover the derivative of $D\Phi$ of Φ along u is given by

$$\begin{aligned}&(D\Phi)(s,t,0,0)(a,b,h,k)\\&= a\cdot u_s(s,t)+b\cdot u_t(s,t)+c\cdot e^{-2\pi i\ell s+\delta(s)}\,[(h+ik)\cdot_{J(u(s,t))}\gamma(s,t)].\end{aligned}$$

This map is complex linear viewed as a map between

$$(\mathbf{C}\times\mathbf{C},i)\to(\mathbf{C}^2,J(u(s,t))),$$

with the identification

$$\mathbf{R}\times\mathbf{R}\to\mathbf{C}^2:(a,b)\to a+ib.$$

Indeed

$$\begin{aligned}&D\Phi(s,t,0)(i(a+ib,h))\\&= D\Phi(s,t,0)(-b,a,ih)\\&= -bu_s(s,t)+au_t(s,t)+(ih)\cdot_{J(u(s,t))}\gamma(s,t)\\&= -bu_s(s,t)+aJ(u(s,t))u_s(s,t)\\&\quad +J(u(s,t))[h\cdot_{J(u(s,t))}\gamma(s,t)]\\&= J(u(s,t))[(a+bJ(u(s,t)))u_s(s,t)\\&\quad +J(u(s,t))[h\cdot_{J(u(s,t))}\gamma(s,t)]]\\&= J(u(s,t))D\Phi(s,t,0)(a+ib,h).\end{aligned}$$

For further references we have to understand the behaviour of

$$(s,t,x,y)\to\frac{1}{c}e^{2\pi\ell s-\delta(s)}D\Phi(s,t,x,y).$$

We compute, suppressing for notational convenience the s and t-dependence

$$\begin{aligned}&\frac{1}{c}e^{2\pi\ell s-\delta(s)}D\Phi(s,t,x,y)(a,b,h,k)\\&= a[r_s+x\gamma_s+yJ(u)\gamma_s+(DJ(u)u_s)\gamma]\\&\quad +a(\delta'(s)-2\pi\ell)[a_0e^{-2\pi i\ell t}+r+x\gamma+yJ(u)\gamma]\\&\quad +b[-2\pi\ell i a_0e^{-2\pi i\ell t}+r_t+x\gamma_t\\&\quad +yJ(u)\gamma_t+(DJ(u)u_t)\gamma]\\&\quad h\gamma+kJ(u)\gamma\end{aligned}$$

Recalling that $a_0 = (1, 0)$, define a smooth family of (t, x, y)-dependent linear maps by

$$\begin{aligned}
&\Psi(t, x, y)(a, b, h, k) \\
&= -2a\pi\ell[(1, 0)e^{-2\pi i\ell t} + (0, x + iy)] \\
&\quad -2b\pi i\ell(1, 0)e^{-2\pi i\ell t} + (0, h + ik) \\
&= -2\pi\ell e^{-2\pi i\ell t}(a + ib, 0) - 2a\pi\ell(0, x + iy) + (0, h + ik).
\end{aligned}$$

Identifying $\mathbf{C}$ with $\mathbf{R} \times \mathbf{R}$ we can represent $\Psi(t, x, y)$ by the matrix

$$\begin{bmatrix}
-2\pi\ell\cos(-2\pi\ell t) & 2\pi\ell\sin(-2\pi\ell t) & 0 & 0 \\
-2\pi\ell\sin(-2\pi\ell t) & -2\pi\ell\cos(-2\pi\ell t) & 0 & 0 \\
-2\pi\ell x & 0 & 1 & 0 \\
-2\pi\ell y & 0 & 0 & 1
\end{bmatrix}. \tag{6.19}$$

By the properties of δ and r we obtain the following

Lemma 6.4.1 *For every multi-index $\beta \in \mathbf{N}^4$ and constant $\sigma > 0$ there exists a constant c_β^σ so that the following holds*

$$|D^\beta[\frac{1}{c}e^{2\pi\ell s-\delta(s)}D\Phi(s, t, x, y) - \Psi(t, x, y)]| \le c_\beta^\sigma e^{-(2\pi-\sigma)s}.$$

We need the following

Lemma 6.4.2 *There exists $s_0 > 0$ and $\varepsilon > 0$ small enough such that the linear maps $D\Phi(s, t, x, y)$ are injective for every $s \in [s_0, \infty)$, $t \in S^1$ and $(x, y) \in [-\varepsilon, \varepsilon] \times [-\varepsilon, \varepsilon]$.*

Proof In view of Lemma 6.4.1 we see that

$$\frac{1}{c}e^{2\pi\ell s-\delta(s)}D\Phi(s, t, x, y)(a, b, h, k)$$

define for s large a linear injective operator. This proves the lemma.

Next define for a number $\tau > 0$ the subset A_τ of $\mathbf{C}^2$ by

$$A_\tau = \{(z_1, z_2)\ |0 < |z_1| \le \tau,\ |z_2| \le \frac{\varepsilon}{2}|z_1|\ \}.$$

We need the following

Proposition 6.4.3 *There exist numbers $s_0^* \ge s_0$, $0 < \varepsilon^* \le \varepsilon$ and $\tau > 0$, where s_0 and ε are as described in Lemma 6.4.2 such that for $g = (z_1, z_2) \in A_\tau$ there exists a constant $s_1 > s_0^*$ so that the mapping degree*

$$d(s_2) = deg(\Phi, [s_0^*, s_2] \times S^1 \times [-\varepsilon, \varepsilon] \times [-\varepsilon, \varepsilon], g)$$

for all $s_2 \geq s_1$ is well-defined and satisfies

$$d(s_2) = \ell \text{ for } s_2 \geq s_1.$$

Proof Pick numbers

$$\begin{gathered} s_0^* \geq s_0, \\ \varepsilon^* \leq \min\{\tfrac{1}{2}, \varepsilon\} \\ \tau > 0, \end{gathered}$$

such that

$$\begin{gathered} |r(s,t)| \leq 10^{-6} \text{ for } s \geq s_0^* \\ |e^{\delta(s)} - 1| \leq 10^{-6} \text{ for } s \geq s_0^* \\ |\gamma(s,t) - (0,1)| \leq 10^{-6} \text{ for } s \geq s_0^* \\ |J(u(s,t))\gamma(s,t) - (0,i)| \leq 10^{-6} \text{ for } s \geq s_0^*. \end{gathered}$$

Next let $g \in A_\tau$ and consider the homotopy Φ_σ defined on

$$[s_0^*, s_2] \times S^1 \times [-\varepsilon, \varepsilon] \times [-\varepsilon, \varepsilon]$$

by

$$\begin{aligned} &\Phi_\sigma(s,t,x,y) \\ &= c \cdot e^{-2\pi\ell s + \sigma\delta(s)} \cdot \\ &\quad [(1,0)e^{-2\pi i\ell t} + \sigma r(s,t) + x((1-\sigma)(0,1) + \sigma\gamma(s,t)) \\ &\quad + y((1-\sigma)(0,i) + \sigma J(u(s,t))\gamma(s,t))]. \end{aligned}$$

If s_2 is large enough there exists no solution of

$$\Phi_\sigma(s,t,x,y) = g$$

with

$$(s,t,x,y) \in \partial([s_0^*, s_2] \times S^1 \times [-\varepsilon, \varepsilon] \times [-\varepsilon, \varepsilon]).$$

By the homotopy invariance of the degree this implies that for s large enough

$$d(s_2) = \deg(\Psi, [s_0^*, s_2] \times S^1 \times [-\varepsilon, \varepsilon] \times [-\varepsilon, \varepsilon], g),$$

where

$$\Psi(s,t,x,y) = c \cdot e^{-2\pi\ell s}[(1,0)e^{-2\pi i\ell t} + (0, x+iy)].$$

Clearly, for s_2 large enough

$$\deg(\Psi, [s_0^*, s_2] \times S^1 \times [-\varepsilon, \varepsilon], g) = \ell.$$

This completes the proof of the proposition.

As a consequence of Lemma 6.4.2 and Proposition 6.4.3 we obtain the following

Theorem 6.4.4 *There exist numbers $s_0 > 0$, $\varepsilon > 0$ and $\tau > 0$ such that the following holds. Let B_τ be defined by*

$$B_\tau = \{(s,t,x,y) \in [s_0, \infty) \times S^1 \times [-\varepsilon, \varepsilon] \times [-\varepsilon, \varepsilon] \mid \Phi(s,t,x,y) \in A_\tau\}.$$

Then the map

$$\Phi | B_\tau;\ B_\tau \to A_\tau$$

is an ℓ-fold covering map and B_τ is connected.

Proof This follows from the degree argument and the fact that the linearisation of Φ at every zero is orientation preserving. Moreover B_τ is compact. We observe that the connecteness assertion is trivial if we replace Φ by Ψ. The assertion follows now from a homotopy argument as in Proposition 6.4.3.

For the following we need a better understanding of the behaviour of Φ for $s \to \infty$. In order to achieve this we need to understand the behaviour of $\gamma(s,t)$. For this we prove first

Lemma 6.4.5 *The map $(s,t) \to \Pi_{(s,t)}(0,1)$ satisfies the following estimate*

$$|D^\beta[(0,1) - \Pi_{(s,t)}(0,1)]| \le c_\beta^\sigma e^{-(2\pi-\sigma)s}.$$

Proof By definition

$$\begin{aligned}&(0,1) - \Pi_{(s,t)}(0,1)\\ &= ((0,1), u_s(s,t))_{(s,t)} |u_s(s,t)|_{(s,t)}^{-2} u_s(s,t).\end{aligned}$$

Using Lemma 6.3.6 we obtain immediately the following estimate

$$|D^\beta[(0,1) - \Pi_{(s,t)}(0,1)]| \le c_\beta^\sigma e^{-(2\pi-\sigma)s}.$$

Define an almost complex structure $\tilde{j}$ on $[0,\infty)\times S^1\times[-\varepsilon,\varepsilon]\times[-\varepsilon,\varepsilon]$ by

$$J(\Phi(s,t,x,y))D\Phi(s,t,x,y)=D\Phi(s,t,x,y)\tilde{j}(s,t,x,y).$$

We note that

$$\tilde{j}(s,t,0,0)=i$$

viewing $[0,\infty)\times S^1$ as $(\mathbf{R}+i\mathbf{R})/(i\mathbf{Z})$. Denote by $j(t,x,y)$ the almost complex structure depending on (t,x,y) and defined by

$$j(t,x,y)=\Psi(t,x,y)^{-1}\circ i\circ\Psi(t,x,y). \tag{6.20}$$

The following estimate is needed

Lemma 6.4.6 *Define for a multi-index $\beta\in\mathbf{N}^4$*

$$D^\beta=\partial_s^{\beta_1}\partial_t^{\beta_2}\partial_x^{\beta_3}\partial_y^{\beta_4}.$$

For $\sigma>0$ and a multi-index β there exists a constant c_β^σ independent of x and y in $[-\varepsilon,\varepsilon]$ such that

$$|D^\beta[\tilde{j}(s,t,x,y)-j(t,x,y)]|\le c_\beta^\sigma e^{-(2\pi-\sigma)s}.$$

Proof Observe that

$$\begin{aligned}&\frac{1}{c}e^{2\pi\ell s-\delta(s)}D\Phi(s,t,x,y)\tilde{j}(s,t,x,y)\\&=J(\Phi(s,t,x,y))\frac{1}{c}e^{2\pi\ell s-\delta(s)}D\Phi(s,t,x,y).\end{aligned}$$

Recall that

$$\frac{1}{c}e^{2\pi\ell s-\delta(s)}D\Phi(s,t,x,y)=\Psi(t,x,y)+\Delta(s,t,x,y),$$

where Δ converges to 0 with all its derivatives, in fact exponentially fast to the order $e^{-(2\pi-\sigma)s}$ for every $\sigma>0$. Moreover

$$J(\Phi(s,t,x,y))=i+K(s,t,x,y)$$

with K and all derivatives converging to 0 with the same exponential rate of convergence as the previous term. Hence we can write

$$\begin{aligned}&(i+K(s,t,x,y))(\Psi(t,x,y)+\Delta(s,t,x,y))\\&=(\Psi(t,x,y)+\Delta(s,t,x,y))\tilde{j}(s,t,x,y),\end{aligned}$$

which implies

$$\begin{aligned}
&\tilde{j}(s,t,x,y)\\
&= (\Psi(t,x,y)+\Delta(s,t,x,y))^{-1}(i+K(s,t,x,y))(\Psi(t,x,y)+\Delta(s,t,x,y))\\
&=: (\Psi(t,x,y)^{-1}+\tilde{\Delta}(s,t,x,y))(i+K(s,t,x,y))(\Psi(t,x,y)+\Delta(s,t,x,y))\\
&= j(t,x,y)+\Gamma(s,t,x,y).
\end{aligned}$$

Here the term Γ converges including all its derivatives to the order $e^{-(2\pi-\sigma)s}$ to 0 (for every $\sigma > 0$). The proof of the lemma is complete.

Next we give the main result towards the intersection theory of two holomorphic curves. It follows immediately from our previous discussion.

Theorem 6.4.7 *Let*

$$w_1 : D \to \mathbf{C}^2 \text{ and } w_2 : D \to \mathbf{C}^2$$

be J non-constant holomorphic disks satisfying

$$w_1(0) = w_2(0) = 0$$

having the same order ℓ at 0. Here J is an almost complex structure defined on a neighborhood of $0 \in \mathbf{C}^2$ such that $J(0) = i$. Define $u(s,t) = w_1(e^{-2\pi(s+it)})$ and $v(s,t) = w_2(e^{-2\pi(s+it)})$. Then there exist immersive coordinates Φ as given in Theorem 6.4.4 associated to u, so that for large s_0 the map v can be given as

$$v(s,t) = \Phi(a_j(s,t), b_j(s,t)),$$

where

$$(a_j, b_j) : [s_0, \infty) \times S^1 \to [0, \infty) \times S^1$$

for $j = 1, \ldots, \ell$ are embeddings, which are mutually different and correspond to the different lifts of a point in A_τ to B_τ (see Theorem 6.4.4). Moreover every b_j admits an representation

$$b_j(s,t) = c_j \cdot e^{-2\pi m_j s+\delta_j(s)}[b_0^j e^{-2\pi(\ell+m_j)it} + r_j(s,t)],$$

where $c_j > 0$ and

$$\begin{aligned}
|D^k \delta_j(s)| &\le c_k e^{-2\pi s}\\
D^\beta r_j(s,t)| &\le c_\beta^\sigma e^{-(2\pi-\sigma)s}.
\end{aligned}$$

Proof The map u has the usual representation around 0 and v can be written similarly as

$$v(s,t) = c' \cdot e^{-2\pi\ell s+\delta_1(s)}\,[a_0^{'}e^{-2\pi i\ell t} + r_1(s,t)].$$

By replacing t by $t+\tau$, where τ is a suitable phase we may assume without loss of generality that $a_0 = a_0^{'} = (1,0)$. For s large enough v is in the image of the immersion Φ. For s large we find for $j = 1,\ldots,\ell$ mutually different maps $(a,b) = (a_j, b_j)$

$$[s_0,\infty)\times S^1 \to \mathbf{R}^2 : (s,t) \to (a(s,t), b(s,t))$$

satisfying

$$\tilde{v}(s,t) = \Phi(a(s,t), b(s,t)).$$

We need the following two estimates, which are immediate corollaries of the bevaviour of u and v for $s \to \infty$.

Lemma 6.4.8 *Let (a,b) be as just constructed. Then the following estimates hold for suitable constants c_0 and t_0:*

$$|D^\beta b(s,t)| \le c_\beta^\sigma e^{-(2\pi-\sigma)s} \tag{6.21}$$
$$|D^\beta[a(s,t) + (2\pi\ell s + c_0, t+t_0)]| \le c_\beta^\sigma e^{-(2\pi-\sigma)s}.$$

By construction of $\tilde{j}$ we know that

$$0 = (a,b)_s + \tilde{j}(a,b)(a,b)_t.$$

This gives

$$\begin{aligned} 0 &= (a_s,b_s) + \tilde{j}(a,0)(a_t,b_t) + (\int_0^1 D_2\tilde{j}(a,\tau b)bd\tau)(a,b)_t \qquad (6.22)\\ &= (a_s + ia_t + \Delta^1 b, b_s + ib_t + \Delta^2 b). \end{aligned}$$

We need to compute Δ^2. In order to do so let us compute $j(t,x,y)$. Recall that by definition

$$j(t,x,y) = \Psi(t,x,y)^{-1}\circ i\circ\Psi(t,x,y).$$

From the definition of $\Psi(t,x,y)$ on easily computes $\Psi(t,x,y)^{-1}$ to be given (in real notation) by the matrix

$$\Psi(t,x,y)^{-1} = \begin{bmatrix} -\frac{1}{2\pi\ell}\cos(-2\pi\ell t) & -\frac{1}{2\pi\ell}\sin(-2\pi\ell t) & 0 & 0 \\ \frac{1}{2\pi\ell}\sin(-2\pi\ell t) & -\frac{1}{2\pi\ell}\cos(-2\pi\ell t) & 0 & 0 \\ -x\cos(-2\pi\ell t) & -x\sin(-2\pi\ell t) & 1 & 0 \\ -y\cos(2\pi\ell t) & -y\sin(-2\pi\ell t) & 0 & 1 \end{bmatrix}.$$

Using this we compute $j(t,x,y)$ to be given by the matrix

$$j(t,x,y) = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 2\pi\ell y & -2\pi\ell x & 0 & -1 \\ -2\pi\ell x & -2\pi\ell y & 1 & 0 \end{bmatrix}.$$

We observe that j only depends on (x,y) and not on t. Therefore we shall write $j(x,y)$ instead of $j(t,x,y)$. From Lemma 6.4.6 we infer that

$$|D^\beta[\tilde{j}(s,t,x,y) - j(t,x,y)]| \leq c_\beta^\sigma e^{-(2\pi-\sigma)s}. \tag{6.23}$$

Now we are in the position to proceed from (6.22). We have to understand

$$(\int_0^1 D_2\tilde{j}(a,\tau b)bd\tau)(a,b)_t,$$

which we view as a linear map in b, i.e. as the linear map

$$h \to (\int_0^1 D_2\tilde{j}(a,\tau b)hd\tau)(a,b)_t.$$

Combining (6.21) and Lemma 6.4.6 we deduce from (6.23), abbreviating

$$\begin{aligned} &\Delta(s,t,h,k)(a,b)_t \\ &= (\int_0^1 D_2\tilde{j}(a,\tau b)(h,k)d\tau)(a,b)_t, \end{aligned}$$

that for fixed $(h,k) = h + ik \in \mathbf{C}$

$$\begin{aligned} &|D^\beta[\Delta(s,t,h,k)(a,b)_t - ((Dj)(x,y)(h,k))((0,1),(0,0))]| \\ &\leq c_\beta^\sigma |(h,k)|e^{-(2\pi-\sigma)s}. \end{aligned}$$

This however precisely means that

$$\begin{aligned}&|D^\beta[\Delta(s,t,h,k)(a,b)_t - (0,0,-2\pi\ell h,-2\pi\ell k)]|\\ &\le c^\sigma_\beta|(h,k)|e^{-(2\pi-\sigma)s}\end{aligned}$$

The component b satisfies the equation

$$b_s + ib_t + \Delta^2 b = 0 \tag{6.24}$$

with $\Delta^2(s,t)$ and its derivatives converging exponentially fast to $-2\pi\ell Id$, i.e.

$$|D^\beta[\Delta^2(s,t) + 2\pi\ell Id]| \le c^\sigma_\beta e^{-(2\pi-\sigma)s}. \tag{6.25}$$

We need the following

Lemma 6.4.9 *Assume there exists a number $s_* > s_0$ such that $b(s_*) = 0$. Then there exists a holomorphic map*

$$\phi : [s_*,\infty)\times S^1 \to [s_0,\infty)\times S^1$$

such that

$$v(s,t) = u(\phi(s,t)) \text{ for } s \ge s^*.$$

Proof By the similarity principle $b(s_*) = 0$ implies that $b(s,t) = 0$ for (s,t) in an open neighborhood of $\{s_*\}\times S^1$. From this it follows immediately by a connectivity argument that $b(s,t) = 0$ for $t \in S^1$ and s large. Consequently

$$v(s,t) = u(a(s,t)) \text{ for } s \ge s_*.$$

An easy calculation gives

$$\begin{aligned}&v_s + J(v)v_t\\ &= Du(a)a_s + J(u(a))Du(a)a_t\\ &= Du(a)[a_s + ia_t].\end{aligned}$$

Since $Du(a(s,t))$ is injective for large s we see that a is holomorphic. Hence we put $\phi = a$.

As a consequence of the previous lemma the two curves have identical image near 0 if there exists a sufficiently large s^* so that $b(s^*) = 0$.

If the curves do not have an identical image near 0 we may assume that $\|b(s)\| > 0$ for $s \geq s_0$. In that case we study as before

$$\eta(s) = \frac{b(s)}{\|b(s)\|}.$$

It satisfies an equation of the kind

$$\eta_s + i\eta_t + \Delta^2\eta + \gamma\eta = 0.$$

Here γ is defined by

$$\gamma(s) = \frac{(b(s), b_s(s))}{(b(s), b(s))}$$

and Δ^2 satisfies exponential decay estimates towards $-2\pi\ell Id$, see (6.25). We need the following

Lemma 6.4.10 *Let γ as just defined. Then there exists a positive integer $m > 0$ such that*

$$lim_{s\to\infty}\, \gamma(s) = -2\pi m.$$

Proof We assume $\|b(s)\| \neq 0$ and abbreviate $\| \| \equiv \| \|_{L^2}$. Introduce the smooth function

$$\eta(s,t) = \frac{b(s,t)}{\|b(s)\|}, \qquad \text{then} \qquad \|\eta(s)\| = 1.$$

Differentiating in s, using that b solves the Eq. (6.24) we obtain

$$\eta_s = -i\eta_t - \Delta^2\eta - \gamma(s)\eta. \tag{6.26}$$

This implies by the definition of γ

$$\|b(s)\| = e^{\int_{s_0}^{s}\gamma(\tau)d\tau}\|b(s_0)\|. \tag{6.27}$$

We have to show that the smooth function γ converges as $s \to \infty$ to a negative integer multiple of 2π. Observe that

$$\gamma(s) = (\eta(s), -i\tfrac{d\eta}{dt}(s) - \Delta^2(s,.)\eta(s)),$$
$$\eta_s + i\eta_t + \Delta^2\eta + \gamma\eta = 0.$$

Differentiating γ gives

$$\gamma_s = 2\|\eta_s\|^2 - (\Delta_s^2\eta(s), \eta(s)) - (\Delta^2(s)\eta_s(s), \eta(s)) \\ +(\Delta^2(s)\eta(s), \eta_s(s)).$$

In view of $\|\eta\| = 1$, we have the estimates:

$$|(\Delta_s^2\eta, \eta)| \le c_\sigma \cdot e^{-(2\pi-\sigma)s}$$
$$|(\Delta^2(s)\eta(s), \eta_s(s))| \le c_\sigma \cdot e^{-(2\pi-\sigma)s}\|\eta_s(s)\|$$
$$|(\Delta^2(s)\eta(s), \eta_s(s))| \le c_\sigma \cdot e^{-(2\pi-\sigma)s}\|\eta_s(s)\|$$

Consequently

$$\gamma_s(s) \ge 2\|\eta_s(s)\| \left[\|\eta_s(s)\| - c_\sigma e^{-(2\pi-\sigma)s}\right] - c_\sigma e^{-(2\pi-\sigma)s}. \tag{6.28}$$

Next we claim that γ is bounded

$$|\gamma(s)| \le C, \quad s \ge s_0 \tag{6.29}$$

for some $C > 0$. Arguing by contradiction we assume that γ is not bounded from above. Then there is a sequence $s_n \to \infty$ such that $\gamma(s_n) \to \infty$. On the other hand if $\gamma(s) \ge \delta > 0$ for all s large, then $\|b(s)\| \to +\infty$ in view of (6.27) which contradicts $\|b(s)\| \to 0$, as $s \to \infty$. Hence there exists another sequence $s_n' \to \infty$ such that $\gamma(s_n') < \delta$ and so the function γ has an "oscillatory" behaviour as $s \to \infty$. The spectrum of $-i\frac{d}{dt}$, say $\sigma(-i\frac{d}{dt})$, satisfies

$$\sigma(-i\frac{d}{dt}) = 2\pi\mathbf{Z}.$$

Hence by the oscillatory behaviour of γ we find a sequence $\tau_n \to \infty$ satisfying $\gamma(\tau_n) = (2n+1)\pi$ and $\gamma'(\tau_n) \le 0$. It then follows from (6.28) that $\|\eta_s(\tau_n)\| \to 0$ as $n \to \infty$. Since

$$\eta_s = -i\frac{d\eta}{dt}(s) - \gamma(s)\eta - \Delta^2(s)\eta,$$

we can estimate

$$\begin{aligned} \|\eta_s(s)\| &\ge \|[-i\frac{d}{dt} + 2\pi\ell - \gamma(s)]\eta\| - c_\sigma e^{-(2\pi-\sigma)s} \\ &\ge \operatorname{dist}(\gamma(s), 2\pi\mathbf{Z}) - c_\sigma e^{-(2\pi-\sigma)s}. \end{aligned} \tag{6.30}$$

Here we have used, that $\|\eta(s)\| = 1$ and standard estimates for self-adjoint operators. We conclude from (6.30) that $\|\eta_s(\tau_n)\| \geq \frac{d}{2} > 0$ contradicting $\|\eta_s(\tau_n)\| \to 0$ as $m \to \infty$. This contradiction shows that γ is indeed bounded from above. The same argument shows that γ is also bounded from below, proving the claim (6.29).

There exists a sequence $s_n \to \infty$ such that $\|\eta_s(s_n)\| \to 0$. Indeed, otherwise, for all large s, $\|\eta_s\| \geq \delta > 0$, hence $\gamma_s \geq \delta^2$ in view of (6.28), and $\gamma(s) \geq \delta^2(s - s_0) + \gamma(s_0)$, so that $\|b(s)\| \to \infty$, in view of (6.27). This contradicts $\|b(s)\| \to 0$. Since γ is bounded, the sequence $\gamma(s_n)$ has a convergent subsequence, $\lim_{n\to\infty} \gamma(s_n) = \lambda$ and we conclude from (6.30) that $\lambda \in 2\pi\mathbf{Z}$. Since γ is bounded, every sequence $\gamma(\tau_n)$, $\tau_n \to \infty$ possesses a convergent subsequence, $\lim_{n\to\infty} \gamma(\tau_n) = \mu$ and we claim that $\mu = \lambda$. Indeed, if f.e. $\mu < \lambda$, then γ has again an oscillatory behaviour and we can pick $\mu < \nu < \lambda$ satisfying $\nu \notin 2\pi\mathbf{Z}$, and a sequence $s_n \to \infty$ satisfying $\gamma(s_n) = \nu$ and $\gamma_s(s_n) \leq 0$. Consequently, in view of (6.28), $\|\eta_s(s_n)\| \to 0$ and hence, in view of (6.30), $\nu \in 2\pi\mathbf{Z}$, contradicting $\nu \notin 2\pi\mathbf{Z}$. We have proved that $\lim_{s\to\infty} \gamma(s) = \lambda$ and $\lambda \in 2\pi\mathbf{Z}$. Clearly $\lambda \leq 0$, since otherwise $\|b(s)\| \to +\infty$. However $\lambda = 0$ is impossible. Indeed, this would imply the estimate

$$\begin{aligned} & c_\sigma \cdot e^{-(2\pi-\sigma)s} \\ \geq\ & \|b(s)\| \\ =\ & \|b(s_0)\| e^{\int_{s_0}^{s} \gamma(\tau)d\tau} \\ \geq\ & \|b(s_0)\| d_\delta e^{-\delta s}. \end{aligned}$$

Since $\delta > 0$ and $\sigma > 0$ can be taken arbitrarily small and $d_\delta > 0$ we can conclude for a suitable choice an inequality of the form

$$\|b(s_0)\| \leq c \cdot e^{-\pi s}$$

for all $s \geq s_0$ which gives $b(s_0) = 0$ contradicting the fact that it was non-zero.

By a boot strap argument used already in the proof of the properties of α and ξ this will imply uniform bounds for η and γ and all their derivatives. Proceeding as before we derive the representation formula

$$b(s,t) = ce^{\int_{s_0}^{s} \gamma(\tau)d\tau} [b_0 e^{-2\pi(\ell+m)t} + \tilde{r}(s,t)], \tag{6.31}$$

with $b_0 \in S^1 \subset \mathbf{C}$. Note the constant $m+\ell$. Indeed, whereas in the previous case the asymptotic operator was $-i\frac{d}{dt}$ it is this time $-i\frac{d}{dt} + 2\pi\ell Id$. A nontrivial eigenvector associated to a negative eigenvalue $-2\pi m$ is of the form

$$t \to b_0 \cdot e^{-2\pi(m+\ell)t}.$$

There is another interesting consequence of the relative normal form (6.31). Let $w : D \to \mathbf{C}^2$ satisfy

$$w_s + J(w)w_t = 0 \text{ and } w(0) = 0,$$

where J is an almost complex structure defined near $0 \in \mathbf{C}^2$. Moreover assume that $J(0) = i$. Assume that the order ℓ is at least 2. Introduce holomorphic polar coordinates and consider

$$u(s,t) = w(e^{-2\pi(s+it)}).$$

There are now $\ell - 1$ nontrivial ways to view u as a graph over u. Take the immersive coordinates Φ associated to u. Pick for large s_0 and t_0 points $(s_1, t_1, x_1, y_1), \ldots, (s_\ell, t_\ell, x_\ell, y_\ell)$ which are mutually different and satisfy

$$\Phi(s_j, t_j, x_j, y_j) = u(s_0, t_0).$$

Without loss of generality we may assume that

$$(s_\ell, t_\ell, x_\ell, y_\ell) = (s_0, t_0, 0, 0).$$

Hence we have $\ell - 1$ nontrivial lifts. According to every lift there is a unique solution of

$$u(s,t) = \Phi(a_j(s,t), b_j(s,t)), \ (a_j(s_0,t_0), b_j(s_0,t_0)) = (s_j, t_j, x_j, y_j).$$

We sum this up as a theorem.

Theorem 6.4.11 *Let J be an almost complex structure defined on a neighborhood of 0 in $\mathbf{C}^2$ satisfying $J(0) = i$. Assume that w is a non-constant J-holomorphic curve satisfying $w(0) = 0$. Define u by*

$$u(s,t) = w(e^{-2\pi(s+it)}).$$

Denote the order of w at 0 by ℓ. If $\ell \geq 2$ the following holds. With the immersive coordinates Φ associated to u there exist $\ell - 1$ nontrivial, mutually disjointy lifts (a_j, b_j), satisfying for s large

$$u(s,t) = \Phi(a_j(s,t), b_j(s,t)).$$

Nontrivial here means that $(a_j(s,t), b_j(s,t)) \neq (s,t,0,0)$ (for large s lets say). These local lifts have the following property. Either b_j vanishes identically or it admits a representation formula

$$b_j(s,t) = c_j \cdot e^{-2\pi m_j s + \delta_j(s)}[b_0^j e^{-2\pi i(\ell + m_j)t} + r_j(s,t)],$$

where r_j and δ_j satisfy the exponential decay estimates

$$|D^k \delta_j(s)| \leq c_k e^{-2\pi s}$$
$$|D^\beta r_j(s,t)| \leq c_\sigma^\beta e^{-(2\pi-\sigma)s}.$$

The proof is similar to the previous theorem.

We also note the following. If b_j vanishes on some non-empty open set it follows from the similarity principle that b_j vanishes identically, which implies that

$$u(s,t) = u(a_j(s,t))$$

for large s. Clearly a_j is holomorphic in this case. Since $j \in \{1, \ldots, \ell - 1\}$ and

$$(a_j(s_0,t_0),0) = (s_j, t_j, x_j, y_j)$$

we deduce that $(s_j, t_j, 0, 0) \neq (s_0, t_0, 0, 0)$, which shows that a_j is not the inclusion map. Hence u has some symmetry and u is not an embedding if restricted to $D_\varepsilon \setminus \{0\}$ for every $\varepsilon > 0$. Summing up

Corollary 6.4.12 *Assume $w : D \to \mathbf{C}^2$ satisfies $w_s + J(w)w_t = 0$, where J is an almost complex structure defined on a neighborhood of $0 \in \mathbf{C}^2$. Moreover let $J(0) = i$. Then the following alternative holds. Either there exists an open neighborhood U of 0 such that $w|(U \setminus \{0\})$ is an embedding, or there exists a biholomorphic map $\sigma : (U,0) \to (V,0)$, where U and V are open neighborhoods of 0, so that $w \circ \sigma = w$ and $\sigma(z) \neq z$ for some point $z \in U$.*

Proof Everything is clear except that as a consequence of our previous discussion σ is not defined at 0. However, by Riemann's removable singularity theorem it can be extended over 0. This extension has to be biholomorphic since the local degree at 0 is obviously one.

In the next section we give the applications of the normal forms. The following proposition will be proved later.

Proposition 6.4.13 *Let $u : (S,i) \to (W,J)$ be J-holomorphic map, where $dim(W) = 2n$ and S is a connected closed Riemann surface and u is non-constant. Then there exists a connected closed Riemann surface (T, j), a somewhere injective J-holomorphic map $v : T \to W$ and a holomorphic map $\phi : S \to T$ such that*

$$u = v \circ \phi.$$

So every J-holomorphic map factors through a somewhere injective one.

6.5 Deforming Holomorphic Curves

As a result of Corollary 6.4.12 we know that a non-constant J-holomorphic disk $w : D \to \mathbf{C}^2$ either induces an embedding

$$w|D_\delta : D_\delta \to \mathbf{C}^2$$

or it is multiply-covered, i.e. there exists a nontrivial biholomorphic map $\sigma : (U, 0) \to (V, 0)$ such that $w = w \circ \sigma$. Nontrivial of course means different from the identity. In the first case we shall say that w is a weak embedding near 0.

The following result will be important.

Theorem 6.5.1 *Assume J is an almost complex structure defined near 0 on $\mathbf{C}^2$ satisfying $J(0) = i$. Let w be a nontrivial J-holomorphic disk satisfying $w(0) = 0$ and $o(w, 0) := \ell \geq 2$. Assume $w : D \setminus \{0\} \to \mathbf{C}^2$ is an embedding. Then given any $\delta \in (0, 1)$ and $\varepsilon > 0$ there exists an immersion $\tilde{w} = \tilde{w}_{\delta,\varepsilon}$,*

$$\tilde{w} : D \to \mathbf{C}^2,$$

such that $\|\tilde{w} - w\|_{C^1} \leq \varepsilon$ and $supp(\tilde{w} - w) \subset D_\delta$. Moreover the self-intersections of $\tilde{w}$ are transversal and have index 1.

Proof We can write w without loss of generality as

$$w(z) = (z^\ell, 0) + h(z).$$

Here h is a smooth function vanishing at 0 so that the map $z \to \frac{h(z)}{z^\ell}$ belongs to class $W^{1,p}$ around 0 for every $2 < p < \infty$ and can be extended by 0 over 0. By replacing D by a smaller disk, we may assume without loss of generality that $pr_1 \circ Tw(z)$ is injective for every $z \in D \setminus \{0\}$. Take a smooth map $\phi : \mathbf{R} \to [0, 1]$ satisfying

$$\phi(s) = 1 \text{ for } |s| \leq \frac{\delta}{2}$$

and

$$\phi(s) = 0 \text{ for } |s| \geq \delta.$$

For $b \in \mathbf{C}$ define w_b by

$$w_b(z) = (z^\ell, 0) + h(z) + (0, b \cdot \phi(|z|)z) = w(z) + b(0, \phi(|z|)z).$$

Note that $w_b(z) = w(z)$ for $|z| \geq \delta$. We compute

$$Tw_b(z)a = Tw(z)a + (0, b\phi(|z|)a) + (0, b\phi'(|z|)\frac{\langle a, z\rangle}{|z|}z),$$

which implies that $pr_1 \circ Tw_b(z) = pr_1 \circ Tw(z)$ is injective for $z \in D \setminus \{0\}$. Clearly $Tw_b(0)$ is injective for $b \neq 0$. Consequently w_b is an immersion for $b \neq 0$. There exists a number $\tau > 0$ such that

$$w_b(z) = w_b(z')$$

for $z, z' \in D$ and $z \neq z'$ implies $|z|, |z'| \leq \frac{\delta}{4}$ provided $|b| \leq \tau$. Indeed, otherwise there exist sequences $z_n^{'}$ and z_n in D and $b_n \to 0$ such that

$$w_{b_n}(z_n) = w_{b_n}(z_n^{'}) \text{ and } z_n \neq z_n^{'}. \tag{6.32}$$

After perhaps taking some subsequences we may assume that

$$z_n \to z_0 \text{ and } z_n^{'} \to z_0^{'}.$$

If $z_0 \neq z_0^{'}$ we obtain passing to the limit

$$w(z_0) = w(z_0^{'}),$$

which contradicts our assumption that w on $D \setminus \{0\}$ is an embedding. Hence $z_0 = z_0^{'}$. Perhaps taking another subsequence we may assume that

$$\frac{z_n - z_n^{'}}{|z_n - z_n^{'}|} \to a \neq 0.$$

Dividing (6.32) by $|z_n - z_n^{'}|$ and passing to the limit we obtain

$$Tw(z_0)a = 0,$$

which implies that $z_0 = 0$. This of course implies that z_n and $z_n^{'}$ are near to 0 once $|b|$ is small. Hence, for $|b|$ small $w_b(z) = w_b(z')$ and $z \neq z'$ is equivalent to

$$w(z) + (0, bz) = w(z') + (0, bz') \text{ and } z \neq z'.$$

Note that a solution (z, z') with $z \neq z'$ automatically satisfies $z \neq 0$ and $z' \neq 0$. Consider the set Σ defined by

$$\Sigma = \{(q, p, b) \in B^* \times B^* \times B' | q \neq p, b \neq 0, w(q) + (0, bq) = w(p) + (0, bp)\}.$$

Here B^* is the open $\frac{\delta}{2}$-disk in $\mathbf{C}$ with $\{0\}$ removed and B' the open τ-disk for τ small with $\{0\}$ removed as well. Σ is a smooth submanifold of dimension 1 in $B \times B \times B'$.

Indeed the map

$$B \times B \times B' \to \mathbf{C}^2$$
$$(z, z', b) \to w(z) - w(z') + b(z - z')$$

has 0 as a regular value. A regular value b for the projection

$$\Sigma \to B' : (z, z', b) \to b$$

has the following property. The points z, z' such that $(z, z', b) \in \Sigma$ is a regular self intersection point of w_b with transversal crossing. Hence the local intersection index is ± 1. By the Sard theorem the set of regular values is residual, in particular dense. Take a small regular b. Then w_b is C^1-close to w. One easily shows that w_b is symplectic for the standard symplectic structure on $\mathbf{C}^2$ on D_δ provided b is small enough. This implies then that the intersection has positive index, i.e. 1. The symplecticity assertion we leave as an exercise.

Let w_b as just constructed and denote by $N(z)$ a section of the normal bundle. For simplicity of notation write $u = w_b$ and denote by u_N the curve obtained from u by pushing it slightly in the direction of N. The intersection number of u and u_N satisfies obviously

$$\text{int}(u, u_N) = \sharp\{(q, p) \in D \times D | q \neq p, u(p) = u(q)\}.$$

On the other hand let n be a nowhere vanishing section of the generalized normal bundle and w_n obtained from w by pushing it slightly. Again we can define an intersection number

$$\text{int}(w, w_n).$$

The two intersection numbers are related. Indeed:

Lemma 6.5.2 *The following identity relating int(u, u_N) and int(w, w_n) holds:*

$$int(u, u_N) = int(w, w_n) + \ell - 1.$$

Here $o(w, 0) = \ell$.

Proof The intersection numbers $\text{int}(u, u_N)$ and $\text{int}(w, w_n)$ are related as follows. Restrict n and N to ∂D and recall that u and w coincide near ∂D, so that we can take the same normal bundle there, say the orthogonal line bundle to the tangent bundle with respect to some Hermitian inner product. Write

$$n(z) = f(z) \cdot_{J(u(z))} N(z) \text{ for } z \in \partial D,$$

where $f : \partial D \to \mathbf{C}^*$ is a smooth function. Then

$$\mathrm{int}(w, w_n) = \mathrm{int}(u, u_N) + \mathrm{wind}(f).$$

Recall that by construction $u = w_b$ was symplectic for the standard form with transversal self-intersections. Hence, it is $\tilde{J}$-holomorphic for some $\tilde{J}$ satisfying $\omega(h, \tilde{J}(u)h) > 0$ for $h \neq 0$ ($\tilde{J}$ is tamed by ω). We may assume that the normal bundle is a complex line bundle for the structure $\tilde{J}$. For z near ∂D we may assume that $\tilde{J}(u(z)) = J(u(z))$. Moreover $\tilde{J}$ and J are homotopic through almost complex structures tamed by ω. Note that the maps

$$A : D \times \mathbf{C}^2 \to u^*T\mathbf{C}^2 : (z, a, b) \to aTu(z)(1) + bN(z)$$

and

$$B : D \times \mathbf{C}^2 \to w^*T\mathbf{C}^2 : (z, a, b) \to a\frac{1}{z^{\ell-1}}Tw(z)(1) + bn(z)$$

are the inverses of complex trivialisations. Since the structures J and $\tilde{J}$ on $u^*T\mathbf{C}^2 = D \times \mathbf{C}^2$ are homotopic we infer that the map

$$\partial D \to \mathcal{GL}(\mathbf{C}^2) : z \to A(z)^{-1} \circ B(z)$$

must be contractible. This is however only the case if the winding number of

$$z \to \det(A(z)^{-1} \circ B(z))$$

is 0. Now observe that

$$A(z)^{-1} \circ B(z) = \begin{bmatrix} \frac{1}{z^{\ell-1}} & 0 \\ 0 & f(z) \end{bmatrix}.$$

Hence, the map $z \to f(z)\frac{1}{z^{-1+\ell}}$ must have winding number 0. This gives, that $\mathrm{wind}(f) = -1 + \ell$. Consequently

$$\mathrm{int}(w, w_n) = \mathrm{int}(u, u_N) + \ell - 1.$$

This completes the proof.

6.6 The Positivity of Intersections

Let us begin by studying a single non-constant J-holomorphic curve

$$w : D \to \mathbf{C}^2.$$

We may assume that J is an almost complex structure defined on an open neighborhood of 0 in $\mathbf{C}^2$ satisfying $J(0) = i$. Assume that

$$w_s + J(w)w_t = 0$$
$$w(0) = 0$$
$$\ell := o(w, 0) \geq 2$$

Moreover suppose that the generalized tangent space at 0, say Γ_u is given by $\Gamma_u = \mathbf{C} \times \{0\}$. We also assume that $w|(D \setminus \{0\})$ is an embedding. We define a second order index $o_2(w, 0)$ as follows. Put

$$u(s, t) = w(e^{-2\pi(s+it)}).$$

Then u can be written for large s as

$$u(s, t) = c \cdot e^{-2\pi\ell s+\delta(s)}[(1, 0)e^{-2\pi\ell it} + r(s, t)].$$

Denote by Φ the immersive coordinates associated to u. Then

$$u(s, t) = \Phi(a_j(s, t), b_j(s, t))$$

for large s, where (a_j, b_j), $j = 1, \ldots, \ell - 1$, denotes different non trivial lifts characterised by the requirement, that for some fixed $t_0 \in S^1$ and the $(a_j(s_0, t_0), b_j(s_0, t_0))$ are mutually different. We have a representation formula for b_j and an order m_j occurring in the formula

$$b_j(s, t) = c_j \cdot e^{-2\pi m_j s+\delta_j(s)}[b_0^j e^{-2\pi i(m_j+\ell)t} + r_j(s, t)].$$

We define the total secondary order $o_2(w, 0)$ as follows.

Definition 6.6.1 Let w be as just described. In particular $\ell := o(w, 0) \geq 2$. The secondary order of w at 0 is defined by

$$o_2(w, 0) = [\sum_{j=1}^{\ell-1} (m_j + \ell)] - \ell + 1$$

In the case $\ell = 1$ we define for notational convenience

$$o_2(w, 0) = 0.$$

Note that $o_2(w, 0)$ satisfies the inequality

$$o_2(w, 0) \geq \ell \cdot (\ell - 1).$$

A nontrivial result is the following.

Theorem 6.6.2 *Assume that J is an almost complex structure defined on some open neighborhood of 0 in $\mathbf{C}^2$ such that $J(0) = i$. Let $w : D \to \mathbf{C}$ be a J-holomorphic curve with $w(0) = 0$ and $o(w, 0) \geq 2$. Assume also that $w : D_\delta \setminus \{0\} \to \mathbf{C}^2$ is an embedding. Then the previously defined secondary order $o_2(w, 0)$ satisfies*

$$o_2(w, 0) \in 2 \cdot \mathbf{N}^*.$$

The proof of the theorem relies on the following proposition, which gives a nice geometric interpretation.

Proposition 6.6.3 *Assume that w is a non-constant J-holomorphic disk as previously described. Suppose that the generalized tangent space at 0 is $\mathbf{C} \times \{0\}$ and the order ℓ at 0 is at least 2. Assume that $w|(D \setminus \{0\})$ is an embedding. Take a nowhere vanishing continuous section of the generalized normal bundle, say $n(z)$. Define for ε, $|\varepsilon|$ small, the section*

$$w_\varepsilon(z) = w(z) + \varepsilon n(z).$$

Then for every $\delta \in (0, 1]$ there exists an $\varepsilon_0(\delta)$ with the following property. The intersection number of $w|D_\delta$ and $w_\varepsilon|D_\delta$ is well-defined provided $\varepsilon \in (0, \varepsilon_0]$. Moreover

$$int(w|D_\delta, w_\varepsilon|D_\delta) = o_2(w, 0) + \ell - 1$$

Further, if w_b is the symplectic immersion obtained from w in Theorem 6.5.1, then the number of local self intersection points near 0 is precisely $\frac{1}{2} \cdot o_2(w, 0)$. More precisely

$$\sharp\{(q, p) | q \neq p,\ q, p \in D_\delta,\ w_b(q) = w_b(p)\} = o_2(w, 0).$$

(Note that on the left-hand side the self-intersection points are counted twice.)

Proof In order to prove the proposition we need some preparation. First of all note that we may assume without loss of generality that $w|(D_\delta \setminus \{0\})$ is an embedding. In fact, without loss of generality

$$w|(D_\delta \setminus \{0\}) : D_\delta \setminus \{0\} \to A_\tau$$

and this map can be lifted in $\ell - 1$ mutually different ways

$$\tilde{w}_j : D_\delta \setminus \{0\} \to B_\tau.$$

For $|\varepsilon|$ consider the intersections of $w(D_\delta)$ and $w_\varepsilon(D_\delta)$. Clearly

$$\{0\} \notin w(D_\delta) \cap w_\varepsilon(D_\delta).$$

Moreover from the normal form it follows immediately that

$$\operatorname{dist}(\{0\}, w(D_\delta) \cap w_\varepsilon(D_\delta)) \geq d \cdot |\varepsilon|^{\frac{1}{\ell+1}},$$

since we assume that $\ell \geq 2$. Here $d > 0$ is a suitable positive constant. The intersection number

$$\operatorname{int}(w|D_\delta, w_\varepsilon|D_\delta)$$

is by excision the same as the intersection number

$$\operatorname{int}(w|A_{d'\cdot|\varepsilon|^{\frac{1}{\ell+1}},\delta}, w_\varepsilon|A_{d'\cdot|\varepsilon|^{\frac{1}{\ell+1}},\delta}),$$

where $A_{a,b}$ is an annulus with inner radius a and outer radius b. Moreover $d' > 0$ is a suitable constant. Having restricted the maps to the annulus we can lift $w_\varepsilon|(A_{d'\cdot|\varepsilon|^{\frac{1}{\ell+1}},\delta})$ to B_τ in $\ell - 1$ different ways. Observe that for $|\varepsilon|$ small enough the images of the annuli under w and w_ε have image in A_τ. Fix any lift for w restricted to the same annulus. Every intersection downstairs corresponds to an intersection of a uniquely determined lift of the w_ε with the lift of w. Moreover the map $\Phi : B_\tau \to A_\tau$ is orientation preserving. Hence, we may compute the intersection indices upstairs by comparing the lift of w with the $\ell - 1$ different lifts of w_ε, both of course restricted to an annulus. Up stairs we may now use the relative normal form and observe that the global index for the j's lift has to be $m_j + \ell$.

Let us assume that $u, v : D \to W$ are solutions of

$$w_s + J(w)w_t = 0,$$

where J is an almost complex structure on the smooth (real) four-manifold W. Assume that u and v are non-constant and $u(0) = v(0) = w_0 \in W$. By the relative normal form the following alternative holds. Either there exists a $\delta \in (0, 1)$ such that

$$u(D_\delta) \subset v(D) \text{ and } v(D_\delta) \subset u(D),$$

or there exists a $\delta \in (0, 1)$ such that

$$u(D_\delta \setminus \{0\}) \cap v(D_\delta \setminus \{0\}) = \emptyset.$$

In the first case we say that u and v have locally an identical image and in the second case we say they have an isolated intersection.

Definition 6.6.4 Assume (W, J) is a (real) four-dimensional almost complex manifold and $u, v : D \to W$ are non-constant holomorphic disks. If $u(0) = v(0)$ and

this intersection is isolated we can define a local index as follows. Working in local coordinates in $\mathbf{C}^2$ in the usual way define a relative local index $o_2((u,0),(v,0))$ by

$$o_2((u,0),(v,0)) = o(u,0)\cdot o(v,0)$$

if the generalized tangent spaces at 0 are different. If the generalized tangent spaces at 0 coincide denote by ℓ and ℓ' the orders of u and v at 0 respectively. Let $\tilde{u}$ be the ℓ'-fold covering of u and $\tilde{v}$ the ℓ-fold covering of v and consider for a fixed lift of u the $\ell\cdot\ell'$ different lifts for v and put

$$o_2((u,0),(v,0)) = \frac{1}{\ell\cdot\ell'}\sum_{j=1}^{\ell\cdot\ell'}[m_j+\ell\cdot\ell'].$$

Here $m_j+\ell\cdot\ell'$ is the negative of the winding number of b_j occurring in the relative normal form.

Similarly to Proposition 6.6.3 one proves

Proposition 6.6.5 *If u and v are holomorphic disks with an isolated intersection at 0, i.e. $u(0)=v(0)$ and $u(z)\neq v(z')$ for $z\neq 0$ and $z'\neq 0$, then*

$$int((u,0),(v,0)) = o_2((u,0),(v,0)).$$

The next result in this section is given by the following theorem.

Theorem 6.6.6 *Assume (W,J) is a (real) four-dimensional almost complex manifold and $u,v : D\to W$ are non-constant holomorphic disks. If $u(0)=v(0)$ and this intersection is isolated then the local intersection index at 0 is at least 1. Moreover, if it is precisely one we have $o(u,0)=o(v,0)=1$ and the intersection is transversal.*

Proof If the generalized tangent spaces coincide the local intersection number is at least 2. If the tangent spaces are different it follows immediately that $o(u,0)=o(v,0)=1$ in view of

$$\text{int}((u,0),(v,0)) = o(u,0)\cdot o(v,0).$$

Recall that

$$o_2(u,p)=0 \text{ if } o(u,p)=1,$$

and if $o(u,p)\geq 2$ let $o_2(u,p)$ be given as in Definition 6.6.1. Assume (W,J) is an almost complex (real) four-dimensional manifold and (S,i) a connected closed Riemann surface. Given a J-holomorphic map $u : S\to W$ we call u somewhere injective if it is non-constant and for every choice of points $q\neq p$ with $u(p)=u(q)$ the intersection point is isolated.

Definition 6.6.7 For a somewhere injective J-holomorphic curve define its self-intersection index by

$$\mathrm{sint}(u) = \frac{1}{2}[\sum_{p\neq q, u(p)=u(q)\}} o_2((u,q),(u,p)) + \sum_{q\in S} o_2(u,q)].$$

Note that by the similarity principle the set of points with $Tu(q) = 0$ is finite, so that the second sum is in fact only taken over a finite number of q's. Moreover the number is an integer and not as one might expect at first glance a half integer. Of course, as a consequence of Theorem 6.6.2 we know that $o_2(w,q)$ is even. The first sum is trivially even. As we shall see u is an embedding iff $\mathrm{sint}(u) = 0$. Moreover, and most interestingly $\mathrm{sint}(u)$ can be identified with a certain expression in topological quantities associated to u.

To be more precise, let us observe that the map u defines a homology class $[u] \in H_2(W, \mathbf{Z})$. The intersection product $[u]\cdot[u]$ is called the self-intersection number. Consider $u^*TW \to S$. The Chern number of u denoted by $c_u \in \mathbf{Z}$ is defined by

$$c_u = c_1(u^*(TW, J))([S]),$$

and the virtual genus $g(u)$ of u by

$$g(u) = 1 + \frac{1}{2}([u]\cdot[u] - c_u).$$

Note that the virtual genus is a purely topological quantity depending on the homology class defined by u and the almost complex structure J. The following theorem relates the virtual genus $g(u)$ and $\mathrm{sint}(u)$.

Theorem 6.6.8 *Let u be a J-holomorphic curve defined on a connected closed Riemann surface (S,i) with image in the almost complex manifold (W,J) of real dimension 4. If u is somewhere injective we have the inequality*

$$g(u) \geq g(S),$$

where $g(S)$ is the usual genus of S. Equality holds iff u is an embedding. Moreover

$$sint(u) = g(u) - g(S).$$

Proof The number $[u]\cdot[u]$ is computed by shifting u into the direction of some vectorfield along u to obtain a new map $v : S \to W$. We may assume that u and v are in general position, so that interections are transversal. The oriented intersection number is then precisely $[u]\cdot[u]$. The above argument does not depend on the deformation v of u. For the proof of our theorem it will be therefore important that we take some very special deformations, so that we are able to relate $[u]\cdot[u]$ with

sint(u). Define the set of singular points $\mathcal{S}_u$ by

$$\mathcal{S}_u = \{q \in S | Tu(q) = 0\}.$$

Passing to local coordinates when we study the behaviour in a neighborhood of a point we may assume that

$$u : D \to \mathbf{C}^2$$
$$u_s + J(u)u_t = 0$$
$$u(0) = 0,$$

where $J(0) = i$. We also may assume that the generalized tangent space at 0, say Γ_u is spanned by $(1, 0)$ (the complex multiplication, of course, being i). We may take as v_ε near 0 the curve $u + (0, \varepsilon n)$, where n is a nowhere vanishing section of the generalized normal bundle. Here $\varepsilon > 0$ is assumed to be very small. The following holds in view of Proposition 6.6.3. For sufficiently small $\delta > 0$ fixed and all $\varepsilon > 0$ small enough we have with $\ell = o(u, 0)$

$$\text{int}(u|D_\delta, v_\varepsilon|D_\delta) = o_2(u, 0) + \ell - 1$$

Given $q \neq p$ with $u(q) = u(p)$ we may assume, since the intersection is isolated and has positive index, that it persists under a small perturbation. So, again assuming that we are in the local situation let $q = 0$ and $u(0) = 0$. We deform u slightly in the direction of its general normal space. Extend all these local deformations to a section η of the generalized normal bundle. We may assume that the zeros of the section are non-degenerate. We deform u into the direction of η to obtain u_η. Let us compute the algebraic number of its zeros. We have $u^*TW = \mathcal{T}_u \oplus \mathcal{N}_u$. We compute the Chern number $c(\mathcal{T}_u)$ of $\mathcal{T}_u$. Observe that Tu is a section of $\text{Hom}_{\mathbf{C}}(TS, \mathcal{T}_u) \to S$. The global index (which is the Chern number) is the sum of the local indices which gives

$$c_1(\text{Hom}_{\mathbf{C}}(TS, \mathcal{T}_u))([S]) = \sum_{q \in \mathcal{S}} [o(u, q) - 1]. \tag{6.33}$$

Indeed

$$Tu(z)(\frac{\partial}{\partial s}) = z^{\ell-1} \cdot_{J(u(z))} e(z),$$

where $e(z)$ is a suitable smooth non-zero section of the generalized tangent bundle (for z close to a singularity z_0), and $s + it$ are holomorphic coordinates near z_0. Now observe that

$$\begin{aligned} &c_1(\text{Hom}_{\mathbf{C}}(TS, \mathcal{T}_u))([S]) \\ &= c_1(TS^* \bigotimes \mathcal{T}_u)([S]) \end{aligned} \tag{6.34}$$

$$
\begin{aligned}
&= c_1(TS^*)([S]) + c_1(\mathcal{T}_u)([S]) = 2g(S) - 2 + c_1(\mathcal{T}_u)([S]) \\
&= -c_1(TS)([S]) + c_1(\mathcal{T}_u)([S]) \\
&= 2g(S) - 2 + c_1(\mathcal{T}_u)([S]).
\end{aligned}
$$

Combining (6.33) and (6.34) we obtain

$$
c_1(\mathcal{T}_u)([S]) = 2 - 2g(S) + \sum_{q \in \mathcal{S}} [o(u,q) - 1].
$$

Consequently

$$
\begin{aligned}
& c_1(\mathcal{N}_u)([S]) \\
&= c_u - c_1(\mathcal{T}_u)([S]) \\
&= c_u - [2 - 2g(S) + \sum_{q \in \mathcal{S}} [o(u,q) - 1]].
\end{aligned}
$$

Hence the algebraic number f of zeros of η is

$$
f = c_u - [2 - 2g(S) + \sum_{q \in \mathcal{S}} [o(u,q) - 1]].
$$

Finally we compute that

$$
\begin{aligned}
& [u] \cdot [u] \\
&= \operatorname{int}(u, u_\eta) \\
&= \sum_{\{(q,p) | q \neq p, u(q) = u(p)\}} o_2((u,q),(u,p)) + \sum_{q \in \mathcal{S}} o_2(u,q) \\
&\quad + f + \sum_{q \in \mathcal{S}} [o(u,q) - 1] \\
&= 2 \cdot \operatorname{sint}(u) + f + \sum_{q \in \mathcal{S}} [o(u,q) - 1] + \\
&= 2 \cdot \operatorname{sint}(u) + c_u + 2g(S) - 2.
\end{aligned}
$$

This gives the formula

$$
\left[1 + \tfrac{1}{2}([u] \cdot [u] - c_u)\right] - g(S) = \operatorname{sint}(u).
$$

Recalling that

$$g(u) = \left[1 + \frac{1}{2}([u] \cdot [u] - c_u)\right]$$

is the virtual genus we obtain finally

$$g(u) - g(S) = \text{sint}(u).$$

We observe that the right-hand side is non-negative. Moreover the right-hand side is precisely 0 iff there are no singularities and no self -intersections, i.e. iff u is an embedding. Observe also that the right hand side is entirely composed of local quantities, whereas the left-hand side consists of global topological quantities.

Next we give an appropriate version for the case of a J-holomorphic curve defined on a disk. Assume therefore that u is a non-constant pseudoholomorphic curve in an almost complex manifold (W, J) where $\dim_{\mathbf{R}}(W) = 4$:

$$u : D \to W$$
$$u_s + J(u)u_t$$

We say that u is an embedding at the boundary if the following holds. There exists a small annulus around the boundary A_ε,

$$A_\varepsilon = \{z \in D |\ 1 - \varepsilon \leq |z| \leq 1\}$$

such that

$$\begin{aligned} &u|A_\varepsilon \text{ is an embedding} \\ &u^{-1}(u(A_\varepsilon)) = A_\varepsilon. \end{aligned} \tag{6.35}$$

For such an embedding at the boundary we define a self-intersection index $\text{sint}(u)$ as follows.

Definition 6.6.9 Let $u : D \to W^4$ be a J-holomorphic disk, which is an embedding near in the boundary. The self-intersection index $\text{sint}(u)$ for an embedding at the boundary is defined by

$$\text{sint}(u) = \frac{1}{2}[\sum_{\{(q,p)|q\neq p,\ u(q)=u(p)\}} o_2((u,q),(u,p)) + \sum_{q\in D} o_2(u,q)].$$

Note that $\text{sint}(u)$ is an integer in view of the fact that $o_2(u, q)$ is always even. Define another quantity $I(u)$ as follows. Consider the pull-back $u^*TW \to D$. It is a complex vector bundle with complex multiplication on the fiber above z given by

$J(u(z))$. Take a nowhere vanishing section a of u^*TW which for $z \in \partial D$ coincides with $Tu(z)(1) = Tu(z)(\frac{\partial}{\partial s})$, i.e.

$$a(z) = Tu(z)(1),\ z \in \partial D.$$

Take another nowhere vanishing section b such that the map

$$D \times \mathbf{C}^2 \to u^*TW : (z, (\alpha, \beta)) \to (u(z), \alpha a(z) + \beta b(z))$$

defines the inverse of a complex trivialisation, where the product $D \times \mathbf{C}^2$ carries the trivial structure i. Now push u slightly in the direction b to obtain the map u_b. Observe that

$$u_b(\partial D) \cap u(D) = \emptyset \text{ and } u_b(D) \cap u(\partial D) = \emptyset$$

since u is an embedding at the boundary in the sense of (6.35). We denote by $I(u)$ half of the oriented intersection number:

$$I(u) := \frac{1}{2}\text{int}(u, u_b).$$

We have the following

Theorem 6.6.10 *Let u be a holomorphic disk in (W^4, J) which is an embedding at the boundary. Then*

$$I(u) = sint(u).$$

Proof The proof is quite similar to the proof of Theorem 6.6.8. We define η to be a nowhere vanishing section of the generalized normal bundle and let u_η be obtained by pushing u slightly in the direction of η. Then

$$\begin{aligned}&\text{int}(u, u_\eta)\\ &= \sum_{q \neq p, u(q)=u(p)} o_2((u, q), (u, p)) + \sum_{q \in \mathcal{S}} o_2(u, q)\\ &\quad + \sum_{q \in \mathcal{S}} [o(u, q) - 1].\end{aligned}$$

We may assume that $\eta(z) = f(z) \cdot_{J(u(z))} b(z)$ for $z \in \partial D$. Similarly to the proof of Theorem 6.6.8 we find that

$$\text{int}(u, u_\eta) = \text{int}(u, u_b) + \text{wind}(f).$$

An easy calculation as in Lemma 6.5.2 shows that

$$\mathrm{wind}(f) = \sum_{q \in \mathcal{S}} [o(u, q) - 1].$$

This gives

$$\begin{aligned} I(u) &= \frac{1}{2}\mathrm{int}(u, u_b) \\ &= \frac{1}{2}\left[\mathrm{int}(u, u_\eta) - \sum_{q \in \mathcal{S}} [o(u, q) - 1] \right] \\ &= \mathrm{sint}(u). \end{aligned}$$

The proof is complete. □

Exercise Find the appropriate generalisation of the previous results for J-holomorphic curves on compact Riemann surfaces with boundary, which are embeddings at the boundary.

The quantity $I(u)$ enjoys a certain homotopy property. Let $(u_\tau, J_\tau)_{\tau \in [0,1]}$ be a smooth family of maps $u_\tau : D \to W^4$ and almost complex structures on W such that every u_τ is an embedding near the boundary in the sense previously defined and in addition we assume that u_τ is J_τ holomorphic near ∂D. Observe that a number $I(u_\tau, J_\tau)$ can be well-defined by taking a nowhere vanishing section a of $u_\tau^*(TW, J_\tau)$ such that $a(z) = Tu_\tau(z)(1)$ for $z \in \partial D$. Then take a second nowhere vanishing section b so that $a(z)$ and $b(z)$ is a basis for $(T_{u(z)}W, J_\tau(u(z)))$. Then push u_τ into the direction of b slightly in order to obtain $u_{\tau,b}$. $I(u_\tau, J_\tau)$ is then defined by

$$I(u_\tau, J_\tau) = \mathrm{int}(u_\tau, u_{\tau,b}).$$

The following lemma follows immediately from the definition.

Lemma 6.6.11 *For a smooth family $\tau \to (u_\tau, J_\tau)$, $\tau \in [0, 1]$, where u_τ is an embedding at the boundary and J_τ-holomorphic near ∂D the map $\tau \to I(u_\tau, J_\tau)$ is constant.*

Useful is the following corollary which we formulate as a theorem.

Theorem 6.6.12 *Let $\tau \to u_\tau$, $\tau \in [0, 1]$, be a smooth family of maps $u_\tau : D \to W^4$ which are embeddings at the boundary and J_τ-holomorphic near ∂D. Assume that u_0 is an embedding and J_0-holomorphic on the whole of D. Moreover assume that u_1 is J_1-holomorphic on D. Then u_1 is an embedding.*

Proof Since u_0 is an embedding we see that $\mathrm{sint}(u_0) = 0$. By assumption the map $\tau \to I(u_\tau, J_\tau)$ is constant. In view of Theorem 6.6.10 we have $I(u_0, J_0) = \mathrm{sint}(u_0) = 0$. Hence

$$\mathrm{sint}(u_1) = I(u_1, J_1) = 0.$$

This implies that u_1 has no self-intersections and no singularities. Hence u_1 is an embedding.

Chapter 7
Local Existence and Global Uniqueness Results



7.1 Introduction

In this chapter we will deal with the local existence of families of embedded pseudoholomorphic disks and their properties. In the following let (W, J) be a four-dimensional manifold equipped with an almost complex structure J. Moreover let $F \subset W$ be a *totally real submanifold*, i.e. $\dim F = 2$ and

$$T_x F \oplus J(x) T_x F = T_x W$$

for $x \in F$ (if $T_x F$ satisfies the above condition we will call it a *totally real subspace* of the complex vector space $(T_x W, J(x))$). We consider the nonlinear Cauchy-Riemann equation

$$\begin{aligned} \overline{\partial}_J u \ := \ \frac{\partial u}{\partial s} + J(u)\frac{\partial u}{\partial t} = 0 \ \text{ on } \mathring{D} \\ u(z) \in F \text{ for } z \in \partial D \end{aligned} \tag{7.1}$$

where $D \subset \mathbf{C}$ is the closed unit disk. First, we will introduce the notion of Maslov index for a solution u of (7.1) which is an embedding.

In Sect. 7.3 we will establish existence for the special case that we are actually interested in: Let (M, λ) be a closed three dimensional manifold with contact form λ and compatible complex structure J on the contact structure $\xi = \ker \lambda$. We consider the almost complex manifold $W = \mathbf{R} \times M$ with the usual almost complex structure $\tilde{J}$ defined by

$$\tilde{J}(t,u)(h,k) \ := \ \big(-\lambda(u)\, k \ , \ J(u)(\pi_\lambda k) \ + \ h \cdot X_\lambda(u) \big)$$

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where $\pi_\lambda : TM \to \xi$ is the projection along the Reeb vector field X_λ and $t, h \in \mathbf{R}$, $k \in T_u M$. Let $F \subset M$ be a two dimensional submanifold of M with an elliptic singular point $e \in F$. We assume that there is an open neighborhood U of e in F so that $U \cap F$ does not contain any other singular point. After a C^0-small deformation of the surface F near the singular point e and a suitable choice of J we will prove the existence of a smooth 1-parameter family of embedded $\tilde{J}$-holomorphic disks

$$\tilde{u}_\tau = (a_\tau, u_\tau) : D \longrightarrow \mathbf{R} \times M \ , \ \tau \in (0, 1)$$

which satisfy the (totally real) boundary condition

$$\tilde{u}_\tau(\partial D) \subset \{0\} \times (F \backslash \{e\})$$

so that the disjoint union

$$\{e\} \cup \left(\bigcup_{0<\tau<1} u_\tau(\partial D) \right)$$

is an open neighborhood of the singular point e in F. Moreover the images of two such disks $\tilde{u}_\tau(D)$ and $\tilde{u}_\rho(D)$ are disjoint whenever $\rho \neq \tau$.

The first result of this kind was proved by E. Bishop [14] in 1965. He considered a real two-dimensional surface in $(\mathbf{C}^2, i)$ with an elliptic singularity $e \in F$ and showed the existence of a 1-parameter family of pairwise disjoint holomorphic disks coming out from the point e with boundaries in F. For further results consider [11, 12] or [92]. Interesting applications and generalizations of this method in symplectic geometry are described in [32]. In contrast to E. Bishop's proof our result is completely elementary since we have the freedom to change the surface F near the singularity e unless we do not create any additional singular points and we may also choose J appropriately. Despite this we will call our disks a *Bishop family*.

In the next sections we will go back to the general setting of an arbitrary almost complex manifold (W, J). We will study the solution set of (7.1) near a given solution u_0 which is an embedding. A first step is to establish a normal form near u_0 so that we may replace W by $\mathbf{C}^2$. Moreover u_0 will look like $z \longmapsto (z, 0)$ and F will be given by some explicit formula. This is done in Sect. 7.4. In order to study the solution set of (7.1) near u_0 we would like to view the Cauchy–Riemann operator $\bar{\partial}_J$ as a smooth map from the Banach manifold $\mathcal{B} = \{u \in W^{1,p}(D, \mathbf{C}^2) \mid u(\partial D) \subset F\}$ into $L^p(D, \mathbf{C}^2)$ and we will consider the linearization $D\bar{\partial}_J(u_0)$ in u_0. This will be the subject of Sect. 7.5. Here we will benefit from the normal form derived in Sect. 7.4.

Then we will prove that there is a 1-parameter family D^τ of disjoint embedded holomorphic disks near $u_0(D)$ with boundary values in F by using the implicit function theorem. Finally we will prove a global uniqueness theorem: If v is a solution of (7.1) and $(D^\tau)_{\tau \in (0,\varepsilon)}$ a family of holomorphic disks as explained above

so that $D^0 \cap v(D) \neq \emptyset$ and $D^\tau \cap v(D) = \emptyset$ for $\tau \in (0, \varepsilon)$ then we must have $v(D) = D^0$ provided all the disks have their boundaries in a pseudoconvex hypersurface $M \subset W$ and lie on the same side of W^- of W. This condition is always satisfied by holomorphic disks in the symplectisation $\mathbf{R} \times M$ with boundary values in a totally real surface $\{0\} \times F \subset \mathbf{R} \times M$. This result enables us to glue families of holomorphic disks which will become important if we discuss disk filling methods later on.

7.2 The Maslov Index for Embedded Pseudoholomorphic Disks

Let $u_0 : D \to W$ be a solution of (7.1) which is an embedding. Consider the pullback bundle $u_0^* TW \to D$. This is a complex vector bundle over the disk and therefore trivializable. Now we would like to obtain a complex trivialization

$$\Gamma : (u_0^* TW, J) \longrightarrow (D \times \mathbf{C}^2, i).$$

One way to do this is the following. We pick a connection ∇ on $TW \to W$. Then

$$\tilde{\nabla}_X Y := \nabla_X Y - \frac{1}{4}(\nabla_{JY} J)X - \frac{1}{4}J((\nabla_Y J)X) - \frac{1}{2}J((\nabla_X J)Y)$$

is again a connection which is compatible with J, i.e.

$$\tilde{\nabla}_X JY = J(\tilde{\nabla}_X Y)$$

(see [77] chapter IX.3). Now choose a basis $(\sigma_1, J(u_0(0))\sigma_1, \sigma_2, J(u_0(0))\sigma_2)$ of $(u_0^* TW)_0$. We obtain a global frame (s_1, s_1', s_2, s_2') by parallel transport with respect to $\tilde{\nabla}$ along radial rays $\gamma \subset D$ starting from the origin. We calculate with $(u_0^* J)(z) := J(u_0(z))\ ;\ k = 1, 2$

$$\tilde{\nabla}_{\dot{\gamma}}((u_0^* J)s_k) = (u_0^* J)\tilde{\nabla}_{\dot{\gamma}} s_k = 0$$

and

$$J(u_0(0))s_k(0) = J(u_0(0))\sigma_k.$$

Hence $s_1' = (u_0^* J)s_1$ and $s_2' = (u_0^* J)s_2$. Now define

$$\Gamma : (u_0^* TW, J) \longrightarrow (D \times \mathbf{C}^2, i)$$

by the assignment

$$\Gamma(s_k(z)) := (z, e_k) \ ; \ k = 1, 2$$

$$\Gamma(J(u_0(z))s_k(z)) := (z, ie_k),$$

where $\{e_1, e_2\}$ is a basis of $\mathbf{C}^2$. If $\mathrm{pr}_2 : D \times \mathbf{C}^2 \to \mathbf{C}^2$ is the projection onto the second factor then

$$\partial D \ni z \longmapsto (\mathrm{pr}_2 \circ \Gamma)(\{z\} \times T_{u_0(z)}F)$$

defines a loop, say $z \longmapsto L_{u_0,F}(z)$, of totally real subspaces in $\mathbf{C}^2$ with respect to the standard complex structure i. Take any complex linear map η such that $\eta(L_{u_0,F}(1)) = \mathbf{R}^2$. Let us denote the manifold of totally real subspaces of $\mathbf{C}^2$ by $\mathcal{T}(2)$. Then consider the homotopy class

$$[\eta L_{u_0,F}] \in \pi_1(\mathcal{T}(2)) =: \pi_1(\mathcal{T}(2), \mathbf{R}^2),$$

which does not depend on the choice of η. Therefore we denote for simplicity the associated class to $\eta L_{u_0,F}$ in $\pi_1(\mathcal{T}(2))$ by $[L_{u_0,F}]$. As explained in Theorem A.3.4 there exists a distinguished group isomorphism $\mu_2 : \pi_1(\mathcal{T}(2)) \to \mathbf{Z}$ defined by

$$\mu_2\left(\left[\,[0, 1] \ni t \longmapsto e^{\pi i t}\mathbf{R} \oplus \mathbf{R}\right]\right) = 1,$$

which is called the *Maslov index*. In the following, assuming F to be fixed, we shall write $\mu(u_0)$ instead of $\mu_2(L_{u_0,F})$. We observe that the definition does not depend on the choices involved.

Indeed, if Γ_1 and Γ_2 are two complex trivializations of u_0^*TW then we have $\Gamma_1 \circ \Gamma_2^{-1} = \mathrm{id}_D \times A$ with $A : D \to \mathrm{GL}(\mathbf{C}^2)$. If we assume that the corresponding loops $L^1_{u_0,F} = AL^2_{u_0,F}$ and $L^2_{u_0,F}$ have endpoint $\mathbf{R}^2$ then $A(1) \in \mathrm{GL}(\mathbf{R}^2) \subset \mathrm{GL}(\mathbf{C}^2)$. Let B_t be a path in $\mathrm{GL}(\mathbf{R}^2)$ with $B_1 = \pm\mathrm{Id}$ and $B_0 = A(1)^{-1}$. Then

$$L_t(z) := B_t A(tz + (1 - t))L^2_{u_0,F}(z)$$

is a homotopy of loops of totally real subspaces between $L_0(z) = L^2_{u_0,F}(z)$ and $L_1(z) = L^1_{u_0,F}(z)$ with $L_t(1) = \mathbf{R}^2$. Hence $L^1_{u_0,F}$ and $L^2_{u_0,F}$ have the same Maslov index. The fact that u_0 solves the differential equation is, of course, irrelevant. If $u : D \to W$ is a continuous map with $u(\partial D) \subset F$, which is homotopic to u_0 as a disk map, sending ∂D to F during the homotopy, then $\mu(u_0) = \mu(u)$.

7.3 Local Existence of a Bishop Family

As we announced in the introduction to this chapter we will prove the following existence theorem:

Theorem 7.3.1 *Let M be a three-dimensional manifold with contact form λ and let $F \subset M$ be a two dimensional submanifold with an isolated elliptic singular point $e \in F$ for the characteristic foliation on F induced by the contact structure* $\ker \lambda$. *Then after a C^0-small perturbation of the surface F near e (keeping e fixed without introducing any other singular points) and a suitable choice of a compatible complex structure J on* $\ker \lambda$ *there exists a smooth family*

$$\tilde{u}_\tau : D \longrightarrow \mathbf{R} \times M \ ; \ \tau \in (0, 1)$$

of embedded disks with the following properties:

- *The above disks satisfy the nonlinear Cauchy Riemann equation*

$$\partial_s \tilde{u}_\tau + \tilde{J}(\tilde{u}_\tau)\partial_t \tilde{u}_\tau = 0.$$

- *They satisfy the (totally real) boundary condition $\tilde{u}_\tau(e^{i\theta}) \in \{0\} \times (F \backslash \{e\})$, $\theta \in [0, 2\pi]$.*
- *$\tilde{u}_\tau(D) \cap \tilde{u}_\rho(D) = \emptyset$ whenever $\rho \neq \tau$.*
- *The disjoint union*

$$\{e\} \cup \left(\bigcup_{0<\tau<1} u_\tau(\partial D) \right)$$

  is an open neighborhood of the singular point e in F.
- *For each $\tau \in (0, 1)$ the Maslov index $\mu(\tilde{u}_\tau)$ equals 2.*

Proof By the results of Chap. 3 we may assume that after a C^0-small perturbation of the surface F near the elliptic singular point e we are in the following situation. The contact manifold is just $\mathbf{R}^3$ with the standard contact form $\lambda = dz + xdy$. Moreover the surface F is given by the set

$$\{(x, y, -\frac{1}{2}xy) \mid (x, y) \in B_\varepsilon(0) \subset \mathbf{R}^2\}$$

with some $\varepsilon > 0$ and e corresponds to the nicely elliptic singular point $(0, 0, 0)$. The contact structure is given by

$$\ker \lambda(x, y, z) = \mathrm{Span}_{\mathbf{R}} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -x \end{pmatrix} \right\}$$

and we define a complex structure on $\ker\lambda$ as follows: For (x, y, z) near $(0, 0, 0)$ we put

$$J(x, y, z) \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} := \begin{pmatrix} 0 \\ 1 \\ -x \end{pmatrix}$$

$$J(x, y, z) \cdot \begin{pmatrix} 0 \\ 1 \\ -x \end{pmatrix} := \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}$$

and we continue J arbitrarily onto the whole manifold. In the above coordinates the nonlinear Cauchy Riemann equation looks as follows:

$$\begin{aligned} \partial_s a - \partial_t z - x\partial_t y &= 0 \\ \partial_s z + \partial_t a - x\partial_t x &= 0 \\ \partial_s x - \partial_t y &= 0 \\ \partial_s y + \partial_t x &= 0. \end{aligned} \tag{7.2}$$

The required boundary condition is

$$\begin{aligned} z(e^{i\theta}) &= -\frac{1}{2}x(e^{i\theta})y(e^{i\theta}) \\ a(e^{i\theta}) &= 0 \end{aligned} \tag{7.3}$$

for $\theta \in [0, 2\pi]$. We leave it as an exercise for the reader to verify that the maps

$$\tilde{u}_\tau : D \longrightarrow \mathbf{R}^4$$

$$\tilde{u}_\tau(s, t) := \left(\frac{\tau^2}{4}(s^2 + t^2 - 1), \tau s, \tau t, -\frac{\tau^2}{2}st\right) \; ; \; \tau > 0 \, , \; s^2 + t^2 \leq 1$$

are solutions of the boundary value problem (7.2) and (7.3). We also have $\tilde{u}_\tau(D) \cap \tilde{u}_\rho(D) = \emptyset$ whenever $\rho \neq \tau$.

It remains to show that the Maslov index $\mu(\tilde{u}_\tau)$ equals 2 for all $\tau > 0$. For this we compute with $(s, t) \in D$, $(h, k_1, k_2, k_3) \in \mathbf{R}^4$

$$\tilde{J}(\tilde{u}_\tau(s, t)) \cdot (h, k_1, k_2, k_3) = (-k_3 - \tau s k_2, -k_2, k_1, h - \tau s k_1)$$

using the definition of J. Define now $\Phi : D \to \mathrm{GL}_4(\mathbf{R})$ by

$$\Phi(s, t) \cdot (h, k_1, k_2, k_3) := (-\tau s k_2 - k_3, k_1, k_2, h).$$

The inverse is given by

$$\Phi(s,t)^{-1}\cdot(h,k_1,k_2,k_3)=(k_3,k_1,k_2,-h-\tau s k_2)$$

so that

$$\begin{aligned}\Phi(s,t)\tilde{J}(\tilde{u}_\tau(s,t))\Phi(s,t)^{-1}\cdot(h,k_1,k_2,k_3)&=\Phi(s,t)\cdot(h,-k_2,k_1,k_3-\tau s k_1)\\&=(-k_3,-k_2,k_1,h).\end{aligned}$$

With the identification

$$\mathbf{R}^4\overset{\approx}{\to}\mathbf{C}^2$$

$$(h,k_1,k_2,k_3)\longmapsto(h+ik_3,k_1+ik_2)$$

we have

$$\Phi(s,t)\tilde{J}(\tilde{u}_\tau(s,t))\Phi(s,t)^{-1}=i$$

hence we may view Φ as a bundle isomorphism between the complex vectorbundle $(\tilde{u}_\tau^*T(\mathbf{R}\times\mathbf{R}^3),\tilde{J}(\tilde{u}_\tau))\approx(D\times\mathbf{R}^4,\tilde{J}(\tilde{u}_\tau))$ and $(D\times\mathbf{C}^2,i)$. Then

$$L:[0,2\pi]\longrightarrow\mathbf{C}^2$$

$$L(\theta):=\Phi(e^{i\theta})\cdot T_{\tilde{u}_\tau(e^{i\theta})}(\{0\}\times F)$$

is a loop of totally real subspaces of $(\mathbf{C}^2,i)$. We calculate

$$\begin{aligned}L(\theta)&=\Phi(e^{i\theta})\cdot\operatorname{Span}_{\mathbf{R}}\left\{\begin{pmatrix}0\\1\\0\\-\frac{\tau}{2}\sin\theta\end{pmatrix},\begin{pmatrix}0\\0\\1\\-\frac{\tau}{2}\cos\theta\end{pmatrix}\right\}\\&=\operatorname{Span}_{\mathbf{R}}\left\{\begin{pmatrix}\frac{\tau}{2}\sin\theta\\1\\0\\0\end{pmatrix},\begin{pmatrix}-\frac{\tau}{2}\cos\theta\\0\\1\\0\end{pmatrix}\right\}\\&\tilde{=}\operatorname{Span}_{\mathbf{R}}\left\{\begin{pmatrix}\frac{\tau}{2}\sin\theta\\1\end{pmatrix},\begin{pmatrix}-\frac{\tau}{2}\cos\theta\\i\end{pmatrix}\right\}\end{aligned}$$

after the identification $\mathbf{R}^4\approx\mathbf{C}^2$. Define now

$$\eta:=\frac{2}{\tau}\begin{pmatrix}-1&0\\i&\frac{\tau}{2}\end{pmatrix}\in\mathrm{GL}_2(\mathbf{C})$$

so that

$$\eta(L(0)) = \mathrm{Span}_{\mathbf{R}} \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\} = \mathbf{R}^2 \subset \mathbf{C}^2$$

and

$$(\eta \circ L)(\theta) = \mathrm{Span}_{\mathbf{R}} \left\{ \begin{pmatrix} -\sin\theta \\ 1 + i\sin\theta \end{pmatrix}, \begin{pmatrix} \cos\theta \\ i(1-\cos\theta) \end{pmatrix} \right\}$$

which is a loop of totally real subspaces of $(\mathbf{C}^2, i)$. We introduce a parameter $\rho \in [0, 1]$ and define

$$(\eta \circ L)_\rho(\theta) = \mathrm{Span}_{\mathbf{R}} \left\{ \begin{pmatrix} -\rho\sin\theta \\ 1 + i\rho\sin\theta \end{pmatrix}, \begin{pmatrix} \cos\theta + (1-\rho)i\sin\theta \\ i\rho(1-\cos\theta) \end{pmatrix} \right\}.$$

Then $(\eta \circ L)_1 = \eta \circ L$ and $(\eta \circ L)_0$ equals the loop

$$[0, 2\pi] \ni \theta \longmapsto \mathrm{Span}_{\mathbf{R}} \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} e^{i\theta} \\ 0 \end{pmatrix} \right\} = e^{i\theta}\mathbf{R} \oplus \mathbf{R}$$

which has Maslov index 2 by the results in Appendix A.3. It remains to verify that the family $\{(\eta \circ L)_\rho | \rho \in [0, 1]\}$ consists of loops of totally real subspaces of $(\mathbf{C}^2, i)$, i.e. the determinant

$$\Delta := \begin{vmatrix} -\rho\sin\theta & \cos\theta + (1-\rho)i\sin\theta \\ 1 + i\rho\sin\theta & i\rho(1-\cos\theta) \end{vmatrix}$$

must be nonzero for all $\rho \in [0, 1]$ and $\theta \in [0, 2\pi]$. Assume that $\Delta = 0$ for some ρ, θ. Then

$$\cos\theta = \rho(1-\rho)\sin^2\theta \tag{7.4}$$

and

$$\rho^2 \sin\theta(1-\cos\theta) + \rho\sin\theta\cos\theta + (1-\rho)\sin\theta = 0. \tag{7.5}$$

Inserting (7.4) into (7.5) we obtain

$$\begin{aligned} 0 &= \left(1 - \rho + \rho^2 + \rho^2(1-\rho)^2\sin^2\theta\right)\sin\theta \\ &=: \left(p_1(\rho) + p_2(\rho)\sin^2\theta\right)\sin\theta. \end{aligned}$$

The case $\sin\theta = 0$ can not occur because of (7.4), so we must have

$$p_1(\rho) + p_2(\rho)\sin^2\theta = 0.$$

But this is also impossible since we have $p_1([0,1]) = [\frac{3}{4}, 1]$ and $p_2([0,1]) = [0, \frac{1}{16}]$ proving that Δ is always nonzero. □

7.4 Normal Form Near an Embedded Pseudoholomorphic Disk

The aim of this section is the following normal form (see Fig. 7.1):

Theorem 7.4.1 *Let $u_0 : D \to W$ be a solution of (7.1) which is an embedding. Then there exists an open neighborhood U of $u_0(D) \subset W$ and a diffeomorphism $\varphi : U \to V \subset \mathbf{C}^2$ onto an open neighborhood of $0 \in \mathbf{C}^2$ so that*

$$(\varphi \circ u_0)(z) = (z, 0) \quad \textit{for all} \quad z \in D.$$

Moreover if $u : D \to W$ is another solution of (7.1) with $u(D) \subset U$ then $w := \varphi \circ u$ satisfies $w_s + \hat{J}(w)\, w_t = 0$ on $\overset{\circ}{D}$ where the almost complex structure

$$\hat{J}(z, w) := T\varphi(\varphi^{-1}(z, w)) \circ J(\varphi^{-1}(z, w)) \circ T\varphi^{-1}(z, w)$$

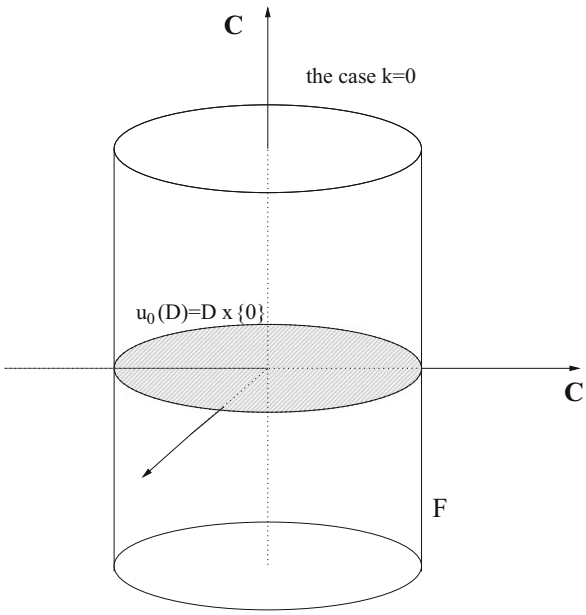


Fig. 7.1 Pseudoholomorphic disk and boundary condition after Theorem 7.4.1

satisfies in addition

$$\hat{J}(z,0) = i \quad \textit{if} \quad |\, z \,| \leq 1$$

and where $w(\partial D) \subset \hat{F} \subset \mathbf{C}^2$ *with*

$$\hat{F} := \{(z, az^{k/2}) \in \mathbf{C}^2 \mid z \in \partial D\,;\; a \in \mathbf{R}\},$$

and

$$k := \mu(u_0) - 2.$$

The proof consists of several steps of incremental improvements. We will start with some standard remarks on embeddings of disks and on extending isotopies. We will follow the presentation of [62] for a moment.

Lemma 7.4.2 *Let* $f : M \to N$ *be a smooth map between smooth finite dimensional manifolds and let* $K \subset M$ *be a compact subset so that*

- $f\,|_K$ *is injective*
- $Df(x) : T_xM \to T_{f(x)}N$ *is injective for all* $x \in K$

Then there is an open neighborhood U *of* K *in* M *so that* $f\,|_U$ *is an embedding*

Proof Each $x \in K$ has a neighborhood $U_x \subset M$ so that $f\,|_{U_x}$ is an embedding by the implicit function theorem. Cover K by finitely many of these U_x. Then $Df(y)$ is injective for all $y \in \bigcup U_x \supset K$. We still have to show that $f|_U$ is injective if U is a sufficiently small neighborhood of K. We argue indirectly and assume there are neighborhoods $(U_k)_{k\in\mathbf{N}}$ of K with $\cap_{k\in\mathbf{N}} U_k = K$, $U_{k+1} \subset \overline{U}_{k+1} \subset U_k$, $\overline{U}_k$ compact and $x_k, y_k \in U_k$ with $x_k \neq y_k$ and $f(x_k) = f(y_k)$.

After choosing a subsequence we may assume that $x_k \to x$; $y_k \to y$ with $x, y \in K$. Since $f|_K$ is injective we have $x = y$, hence $x_k, y_k \in U_x$ for k large enough. Now $x_k \neq y_k$ and $f(x_k) = f(y_k)$ contradict the fact that $f\,|_{U_x}$ is an embedding. This shows that $f|_U$ is injective. Shrinking the set U if necessary we may assume it is an embedding. □

Lemma 7.4.3 *Let* M, N *be smooth manifolds,* $\partial N = \emptyset$ *and let* $K \subset M$ *be a compact submanifold. Assume* $f : M \to N$ *is an embedding and* $G : M \times \mathbf{R} \to N$ *an isotopy of* f*, i.e. all the* $G(\,.\,,t)$ *are embeddings and* $G(\,.\,,0) = f$.

Then there is a diffeotopy $H : N \times \mathbf{R} \to N$ *extending* G *over* K*, i.e. each* $H(\,.\,,t)$ *is a diffeomorphism and*

$$H(f(x),t) = G(x,t) \quad \textit{for} \quad x \in K\,;\; t \in [0,1].$$

Proof We define $\hat{G} : M \times \mathbf{R} \to N \times \mathbf{R}$ by $\hat{G}(x,t) := (G(x,t),t)$. Let us consider the following vector field X on the submanifold $\hat{G}(M \times \mathbf{R}) \subset N \times \mathbf{R}$:

$$\begin{aligned} X(G(x,t),t) &= \left(DG(x,t) \cdot \frac{\partial}{\partial t}, 1 \right) \\ &=: (\xi(G(x,t),t),1) \in T_{G(x,t)}N \times \mathbf{R}. \end{aligned}$$

Here ξ is a smooth map $\xi : \hat{G}(M \times \mathbf{R}) \to TN$. X just consists of the tangent vectors to the curves

$$\begin{aligned} \gamma_x : \mathbf{R} &\longrightarrow N \times \mathbf{R} \\ t &\longmapsto \hat{G}(x,t) = (G(x,t),t) \end{aligned}$$

where $x \in M$. Now we would like to extend X to $N \times \mathbf{R}$ in the following way:

- The $\mathbf{R}$–component of X, i.e. the image of X under the differential of the projection $N \times \mathbf{R} \to \mathbf{R}$ should be $1 \equiv \frac{\partial}{\partial t}$
- X equals $\frac{\partial}{\partial t}$ outside an open neighborhood U of $\hat{G}(K \times [0,1])$ with compact closure

Before we turn to the construction of the extension let us note that X defines a global flow

$$\begin{aligned} \Phi : \mathbf{R} \times (N \times \mathbf{R}) &\longrightarrow N \times \mathbf{R} \\ (t,(x,\tau)) &\longmapsto \Phi_t(x,\tau) \end{aligned}$$

with the following properties:

- Φ_t maps $N \times \{\tau\}$ onto $N \times \{t+\tau\}$ since $\mathrm{pr}_2(\Phi_t(x,\tau)) = \tau + t$.
- Outside U we have $\Phi_t(x,\tau) = (x, \tau + t)$.
- For $x \in K$ and $t \in [0,1]$ we have $\Phi_t(f(x),0) = \gamma_x(t) = (G(x,t),t)$.

Hence

$$\begin{aligned} H : N \times \mathbf{R} &\longrightarrow N \\ H(x,t) &:= \mathrm{pr}_1(\Phi_t(x,0)) \end{aligned}$$

is the desired diffeotopy.

So we are left with the extension of the vector field X. Obviously it is sufficient to extend the map ξ smoothly so that it has compact support and we are done if we can do this locally. For $\varepsilon > 0$ small we cover $\hat{G}(K \times [-\varepsilon, 1+\varepsilon])$ by finitely many sets U_i which are open in $N \times \mathbf{R}$ and diffeomorphic to an open neighborhood of zero in $\mathbf{R}^{n+1}$ so that $U_i \cap \hat{G}(K \times \mathbf{R})$ looks like

$$\{(x_1, \ldots, x_n, t) \in \mathbf{R}^{n+1} \mid x_{k+1} = \cdots = x_n = 0\}$$

or

$$\{(x_1, \ldots, x_n, t) \in \mathbf{R}^{n+1} \mid x_{k+1} = \cdots = x_n = 0\,;\ x_k \geq 0\}$$

and $TN\mid_{\xi(U_i)} \approx U_i \times \mathbf{R}^n$. Then ξ corresponds to a smooth map

$$\begin{aligned} \mathbf{R}^k \times \mathbf{R} &\longrightarrow \mathbf{R}^n \times \mathbf{R}^n \\ (x_1, \ldots, x_k, t) &\longmapsto ((x_1, \ldots, x_k, 0, \ldots, 0), \tilde{\xi}(x_1, \ldots, x_k, t)) \end{aligned}$$

(or $\mathbf{R}^k_+ \times \mathbf{R} \to \mathbf{R}^n \times \mathbf{R}^n$ respectively). We extend the first component by the identity of $\mathbf{R}^n$, the second component by $\tilde{\xi}(x_1, \ldots, x_k, t) + (0, \ldots, 0, x_{k+1}, \ldots, x_n)$ and in the case $\mathbf{R}^k_+$ we find also a smooth extension for $x_k < 0$ (see for example [106] Chapter VI 3.2. Theorem 5). Finally we glue the local expressions together by using a subordinate partition of unity for $\{U_i\}$ and we are done. □

Lemma 7.4.4 *Let M be a connected oriented n-manifold without boundary and $f, g : D^n \to M$ embeddings of the n-disk which are both orientation preserving or orientation reversing. Then f and g are isotopic. Moreover we can realize this as a diffeotopy with compact support.*

Proof M is connected, so the embeddings $f\mid_{\{0\}}$, $g\mid_{\{0\}}\colon \{0\} \to M$ are isotopic and by Lemma 7.4.3 diffeotopic. Hence we may assume that $f(0) = g(0)$. Let (ϕ, U) be a chart of M near $f(0)$ with $\phi(U, f(0)) = (\mathbf{R}^n, 0)$. We can isotope f and g radially to embeddings with image in U; take for example

$$(x, t) \longmapsto f((1 - t + t\varepsilon)x)$$

with $x \in D^n, 0 \leq t \leq 1$ and $\varepsilon > 0$ small. So we may assume that $f(D^n) \cup g(D^n) \subset U$. It is enough to show that $\phi f, \phi g : D^n \to \mathbf{R}^n$ are isotopic. Assume that ϕf and ϕg preserve orientation by choosing ϕ carefully. Using

$$(x, t) \longmapsto \begin{cases} t^{-1}h(tx)\ ;\ 0 < t \leq 1 \\ Dh(0)x \quad\ ;\ t = 0 \end{cases}$$

with $h = \phi f$ or ϕg we may assume that ϕf and ϕg are isotopic to restrictions of linear maps. Now

$$\operatorname{sign} \det(\phi f) = \operatorname{sign} \det(\phi g),$$

so we find a path $A_t \in \mathrm{GL}(\mathbf{R}^n)$ with $A_0 = \phi f$ and $A_1 = \phi g$ which gives the desired isotopy. By Lemma 7.4.3 this extends to a diffeotopy. □

Remarks

- If we consider D^k instead of D^n with $k < n$ we can proceed as before and find a path $A_t \in \mathrm{GL}(\mathbf{R}^n)$ with $A_0 \mid \mathbf{R}^k = \phi f$ and $A_1 \mid \mathbf{R}^k = \phi g$. So the lemma is also valid in this case.
- If $\partial M \neq \emptyset$ we can isotop f and g so that the image lies in $M \backslash \partial M$ by using a collar. Then we can carry out the proof as before.

The following proposition shows that there are local coordinates in a neighborhood of the image of the embedded pseudoholomorphic disk u_0, so that this disk is 'flat' and the almost complex structure along the disk is standard.

Proposition 7.4.5 *There exists an open neighborhood U of $u_0(D) \subset W$ and a diffeomorphism $\sigma : U \to V \subset \mathbf{C}^2$ onto an open neighborhood of $0 \in \mathbf{C}^2$ so that*

$$(\sigma \circ u_0)(z) \;=\; (z, 0) \in \mathbf{C}^2 \;\; \textit{for} \;\; \mid z \mid \leq 1$$

and the almost complex structure

$$J'(z, w) \;:=\; T\sigma(\sigma^{-1}(z, w)) \circ J(\sigma^{-1}(z, w)) \circ T\sigma^{-1}(z, w).$$

satisfies $J'(z, 0) = i$ for $\mid z \mid \leq 1$. In addition $\mu(\sigma \circ u_0) = \mu(u_0)$.

Proof The proof works roughly as follows. First we extend the embedded disk $u_0(D)$ over the boundary and make it flat by using a tubular neighborhood construction. Then we have to modify the construction to achieve $(\sigma \circ u_0)(z) = (z, 0)$ and $J'(z, 0) = i$ for $\mid z \mid \leq 1$. Cover $u_0(\partial D)$ with finitely many charts

$$\phi_i : U_i \to \mathbf{R}^4 \approx \mathbf{C}^2 \;\; (0 \leq i \leq N)$$

so that

$$\phi_i(u_0(D) \cap U_i) \;=\; \{(s, t, 0, 0) \in \mathbf{R}^4 \mid \; t \leq 0\}$$

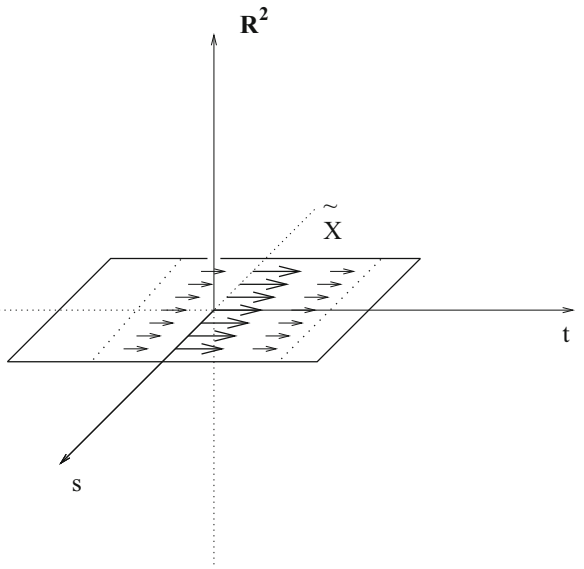
and choose a subordinate partition of unity $\{\beta_i\}_{1 \leq i \leq N}$ so that the supports of the functions β_i cover $u_0(\partial D)$. Take a cut–off function $\beta_\varepsilon : \mathbf{R} \to [0, 1]$ with $\beta_\varepsilon(t) \equiv 1$ for $\mid t \mid \leq \varepsilon$ and $\beta_\varepsilon(t) \equiv 0$ for $\mid t \mid \geq 2\varepsilon$. Define the vector field

$$\tilde{X}(x_1, \ldots, x_4) \;:=\; \beta_\varepsilon(x_2^2 + x_3^2 + x_4^2) \cdot (0, 1, 0, 0)$$

(see Fig. 7.2) and put

$$X(w) \;:=\; \sum_{i=1}^{N} \beta_i(w) \; D\phi_i^{-1}(\phi_i(w)) \; \tilde{X}(\phi_i(w)).$$

Fig. 7.2 The vector field $\tilde{X}$



This is a vector field defined near $u_0(\partial D)$ with compact support which we continue onto the whole manifold W. X does not vanish and is pointing outward along the boundary $u_0(\partial D)$. Now let φ be the time T–map of X for some $T > 0$ and define

$$\begin{aligned} F : D &\longrightarrow W \\ z &\longmapsto \varphi(u_0(z)) \end{aligned}$$

which is clearly smooth and injective. Moreover $DF(z)$ is injective since u_0 is an embedding, so it is an embedding because D is compact. Considering local coordinates we see that $u_0(D)$ is a proper subset of $F(D)$ since $\tilde{X}$ is pointing outward and does not vanish along $u_0(\partial D)$.

Then we can find a tubular neighborhood of $F(\overset{\circ}{D})$, i.e. there is an embedding $\tau : \nu \to W$ where $\nu \to F(\overset{\circ}{D})$ is the normal bundle so that $\tau \mid_{F(\overset{\circ}{D})}$ is the inclusion. Since $F(\overset{\circ}{D})$ is contractible we may assume that $\nu = F(\overset{\circ}{D}) \times \mathbf{R}^2$ so we have a diffeomorphism

$$\begin{aligned} \Psi : \overset{\circ}{D} \times \mathbf{R}^2 &\longrightarrow U \subset W \\ (z, \xi) &\longmapsto \tau(F(z), \xi) \end{aligned}$$

where U is an open neighborhood of $F(\overset{\circ}{D})$ in W. We see that $\Psi^{-1} \circ u_0$ has image in $\overset{\circ}{D} \times \{0\}$ and even $(\Psi^{-1} \circ u_0)(z) = (z, 0)$ if $\mid z \mid \leq r < 1$ for some r.

Now we have to arrange $w_0'(z) := (\Psi^{-1} \circ u_0)(z) = (z, 0)$ for all $z \in D$. If we project w_0' onto the first component of $\mathbf{R}^2 \times \mathbf{R}^2 \approx \mathbf{C}^2$ we get an embedding $w_0' : D \to \mathbf{R}^2$ with $w_0'(z) = z$ for $| z | \leq r < 1$. By Lemma 7.4.4 we find a diffeotopy $H_t : \mathbf{R}^2 \to \mathbf{R}^2$ so that $H_1 |_D \equiv w_0'$ and $H_0 \equiv \mathrm{Id}$. Define now a diffeomorphism Φ by

$$\begin{aligned} \Phi : \mathbf{C}^2 &\longrightarrow \mathbf{C}^2 \\ (z_1, z_2) &\longmapsto (H_1^{-1}(z_1), z_2). \end{aligned}$$

Writing $w := \Phi \circ \Psi^{-1} \circ u$ for a solution u of (7.1) with image in U we obtain

$$w : D \longrightarrow \mathbf{C}^2,$$

$$0 = w_s + \hat{J}(w) w_t \quad \text{on } \overset{\circ}{D},$$

$$w(\partial D) \subset \hat{F} := (\Phi \circ \Psi^{-1})\,(F \cap U) \subset \mathbf{C}^2,$$

where $\hat{J}$ is obtained by conjugating J with the differential of the diffeomorphism $\sigma' := \Phi \circ \Psi^{-1}$. Moreover

$$w_0(z) := (\sigma' \circ u_0)(z) = (z, 0).$$

Now we only have to achieve $\hat{J}(z, 0) = i$ for $| z | \leq 1$. We have

$$\hat{J}(\sigma'(m)) \circ T\sigma'(m) = T\sigma'(m) \circ J(m) \;\; ; \;\; m \in U.$$

We observe that we also have

$$T\sigma'(u_0(z)) \circ J(u_0(z))\, h = i \circ T\sigma'(u_0(z))\, h$$

where $h \in T_{u_0(z)}(u_0(D))$. Indeed, choose $k \in T_z D$ with $Tu_0(z)\, k = h$. Then

$$\begin{aligned} T\sigma'(u_0(z))\, J(u_0(z))\, h &= T\sigma'(u_0(z))\, J(u_0(z))\, Tu_0(z)\, k \\ &= T\sigma'(u_0(z))\, Tu_0(z)\, ik \\ &= (ik, 0) \\ &= i\,(k, 0) \\ &= i\, T\sigma'(u_0(z))\, Tu_0(z)\, k \\ &= i\, T\sigma'(u_0(z))\, h. \end{aligned}$$

Hence $\hat{J}$ acts in the direction of the tangent space of $D \times \{0\}$ as multiplication by i, i.e. $\hat{J}(z, 0)(h, 0) = (ih, 0)$ for $z \in D$ and $(h, 0) \in \mathbf{C} \times \{0\}$. We shall now modify

σ' slightly so that on D the almost complex structure $\hat{J}$ becomes multiplication by i. Find a smooth map $\phi : \mathbf{C} \to \mathrm{GL}_{\mathbf{R}}(\mathbf{C}^2)$ with

$$\phi(z)(h,0) = (h,0) \text{ for all } (h,0) \in \mathbf{C} \times \{0\} \text{ and } z \in \mathbf{C}$$

and in addition

$$\phi(z)\hat{J}(z,0) = i\phi(z) \text{ for all } z \in D.$$

Then define $\sigma : U \to \mathbf{C}^2$ by

$$\sigma(m) = \phi(\mathrm{pr}_1\sigma'(m))\,\sigma'(m).$$

We observe that σ is an embedding on $u_0(D)$ satisfying $\sigma(u_0(z)) = (z,0)$. We see also that $T\sigma(m)h = \phi(\mathrm{pr}_1\sigma'(m))\,T\sigma'(m)\,h$ for $m = u_0(z)$ with $|z| \leq 1$ (so $T\sigma(m) : T_mW \to \mathbf{C}^2$ is an isomorphism). This is true because of the following: Let γ be a path in W with $\gamma(0) = m$ and $\dot{\gamma}(0) = h \in T_mW$. Then

$$\begin{aligned}
&\left[D\phi(\mathrm{pr}_1\sigma'(m))\,\mathrm{pr}_1 T\sigma'(m)h\right]\big(\sigma'(m)\big)\\
&= \left[\frac{d}{dt}\phi(\mathrm{pr}_1\sigma'(\gamma(t)))\Big|_{t=0}\right]\big(\sigma'(m)\big)\\
&= \frac{d}{dt}\Big(\phi(\mathrm{pr}_1\sigma'(\gamma(t)))\,\sigma'(m)\Big)\Big|_{t=0}\\
&= \frac{d}{dt}\sigma'(m)\Big|_{t=0}\\
&= 0.
\end{aligned}$$

By Lemma 7.4.2, the map σ restricted to a sufficiently small neighborhood of $u_0(D)$ is a diffeomorphism onto its image. We will denote this neighborhood again by U. Let V be the image of U and define an almost complex structure J' on V by

$$J' \circ T\sigma = T\sigma \circ J.$$

We compute on $u_0(D)$ with $\sigma(m) := \sigma(u_0(z)) = (z,0)$ and $h \in T_mW$

$$\begin{aligned}
(T\sigma)(m)J(m)h &= \left[D\phi(z)\,\mathrm{pr}_1 T\sigma'(m)(J(m)h)\right]\big(\sigma'(m)\big) +\\
&\quad +\phi(z)T\sigma'(m)J(m)h\\
&= \phi(z)T\sigma'(m)J(m)h\\
&= \phi(z)\hat{J}(z,0)T\sigma'(m)\,h
\end{aligned}$$

$$= i\,\phi(z) T\sigma'(m)\,h$$
$$= i\,T\sigma(m)\,h.$$

Note that $w_0 = \sigma \circ u_0$ is the map $z \longmapsto (z, 0)$. Since the derivative of σ induces a complex bundle isomorphism $(u_0^* TW, J) \to ((\sigma \circ u_0)^* T\mathbf{C}^2, J')$ we have $\mu(u_0) = \mu(\sigma \circ u_0)$. □

Hence near $u_0(D)$ we can rewrite Eq. (7.1) with F replaced by $\sigma(F \cap U)$ and J by J'. In order to simplify notation, we keep the symbols and just note that without loss of generality we may study instead of (7.1)

$$\frac{\partial u}{\partial s} + J(u)\frac{\partial u}{\partial t} = 0 \ \text{ for } u : D \to \mathbf{C}^2, \tag{7.6}$$

where J is an almost complex structure defined on some open neighborhood U of $D \times \{0\} \subset \mathbf{C}^2$ satisfying $J(z, 0) = i$ for $|z| \leq 1$. Moreover the following boundary conditions should hold

$$u(z) \in F \text{ for } z \in \partial D,$$

where F is a piece of a totally real surface for J containing $\partial D \times \{0\}$. We will study this boundary value problem near the obvious solution $u_0(z) = (z, 0)$. The next result improves Proposition 7.4.5 as follows: By changing coordinates, we can achieve that the boundary condition is tangent to the one in Theorem 7.4.1 along $\partial D \times \{0\}$.

Proposition 7.4.6 *There is a diffeomorphism $\Gamma : U \to V$ between open neighborhoods U, V of $D \times \{0\}$ in $\mathbf{C}^2$ leaving $D \times \{0\}$ fixed so that for any solution u of (7.6) with image in U the map $w := \Gamma \circ u$ satisfies $w(\partial D) \subset \hat{F}$ where $\hat{F} \subset \mathbf{C}^2$ is a piece of a totally real surface containing $\partial D \times \{0\}$ and*

$$T_{(z,0)}\hat{F} = iz\mathbf{R} \oplus z^{k/2}\mathbf{R}$$

with $k = \mu(w_0) - 2$. The induced almost complex structure

$$\hat{J}(z, w) = T\Gamma(\Gamma^{-1}(z, w)) \circ J(\Gamma^{-1}(z, w)) \circ T\Gamma^{-1}(z, w)$$

satisfies

$$\hat{J}(z, 0) = i \ \text{ for } \ |\, z \,| \leq 1.$$

Before we begin with the proof we remark first that

$$L(z) := T_{w_0(z)}F = T_{(z,0)}F \subset \mathbf{C}^2 \ ; \ z = e^{2\pi i t} \in \partial D$$

is a loop of totally real subspaces of $(\mathbf{C}^2, i)$ since $J(z, 0) = i$ for $|\, z \,| \leq 1$.

Put $\ell_1(z) := iz\mathbf{R} \oplus \{0\}$. We have $\ell_1(z) \subset L(z)$ for all $z \in \partial D$. Now choose a one dimensional line $\ell_2(z)$ which complements $\ell_1(z)$ in $L(z)$, i.e. $L(z) = \ell_1(z) \oplus \ell_2(z)$. $\ell_2(z)$ is never contained in $\mathbf{C} \times \{0\}$ because $L(z)$ is totally real, so $K(z) := \mathrm{pr}_2(\ell_2(z)) \neq \{0\}$ for all $z \in \partial D$.

Lemma 7.4.7 *Let L and K be as described above. Then $\mu_2(L) = \mu_1(K) + 2$ (The index to μ indicates the dimension).*

Proof of Lemma 7.4.7 Define for $\tau \in [0, 1]$ the following homotopy in $\mathcal{T}(2)$. Pick an arc of unit vectors $t \to x(t)$ in $\mathbf{C}^2$ such that $x(t)$ spans $\ell_2(e^{2\pi it})$. Then $x(0) = \pm x(1)$. Denote the components of x by x_1, x_2 and recall that $x_2(t) \neq 0$ for all $t \in [0, 1]$. Define $L_\tau(z) = \ell_1(z) \oplus \mathbf{R}(\tau x_1(t), x_2(t))$, where $z = e^{2\pi it}$ and $\tau \in [0, 1]$. One shows by a straightforward argument that this is a direct sum and that L_τ is a loop of totally real subspaces of $(\mathbf{C}^2, i)$. Observe that $(0, x_2(t))$ spans $K(z)$. Then we compute by using Proposition A.3.6

$$\mu_2(L) = \mu_2(L_1) = \mu_2(L_0) = \mu_1(\ell_1) + \mu_1(K) = 2 + \mu_1(K).$$

□

Proof of Proposition 7.4.6 $K(t) := K(e^{2\pi it}) = \mathbf{R} \cdot x_2(t)$ is a loop of totally real subspaces of $(\mathbf{C}, i)$ with starting point $\mathbf{R} \cdot x_2(0)$ and Maslov index k. By the results in Appendix A.3 there are loops $(K_\tau)_{\tau \in [0,1]}$ of totally real subspaces of $(\mathbf{C}, i)$ with

$$K_\tau(0) = \mathbf{R} \cdot x_2(0) = K_\tau(1),$$

$$K_1(t) = K(t),$$

$$K_0(t) = \mathbf{R} \cdot x_2(0) e^{\pi ikt}$$

and

$$A_\tau(t) \in \mathrm{GL}(\mathbf{C}) = \mathbf{C}\backslash\{0\} \text{ with } A_\tau(t) \cdot \mathbf{R} = K_\tau(t).$$

We can choose A_τ and K_τ to depend smoothly on τ and $A_0(t) = x_2(0)e^{\pi ikt}$. Define

$$B_\tau(t) := (e^{-\pi ik} A_{\tau(1-t)}(1) A_{\tau(1-t)}(0)^{-1}) e^{-\pi ikt} A_\tau(t).$$

Then $B_\tau(t)(\mathbf{R} \cdot e^{\pi ikt}) = K_\tau(t)$ since $e^{-\pi ik} A_{\tau(1-t)}(1) A_{\tau(1-t)}(0)^{-1} \in \mathrm{GL}(\mathbf{R})$ and $B_\tau(0) = B_\tau(1)$. Hence we obtain loops

$$B_\tau : \partial D \longrightarrow \mathrm{GL}(\mathbf{C})$$

with $B_\tau(z)(\mathbf{R} \cdot z^{k/2}) = K_\tau(z)$ where $z = e^{2\pi it}$ and $B_\tau(z) := B_\tau(t)$. Define now

$$\Phi_\tau(t) := \begin{pmatrix} 1 & -\tau x_1(t) x_2(t)^{-1} \\ 0 & B_\tau(t)^{-1} \end{pmatrix} \in \mathrm{GL}(\mathbf{C}^2).$$

We observe

$$\begin{aligned}\Phi_1(t)(L(z)) &= \Phi_1(t)(iz\mathbf{R} \times \mathbf{R} \cdot (x_1(t), x_2(t))) \\ &= iz\mathbf{R} \times z^{k/2}\mathbf{R},\end{aligned}$$

$$\Phi_\tau(h, 0) = (h, 0) \text{ for all } h \in \mathbf{C}$$

and

$$\Phi_0(t) = \begin{pmatrix} 1 & 0 \\ 0 & x_2(0)^{-1} \end{pmatrix}.$$

Because of $-\tau x_1(0)x_2(0)^{-1} = -\tau x_1(1)x_2(1)^{-1}$ we obtain a family of loops $\Phi_\tau : \partial D \to \mathrm{GL}(\mathbf{C}^2)$ where $\Phi := \Phi_1$ maps $L(z)$ onto $iz\mathbf{R} \times z^{k/2}\mathbf{R}$ and Φ_0 is constant.

Now we would like to continue Φ smoothly onto $\mathbf{C}$. Take a smooth function $\beta : [0, +\infty) \to [0, 1]$ with the properties

$$\beta(t) \equiv 0 \text{ for } t \leq \frac{1}{2},$$

$$\beta(t) \equiv 1 \text{ for } t \geq 1,$$

$$\beta'(t) > 0 \text{ for } \frac{1}{2} < t < 1.$$

Then we define the smooth map

$$\tilde{\Phi} : \mathbf{C} \longrightarrow \mathrm{GL}(\mathbf{C}^2)$$

$$\tilde{\Phi}(z) := \begin{cases} \Phi_{\beta(|z|)}(|\ z\ |^{-1}\ z) & \text{for } |\ z\ | > 0 \\ \Phi_0(0) & \text{for } z = 0 \end{cases}$$

which satisfies $\tilde{\Phi}(z)(h, 0) = (h, 0)$ for all $h \in \mathbf{C}$ and $\tilde{\Phi}(z)(L(z)) = iz\mathbf{R} \times z^{k/2}\mathbf{R}$ for $z \in \partial D$. Now we can define the required diffeomorphism:

$$\Gamma : \mathbf{C}^2 \longrightarrow \mathbf{C}^2$$

$$\Gamma(z_1, z_2) := \tilde{\Phi}(z_1)(z_1, z_2).$$

We see that for $z \in \mathbf{C}$

$$\Gamma(z, 0) = \tilde{\Phi}(z)(z, 0) = (z, 0).$$

Let us calculate the derivative

$$D\Gamma(z,0)(h,k) = (D\tilde{\Phi}(z)(z,0))(h,k) + \tilde{\Phi}(z)(h,k).$$

Now the map $w \longmapsto \tilde{\Phi}(w)(z,0)$ is constant ($\equiv (z,0)$), so we have

$$D\Gamma(z,0)(h,k) = \tilde{\Phi}(z)(h,k).$$

Hence the derivative has full rank on $\mathbf{C} \times \{0\}$ and

$$\begin{aligned} D\Gamma(z,0)T_{(z,0)}F &= \tilde{\Phi}(z)(L(z)) \\ &= iz\mathbf{R} \times z^{k/2}\mathbf{R} \end{aligned}$$

for $z \in \partial D$. Using Lemma 7.4.2 we find an open neighborhood U of $D \times \{0\}$ in $\mathbf{C}^2$ so that $\Gamma\,|_U$ is a diffeomorphism onto its image.

It remains to consider the almost complex structure on $\mathbf{C}^2$ defined by conjugation of J with the derivative of Γ (this is the only place where we use that Φ has image in the complex linear maps of $\mathbf{C}^2$). We calculate

$$\begin{aligned} \hat{J}(z,0) &= T\Gamma(\Gamma^{-1}(z,0)) \circ J(\Gamma^{-1}(z,0)) \circ T\Gamma^{-1}(z,0) \\ &= T\Gamma(z,0) \circ i \circ T\Gamma^{-1}(z,0) \\ &= i \end{aligned}$$

for $|\,z\,| \leq 1$ which finishes the proof of the proposition. □

Proof of Theorem 7.4.1 Using Propositions 7.4.5 and 7.4.6 we may assume that we are in the following situation. We study maps $u : D \to \mathbf{C}^2$ with image near $D \times \{0\} \subset \mathbf{C}^2$ that satisfy

$$\frac{\partial u}{\partial s} + J(u)\frac{\partial u}{\partial t} = 0 \text{ on } \overset{\circ}{D}$$

and

$$u(\partial D) \subset F$$

where J is an almost complex structure defined on some open neighborhood of $D \times \{0\} \subset \mathbf{C}^2$ satisfying $J(z,0) = i$ for $|\,z\,| \leq 1$ and F is a piece of a totally real surface for J containing $\partial D \times \{0\}$. Moreover we have

$$L(z) := T_{(z,0)}F = iz\mathbf{R} \times z^{k/2}\mathbf{R}$$

where $k = \mu(u_0) - 2$ and u_0 is the obvious solution $u_0(z) = (z,0)$.

We have to find a diffeomorphism $\Sigma : U \to V$ between open neighborhoods of $D \times \{0\} \subset \mathbf{C}^2$ satisfying the following conditions:

- $\Sigma(z, 0) = (z, 0)$ for $| z | \leq 1$,
- $\Sigma(U \cap F) = V \cap \{(z, az^{k/2}) \in \mathbf{C}^2 \mid z \in \partial D\ ;\ a \in \mathbf{R}\} =: V \cap \hat{F}$,
- the almost complex structure $\hat{J}$ obtained by conjugation with the derivative of Σ satisfies $\hat{J}(z, 0) = i$ for $| z | \leq 1$.

Choosing some metrics on F and $\hat{F}$ the exponential maps induce smooth maps from

$$\begin{aligned} \nu &:= \{(0, az^{k/2}) \in \mathbf{C}^2 \mid a \in (-\varepsilon, \varepsilon)\ ,\ z \in \partial D\} \\ &\subset TF\ |_{\partial D \times \{0\}} = T\hat{F}\ |_{\partial D \times \{0\}} \end{aligned}$$

onto some open neighborhoods of $\partial D \times \{0\}$ in F and $\hat{F}$ respectively which have full rank at the zero section. Using Lemma 7.4.2 we get a diffeomorphism

$$\phi : W \longrightarrow \hat{W}$$

where W and $\hat{W}$ are open neighborhoods of $\partial D \times \{0\}$. If we restrict ϕ to a compact neighborhood $W_0 \subset W \subset F$ of $\partial D \times \{0\}$ we will get embeddings

$$j : W_0 \hookrightarrow \mathbf{C}^2 \quad \text{(inclusion)},$$

$$i := \tilde{j} \circ \phi : W_0 \longrightarrow \mathbf{C}^2 \quad (\tilde{j} \text{ is the inclusion of } \phi(W_0) \text{ into } \mathbf{C}^2)$$

of a compact piece of the surface F with $j(W_0) \subset F$, $i(W_0) \subset \hat{F}$ and $i\ |_{\partial D \times \{0\}}$ is the inclusion. These embeddings are isotopic by

$$h : [0, 1] \times W_1 \longrightarrow \mathbf{C}^2$$

$$(t, z) \longmapsto t\, i(z) + (1 - t)\, j(z)$$

where $W_1 \subset W_0$ is again a smaller piece of F. This is clear because of Lemma 7.4.2 applied to each $z \longmapsto h_t(z)$ using the fact that the derivative of h_t is injective along $\partial D \times \{0\}$ and $h_t(z) = j(z) = z$ for $z \in \partial D \times \{0\}$.

Now take an open neighborhood U of $D \times \{0\}$ in $\mathbf{C}^2$ with $F \cap U \subset W_0$. Then we can extend the isotopy h to a diffeotopy

$$H : [0, 1] \times U \longrightarrow U$$

with

$$H_t(z, 0) = (z, 0) \ \text{ for } \ z \in \partial D$$

because of Lemma 7.4.3. Taking a careful look at the proof of Lemma 7.4.3 we see that we can also arrange

$$H_1(z,0) = (z,0) \text{ and } DH_1(z,0) = \mathrm{Id} \text{ for } |z| \leq 1.$$

Now we choose $\Sigma := H_1$ and the theorem is proved. □

7.5 Implicit Function Theorem Near a Pseudoholomorphic Disk

By the results of the previous section the investigation of the solution set of (7.1) near an embedded solution u_0 reduces to the following problem:

$$\begin{aligned} u &: D \longrightarrow \mathbf{C}^2, \\ \bar{\partial}_J u &:= \frac{\partial u}{\partial s} + J(u)\frac{\partial u}{\partial t} = 0 \text{ on } \overset{\circ}{D}, \\ u(\partial D) &\subset F, \end{aligned} \tag{7.7}$$

where J is an almost complex structure satisfying $J(z,0) = i$ for $|z| \leq 1$ and

$$F = \{(z, az^{k/2}) \in \mathbf{C}^2 \mid z \in \partial D \,;\, a \in \mathbf{R}\}$$

is totally real for J. We are interested in solutions u near the obvious solution $u_0(z) = (z,0)$ if $\mu(u_0) = 2$, i.e. $k = 0$. In order to understand the solution set of (7.7) we have to view $\bar{\partial}_J$ as a smooth map between appropriate Banach manifolds.

Proposition 7.5.1 *Let $2 < p < \infty$ and U be a sufficiently small neighborhood of $D \times \{0\} = u_0(D) \subset \mathbf{C}^2$. Then*

$$\mathcal{B} := \{u \in W^{1,p}(D, \mathbf{C}^2) \mid u(\partial D) \subset F \,;\, u(D) \subset U\}$$

is a Banach submanifold of $W^{1,p}(D, \mathbf{C}^2)$ admitting an atlas that consists of one chart only. The tangent space at $u \in \mathcal{B}$ is given by

$$T_u\mathcal{B} = \{f \in W^{1,p}(D, \mathbf{C}^2) \mid f(z) \in T_{u(z)}F \text{ for } z \in \partial D\}.$$

Before we begin with the proof we will need the following lemma:

Lemma 7.5.2 *There is a metric on $\mathbf{C}^2$ so that F is totally geodesic.*

Proof Let us first construct a metric g on $\mathbf{C}^* \times \mathbf{C} := \mathbf{C}\backslash\{0\} \times \mathbf{C}$ so that F is totally geodesic, i.e. the induced metric $\tilde{g}$ on F satisfies the following. If γ is a geodesic in

$\mathbf{C}^* \times \mathbf{C}$ with respect to g and $\gamma(0) = x \in F$; $\dot{\gamma}(0) \in T_x F$ then γ is also a geodesic in F with respect to $\tilde{g}$.

Take the Euclidean metric $\langle .\,, .\rangle$ on $(\mathbf{R}^2\backslash\{0\}) \times \mathbf{R}^2$ and define the following diffeomorphism

$$\phi : \mathbf{C}^* \times \mathbf{C} \longrightarrow \mathbf{C}^* \times \mathbf{C}$$

$$(v, w) \longmapsto \left(\frac{1}{\bar{v}}, \bar{w}\frac{v^{k/2}}{\bar{v}^{k/2}}\right).$$

We calculate

$$\phi(\phi(v, w)) = (v, w)$$

and we have $\phi(v, w) = (v, w)$ if and only if $v = \bar{v}^{-1}$ and $w = \bar{w}v^{k/2}\bar{v}^{-k/2}$, i.e. $\mid v \mid^2 = 1$ and $w\bar{v}^{k/2} = wv^{k/2} \in \mathbf{R}$ which is equivalent to $(v, w) \in F$. Moreover $D\phi(v, w)$ induces the identity on $T_{(v,w)}F$. Now we put for $x \in \mathbf{C}^* \times \mathbf{C}$

$$g(x)(h, k) := \frac{1}{2}(\langle h, k\rangle + \langle D\phi(x)h, D\phi(x)k\rangle).$$

We note that ϕ is an isometry for this metric. Assume now that F is not totally geodesic. Then we find a geodesic $\gamma \subset \mathbf{C}^* \times \mathbf{C}$ with $\gamma(0) = x \in F$ and $\dot{\gamma}(0) = \xi \in T_x F$ which is leaving F, i.e. $\gamma(t) \in \mathbf{C}^* \times \mathbf{C}\backslash F$ for $t > 0$. Now $\phi \circ \gamma$ is also a geodesic which differs from γ since $(\gamma \circ \phi)(t) \neq \gamma(t)$ for $t > 0$. On the other hand we have $\phi(\gamma(0)) = \phi(x) = x$ and

$$\frac{d}{dt}(\phi \circ \gamma)(t)\bigg|_{t=0} = D\phi(x)\xi = \xi$$

which is impossible since for small t the geodesic is uniquely determined by x and ξ. We obtain the desired metric g' on $\mathbf{C}^2$ as follows. Let

$$U := (\mathbf{C}\backslash B_{\varepsilon/2}(0)) \times \mathbf{C} \;\; ; \;\; V := B_\varepsilon(0) \times \mathbf{C}$$

and (φ_U, φ_V) a subordinate partition of unity with $\varphi_U \equiv 1$ on $\partial D \times \mathbf{C}$. Take

$$g' := \varphi_U g + \varphi_V \langle .\,, .\rangle.$$

Since F is contained in $\partial D \times \mathbf{C} \subset U$ it is clearly totally geodesic. □

Remark Let (M, g) be a Riemannian manifold (without boundary) and F a compact submanifold. Then M carries a metric $\tilde{g}$ so that F is totally geodesic with respect to $\tilde{g}$. This can be seen as follows. Choose a tubular neighborhood of F in M, i.e. we have a vectorbundle $E \to F$ with $\dim E = \dim M$ and an embedding $i : E \to M$

which is the identity on the zero section (here we identified the zero section with F). Choose a Riemann metric h on E which induces g on F. Consider the map

$$\phi : E \longrightarrow E$$

$$x \longmapsto -x$$

and define the following metric $\tilde{g}$ on E:

$$\tilde{g} := \frac{1}{2}(h + h \circ (T\phi \times T\phi)).$$

Then ϕ is an isometry for this metric. Using the same uniqueness argument as in the proof of the previous lemma one shows that a geodesic for (E, h) which starts in F tangent to F must remain in F already.

Before we begin with the proof of Proposition 7.5.1, let us note the following lemma, which we will use for showing that $\mathcal{B} \subset W^{1,p}(D, \mathbf{C}^2)$ is a Banach–submanifold.

Lemma 7.5.3 *Let $f : U \subset E \to F$ be a smooth map, where E, F are Banach spaces and U is an open subset. Assume further that there is $x \in U$ so that $Df(x)$ is injective with closed range $R(Df(x))$ and $F = R(Df(x)) \oplus F_2$ for some closed subspace $F_2 \subset F$. Then there are two open sets $V \subset F$ and $W \subset E \oplus F_2$, where $f(x) \in V$, and a diffeomorphism $\phi : V \to W$ so that*

$$(\phi \circ f)(x) = (x, 0) \text{ for all } x \in W \cap (E \times \{0\}) \subset E.$$

Proof The proof is a simple application of the inverse function theorem. We define

$$g : U \times F_2 \subset E \times F_2 \longrightarrow F = R(Df(x)) \oplus F_2$$

by

$$g(x, y) := f(x) + (0, y).$$

Then $Dg(x, 0) = (Df(x), \mathrm{Id}_{F_2}) : E \oplus F_2 \to F$ is a bijective continuous linear map and the inverse is also continuous by the open mapping theorem. Hence the inverse function theorem applies and the desired diffeomorphism ϕ is the local inverse of g. □

Proof of Proposition 7.5.1 Because of Lemma 7.5.2 we may choose a metric on $\mathbf{C}^2$ so that F is totally geodesic. Let $\exp : \mathbf{C}^2 \times W \to \mathbf{C}^2$ be the exponential map of the associated Levi-Civita connection where W is some open neighborhood of $(0, 0)$ in $\mathbf{C}^2$. Consider

$$V := \{f \in W^{1,p}(D, \mathbf{C}^2) \mid f(z) \in T_{(z,0)}F = iz\mathbf{R} \oplus z^{k/2}\mathbf{R} \text{ for } z \in \partial D\}$$

First we remark that this definition makes sense since $W^{1,p}(D, \mathbf{C}^2) \subset C^0(D, \mathbf{C}^2)$ by the Sobolev embedding theorem (see [55]). We also note that V is a Banach space. Indeed, if $(f_n) \subset V$ is a sequence converging in $W^{1,p}$ to some f then f_n, $f \in C^0(D, \mathbf{C}^2)$ and $f_n \to f$ in C^0 by the Sobolev embedding theorem, so we also have $f(z) \in T_{(z,0)}F$ for $z \in \partial D$ and therefore $f \in V$.

We can choose a small ball $B \subset V$ with respect to the $W^{1,p}$-norm around zero so that the images of all these $f \in B$ are contained in some ball $B_\delta(0) \subset \overline{B_\delta(0)} \subset W$. This is true because of $\|f\|_{C^0} \leq c\|f\|_{W^{1,p}}$ (Sobolev embedding theorem). Hence for $f \in B$ we can define the map

$$\exp(f) : D \longrightarrow \mathbf{C}^2$$

$$z \longmapsto \exp((z, 0), f(z)).$$

We note that $\exp|_{D\times\{(0,0)\}}$ is just the obvious embedding $D \times \{0\} \hookrightarrow \mathbf{C}^2$. By Lemma 7.4.2 there is an open neighborhood W' of $D \times \{(0, 0)\} \subset \mathbf{C}^2 \times \mathbf{C}^2$ so that $\exp|_{W'}$ is an embedding. Choosing W small enough we may assume that $D \times W \subset W'$. Hence $\exp(f_1) = \exp(f_2)$ implies $f_1 = f_2$ for $f_1, f_2 \in B$. Since $\exp|_{D\times\overline{B_R(0)}}$ and its derivative are bounded, $\exp(f)$ is again of class $W^{1,p}$ and for U small enough we even have

$$\{\exp(f) \mid f \in B\} = \mathcal{B}$$

because F is totally geodesic. So we have a map

$$\Phi : B \longrightarrow \mathcal{B} \subset W^{1,p}(D, \mathbf{C}^2)$$

$$f \longmapsto \exp(f).$$

One verifies directly that Φ is smooth and

$$(D\Phi(f) \cdot h)(z) = D_2 \exp((z, 0), f(z)) \cdot h(z)$$

for $f \in B$; $h \in V$. We would like to show that $D\Phi(0)$ is injective and the image splits, i.e. the image is closed and has a closed complement. Then (after choosing B and U smaller) $\mathcal{B} = \Phi(B)$ is a Banach submanifold of $W^{1,p}(D, \mathbf{C}^2)$ and Φ is a global chart. We have $D_2 \exp((z, 0), 0) = \mathrm{Id}_{\mathbf{C}^2}$, so $D\Phi(0) \cdot h = h$ for $h \in V$. Defining

$$X_z := z\mathbf{R} \oplus iz^{k/2}\mathbf{R} \text{ for } z \in \partial D$$

we have $\mathbf{C}^2 = X_z \oplus T_{(z,0)}F$ and we see that the Banach space

$$V' := \{f \in W^{1,p}(D, \mathbf{C}^2) \mid f(z) \in X_z \text{ for } z \in \partial D\}$$

is a topological complement of V in $W^{1,p}(D, \mathbf{C}^2)$. We are left with the calculation of the tangent space of $\mathcal{B}$ at some point $u \in \mathcal{B}$. Elements in $T_u\mathcal{B}$ are of the form

$$\frac{d}{dt}\gamma_t\bigg|_{t=0} \in W^{1,p}(D, \mathbf{C}^2)$$

where $t \longmapsto \gamma_t$ is a smooth path in $\mathcal{B} \subset W^{1,p}(D, \mathbf{C}^2)$ with $\gamma_0 = u$. Pick $z \in \partial D$. Then $t \longmapsto \gamma_t(z)$ is a path in F and

$$\frac{d}{dt}\gamma_t(z)\bigg|_{t=0} \in T_{u(z)}F$$

which finishes the proof of the proposition. □

Proposition 7.5.4 *The Cauchy Riemann operator*

$$\bar{\partial}_J : \mathcal{B} \longrightarrow L^p(D, \mathbf{C}^2)$$

$$u \longmapsto u_s + J(u)\, u_t$$

is a smooth map. The derivative at u_0

$$T := D\bar{\partial}_J(u_0) : T_{u_0}\mathcal{B} \longrightarrow L^p(D, \mathbf{C}^2)$$

can be written as

$$T(h, k) = (h_s + ih_t + ak, k_s + ik_t + bk)$$

where $(h, k) \in T_{u_0}\mathcal{B}$ *and* a, b *are smooth maps from* D *into* $\mathcal{L}_{\mathbf{R}}(\mathbf{C})$.

Proof We consider the operator

$$\bar{\partial}_J : \{u \in W^{1,p}(D, \mathbf{C}^2) \mid u(D) \subset U\} \longrightarrow L^p(D, \mathbf{C}^2)$$

$$u \longmapsto u_s + J(u)\, u_t.$$

It suffices to show that this is a smooth map. The derivatives

$$\frac{\partial}{\partial s}, \frac{\partial}{\partial t} : W^{1,p}(D, \mathbf{C}^2) \longrightarrow L^p(D, \mathbf{C}^2)$$

are linear and continuous and therefore smooth. The map $u \longmapsto J(u)$ has image in $W^{1,p}(D, \mathbf{C}^2)$, and it is smooth because J is smooth and its derivatives are bounded on some closed ball containing U. Moreover the product of the $W^{1,p}$-map $z \longmapsto J(u(z))$ with the L^p-map $z \longmapsto \frac{\partial u}{\partial t}(z)$ is again in L^p. Indeed, if $\alpha \in W^{1,p}(D)$,

$\beta \in L^p(D)$ and $p > 2$ then

$$\int_D \mid \alpha\beta \mid^p \leq \|\alpha\|^p_{C^0} \|\beta\|^p_{L^p} \leq c\|\alpha\|^p_{W^{1,p}} \|\beta\|^p_{L^p} \ ; c > 0$$

by the Sobolev embedding theorem (see [6, 55]). Now

$$W^{1,p}(D) \times L^p(D) \longrightarrow L^p(D)$$

$$(\alpha, \beta) \longmapsto \alpha\beta$$

is a bilinear continuous map and therefore smooth. This proves that $\bar{\partial}_J$ is smooth.

Let us calculate the derivative at $u_0(z) = (z, 0)$. Take

$$f = (h, k) \in T_{u_0}\mathcal{B}.$$

Then

$$\begin{aligned} D\overline{\partial}_J(u_0)(h,k) &= T(h,k) \\ &= (h_s, k_s) + J(u_0)(h_t, k_t) + [DJ(u_0)(h,k)] \cdot (i,0) \\ &= (h_s + ih_t, k_s + ik_t) + [D_1 J(u_0)\, h] \cdot (i,0) + \\ &\quad + [D_2 J(u_0)\, k] \cdot (i,0) \\ &= (h_s + i\, h_t, k_s + i\, k_t) \ + \ [D_2 J(u_0)\, k] \cdot (i,0) \\ &=: (h_s + i\, h_t + ak, k_s + i\, k_t + bk). \end{aligned}$$

Here $a, b \in C^\infty(D, \mathcal{L}_{\mathbf{R}}(\mathbf{C}))$. We have used the fact that $J(u_0(z)) = J(z,0) = i$ and

$$(D_1 J(u_0) \cdot h)(z) = \frac{d}{dt} J(h_t(z), 0)\Big|_{t=0} = \frac{d}{dt} i\Big|_{t=0} = 0$$

where $t \longmapsto (h_t, 0)$ is a smooth path in $\mathcal{B}$ with $h_0(z) = z$ and $\frac{d}{dt} h_t(z) \mid_{t=0} = h$. □

Let us write $L(z) = T_{(z,0)}F$ for $z \in \partial D$. Then L is a loop of totally real subspaces of $(\mathbf{C}^2, i)$. The following is a special case of Theorem A.4.1.

Proposition 7.5.5 *The operator T is Fredholm and its index is given by*

$$ind(T) = \mu_2(L) + 2.$$

□

Now we can prove one main theorem of this section: A criterion for surjectivity of the operator $T = D\bar{\partial}_J(u_0)$ in terms of the Maslov index $\mu_2(L)$. This result

however does not require the special boundary condition $L(z) = iz\mathbf{R} \oplus z^{k/2}\mathbf{R}$, so let $2 < p < \infty$ and assume L is a smooth arc in $\mathcal{T}(2)$ with Maslov index $\mu_2(L)$ such that $L(z) \supset iz\mathbf{R} \times \{0\}$ for all $z \in \partial D$. Consider the linear operator T defined by

$$T(h,k) = (h_s + ih_t + ak, k_s + ik_t + bk)$$

with domain V_L^p given by

$$V_L^p = \{f \in W^{1,p}(D, \mathbf{C}^2) | f(z) \in L(z) \text{ for all } z \in \partial D\}.$$

The operator has range in $L^p = L^p(D, \mathbf{C}^2)$. We assume $a, b \in L^\infty(D, \mathcal{L}_{\mathbf{R}}(\mathbf{C}))$, which is a slight generalization of the assumptions that we had before. The following theorem holds:

Theorem 7.5.6 *If $\mu_2(L) \geq 1$ then the operator T is surjective and its kernel is precisely $(\mu_2(L) + 2)$-dimensional. If moreover $\mu_2(L) = 2$ then every element $(h,k) \in \ker(T)$ with $k(z) \neq 0$ for some $z \in D$ satisfies $k(z) \neq 0$ for every $z \in D$.*

Proof We rely on the results of Appendices A.4 and A.5. Consider first elements of the form $(h, 0)$ in $\ker(T)$. These correspond precisely (via $h \longmapsto (h, 0)$) to the solutions of the problem $h_s + ih_t = 0$ on D with boundary conditions $h(z) \in iz\mathbf{R}$. One can compute all these solutions. Set

$$h(z) = \sum_{k=0}^{\infty} a_k z^k$$

and take $z = e^{it} \in \partial D$. Then

$$\frac{h(z)}{iz} = \frac{1}{i} \sum_{k=0}^{\infty} a_k e^{i(k-1)t}$$

must be real because of the boundary condition. This is exactly the case if $a_0 = -\bar{a}_2$, $a_1 = -\bar{a}_1$ and $a_k = 0$ for $k \geq 3$. So h is given by

$$h(z) = \alpha + isz - \bar{\alpha} z^2$$

where $\alpha \in \mathbf{C}$ and $s \in \mathbf{R}$. Let $N \subset \ker(T)$ be the three-dimensional subspace with elements of the form $(h, 0)$ as just discussed. Let M be an algebraic complement of N in $\ker(T)$. Denote by $\ell_2(z)$ a smooth loop of 1-dimensional real subspaces of $\mathbf{C}^2$ complementing $\ell_1(z) = iz\mathbf{R} \times \{0\}$ in $L(z)$. Let K be the loop in $\mathcal{T}(1)$ obtained by projecting $\ell_2(z)$ onto the second component of $\mathbf{C}$ (here $\mathcal{T}(1)$ is the space of

totally real subspaces of $\mathbf{C}$, i.e. the space of real one dimensional subspaces). By Lemma 7.4.7 we have the formula

$$\mu_1(K) = \mu_2(L) - 2.$$

Consider the operator T_0 with domain V_K^p and range in $L^p(D, \mathbf{C})$ defined by $T_0(k) = k_s + ik_t + bk$. It has been proved in Theorem A.4.1 that the index is $\mu_1(K) + 1$. We have by assumption $\mu_2(L) \geq 1$, and $\mu_1(K) \geq -1$ by Lemma 7.4.7. Then we conclude from Theorem A.5.5 that

$$\dim \ker T_0 \;=\; \mu_1(K) + 1 \;=\; \mu_2(L) - 1.$$

Moreover T_0 is surjective because its index is $\mu_1(K) + 1$. The map $M \to \ker(T_0) : (h, k) \longmapsto k$ is well-defined because $(h, k)(z) \in L(z)$ for $z \in \partial D$ implies $k(z) \in K(z) = \mathrm{pr}_2 \ell_2(z)$ and obviously injective since M complements N. We infer that

$$\dim \ker(T) = \dim(N) + \dim(M) \leq 3 + \mu_2(L) - 1 = \mu_2(L) + 2.$$

Since $\mathrm{ind}(T) = \mu_2(L) + 2$ we deduce that T is surjective. Moreover if $\mu_2(L) = 2$ we must have $\mu_1(K) = 0$ and it follows that $\dim \ker(T_0) = 1$. By Theorem A.5.4 this kernel is spanned by a nowhere vanishing function k_0. Since obviously $\dim(M) = 1$ in this case, the proof of the theorem is completed. □

Returning to our geometric model, we conclude that the linearization of the Cauchy Riemann operator at an embedded pseudoholomorphic disk is surjective provided $\mu_2(L) \geq 1$ or equivalently $\mu_1(K) \geq -1$.

The situation near an embedded pseudoholomorphic disk is the following: The embedded disk is given by $u_0 : D \to \mathbf{C}^2$, $u_0(z) = (z, 0)$ and the totally real surface F is

$$\{(z, az^{k/2}) \in \mathbf{C}^2 \mid z \in \partial D \;;\; a \in \mathbf{R}\}$$

where $k = \mu_2(L) - 2$. Let us write L for the loop

$$\partial D \ni z \longmapsto T_{u_0(z)} F = iz\mathbf{R} \oplus z^{k/2}\mathbf{R}$$

of totally real subspaces in $\mathbf{C}^2$. We assume that $\mu_2(L) = 2$, i.e. $F = \partial D \times \mathbf{R}$. This implies by the previous discussion that the linearization of the Cauchy Riemann operator is surjective and has index four. Moreover the kernel has a three-dimensional subspace N generated by the (three dimensional) group of Möbius transformations acting on the space of pseudoholomorphic disks (we will explain this later on). Any algebraic complement of N in the kernel however is generated by an element (h, k) with $k(z) \neq 0$ for all $z \in D$. This latter observation will be crucial in the following.

Recall the definition of the Banach manifold $\mathcal{B}$. A map u belongs to $\mathcal{B}$ if and only if $u \in W^{1,p}(D, \mathbf{C}^2)$ with $u(D) \subset U$ and $u(\partial D) \subset F$. Here U is an open neighborhood of $D \times \{0\}$, and $2 < p < \infty$. The tangent space of $\mathcal{B}$ at u_0 is given by

$$T_{u_0}\mathcal{B} = \{f \in W^{1,p}(D, \mathbf{C}^2) | f(z) \in T_{(z,0)}F = iz\mathbf{R} \oplus \mathbf{R} \text{ for } z \in \partial D\}.$$

The linearization T of the Cauchy Riemann operator is given by

$$T(h, k) = (h_s + ih_t + ak, k_s + ik_t + bk).$$

Here $a, b \in C^\infty(D, \mathcal{L}_{\mathbf{R}}(\mathbf{C}))$. The Cauchy Riemann operator

$$\bar{\partial}_J : \mathcal{B} \longrightarrow L^p(D, \mathbf{C}^2)$$

$$u \to u_s + J(u)u_t$$

has the particular feature that $J(z, 0) = i$ for $|z| \leq 1$. Elliptic regularity theory implies that every solution of the nonlinear equation is smooth.

Now we would like to study the solution set $\bar{\partial}_J^{-1}(0) \subset \mathcal{B}$ near u_0 using the implicit function theorem. For this purpose we work with the chart Φ that we constructed in the proof of Proposition 7.4.5. Remember that $\Phi : T_{u_0}\mathcal{B} \supset B \to \mathcal{B}$; $\Phi(f) = \exp(f)$ satisfies $\Phi(0) = u_0$ and $D\Phi(0) = \mathrm{Id}_{T_{u_0}\mathcal{B}}$. Since $\ker D\bar{\partial}_J(u_0)$ is finite dimensional we find a topological complement X in $T_{u_0}\mathcal{B}$.

For $\varepsilon > 0$ sufficiently small we denote by $B_\varepsilon(0)$ the ball of radius ε centered at 0 in $\ker D\bar{\partial}_J(u_0)$ and we define the smooth map

$$F : B_\varepsilon(0) \times X \longrightarrow L^p(D, \mathbf{C}^2)$$

$$(\eta, f) \longmapsto \bar{\partial}_J(\Phi(\eta + f)).$$

We observe that $F(0, 0) = \bar{\partial}_J(u_0) = 0$ and for $(\zeta, g) \in \ker D\bar{\partial}_J(u_0) \times X$

$$\begin{aligned} DF(0, 0)(\zeta, g) &= D\bar{\partial}_J(u_0)\, D\Phi(0) \cdot [\zeta + g] \\ &= D\bar{\partial}_J(u_0)\zeta + D\bar{\partial}_J(u_0)\, g \\ &= D\bar{\partial}_J(u_0)\, g. \end{aligned}$$

Since $D_2F(0, 0) : X \to L^p(D, \mathbf{C}^2)$; $g \longmapsto D\bar{\partial}_J(u_0)g$ is an isomorphism we find by the implicit function theorem a neighborhood X' of zero in X, $\varepsilon' \leq \varepsilon$ and a unique smooth map

$$\alpha : B_{\varepsilon'}(0) \longrightarrow X'$$

so that

$$F(\eta, \alpha(\eta)) = 0 \text{ and } \alpha(0) = 0.$$

This shows that $\bar\partial_J^{-1}(0)$ is a four dimensional submanifold of $\mathcal{B}$ covered by one coordinate chart:

$$B_{\varepsilon'}(0) \widetilde{\longrightarrow} \bar\partial_J^{-1}(0)$$

$$\eta \longmapsto u(\eta) := \Phi(\eta + \alpha(\eta))$$

(after maybe shrinking the set U in the definition of $\mathcal{B}$ further). Fix now an element $(h_0, k_0) \in \ker D\bar\partial_J(u_0)$ with $k_0(z) \neq 0$ for all $z \in D$ (and $|k_0(z)| \leq 1$). Define

$$u : (-\varepsilon', \varepsilon') \times D \longrightarrow \mathbf{C}^2$$

$$(\tau, z) \longmapsto \Phi\Big(\tau(h_0, k_0) + \alpha(\tau(h_0, k_0))\Big)(z).$$

We note that $\tau \longmapsto u(\tau, .)$ is a smooth path in $\mathcal{B}$ with $u(0, z) = u_0(z) = (z, 0)$ and $\bar\partial_J u(\tau, .) = 0$. By elliptic regularity the maps $z \longmapsto u(\tau, z)$ are smooth.

Let us calculate the derivative $Du(0, z)$:

$$\begin{aligned} Du(0, z)(\sigma, \beta) &= (\beta, 0) + \sigma \frac{d}{d\tau} \Phi\Big(\tau(h_0, k_0) + \alpha(\tau(h_0, k_0))\Big)(z)\Big|_{\tau=0} \\ &= (\beta, 0) + \sigma(h_0(z), k_0(z)) + \sigma\big[D\alpha(0)(h_0, k_0)\big](z). \end{aligned}$$

Now we have

$$0 = \frac{d}{d\tau} F(\tau\eta, \alpha(\tau\eta))\Big|_{\tau=0} = DF(0, 0)(\eta, D\alpha(0)\eta) = D\bar\partial_J(u_0)D\alpha(0)\eta$$

which implies $D\alpha(0) \equiv 0$, hence

$$Du(0, z)(\sigma, \beta) = (\beta + \sigma h_0(z), \sigma k_0(z)) \neq (0, 0) \text{ for } \sigma, \beta \neq 0\ ,\ z \in D.$$

Since $u(0, z) = (z, 0)$ we infer by Lemma 7.4.2 that u is an embedding for ε' sufficiently small. Hence we have proved the following (see Fig. 7.3):

Proposition 7.5.7 *Under the assumptions described before there exists a smooth embedding $u : (-\varepsilon, \varepsilon) \times D \to \mathbf{C}^2$ such that with $u(\tau)(z) = u(\tau, z)$ we have*

$$\begin{aligned} u(\tau)(z) &\in F \text{ for all } z \in \partial D, \\ \bar\partial_J u(\tau) &= 0, \\ u(0)(z) &= (z, 0). \end{aligned}$$

□

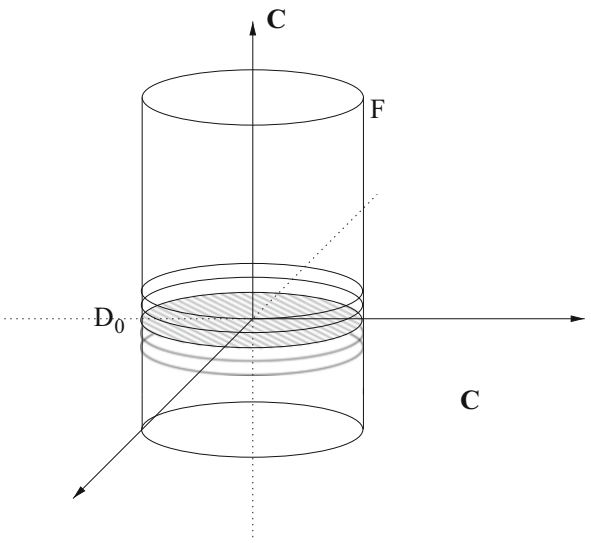


Fig. 7.3 Applying the implicit function theorem, Proposition 7.5.7

In the following we will study the uniqueness properties of this map. There are of course different possible choices for (h_0, k_0) which yield different families of pseudoholomorphic disks. It turns out that the map u above is unique up to some trivial modifications. Denote the group of biholomorphic automorphisms of the disk by $Aut(D)$. This group is given by

$$Aut(D) = \left\{ z \longmapsto \phi \frac{z - w}{\bar{w}z - 1} \;\middle|\; \phi \in S^1 \,,\; w \in \overset{\circ}{D} \right\}$$

and a straightforward calculation shows that its tangent space at the identity map is given by

$$T_{Id} Aut(D) = \{ h : D \to \mathbf{C} \,|\, h(z) = \alpha + isz - \bar{\alpha} z^2 \,,\; \alpha \in \mathbf{C} \,,\; s \in \mathbf{R} \}$$

which we identify with

$$\{ (h, 0) \in \ker D\bar{\partial}_J (u_0) \}$$

(see the proof of Theorem 7.5.6). The path $\Lambda := \{ u(\tau) \in \bar{\partial}_J^{-1}(0) \mid |\tau| \leq \varepsilon \}$ constructed in Proposition 7.5.7 is a 1–dimensional submanifold of $\bar{\partial}_J^{-1}(0)$ with tangent space $T_{u_0}\Lambda$ given by $\mathbf{R} \cdot (h_0, k_0)$. Using the smooth group action of $Aut(D)$ on the solution space $\bar{\partial}_J^{-1}(0)$, we define:

$$A : \Lambda \times Aut(D) \longrightarrow \bar{\partial}_J^{-1}(0)$$

$$(u(\tau), \sigma) \longrightarrow u(\tau) \circ \sigma.$$

The derivative at (u_0, Id)

$$DA(u_0, Id) : T_{u_0}\Lambda \times T_{Id}Aut(D) \longrightarrow \ker D\bar{\partial}_J(u_0)$$

is given by

$$(s(h_0, k_0), \eta) \longmapsto s(h_0, k_0) + (\eta, 0),$$

which is an isomorphism. The inverse function theorem then implies that A is a local diffeomorphism near (u_0, Id). This means that after having chosen a 1–parameter family of pseudoholomorphic disks $u(\tau)$, the remaining three dimensions of the solution space $\bar{\partial}_J^{-1}(0)$ correspond to biholomorphic reparametrizations of the domain D. More precisely, assume u_1 and u_2 are families of pseudoholomorphic disks as in Proposition 7.5.7 with $u_1(\tau) = \Phi(\tau(h_0, k_0) + \alpha(\tau(h_0, k_0)))$. Then we can find a smooth path $v(\tau')$ in Λ which differs from $u_1(\tau)$ only by parameterization of the 1–dimensional manifold Λ and a smooth path $\sigma_{\tau'}$ in $Aut(D)$ so that

$$u_2(\tau') = A(v(\tau'), \sigma_{\tau'}) = v(\tau') \circ \sigma_{\tau'}.$$

We summarize:

Proposition 7.5.8 *Given smooth embeddings u_1 and u_2 as in Proposition 7.5.7 there exist $\varepsilon > 0$, a smooth diffeomorphism $\phi : (-\varepsilon, \varepsilon) \to \phi((-\varepsilon, \varepsilon))$ satisfying $\phi(0) = 0$, and a smooth map $(-\varepsilon, \varepsilon) \ni \tau \longmapsto \sigma_\tau$, where σ_τ is a biholomorphic self map of the disk, with $\sigma_0 = Id_D$, such that*

$$u_1(\tau, \sigma_\tau(z)) = u_2(\phi(\tau), z)$$

for all $(\tau, z) \in (-\varepsilon, \varepsilon) \times D$. □

So in other words there is up to parametrisation a unique disk family $\tau \longmapsto D_\tau$, with $D_0 = D \times \{0\}$, determined by the first order elliptic boundary value problem. Here $D_\tau = u(\tau, D)$. Summarizing the previous two propositions and Theorem 7.4.1 we get:

Corollary 7.5.9 *Let (W, J) be an almost complex four-manifold and let $F \subset W$ be a totally real submanifold. Moreover let u_0 be an embedded pseudoholomorphic disk with boundary values in F and Maslov index $\mu(u_0) = 2$.*

Then there exists a smooth embedding $u : (-\varepsilon, \varepsilon) \times D \to W$ such that with $u(\tau)(z) = u(\tau, z)$ we have

$$u(\tau)(z) \in F \text{ for all } z \in \partial D,$$
$$\bar{\partial}_J u(\tau) = 0,$$
$$u(0)(z) = u_0(z).$$

Moreover the associated disk family $\tau \longmapsto u(\tau)(D)$ is unique up to parametrisation (biholomorphic parameterization of D and smooth parameterization of $(-\varepsilon, \varepsilon)$). □

7.6 Global Uniqueness for Families of Pseudoholomorphic Disks

Starting with an embedded solution of the Cauchy Riemann equation, having Maslov index $\mu(u_0) = 2$ we have seen that all close by solutions are given by the implicit function theorem. Moreover the images of two nearby holomorphic disks are either identical or disjoint. We refer to this situation as local existence and local uniqueness for disk families.

In this section we will study the following more global question. Assume u_0 is as just described and v is another embedded solution of the boundary value problem, which is not necessarily close by such that the associated disk D_v intersects $D_0 = D_{u_0}$. We shall show that under some geometric assumption on F, namely that F is a subset of a pseudoconvex three-dimensional submanifold M of W, the disk D_v will necessarily intersect all D_τ for $\tau \in (-\varepsilon, 0)$, for some small $\varepsilon > 0$ unless $D_v = D_0$, provided the disks lie on the same side of M in W.

This has the following striking consequence. Assume that $(D^1_\tau)_{\tau\in(-\varepsilon,0]}$ and $(D^2_\tau)_{\tau\in[0,\varepsilon)}$ are two disk families associated to the same elliptic boundary value problem such that $D^1_{-\tau_1} \cap D^2_{\tau_0} = \emptyset$ for all $\tau_0, \tau_1 > 0$ but $D^1_0 \cap D^2_0 \neq \emptyset$. We will show that necessarily $D^1_0 = D^2_0$. Hence, after changing the parameterisation of the disks, we can glue this two disk families to obtain a larger disk family.

Assume (W, J) is an almost complex four-manifold and $F \subset W$ is a totally real submanifold. We assume that $u_0 : D \to W$ is an embedded holomorphic disk with boundary in F, i.e.

$$\bar\partial_J u_0 = 0 \text{ on } \overset{\circ}{D} \text{ and } u_0(\partial D) \subset F.$$

Suppose also that the Maslov index satisfies

$$\mu(u_0) = 2.$$

Denote by $(D_\tau)_{\tau\in(-\varepsilon,\varepsilon)}$ the local disk family associated to u_0, with $D_0 = u_0(D)$. Let $v : D \to W$ be another embedded solution of the boundary value problem. We will not assume that the associated disk D^v is close to our disk family. Instead we assume the following:

$$D_0 \cap D^v \neq \emptyset,$$
$$D_\tau \cap D^v = \emptyset \text{ for } \tau \in (-\varepsilon, 0).$$

Our aim is to show that the disks D_0 and D^v must be identical under some reasonable assumptions. If there are no isolated intersections between D_0 and D^v then we conclude using Theorem 5.1.3 that the disks are already identical (using also the fact that our pseudoholomorphic disks are embedded). We will investigate the case where all intersection points of D_0 with D^v are isolated and we will formulate additional assumptions that forbid their existence. Later on, we will show that we can always satisfy these assumptions. First of all we reduce our problem to the normal form given in Sect. 7.4. We may assume therefore that

$$u_0 : D \to \mathbf{C}^2 \,,\ u_0(z) = (z, 0) \ \text{ for all } \ z \in \partial D$$

and

$$F = \partial D \times \mathbf{R} \subset \mathbf{C}^2.$$

The almost complex structure J, which we assume to be defined on the whole $\mathbf{C}^2$ satisfies $J(z, 0) = i$ for all $z \in \mathbf{C}^2$ with $|z| \leq 1$. The totally real loop L defined by $L(z) = T_{(z,0)}F$ is just $iz\mathbf{R} \oplus \mathbf{R}$. The differential equation we are going to study is

$$\bar{\partial}_J u = 0 \ \text{ on } \ \overset{\circ}{D} \ \text{ and } u(\partial D) \subset F.$$

The solutions near u_0 define a disk family (D_τ). Moreover assume $v : A \to \mathbf{C}^2$ is a solution of $\bar{\partial}_J v = 0$ on A, where A is a connected open subset of D. If $\partial A = \partial D \cap A$ is nonempty we assume that $v(\partial A) \subset \partial D \times \mathbf{R}$. Moreover we assume that $v(A) \cap D_0 \neq \emptyset$ and $v(A) \cap D_\tau = \emptyset$ for all $\tau \in (-\varepsilon, 0)$. By our assumption there are only a finite number of intersection points of $v(A)$ with D_0. Pick $z \in A$ with $v(z) \in D_0$. Assume first that $v(z) \notin u_0(\partial D) = \partial D \times \{0\}$. Then z does not belong to the boundary of D and we have an interior intersection of the embedded holomorphic curve u_0 with v (see Fig. 7.4). The local intersection number is positive, so that $v(A)$ would also intersect neighboring disks of D_0 (see Chap. 6) which contradicts our assumption $v(A) \cap D_\tau = \emptyset$ for $-\varepsilon < \tau < 0$. Hence we conclude that $v(A) \cap D_0 \subset \partial D \times \{0\}$, i.e. the piece $v(A)$ of the pseudoholomorphic disk D^v has to intersect D_0 at its boundary. Now comes the first crucial assumption (see Fig. 7.5):

$$\begin{aligned} v(A \setminus \partial A) \cap F &= v(A \setminus \partial A) \cap (\partial D \times \mathbf{R}) \\ &= \emptyset. \end{aligned} \tag{7.8}$$

In other words, interior points of A do not get mapped to F. Hence $v(A \setminus \partial A) \cap D_0 = \emptyset$ and therefore $v(\partial A) \cap D_0 \neq \emptyset$, i.e. if there is an isolated intersection then it must be a common boundary point of D^v and D_0. Hence there exists a point $z_0 \in \partial D$ such that

$$v(z_0) = u_0(z_0) = (z_0, 0).$$

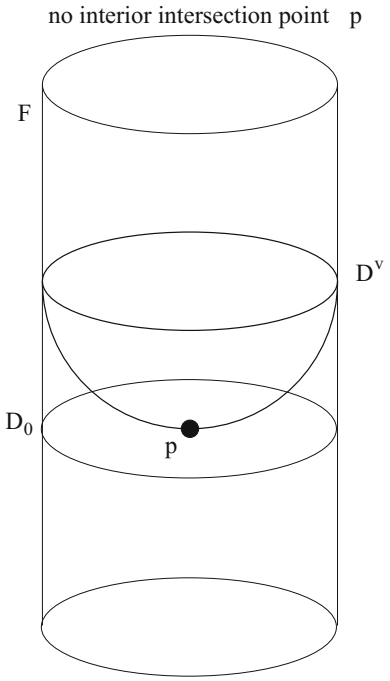


Fig. 7.4 No interior intersections by positivity of intersection

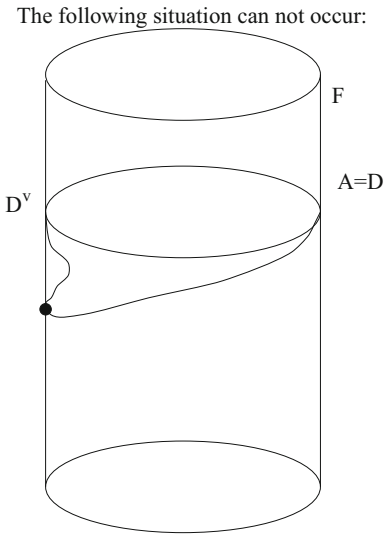


Fig. 7.5 Scenario excluded by assumption (7.8)

We have $Dv(z_0)i = i\,Dv(z_0)$ since $J(v(z_0)) = J(z_0, 0) = i$. Now

$$\begin{aligned} Dv(z_0)(T_{z_0}\partial D) &= Dv(z_0) \cdot \mathbf{R} i z_0 \\ &= i z_0 \mathbf{R} \times \{0\} \\ &\subset T_{(z_0,0)} F \end{aligned}$$

since by our assumption $D^v \cap D_\tau = \emptyset$ for $\tau \in (-\varepsilon, 0)$. Hence

$$\begin{aligned} Dv(z_0) \cdot \mathbf{R} z_0 &= -i\,Dv(z_0) \cdot i\mathbf{R} z_0 \\ &= \mathbf{R} z_0 \times \{0\}. \end{aligned}$$

This implies that $Dv(z_0)$ and $Du_0(z_0)$ have the same range, namely $\mathbf{C} \times \{0\}$. If $B \subset D$ is a suitable open neighborhood of z_0 then we may represent the set $v(B)$ as a graph over a subset of $\mathbf{C} \subset \mathbf{C} \times \{0\}$. Denote the projection $\mathbf{C}^2 \to \mathbf{C}$ onto the first factor by pr_1. If B is small enough we conclude from assumption (7.8) that $\mathrm{pr}_1(v(B))$ is either a subset of D or a subset of $\mathbf{C} \setminus \overset{\circ}{D}$. Here comes the second assumption (see Fig. 7.6) we need to impose:

$$\text{We assume that } \mathrm{pr}_1(v(B)) \subset D. \tag{7.9}$$

Our aim is to show that also the first picture in Fig. 7.6 is impossible, so that the only remaining possibility is $D^v = D_0$. Decomposing v into its components, we write $v(z) = (a(z), b(z))$ for $z \in B$ and note that $|a(z)| \leq 1$ for $z \in B$ because of (7.9)

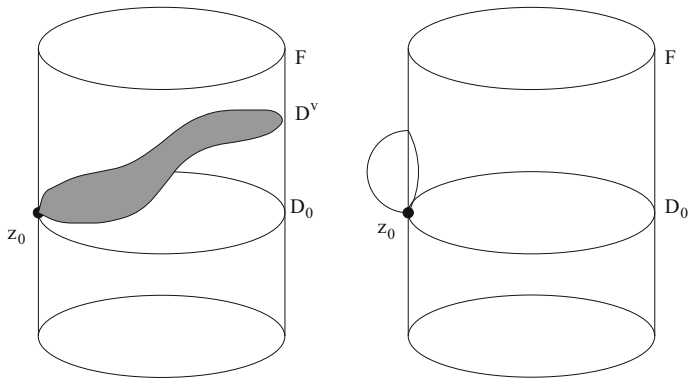


Fig. 7.6 Assumption (7.9) and the only remaining intersection scenario

$b(z_0) = 0$ and $Db(z_0) = 0$ since the range of $Dv(z_0)$ equals $\mathbf{C} \times \{0\}$. We compute over B

$$\begin{aligned} 0 &= (a_s, b_s) + J(a,b)(a_t, b_t) \\ &= (a_s, b_s) + \left(J(a,0) + \int_0^1 D_2 J(a, \tau b) b \, d\tau \right)(a_t, b_t) \\ &= (a_s + i a_t + \alpha b, b_s + i b_t + \beta b). \end{aligned} \tag{7.10}$$

Here $(\alpha x, \beta x) = (\int_0^1 D_2 J(a, \tau b) x \, d\tau)(a_t, b_t)$. Now we take the projection of equation (7.10) onto the second coordinate and obtain

$$0 = b_s + i b_t + \beta b \ \text{ where } \ b(\partial B) \subset \mathbf{R}. \tag{7.11}$$

Using the Similarity principle (see Appendix A.5) we conclude that we would have $b \equiv 0$ if the ∞-jet of b vanished at z_0. Assuming without loss of generality that B is connected, this would imply

$$v(z) = (a(z), 0) = (u_0 \circ a)(z)$$

for all $z \in B$, where a is holomorphic, but then z_0 would not be an isolated intersection point. Hence we are done if we can show that the ∞–jet of b vanishes at z_0. Arguing indirectly, we assume that the ∞-jet of b does not vanish at z_0. If we replace B by a smaller neighborhood B' of z_0 in D and we apply a biholomorphic map

$$\psi : (D^+, (-1,1), 0) \longrightarrow (B', \partial B', z_0),$$

then we may assume that b is defined on the upper half disk D^+. Moreover we may assume that $b(0) = 0$, $Db(0) = 0$, $b((-1,1)) \subset \mathbf{R}$ and $b \neq 0$ on $D^+ \backslash \{0\}$ taking into account that z_0 is an isolated intersection point.

Because of the Similarity principle we find a holomorphic map $\sigma : D^+_{\varepsilon_0} \to \mathbf{C}$ on some smaller half disk $D^+_{\varepsilon_0}$ and a map $\Phi \in \cap_{2<p<\infty} W^{1,p}(D^+_{\varepsilon_0}, \mathrm{GL}(\mathbf{C}))$ with $\Phi(0) = 1$ and $\Phi((-\varepsilon_0, \varepsilon_0)) \subset (0, +\infty)$ such that

$$b(z) = \Phi(z)\sigma(z) \ \text{ on } \ D^+_{\varepsilon_0}.$$

We have $\sigma((-\varepsilon_0, \varepsilon_0)) \subset \mathbf{R}$ since $\Phi((-\varepsilon_0, \varepsilon_0))$, $b((-\varepsilon_0, \varepsilon_0)) \subset \mathbf{R}$. Let us write

$$\sigma(z) = \sum_{k=k_0}^{\infty} a_k z^k$$

with $k_0 \geq 2$ where we assume that $a_{k_0} \neq 0$. If k_0 was odd then $\sigma\,|_{(-\varepsilon_0,\varepsilon_0)}$ would change its sign near 0. This would imply $D^v \cap D_\tau \neq \emptyset$ for some $\tau \in (-\varepsilon, 0)$ which contradicts our assumption. Hence we may assume that k_0 is even. Then either $b((-\varepsilon_0, \varepsilon_0)) \subset [0, \infty)$ or $b((-\varepsilon_0, \varepsilon_0)) \subset (-\infty, 0]$; let us assume without loss of generality that $b((-\varepsilon_0, \varepsilon_0)) \subset [0, \infty)$. Denote by $\partial D^+_{\varepsilon_0}$ the set theoretic boundary of $D^+_{\varepsilon_0}$. Then

$$0 \notin b(\overline{D^+_{\varepsilon_0}} \backslash \{0\})$$

and $-\delta \notin b(\partial D^+_{\varepsilon_0})$ for $\delta > 0$ small. Then the Brouwer degree $\deg(b, D^+_{\varepsilon_0}, -\delta)$ is well defined (see the end of this chapter for a brief outline) and coincides with $\deg(\sigma, D^+_{\varepsilon_0}, -\delta)$ because we can remove $\Phi(z)$ by a simple homotopy argument. Now we continue σ analytically onto D_{ε_0} by reflection. Then

$$\begin{aligned} k_0 &= \deg(D_{\varepsilon_0}, \sigma, 0) \\ &= \deg(D_{\varepsilon_0}, \sigma, -\delta) \\ &= \deg(D^-_{\varepsilon_0}, \sigma, -\delta) + \deg(D^+_{\varepsilon_0}, \sigma, -\delta) \\ &= 2\deg(D^+_{\varepsilon_0}, \sigma, -\delta) \end{aligned}$$

and therefore

$$\deg(D^+_{\varepsilon_0}, b, -\delta) = k_0/2 \tag{7.12}$$

if $\delta > 0$ is small enough. Write $u_\tau = (a_\tau, b_\tau)$ and assume that u_τ is defined on $D^+_{\varepsilon_0} \subset D^+$ by reparameterizing the domain with a biholomorphic map ψ as before. We know that $a_\tau(D^+_{\varepsilon_0}) \subset D$ and that $a_\tau \to \psi$ in C^∞ as $\tau \nearrow 0$. Moreover $b_\tau(z) \neq 0$ for all $z \in D^+_{\varepsilon_0}$ and $b_\tau \to 0$ in C^∞ as $\tau \nearrow 0$. We also have $b_\tau(-\varepsilon_0, \varepsilon_0) \subset (-\infty, 0)$ for $\tau \in (-\varepsilon, 0)$ (see Fig. 7.7). Consider the intersection problem $D^v \cap D_\tau$ for $\tau < 0$ near the point z_0, the intersection point of D^v and D_0. This is equivalent to

$$\begin{aligned} a(z) &= a_\tau(z') \\ b(z) &= b_\tau(z') \end{aligned} \tag{7.13}$$

where $z' \in \overline{D^+_{\varepsilon_0}}$ and $z \in \overline{D^+_{\varepsilon_1}}$ for some sufficiently small $\varepsilon_1 > 0$. Of course, we know that this problem does not have any solution by assumption. Since a_τ is C^∞- close to ψ we see that $a_\tau : D^+_{\varepsilon_0} \to D$ is a (local) diffeomorphism onto its image which is some neighborhood of z_0 in D. Hence Eqs. (7.13) can be rewritten as

$$b(z) = b_\tau(a_\tau^{-1}(a(z))) \tag{7.14}$$

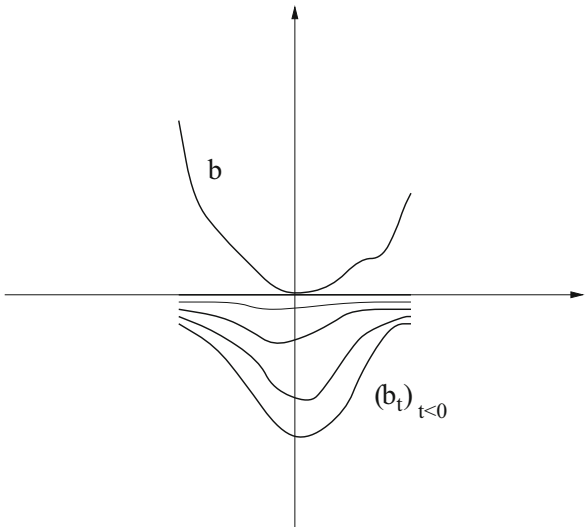


Fig. 7.7 The second components b, b_t of the disks v and u_t

with $z \in \overline{D_{\varepsilon_1}^+}$. Consider the homotopy

$$\Lambda : [0, 1] \times \overline{D_{\varepsilon_1}^+} \longrightarrow \mathbf{C}$$

$$\Lambda(\theta, z) := b(z) + \theta\delta - (1 - \theta)b_\tau(a_\tau^{-1}(a(z))).$$

For τ and δ small enough we have $0 \notin \Lambda([0, 1] \times \partial D_{\varepsilon_1}^+)$. Let us check this quickly:

$$0 \in \Lambda(\{0\} \times \partial D_{\varepsilon_1}^+) \text{ for some } \tau < 0$$

is equivalent to

$$0 = b(z) - b_\tau(a_\tau^{-1}(a(z))),$$

but we said above that this problem has no solution for any $\tau < 0$. This also holds for $\theta \leq \theta_0$ with a suitable number $\theta_0 \in (0, 1)$. For $\theta_0 \leq \theta \leq 1$, $z \in \partial D_{\varepsilon_1}^+$ and $\tau = 0$ we have

$$\Lambda(\theta, z) = b(z) + \theta\,\delta$$

since $b_0 \equiv 0$, but we saw before that this expression is nonzero for all $\theta \in [\theta_0, 1]$ if $\delta > 0$ is sufficiently small and $z \in \partial D_{\varepsilon_1}^+$. Then it is also nonzero if $\theta_0 \leq \theta \leq 1$ and $\tau < 0$ with $|\ \tau\ |$ small. Using the homotopy invariance of the degree, Eqs. (7.12)

and (7.13), we obtain the contradiction

$$\begin{aligned} 0 &= \deg(\Lambda(0,\ .\ ), D_{\varepsilon_1}^+, 0) \\ &= \deg(\Lambda(1,\ .\ ), D_{\varepsilon_1}^+, 0) \\ &= \deg(b+\delta, D_{\varepsilon_1}^+, 0) \\ &= \deg(b, D_{\varepsilon_1}^+, -\delta) \\ &= \frac{k_0}{2}. \end{aligned}$$

Hence we have proved the following:

Proposition 7.6.1 *Let J be an almost complex structure on $\mathbf{C}^2$ with $J(z,0) = i$ for $|\ z\ | \leq 1$ and let*

$$(D_\tau)_{\tau\in(-\varepsilon,\varepsilon)} = (u(\tau, D))_{\tau\in(-\varepsilon,\varepsilon)}$$

be the family of holomorphic disks with $u(\tau, \partial D) \subset \partial D \times \mathbf{R}$ near $u(0,z) = (z,0)$ given by the implicit function theorem. Assume that $A \subset D$ is an open and connected subset and that $v : A \longrightarrow \mathbf{C}^2$ is an embedded solution of

$$\bar{\partial}_J v(z) = 0 \ \text{ on } \ A \text{ and } \ v(\partial A) \subset \partial D \times \mathbf{R} \ \text{ if } \ \partial A = \partial D \cap A \neq \emptyset$$

Moreover

- $v(A) \cap D_0 \neq \emptyset$
- $v(A) \cap D_\tau = \emptyset$ *for* $\tau \in (-\varepsilon, 0)$
- v *satisfies the assumptions* (7.8) *and* (7.9)

Then $v(A)$ is contained in D_0. □

At the moment, we do not know whether the assumptions (7.8) and (7.9) are realistic (fortunately, they are, as we will see in a moment). We apply the previous result to the following situation. Assume (W, J) is an almost complex four-manifold. Assume $M \subset W$ is a three-dimensional orientable submanifold of W dividing W into two disjoint parts, which we denote by $W^{\pm}$,i.e. $M = \overline{W^-} \cap \overline{W^+}$ and $W = M \cup W^+ \cup W^-$. Assume that M is pseudoconvex with respect to W^-. Let $F \subset M$ be a smooth two dimensional submanifold. We consider the following boundary value problem:

$$w : D \to W^- \cup M$$

$$w_s + J(w) w_t = 0 \text{ on } \overset{\circ}{D} \tag{7.15}$$

and

$$w(\partial D) \subset F.$$

The following theorem holds:

Theorem 7.6.2 *Let (W, J), M and F be as described above, where M is pseudoconvex with respect to W^-. Let u and v be two embedded solutions of the boundary value problem (7.15) with image in $W^- \cup M$. Assume that $\mu(u) = 2$, $v(D) \cap W^- \neq \emptyset$, $u(D) \cap W^- \neq \emptyset$ and $u(D) \cap v(D) \neq \emptyset$. Denote by $(D_\tau)_{\tau\in(-\varepsilon,\varepsilon)}$ the local disk family near $D_0 = u(D)$. If $v(D) \cap D_\tau = \emptyset$ for all $\tau < 0$ near 0 then $u(D) = v(D)$.*

Proof First note that we have $v(D\backslash\partial D) \subset W^-$ because of Theorem 5.2.1 and the assumption $v(D)\cap W^- \neq \emptyset$. It follows that $v(D\backslash\partial D)\cap F = \emptyset$. In order to prove the theorem we have to verify the assumptions (7.8) and (7.9). They were formulated for the model equation in $\mathbf{C}^2$ for a proper choice of coordinates. We study the equation

$$\begin{gathered} u : D \longrightarrow \mathbf{C}^2 \\ u_s + J(u)u_t = 0 \ \text{ on } \overset{\circ}{D} \\ u(\partial D) \subset \partial D \times \mathbf{R}. \end{gathered} \tag{7.16}$$

Here $J(z, 0) = i$ for all $|z| \leq 1$. Moreover $u_0(z) = (z, 0)$ is a solution. The solutions of (7.16) near u_0 define a disk family D_τ. We assume that moreover an embedding $v : A \to \mathbf{C}^2$ is given with $A \subset D$ and $v(\partial A) \subset \partial D \times \mathbf{R}$. Moreover $v(A)$ and $D_0 = D \times \{0\}$ intersect at a boundary point. Since u_0 and v come from maps with image in $W^- \cup M$ we have $v(A \setminus \partial A) \cap F = \emptyset$. This is the first assumption (7.8).

Next consider an embedded arc in W, say $\gamma : [-1, 1] \to W$, satisfying $\gamma(t) = u_0((1+t)z)$ for $t \in [-1, 0]$, where z is any point in ∂D. For $\varepsilon \in (0, 1)$ small we have as a consequence of the pseudoconvexity assumption $\gamma(0, \varepsilon) \in W^-$. Going back to our model we may assume that we have a piece of a pseudoconvex hypersurface $\tilde{M} \subset \mathbf{C}^2$, containing $\partial D \times (-\varepsilon, \varepsilon)$ for some small $\varepsilon > 0$. Since $\overset{\circ}{D} \times \{0\} \subset \tilde{W}^-$ we infer from the above discussion that $\{(z, 0) \in \mathbf{C}^2 | 1 < |z| < 1 + \varepsilon\} \subset \tilde{W}^+$. Here $\tilde{W}^\pm$ denote the local components in the model $\tilde{M}$. Since $v(A) \subset \tilde{W}^- \cup \tilde{M}$, $a(\partial A) \subset \partial D$, where $v = (a, b)$ we deduce that $a(A) \subset D$, perhaps after replacing A by some smaller set. This shows that (7.9) holds. □

Summing up we have proved the following for the nonlinear Cauchy Riemann equation in the symplectisation:

Assume (M, λ) is a three-manifold equipped with a contact form λ. Let J be an admissible complex multiplication for the associated contact structure $\xi \to M$. Denote by $\tilde{J}$ the associated translation invariant almost complex structure on $\mathbf{R} \times M$.

Let $F \subset M$ be a totally real surface. Consider the nonlinear boundary value problem

$$\begin{aligned} \tilde{u}_s + \tilde{J}(\tilde{u})\tilde{u}_t = 0 \ \text{ on } \mathring{D} \\ \tilde{u}(\partial D) \subset F = \{0\} \times F. \end{aligned} \tag{7.17}$$

Then we have the following result.

Theorem 7.6.3 *Let $\tilde{u_0}$ and $\tilde{v}$ be embedded solutions of (7.17) such that $\mu(\tilde{u_0}) = 2$. Let $(D_\tau)_{\tau\in(-\varepsilon,\varepsilon)}$ be the local disk family near $D_0 = \tilde{u_0}(D)$. Assume that $\tilde{v}(D) \cap D_0 \neq \emptyset$ and $\tilde{v}(D) \cap D_\tau = \emptyset$ for all $\tau \in (-\varepsilon, 0)$. Then there exists a biholomorphic transformation $\phi : D \to D$ with $\tilde{v} = \tilde{u_0} \circ \phi$. In particular $\tilde{v}(D) = D_0$.* □

This uniqueness result is crucial whenever we have to deal with more than one family of pseudoholomorphic disks where we need to know whether they can collide or not. This will be necessary for the proof of the Weinstein conjecture in the case $\pi_2(M) \neq 0$.

A quick outline of the Brouwer degree Let $\Omega \subset \mathbf{R}^n$ be an open bounded set and let $f : \overline{\Omega} \to \mathbf{R}^n$ be a continuous function. Assume moreover that $y \in \mathbf{R}^n \backslash f(\partial\Omega)$. The Brouwer degree associates to each triple (f, Ω, y) as above an integer $d(f, \Omega, y)$ satisfying the following conditions:

1. $d(\mathrm{id}, \Omega, y) = 1$ and $d(f, \emptyset, y) = 0$ for $y \in \Omega$.
2. If $\Omega_1, \Omega_2 \subset \Omega$ are disjoint open subsets and $y \notin f(\overline{\Omega}\backslash(\Omega_1 \cup \Omega_2))$ then

$$d(f, \Omega, y) = d(f, \Omega_1, y) + d(f, \Omega_2, y).$$

3. Homotopy property: If $h : [0, 1] \times \Omega \to \mathbf{R}^n$ and $y : [0, 1] \to \mathbf{R}^n$ are continuous and $y(t) \notin h(t, \partial\Omega)$ then $d(h(t, \ .\ ), \Omega, y(t))$ is independent of t.
4. $d(f, \Omega, y) \neq 0$ implies $f^{-1}(y) \neq \emptyset$.

The reason for the introduction of the Brouwer degree is the following: We want to show existence of solutions $x \in \Omega$ of $f(x) = y$. The aim is to homotope the problem to a simpler one and show that the Brouwer degree is nonzero. The theorem is that there is a Brouwer degree and that it is uniquely determined by the above conditions (see [25, 66]). The idea of the existence proof is the following: If we assume for the moment that f is C^1 and y is a regular value of f with $y \notin f(\partial\Omega)$ then the set $f^{-1}(y) \subset \Omega$ consists of finitely many points and $\det Df(x) \neq 0$ for each $x \in f^{-1}(y)$. Then one defines

$$d(f, \Omega, y) := \sum_{x\in f^{-1}(y)} \mathrm{sign}(\det Df(x)).$$

The Brouwer degree for continuous maps is defined by using an approximation argument. There is also a purely topological definition of the Brouwer degree (see [62]).

The following properties are consequences of the ones above:

1. Excision property: If $\Omega' \subset \Omega$ is an open subset so that $y \notin f(\overline{\Omega}\backslash\Omega')$ then $d(f, \Omega, y) = d(f, \Omega', y)$.
2. $d(f, \Omega, y) = d(g, \Omega, y)$ if $f|_{\partial\Omega} \equiv g|_{\partial\Omega}$.

Chapter 8
Bubbling-off in Families of Pseudoholomorphic Disks



In this chapter, let (M, λ) be a three dimensional manifold with contact form λ, and let $F \subset M$ be an embedded surface. Denote by Σ_F the set of singular points on the surface, i.e.

$$\Sigma_F := \{p \in F \mid \ker \lambda(p) = T_p F\}$$

and by F^* the surface without its singular points. We consider families of pseudoholomorphic disks

$$\tilde{u}_\tau = (a_\tau, u_\tau) : D \to \mathbf{R} \times M \ , \ \tau \in [0, \tau_0)$$

$$\partial_s \tilde{u}_\tau + \tilde{J}(\tilde{u}_\tau)\partial_t \tilde{u}_\tau \ = \ 0$$

so that

$$\tilde{u}_\tau(\partial D) \subset \{0\} \times F^* \text{ and } \inf{}_{0 \le \tau < \tau_0} \text{dist}(u_\tau(\partial D), \Sigma_F) > 0.$$

Compactness or Noncompactness of a family $\{\tilde{u}_\tau\}$ depends on the existence of a uniform bound on $\nabla \tilde{u}_\tau$. If ϕ is a conformal automorphism of the unit disk then $\tilde{u}_\tau \circ \phi$ is also a solution of the above boundary value problem. All such automorphisms ϕ are of the form

$$\phi(z) = e^{i\alpha} \frac{a - z}{1 - \bar{a} z},$$

where $0 \le \alpha \le 2\pi$ and $a \in \overset{\circ}{D}$. Choosing a close to the boundary of the disk, we can make $\|\nabla \phi\|_{C^0(D)}$ as large as we wish. Hence we should fix suitable parameterizations of the disk when considering $\sup_{0 \le \tau < \tau_0} \|\nabla \tilde{u}_\tau\|_{C^0(D)}$. It will be

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convenient later on to use the following number:

$$e(\tilde{u}_\tau) := \inf_{\phi \in Aut(D)} \|\nabla(\tilde{u}_\tau \circ \phi)\|_{C^0(D)}. \tag{8.1}$$

We can always reparametrize a pseudoholomorphic disk conformally so that the infimum is attained. We say that such a disk is *nicely parameterized*. We will investigate the behavior of a pseudoholomorphic disk family in the case where there is a τ–uniform bound on $\|\nabla \tilde{u}_\tau\|_{C^0(D)}$ (maybe after conformally rescaling the domain); then we expect convergence in C^∞ to some pseudoholomorphic disk $\tilde{u}_{\tau_0}$ as $\tau \to \tau_0$. In the other case, where $e(\tilde{u}_\tau) \to \infty$, we will show that a non constant pseudoholomorphic plane with finite energy will bubble off providing existence of a periodic orbit of the Reeb vector field by the results in Chap. 4.

8.1 Energy Estimates and Compactness

The following results shows that a τ–uniform gradient bound implies compactness of a family $(\tilde{u}_\tau)_{0 \le \tau < \tau_0}$ of pseudoholomorphic disks. We start with a lemma which follows easily from earlier results.

Lemma 8.1.1 *Let (M, λ) be a contact manifold and let F be an embedded surface in M. Assume that*

$$\tilde{u}_\tau = (a_\tau, u_\tau) : D \longrightarrow \mathbf{R} \times M \ , \ \tau \in [0, \tau_0)$$

is a smooth family of pseudoholomorphic disks satisfying the boundary condition $\tilde{u}_\tau(\partial D) \subset \{0\} \times F^*$ and $\inf_{0 \le \tau < \tau_0} dist(u_\tau(\partial D), \Sigma_F) > 0$. Assume furthermore that

$$sup_{0 \le \tau < \tau_0} \|\nabla \tilde{u}_\tau\|_{C^0(D)} < \infty. \tag{8.2}$$

Then for every sequence $\tau_k \to \tau_0$ there exists a subsequence (also denoted by τ_k for simplicity) and a pseudoholomorphic disk $\tilde{u}_{\tau_0}$ with $\tilde{u}_{\tau_0}(\partial D) \subset \{0\} \times F^*$ so that $\tilde{u}_{\tau_k} \to \tilde{u}_{\tau_0}$ in $C^\infty(D)$ as $k \to \infty$.

Proof This is an obvious consequence of the estimates in Chap. 4.3 and Appendix A.2. We then have τ–uniform C^∞ bounds. Then we conclude by the Ascoli–Arzela theorem for every sequence $\tau_k \to \tau_0$ there is a C^∞–convergent subsequence of $\tilde{u}_{\tau_k}$. □

In our situation, the disk family $(\tilde{u}_\tau)$ has special features which improve the above lemma such as

- all disks have Maslov index 2
- the disks in the family have pairwise disjoint images
- they are all embeddings
- they originate from local Bishop disks

Proposition 8.1.2 *Let $(\tilde{u}_\tau)$ be a family of pseudoholomorphic disks as in Lemma 8.1.1 above, but assume in addition that the family is given by the implicit function theorem, Corollary 7.5.9. Then $\tilde{u}_\tau \to \tilde{u}_{\tau_0}$ as $\tau \to \tau_0$, and $\tilde{u}_{\tau_0}$ is an embedding.*

Proof Pick a sequence $\tau_k \to \tau_0$ as in Lemma 8.1.1 above. Since all the disks $\tilde{u}_{\tau_k}$ have Maslov–index equals 2 and zero self–intersection number, the same applies to $\tilde{u}_{\tau_0}$. Hence $\tilde{u}_{\tau_0}$ is an embedded pseudoholomorphic disk where we may apply the implicit function theorem, Corollary 7.5.9. Then all solutions which are C^∞–close to $\tilde{u}_{\tau_0}$ lie on a 1–dimensional branch of solutions passing through $\tilde{u}_{\tau_0}$, hence for all sequences τ_k the limit has to be the same. □

Let (M, λ) be a three dimensional contact manifold and let $F \subset M$ be an embedded surface which is either diffeomorphic to a sphere with precisely two nicely elliptic singular points (a positive elliptic one e^+ and a negative elliptic one e^-) or F is diffeomorphic to a disk with Legendrian boundary having exactly one elliptic singular point $e \in \overset{\circ}{F}$. We denote by F^* the surface F without its singular point(s). We study pseudoholomorphic disks $\tilde{u} = (a, u) : D \to \mathbf{R} \times M$:

$$\partial_s \tilde{u} + \tilde{J}(\tilde{u})\partial_t \tilde{u} = 0 \tag{8.3}$$

with the totally real boundary condition

$$\tilde{u}(\partial D) \subset \{0\} \times F^* \,,\ \inf{}_{0 \le \tau < \tau_0} \mathrm{dist}(u_\tau(\partial D), \Sigma_F) > 0 \tag{8.4}$$

and the additional assumption that the loop $u|_{\partial D}$ winds around e or e^+ exactly once. The following lemma shows that there is a bound on the energy depending only on the contact form λ and the surface F.

Lemma 8.1.3 *Let F be a surface as described above and let $\tilde{u}$ be a solution of Eq. (8.3) satisfying the boundary condition (8.4) and the above assumption on the winding number. Then there exists a constant $C = C(\lambda, F) > 0$ (not depending on $\tilde{u}$) so that*

$$E(\tilde{u}) = \sup_{\phi \in \Sigma} \int_D \tilde{u}^* d(\phi\lambda) \le C.$$

Proof We recall that $\Sigma = \{\phi \in C^\infty(\mathbf{R}, [0,1]) \,|\, \phi' \ge 0\}$ and that the 1–form $(u|_{\partial D})^*\lambda$ never vanishes by Corollary 5.2.2. We may estimate

$$\begin{aligned} 0 \le \int_D \tilde{u}^* d(\phi\lambda) &= \int_{\partial D} \tilde{u}^*(\phi\lambda) \\ &= \phi(0) \int_{\partial D} u^*\lambda \\ &\le \int_{\partial D} u^*\lambda. \end{aligned}$$

The curve $\gamma = u(\partial D) \subset F$ is embedded and divides the surface F into two parts F_1 and F_2. If σ is an volume form on F then we can write $d\lambda = f\sigma$ for a suitable function f and we define

$$\int_F |d\lambda| := \int_F |f|\sigma.$$

By Stokes' theorem we have

$$0 = \int_F d\lambda = \int_{F_1} d\lambda + \int_{F_2} d\lambda,$$

hence

$$\begin{aligned}\int_{\partial D} u^*\lambda &= \int_\gamma \lambda \\ &= \left| \int_{F_1} d\lambda \right| \\ &\leq \frac{1}{2} \int_F |d\lambda|.\end{aligned}$$

□

The following estimate shows that the $d\lambda$–energy is bounded away from zero.

Proposition 8.1.4 *Let $\tilde{u}_\tau$ be a family of pseudoholomorphic disks as before, i.e. $\tilde{u}_\tau$ satisfies Eq. (8.3), the boundary condition (8.4) and the assumption on the winding number. We assume in addition that there is a compact subset $K \subset M$ so that $u_\tau(D) \subset K$ for all $0 \leq \tau < \tau_0$. Then there exists a constant $c > 0$ independent of τ so that*

$$\int_D u_\tau^* d\lambda \geq c.$$

Before we start with the proof, we have to insert the following lemma which is a boundary version of Proposition 4.4.2.

Lemma 8.1.5 *Let*

$$\tilde{u} = (a, u) : H^+ := \{z \in \mathbf{C} \,|\, Im(z) \geq 0\} \to \mathbf{R} \times M$$

be a solution of $\partial_s \tilde{u} + \tilde{J}(\tilde{u})\partial_t \tilde{u} = 0$ satisfying the boundary condition $\tilde{u}(\partial H^+) \subset \{0\} \times F^*$ and the finite energy condition $E(\tilde{u}) < +\infty$. If also $\int_{H^+} u^* d\lambda = 0$ then $\tilde{u}$ must be constant.

Proof Because of $\int_{H^+} u^* d\lambda = 0$ we have $\triangle a = 0$. Defining $f : H^+ \to \mathbf{R}$ by

$$f(s,t) := \int_0^t \partial_s a(s,\tau) d\tau - \int_0^s \partial_t(\sigma, 0) d\sigma$$

we see that $\Phi := a + if : H^+ \to \mathbf{C}$ is a holomorphic function with $\Phi(\partial H^+) \subset i\mathbf{R}$. It becomes an entire function after reflecting it as follows:

$$\Phi(s+it) := \begin{cases} \Phi(s+it) & \text{if } t \geq 0 \\ -\bar{\Phi}(s-it) & \text{if } t < 0 \end{cases}.$$

Case 1: $|\nabla\Phi|$ is bounded.
Since Φ is an entire function, it must be an affine function of the form $\Phi(z) = i\alpha z + i\beta$, where α and β are real numbers, in particular we have $a(s,t) = -\alpha t$ and $f(s,t) = \alpha s + \beta$. Assume now that $\alpha \neq 0$. Choosing a suitable non constant function $\phi \in \Sigma$ we obtain a contradiction to the finite energy condition like in the proof of Proposition 4.4.2. Hence $\alpha = 0$, i.e. $a \equiv 0$, but this implies also that $\tilde{u}$ is constant.
Case 2: $|\nabla\Phi|$ is unbounded.
Applying Lemma 4.4.4, we find $\varepsilon_k \searrow 0$, $z_k \in \mathbf{C}$ so that

$$\varepsilon_k R_k := \varepsilon_k |\nabla\Phi(z_k)| \to +\infty$$

and

$$|\nabla\Phi(z)| \leq 2$$

if $|z - z_k| \leq \varepsilon_k$. Defining the entire functions

$$\Phi_k(z) := \Phi(z_k + zR_k^{-1}) - \Phi(z_k),$$

we see that they satisfy

$$|\nabla\Phi_k(0)| = 1\ ,\ \Phi_k(0) = 0$$

and

$$|\nabla\Phi_k(z)| \leq 2$$

whenever $z \in B_{\varepsilon_k R_k}(0)$. Using the Cauchy integral formula to obtain C^∞_{loc}–bounds uniform in k and applying the Ascoli–Arzela theorem, we see that a subsequence of (Φ_k) converges in C^∞_{loc} to an entire function Ψ, which is not constant and has bounded gradient. By Liouville's theorem Ψ must be an affine function, actually $\Psi(z) = \alpha z$ with some nonzero $\alpha \in \mathbf{C}$.

We estimate with $\phi_k(s) := \phi(s - \operatorname{Re}\Phi(z_k))$ ($\phi \in \Sigma$), $\tau_\phi(s,t) = \phi'(s)ds \wedge dt$ and $\Omega_k := \{z \in \mathbf{C} \mid \operatorname{Im}(z) \geq -t_k R_k\}$:

$$\begin{aligned}\int_{\Omega_k} (\Phi_k|_{\Omega_k})^* \tau_\phi &= \int_{H^+} (\Phi|_{H^+})^* \tau_{\phi_k} \\ &= \int_{H^+} \phi_k'(a) da \wedge df \\ &= \int_{H^+} \tilde{u}^* d(\phi_k \lambda) \\ &\leq E(\tilde{u}).\end{aligned}$$

In the case $t_k R_k \to +\infty$ we conclude that

$$+\infty = \int_{\mathbf{C}} \tau_\phi = \int_{\mathbf{C}} \Psi^* \tau_\phi \leq E(\tilde{u}),$$

which is a contradiction. In the case $-t_k R_k \to -l \in (-\infty, 0]$ we pick any $l' < l$ and we observe that for every compact subset K of $H_{-l'} := \{z \in \mathbf{C} | \operatorname{Im}(z) \geq -l'\}$ there is a number k_0 so that $K \subset \Omega_k$ whenever $k \geq k_0$. From this we conclude that

$$\int_{H_{-l'}} \Psi^* \tau_\phi \leq E(\tilde{u}).$$

Since $\Psi(H_{-l'})$ is again a half plane in $\mathbf{C}$, we infer that

$$\int_{H_{-l'}} \Psi^* \tau_\phi = \int_{\Psi(H_{-l'})} \tau_\phi = +\infty.$$

This contradiction shows that case 2. cannot occur and therefore completes the proof of the lemma. □

Proof of Proposition 8.1.4 Let us argue by contradiction and assume that $\int_D u_\tau^* d\lambda \to 0$ as $\tau \to \tau_0$. If we have a uniform gradient bound $\sup_{0 \leq \tau < \tau_0} \|\nabla \tilde{u}_\tau\|_{C^0(D)}$ (after conformally rescaling the disk D) then the sequence $\tilde{u}_\tau$ would converge in C^∞ to some $\tilde{u}_{\tau_0}$ with $u_{\tau_0}(\partial D) \subset F^*$ and $\int_D u_{\tau_0}^* d\lambda = 0$. Hence

$$\pi_\lambda \partial_s u_{\tau_0} \equiv \pi_\lambda \partial_t u_{\tau_0} \equiv 0.$$

Moreover, the $\mathbf{R}$–component a_{τ_0} is harmonic in view of

$$\Delta a_{\tau_0}\, ds \wedge dt = -d(da_{\tau_0} \circ i) = u_{\tau_0}^* d\lambda \equiv 0,$$

has zero boundary values and is therefore constant implying that $\tilde{u}_{\tau_0}$ has to be constant as well in contradiction to the fact that it has nonzero winding number around the elliptic singularity. Let us now derive a contradiction in the case where $e(\tilde{u}_\tau) \to \infty$ as $\tau \to \tau_0$. We pick sequences $\tau_k \to \tau_0$ and $z_k \in D$ so that

$$R_k := |\nabla \tilde{u}_{\tau_k}(z_k)| \longrightarrow \infty$$

as $k \to \infty$. We may assume that the sequence z_k converges to some point $z_0 \in D$. We now launch the usual bubbling off argument: By Lemma 4.4.4, we may find a sequence of positive numbers $\varepsilon_k \searrow 0$ so that, after maybe changing the sequence z_k slightly,

- $R_k \varepsilon_k \to +\infty$,
- $|\nabla \tilde{u}_{\tau_k}(z)| \leq 2R_k$ if $|z - z_k| \leq \varepsilon_k$.

We then define rescaled pseudoholomorphic curves

$$\tilde{u}_k(z) = (a_k(z), u_k(z)) := \left(a_{\tau_k}\left(\frac{z}{R_k} + z_k\right) - a_{\tau_k}(z_k), u_{\tau_k}\left(\frac{z}{R_k} + z_k\right)\right)$$

for $z \in B_{R_k}(-R_k z_k)$, which satisfy

$$|\nabla \tilde{u}_k(0)| = 1,$$

$$|\nabla \tilde{u}_k(z)| \leq 2$$

if $z \in \Omega_k := B_{\varepsilon_k R_k}(0) \cap B_{R_k}(-R_k z_k)$ and

$$\int_{B_{R_k}(-R_k z_k)} u_k^* d\lambda = \int_D u_{\tau_k}^* d\lambda \longrightarrow 0$$

as $k \to \infty$. There are two cases that we have to consider. The first one is $R_k \operatorname{dist}(z_k, \partial D) \to \infty$. We then have $\bigcup_k B_{R_k}(-R_k z_k) = \mathbf{C}$ and therefore also $\bigcup_k \Omega_k = \mathbf{C}$. Indeed, if $z \in \mathbf{C}$ then we have $|z| \leq R_k - |R_k z_k|$ for large k because of

$$R_k \operatorname{dist}(z_k, \partial D) = R_k(1 - |z_k|) \to \infty.$$

We conclude $|z + R_k z_k| \leq |z| + |R_k z_k| \leq R_k$, i.e. $z \in B_{R_k}(-R_k z_k)$. By the results in Chap. 4.3, we obtain on any compact subset K of the complex plane k–uniform bounds for all derivatives of $\tilde{u}_k$. We note that we also have k–uniform $C^0(K)$ bounds since $a_k(0) = 0$ and all the u_k have image in a compact subset of M by assumption. Applying the Ascoli–Arzela theorem, a subsequence of $\tilde{u}_k$ converges in C^∞_{loc} to a pseudoholomorphic plane

$$\tilde{u} : \mathbf{C} \longrightarrow \mathbf{R} \times M$$

satisfying $|\nabla\tilde{u}(0)| = 1$, $\int_{\mathbf{C}} u^* d\lambda = 0$ and $E(\tilde{u}) < \infty$ since

$$\begin{aligned}\sup_{\phi\in\Sigma}\int_K \tilde{u}_k^* d(\phi\lambda) &\le \sup_{\phi\in\Sigma}\int_{B_{R_k}(-R_k z_k)} \tilde{u}_k^* d(\phi\lambda)\\ &= \sup_{\phi\in\Sigma}\int_D \tilde{u}_{\tau_k}^* d(\phi\lambda)\\ &\le C,\end{aligned}$$

where C is a positive constant only depending on the contact form λ and the surface F; the existence is guaranteed by Lemma 8.1.3. Take the limit $k \to \infty$ and then the supremum over all compact $K \subset \mathbf{C}$ to obtain $E(\tilde{u}) < \infty$. Proposition 4.4.2 implies that $\tilde{u}$ is constant which is a contradiction to $|\nabla\tilde{u}(0)| = 1$. Hence the case $R_k \operatorname{dist}(z_k, \partial D) \to \infty$ is impossible. The other case is $R_k \operatorname{dist}(z_k, \partial D) \to l \in [0, \infty)$. By rotating the disk D we may assume that $z_k \to z_0 = -i$ so that $R_k|z_k + i| \to l$. Let $H^l := \{z \in \mathbf{C} \,|\, \operatorname{Im}(z) \ge l\}$. Pick any point z in the interior of H^l, i.e. $z \in H^{l+\delta}$ for some $\delta > 0$. If k is large then we have $|z - R_k i| \le R_k - l - \delta$ so that

$$|z + R_k z_k| \le R_k - l - \delta + |R_k i + R_k z_k| \le R_k.$$

Choosing k so large that in addition $|z| \le R_k \varepsilon_k$, we have shown that $z \in \Omega_k$. We then rescale conformally as before and obtain finally in the limit a pseudoholomorphic half–plane

$$\tilde{u} : H^l \longrightarrow \mathbf{R} \times M$$

satisfying $|\nabla\tilde{u}(0)| = 1$, $\int_{H^l} u^* d\lambda = 0$, $\tilde{u}(\partial H^l) \subset \{0\} \times F^*$ and $E(\tilde{u}) < \infty$. On the other hand, $\tilde{u}$ has to be constant by Lemma 8.1.5 (it is of course irrelevant that we have H^l instead of H^+). This contradiction shows that our initial assumption $\int_D u_\tau^* d\lambda \to 0$ as $\tau \to \tau_0$ was false. □

8.2 Bubbling at the Boundary

Let

$$\tilde{u}_\tau = (a_\tau, u_\tau) : D \to \mathbf{R} \times M\ ,\ \tau \in [0, \tau_0)$$

be a family of pseudoholomorphic disks as in Lemma 8.1.1, but instead of assuming the gradient bound (8.2) we rather assume that the gradient blows up:

$$\sup_{0\le\tau<\tau_0} e(\tilde{u}_\tau) = +\infty.$$

We assume in addition that all the $\tilde{u}_\tau$ are embedded and that the curve

$$u_\tau|_{\partial D} : \partial D \longrightarrow F^*$$

bounds a disk in F which contains exactly one elliptic singular point e; then the winding number with respect to the elliptic singularity equals ± 1. We then fix parameterizations of the disk D as follows: We parameterize the leaves of the characteristic foliation approaching e by $(l_\alpha)_{\alpha \in S^1}$ and demand

$$u_\tau(1) \in l_1 \ , \ u_\tau(i) \in l_i \ , \ u_\tau(-1) \in l_{-1}. \tag{8.5}$$

The following proposition excludes the possibility of something else than a pseudo-holomorphic plane arising from the bubbling off argument in the next section.

Proposition 8.2.1 *Let $\tilde{u}_\tau$ be a family of pseudoholomorphic disks as described above satisfying also the normalization condition (8.5). If $\tau_k \to \tau_0$ and $(z_k)_{k \in \mathbf{N}} \subset D$ are sequences so that $R_k := |\nabla \tilde{u}_{\tau_k}(z_k)| \to \infty$ then the sequence $R_k \, dist(z_k, \partial D)$ is unbounded.*

Proof Arguing indirectly, we assume that the sequence $R_k \, \mathrm{dist}(z_k, \partial D)$ converges to some $\rho \in [0, \infty)$. Let us write $\tilde{u}_k$ instead of $\tilde{u}_{\tau_k}$. Take a sequence $\varepsilon_k \searrow 0$ so that $R_k \varepsilon_k \to \infty$. Using Lemma 4.4.4, we may assume in addition that

$$|\nabla \tilde{u}_k(z)| \leq 2R_k \tag{8.6}$$

for all $z \in D$ with $|z - z_k| \leq \varepsilon_k$. Define the measure $\mu = \frac{1}{2\pi} d\theta$ on S^1, so that $\mu(S^1) = 1$. The point z_0 is contained in at most two of the following segments in ∂D:

$$\sigma_1 := \{e^{i\phi} \,|\, 0 \leq \phi \leq \frac{\pi}{2}\},$$

$$\sigma_2 := \{e^{i\phi} \,|\, \frac{\pi}{2} \leq \phi \leq \pi\},$$

$$\sigma_3 := \{e^{i\phi} \,|\, \pi \leq \phi \leq 2\pi.\}$$

Let $S \in \{\sigma_1, \sigma_2, \sigma_3\}$ be a segment not containing z_0. Then

$$\mu(\{\alpha \in S^1 \,|\, u_k(S) \cap l_\alpha \neq \emptyset\}) \geq \frac{1}{4} \text{ for all } k \tag{8.7}$$

because of the normalization condition (8.5). If S' is any segment in ∂D containing z_0 and disjoint from S, then we define

$$m_k(S') := \mu(\{\alpha \in S^1 \,|\, u_k(S') \cap l_\alpha \neq \emptyset\})$$

which measures the number of leaves of the characteristic foliation hit by the image of the segment S' under u_k. The sets

$$\{\alpha \in S^1 \mid u_k(S) \cap l_\alpha \neq \emptyset\}$$

and

$$\{\alpha \in S^1 \mid u_k(S') \cap l_\alpha \neq \emptyset\}$$

are disjoint because the loop $u_k|\partial D : \partial D \to F^*$ intersects each leaf l_α of the characteristic foliation exactly once. Then

$$\begin{aligned} 1 &\geq \mu\Big(\{\alpha \in S^1 \mid u_k(S) \cap l_\alpha \neq \emptyset\} \cup \{\alpha \in S^1 \mid u_k(S') \cap l_\alpha \neq \emptyset\}\Big) \\ &\geq \frac{1}{4} + m_k(S'). \end{aligned} \tag{8.8}$$

We will now construct a sequence of segments S_k containing z_0, shrinking in length to zero and disjoint from S, so that $m_k := m_k(S_k) > \frac{3}{4}$ for sufficiently large k contradicting inequality (8.8) and therefore proving the proposition.

The biholomorphic map on the upper half plane $H^+ \subset \mathbf{C}$

$$\psi_\varepsilon : H^+ \longrightarrow D \backslash \{-z_0\}$$

$$\psi_\varepsilon(z) := z_0 \frac{i - \varepsilon z}{i + \varepsilon z} \ , \ \varepsilon > 0$$

maps the unit upper half disk D^+ onto a small neighborhood of the point z_0 in D. We also choose $\varepsilon > 0$ so small that $\psi_\varepsilon([-1, +1]) \cap S = \emptyset$. Considering $\tilde{u}_k \circ \psi_\varepsilon$ instead of $\tilde{u}_k$, we may assume that all the maps $\tilde{u}_k$ are defined on the unit upper half disk D^+. So we have

$$\tilde{u}_k : D^+ \longrightarrow \mathbf{R} \times M$$

$$\partial_s \tilde{u}_k + \tilde{J}(\tilde{u}_k)\partial_t \tilde{u}_k = 0,$$

$$\tilde{u}_k([-1, +1]) \subset \{0\} \times F^* \ , \ \inf_k \operatorname{dist}(u_k([-1, +1]), \Sigma_F) > 0,$$

$$E(\tilde{u}_k) \leq C < +\infty.$$

Moreover, there are sequences $\varepsilon_k \searrow 0$ and $(z_k)_{k \in \mathbf{N}} \subset D^+$, $z_k \to 0$ so that

$$R_k := |\nabla \tilde{u}_k(z_k)| \to \infty \ , \ R_k \varepsilon_k \to \infty,$$

$$\operatorname{Im}(z_k)\, R_k \to \rho \in [0, \infty) \tag{8.9}$$

and

$$|\nabla \tilde{u}_k(z)| \leq cR_k \ \text{ if } z \in D^+ \text{ with } |z - z_k| \leq \varepsilon_k, \tag{8.10}$$

where c is a positive constant replacing '2' in Eq. (8.6). Note that the number ρ in (8.9) is a different one than the previous ρ due to the change of the domain to D^+. If k is sufficiently large we may assume that

$$|\mathrm{Re}(z_k)| \leq \frac{1}{2}.$$

We define rescaled pseudoholomorphic maps $\tilde{v}_k = (b_k, v_k)$ on $D^+_{\frac{R_k}{2}} := \{z \in \mathbf{C} \,|\, |z| \leq \frac{R_k}{2}\ ,\ \mathrm{Im}(z) \geq 0\}$ by

$$\tilde{v}_k(z) := \left(a_k\left(\mathrm{Re}(z_k) + \frac{z}{R_k}\right) - a_k(\mathrm{Re}(z_k)), u_k\left(\mathrm{Re}(z_k) + \frac{z}{R_k}\right)\right).$$

Defining $\tilde{z}_k := i R_k \,\mathrm{Im}(z_k)$, we see that

$$|\nabla \tilde{v}_k(\tilde{z}_k)| = 1$$

and $\tilde{z}_k \to i\rho$ in view of (8.9). Moreover, by (8.10),

$$|\nabla \tilde{v}_k(z)| \leq c \text{ if } z \in \Omega_k := D^+_{\frac{R_k}{2}} \cap B_{R_k\varepsilon_k}(\tilde{z}_k).$$

If $K \subset H^+ := \{z \in \mathbf{C} \,|\, \mathrm{Im}(z) \geq 0\}$ is any compact subset then $K \subset \Omega_k$ if k is large enough, hence we obtain k–uniform bounds on K for all derivatives of $\tilde{v}_k$ by the results in Chap. 4.3. There are also uniform $C^0(K)$–bounds since the images of all the maps v_k lie in a compact subset of M and $b_k(0) = 0$. Applying the Ascoli–Arzela theorem and passing to a suitable subsequence we obtain convergence in C^∞_{loc} to a pseudoholomorphic half plane $\tilde{v} = (b, v)$ with the properties:

$$|\nabla \tilde{v}(i\rho)| = 1\ ,\ |\nabla \tilde{v}(z)| \leq c,\ ,\ E(\tilde{v}) < \infty$$

and

$$\tilde{v}(\partial H^+) \subset \{0\} \times F^*\ ,\ \mathrm{dist}(v(\partial H^+), \Sigma_F) > 0.$$

We will now show that

$$\mu(\{\alpha \in S^1 \,|\, v(\partial H^+) \cap l_\alpha \neq \emptyset\}) = 1, \tag{8.11}$$

i.e. the boundary curve $v(\partial H^+)$ hits almost every leaf l_α of the characteristic foliation. Consider the biholomorphic map

$$\psi : H^+ \longrightarrow D\backslash\{-1\}$$

$$\psi(z) := \frac{i-z}{i+z}$$

and the pseudoholomorphic map $\tilde{v} \circ \psi$ on the punctured disk $D\backslash\{-1\}$ which has finite energy and therefore extends smoothly to the disk D by the removable singularity theorem. By Corollary 5.2.2, the loop along ∂D in F^* is transverse to the characteristic foliation and therefore hits all leaves l_α proving (8.11). We now finish the proof as follows: Pick any number $0 < \delta < \frac{1}{8}$. Then we can find $R > 0$ so that

$$\mu(\{\alpha \in S^1 \mid v([-R, R]) \cap l_\alpha \neq \emptyset\}) \geq 1 - \delta$$

by (8.11). Because the sequence $\tilde{v}_k$ converges to $\tilde{v}$ in C^∞_{loc} we can find a number $k_0 = k_0(\delta, R) > 0$ so that still

$$\mu(\{\alpha \in S^1 \mid v_k([-R, R]) \cap l_\alpha \neq \emptyset\}) \geq 1 - 2\delta$$

for all $k \geq k_0$. We scale back to $\tilde{u}_k$ and define

$$S_k := [\mathrm{Re}(z_k) - \frac{R}{R_k}, \mathrm{Re}(z_k) + \frac{R}{R_k}]$$

so that

$$v_k([-R, R]) = u_k(S_k)$$

and

$$\left|\mathrm{Re}(z_k) - \frac{R}{R_k}\right| \longrightarrow 0 \text{ as } k \to \infty.$$

The segments S_k satisfy

$$m_k \geq 1 - 2\delta > \frac{3}{4}$$

whenever $k \geq k_0$. □

Remark Starting with a sequence $\delta_k \searrow 0$, we obtain segments on ∂D centered at z_0 and shrinking in length to zero so that $m_k \to 1$, i.e. the curves $u_k|_{S_k}$ have to hit more and more leaves of the characteristic foliation. Our argument does not prohibit the existence of a bubbling point z_0 at the boundary if $R_k \,\mathrm{dist}(z_k, \partial D)$ is unbounded.

However, H. Hofer, K. Wysocki and E. Zehnder showed in [67] that there cannot be any bubbling point at the boundary.

8.3 Existence of a Finite Energy Plane

We will now show that explosion of the gradient in a family of pseudoholomorphic disks (satisfying the normalization condition (8.5)) leads to a non constant pseudoholomorphic plane with finite energy and therefore to the existence of a periodic orbit of the Reeb vector field.

Theorem 8.3.1 *Let $\tilde{u}_\tau$ be a family of pseudoholomorphic disks, i.e. $\tilde{u}_\tau$ satisfies Eq. (8.3), the normalization condition (8.5), the boundary condition (8.4) and the assumption on the winding number that $u_\tau|_{\partial D}$ winds exactly once around the elliptic singularity. We assume in addition that there is a compact subset $K \subset M$ so that $u_\tau(D) \subset K$ for all $0 \leq \tau < \tau_0$. Moreover, we assume that*

$$\sup_{0\leq\tau<\tau_0} e(\tilde{u}_\tau) = +\infty.$$

Then there exists a non constant pseudoholomorphic plane $\tilde{v} = (b, v) : \mathbf{C} \to \mathbf{R}\times M$ with $E(\tilde{v}) < +\infty$ and $v(\mathbf{C}) \subset K$.

Proof We pick a sequence $(\tilde{u}_k)$ out of our family $(\tilde{u}_\tau)$, i.e. $e(\tilde{u}_k) = \|\nabla\tilde{u}_k\|_{C^0(D)} \to \infty$ as $k \to \infty$. We know by Lemma 8.1.3 and Proposition 8.1.4 that there are positive constants c, C not depending on k so that

$$c \leq E(\tilde{u}_k) \leq C.$$

We pick a sequence $(z_k)_{k\in\mathbf{N}} \subset D$ so that $R_k := |\nabla\tilde{u}_k(z_k)| \to \infty$ as $k \to \infty$. We may assume after passing to some subsequence that $z_k \to z_0$ and we know by Proposition 8.2.1 that $R_k \operatorname{dist}(z_k, \partial D) \to \infty$. Take a sequence $\varepsilon_k \searrow 0$ so that $R_k\varepsilon_k \to \infty$. Using Lemma 4.4.4, we may assume in addition that

$$|\nabla\tilde{u}_k(z)| \leq 2R_k$$

for all $z \in D$ with $|z - z_k| \leq \varepsilon_k$. We define for $z \in B_{R_k}(-R_k z_k)$ the pseudoholomorphic maps $\tilde{v}_k = (b_k, v_k)$ by

$$\tilde{v}_k(z) := \left(a_k\left(z_k + \frac{z}{R_k}\right) - a_k(z_k), u_k\left(z_k + \frac{z}{R_k}\right)\right),$$

so that

$$b_k(0) = 0 \text{ and } |\nabla\tilde{v}_k(0)| = 1.$$

The domains $\Omega_k := B_{R_k}(-R_k z_k) \cap B_{\varepsilon_k R_k}(0)$ satisfy $\bigcup_k \Omega_k = \mathbf{C}$ and we have

$$|\nabla \tilde{v}_k(z)| \leq 2 \text{ on } \Omega_k.$$

By the C^∞_{loc}–bounds from Chap. 4.3 and the Ascoli–Arzela theorem, we conclude that a suitable subsequence of $\tilde{v}_k$ converges in C^∞_{loc} to a pseudoholomorphic plane

$$\tilde{v} = (b, v) : \mathbf{C} \longrightarrow \mathbf{R} \times M$$

which is not constant in view of $|\nabla \tilde{v}(0)| = 1$. We have for all compact $K \subset \mathbf{C}$ and sufficiently large k:

$$\begin{aligned} \sup_{\phi \in \Sigma} \int_K \tilde{v}_k^* d(\phi\lambda) &\leq \sup_{\phi \in \Sigma} \int_{B_{R_k}(-R_k z_k)} \tilde{v}_k^* d(\phi\lambda) \\ &= \sup_{\phi \in \Sigma} \int_D \tilde{u}_k^* d(\phi\lambda) \\ &= E(\tilde{u}_k) \leq C. \end{aligned}$$

Taking the limit $k \to \infty$ and then the supremum over all compact $K \subset \mathbf{C}$ yields $E(\tilde{v}) < \infty$. Moreover, the image of v lies in a compact subset of M since this is true for all the maps v_k. □

Chapter 9
Disk Filling Methods and Applications



9.1 The Weinstein Conjecture for Overtwisted Contact Forms

In this section we will prove the following theorem due to H. Hofer [64]:

Theorem 9.1.1 *Let (M, λ) be a closed three dimensional contact manifold with overtwisted contact structure* $\ker \lambda$. *Then there exists a contractible periodic orbit for the Reeb vector field X_λ.*

Proof We first pick an overtwisted disk $F \subset M$; and we may assume by Theorem 3.5.7 that F is a nice overtwisted disk, i.e. F has exactly one singular point e which is nicely elliptic and lies in the interior of F. We then apply Theorem 7.3.1 and find, after a C^0–small perturbation of F near e, a $d\lambda$–compatible complex structure $J : \xi \to \xi$ and a family of embedded pseudoholomorphic disks

$$\tilde{u}_\tau : D \longrightarrow \mathbf{R} \times M \ , \ \tau \in [0, \tau_0)$$

with $\mu(\tilde{u}_\tau) = 2$ satisfying the boundary condition $\tilde{u}_\tau(\partial D) \subset \{0\} \times F^*$. Using the implicit function theorem, Corollary 7.5.9, we continue the local family further. We then take a maximal family $(\tilde{u}_\tau)_{0 \leq \tau < \tau_0}$ of embedded pseudoholomorphic disks obtained by extension of the local family near e we started with, using the implicit function theorem. The loops $u_\tau(\partial D)$ never intersect ∂F in view of Corollary 5.2.2 because they are transverse to the characteristic foliation and ∂F is a leaf of the characteristic foliation. If $e(\tilde{u}_\tau)$ as defined in (8.1), Chap. 8, always remained bounded, then we could continue the family $(\tilde{u}_\tau)$ further beyond τ_0, using the compactness result Proposition 8.1.2 and the implicit function theorem Corollary 7.5.9 in contradiction to the assumption that the family $(\tilde{u}_\tau)_{0 \leq \tau < \tau_0}$ was the largest family of disks obtained from the local family by the implicit function theorem. Hence $e(\tilde{u}_\tau) \to +\infty$ as $\tau \to \tau_0$. We then conclude using Theorem 8.3.1

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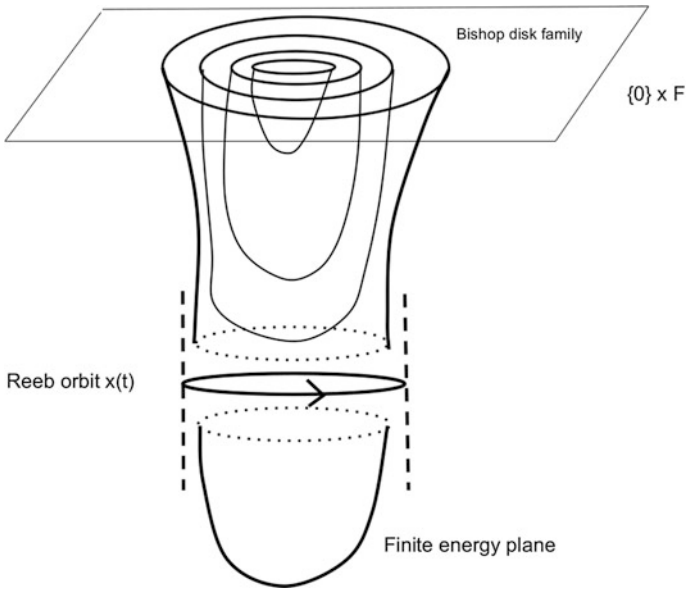


Fig. 9.1 Bubbling in a Bishop disk family. A picture in $\mathbf{R} \times M$

that there exists a non constant pseudoholomorphic plane with finite energy which implies existence of a contractible periodic orbit of the Reeb vector field by Theorem 4.4.1. □

Figure 9.1 depicts a family of Bishop disks emanating from an elliptic singular point on F and breaking up into a disk with a negative puncture asymptotic to a periodic Reeb orbit and a finite energy plane asymptotic to the same Reeb orbit. This configuration is also called a holomorphic building of height 2 (see [2, 15]). More complicated scenarios are of course possible such as several punctures, multiple Reeb orbits such as depicted in Fig. 9.2 and a larger height holomorphic building. The analysis in this book only recovers one of the finite energy planes at the bottom of each of the figures. The intricate holomorphic buildings are analyzed in [2, 15].

9.2 The Weinstein Conjecture for Tight Forms if $\pi_2(M) \neq 0$

The following theorem is also due to H. Hofer [64]:

Theorem 9.2.1 *Let (M, λ) be a closed three dimensional contact manifold with tight contact form and $\pi_2(M) \neq 0$. Then there exists a contractible periodic orbit of the Reeb vector field X_λ.*

Proof By Theorem 3.5.6 there exists an embedded two dimensional sphere in M representing a nontrivial homotopy class in $\pi_2(M)$ so that the characteristic foliation has only two singular points e_+ and e_-. These singular points are nicely elliptic with

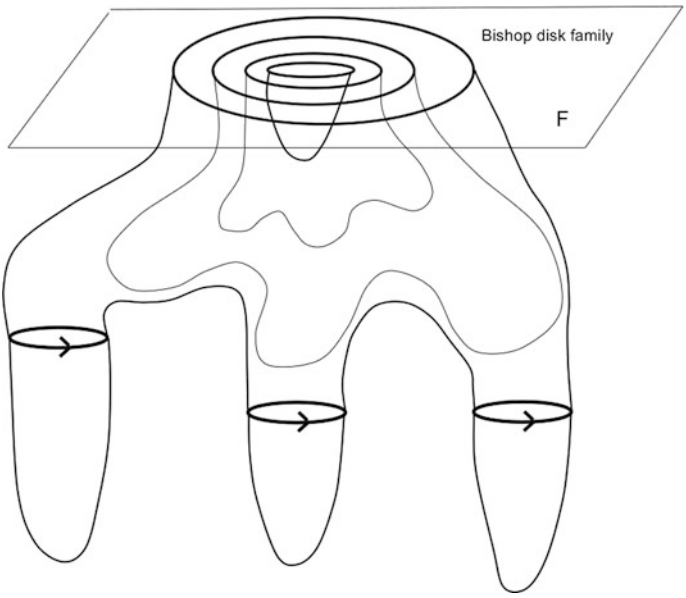


Fig. 9.2 Bubbling in a Bishop disk family in M

opposite signs and the leaves of the characteristic foliation run from one elliptic singularity into the other. Near each elliptic singularity there is a local family of embedded pseudoholomorphic disks with boundaries in $\{0\} \times F^*$ and Maslov index equal to 2 (Theorem 7.3.1). Denote these local disk families by $(\tilde{u}_\tau^+)$ and $(\tilde{u}_\tau^-)$. Their images are all pairwise disjoint. We now continue the disks from the $(+)$–family further using the implicit function theorem, Corollary 7.5.9. By Theorem 7.6.3 a member of the $(+)$–family can only intersect a member of the $(-)$–family if the two disk families are identical. But then we would have constructed an embedding

$$\Phi : (-1, +1) \times D \longrightarrow \mathbf{R} \times M$$

with $\Phi(\pm 1, z) = (0, e_\pm)$ for all $z \in D$ so that

$$\Phi(t, z) \in \{0\} \times F^* \text{ if } (t, z) \in [-1, +1] \times \partial D.$$

By collapsing top and bottom of the full cylinder $[-1, 1] \times D$ we obtain a homeomorphism from the three dimensional ball B^3 into $\mathbf{R} \times M$ so that ∂B^3 is mapped onto the surface $\{0\} \times F$. This contradicts the assumption that F is a non contractible sphere in M. Hence disks from the $(+)$–family can never intersect with disks from the $(-)$–family. We take the maximal family of pseudoholomorphic disks $(\tilde{u}_\tau^+)_{0 \leq \tau < \tau_0}$ obtained from the local family near e_+ by using the implicit function theorem. The boundary loops $u_\tau^+(\partial D) \subset F^*$ then have to stay away from a suitable open neighborhood of e_- in F. Then we argue exactly as in the overtwisted

case: If $e(\tilde{u}_\tau^+)$ always remained bounded, then we could continue the family $(\tilde{u}_\tau^+)$ further beyond τ_0, using the compactness result Proposition 8.1.2 and the implicit function theorem Corollary 7.5.9 in contradiction to the assumption that the family $(\tilde{u}_\tau^+)_{0\le\tau<\tau_0}$ was the largest family of disks obtained from the local family by the implicit function theorem. Hence $e(\tilde{u}_\tau^+) \to +\infty$ as $\tau \to \tau_0$. We then conclude using Theorem 8.3.1 that there exists a non constant pseudoholomorphic plane with finite energy which implies existence of a contractible periodic orbit of the Reeb vector field by Theorem 4.4.1. □

9.3 The Weinstein Conjecture for the Three-Sphere

Using the classification theorem for tight contact structures on the three dimensional sphere by Y. Eliashberg [33] (Theorem 2.1.1.) and an existence result for periodic orbits of Hamiltonian systems on star shaped energy hypersurfaces by P. Rabinowitz [97] we prove the following result by H. Hofer [64]:

Theorem 9.3.1 *Let λ be any contact form on the three dimensional sphere S^3. Then there exists a periodic orbit for the Reeb vector field X_λ.*

Proof By Y. Eliashberg's classification result there is a function $f : S^3 \to (0,\infty)$ and a diffeomorphism $\Psi : S^3 \to S^3$ so that $\Psi^*\lambda = f\lambda_0$, where λ_0 is the standard contact form, i.e. it is the restriction of the 1–form

$$\frac{1}{2}\sum_{k=1}^{2}(x_k dy_k - y_k dx_k)$$

to the three dimensional sphere. Then the Reeb flows for (S^3,λ) and $(S^3, f\lambda_0)$ are conjugate, hence it suffices to show existence of a periodic orbit for the Reeb vector field $X_{f\lambda_0}$. The hypersurface

$$S := \{\sqrt{f(z)}\, z \in \mathbf{R}^4 \mid z \in S^3\}$$

bounds a star shaped domain in $\mathbf{R}^4$. The diffeomorphism

$$\Phi : S^3 \longrightarrow S$$

$$z \longmapsto \sqrt{f(z)}\, z$$

satisfies $\Phi^*(\lambda_0|_S) = f\,\lambda_0|_{S^3}$. Indeed, we compute with $\sigma(z) := \sqrt{f(z)}$ and $h \in T_zS^3$

$$\begin{aligned}(\Phi^*\lambda_0)_z h &= \lambda_0(\sigma(z)z)[(D\sigma(z)h)\, z] + \lambda_0(\sigma(z)z)[\sigma(z)h]\\ &= \sigma^2(z)\,\lambda_0(z)h.\end{aligned}$$

The hypersurface S is of contact type, hence for any Hamiltonian $H : S \to \mathbf{R}$ having S as its regular zero level set, the orbits of the Hamiltonian vector field $X_H|_S$ and the Reeb vector field X_{λ_0} on S coincide up to parameterization (Proposition 1.2.1). Invoking P. Rabinowitz' result, we obtain existence of a periodic orbit of the Hamiltonian vector field $X_H|_S$ on S which completes the proof. □

9.4 Bennequin's Theorem

We will now use pseudoholomorphic curves to prove a result due to D. Bennequin [13] which states that the standard contact structure on $\mathbf{R}^3$ is tight. In this section we consider $\mathbf{R}^3$ endowed with the standard contact form $\lambda = dz + xdy$. We assume that $F \subset \mathbf{R}^3$ is an embedded compact surface with finitely many singular points. The contact structure is given by

$$\ker \lambda(x, y, z) = \mathrm{Span}_{\mathbf{R}} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -x \end{pmatrix} \right\}$$

and we define a complex structure on $\ker \lambda$ as follows: For $(x, y, z) \in \mathbf{R}^3$ we define

$$J(x, y, z) \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} := \begin{pmatrix} 0 \\ 1 \\ -x \end{pmatrix}$$

$$J(x, y, z) \cdot \begin{pmatrix} 0 \\ 1 \\ -x \end{pmatrix} := \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}.$$

We define an almost complex structure $\tilde{J}$ on the symplectisation $\mathbf{R}^4$ in the usual way. We would like to obtain an a priori C^0–bound for the $\mathbf{R}^3$–components u of pseudoholomorphic disks

$$\tilde{u} = (a, u) : D \longrightarrow \mathbf{R} \times \mathbf{R}^3$$

which satisfy the boundary condition

$$\tilde{u}(\partial D) \subset \{0\} \times F^*.$$

In the above coordinates the nonlinear Cauchy Riemann equation looks as follows (see Chap. 7.3):

$$\partial_s a - \partial_t z - x \partial_t y = 0$$
$$\partial_s z + \partial_t a - x \partial_t x = 0$$

$$\partial_s x - \partial_t y = 0$$
$$\partial_s y + \partial_t x = 0.$$

We have

$$a(e^{i\theta}) \equiv 0$$

and $u(e^{i\theta}) = (x, y, z)(e^{i\theta})$ lies on the surface F. Take a ball $B_R(p_0)$ in $\mathbf{R}^3$ containing the surface F with radius half the diameter of F. We derive by a straightforward calculation that the components x, y, z are all harmonic. Then the function $(s, t) \longmapsto |u(s, t) - p_0|^2$ on D satisfies $\Delta|u - p_0|^2 \geq 0$. By the maximum principle $|u - p_0|^2$ assumes its maximum on the boundary, but this implies that

$$\sup_D |u - p_0|^2 \leq R^2$$

since this inequality holds on ∂D due to the boundary condition. We summarize

Proposition 9.4.1 *Consider* $(\mathbf{R}^3, \lambda = dz + xdy)$ *and let* F *be a compact surface in* $\mathbf{R}^3$. *Assume that*

$$\tilde{u} = (a, u) : D \longrightarrow \mathbf{R} \times \mathbf{R}^3$$

satisfies

$$\partial_s \tilde{u} + \tilde{J}(\tilde{u})\partial_t \tilde{u} = 0,$$

where $\tilde{J}$ *is defined as above. Assume also that* $\tilde{u}$ *satisfies the boundary condition*

$$\tilde{u}(\partial D) \subset \{0\} \times F^*.$$

If $B_R(p_0) \subset \mathbf{R}^3$ *is a ball with* $R = \frac{1}{2}\, diam(F)$ *containing* F *then*

$$\sup_{(s,t)\in D} |u(s, t) - p_0| \leq R^2.$$

□

Remark There is no bound on the component a of $\tilde{u}$. We have only $\Delta a = (\partial_t y)^2 + (\partial_t x)^2 \geq 0$ and $a|_{\partial D} \equiv 0$, hence $a \leq 0$ by the maximum principle. If we have a family $\tilde{u}_\tau = (a_\tau, u_\tau)$ of pseudoholomorphic disks where bubbling occurs, we will have $\sup_\tau \|a_\tau\|_{C^0(D)} = +\infty$.

We now prove D. Bennequin's theorem:

Theorem 9.4.2 *The contact structure on* $\mathbf{R}^3$ *induced by the standard contact form* $\lambda = dz + xdy$ *is tight.*

Proof We argue indirectly and assume that there is an overtwisted disk F. We may assume also that the overtwisted disk is nice by Theorem 3.5.7, hence there is only one singular point e. We then proceed exactly as in the proof of the Weinstein conjecture for overtwisted closed contact manifolds M. We only needed compactness of M in order to have a priori C^0–bounds for the M–components of pseudoholomorphic disks. Our contact manifold $\mathbf{R}^3$ is not compact but we have derived the required bounds directly in Proposition 9.4.1. We start with a family of pseudoholomorphic disks near e with boundary condition $\{0\} \times F^*$. The same bubbling off argument as in the proof of the Weinstein conjecture then yields a periodic orbit of the Reeb vector field. The Reeb vector field of λ is given by the constant vector field $\frac{\partial}{\partial z}$, hence there are no periodic orbits. This contradiction shows that our original assumption, the existence of an overtwisted disk, must have been false. □

Appendix A
Cauchy Riemann Type Operators

A.1 The Cauchy Riemann Operator on S^2

Denote by $z = s + it$ the coordinates on $\mathbf{C}$. The Cauchy-Riemann operator $\bar{\partial}$ is the linear operator defined by

$$\bar{\partial} : C^\infty(\mathbf{C}, \mathbf{C}^n) \to C^\infty(\mathbf{C}, \mathbf{C}^n)$$

$$\bar{\partial} w = \frac{\partial w}{\partial s} + i \frac{\partial w}{\partial t}.$$

The aim of this section is to solve the equation

$$\bar{\partial} w = g,$$

where g is a smooth $\mathbf{C}^n$–valued function with compact support on $\mathbf{C}$. We denote by $C_0^\infty(\mathbf{C}, \mathbf{C}^n)$ the vector space of all $\mathbf{C}^n$-valued compactly supported smooth functions on $\mathbf{C}$.

For $\varepsilon > 0$ and $u \in C_0^\infty(\mathbf{C}, \mathbf{C}^n)$ we define a map $A_\varepsilon u : \mathbf{C} \to \mathbf{C}^n$ by

$$(A_\varepsilon u)(z) = -\frac{1}{2\pi} \int_{|\xi| \geq \varepsilon} \frac{u(z + \xi)}{\xi} ds\, dt,$$

with $\xi = s + it$.

Let $e \in \mathbf{C}$ be a vector of unit length and $h \in \mathbf{R} \setminus \{0\}$. For $u \in C_0^\infty(\mathbf{C}, \mathbf{C}^n)$ we consider the difference quotient $d_h u$ defined by

$$(d_h u)(z) = \frac{u(z + he) - u(z)}{h}$$

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We compute

$$\begin{aligned}(d_h A_\varepsilon u)(z) &= -\frac{1}{2\pi h}\int_{|\xi|\geq\varepsilon}\frac{u(z+he+\xi)-u(z+\xi)}{\xi}ds\,dt\\ &= -\frac{1}{2\pi}\int_{|\xi|\geq\varepsilon}\frac{(d_h u)(z+\xi)}{\xi}ds\,dt\\ &= (A_\varepsilon d_h u)(z).\end{aligned}$$

Therefore

$$d_h A_\varepsilon = A_\varepsilon d_h.$$

Let us show first that $A_\varepsilon u$ converges to some function Au in $C^0(\mathbf{C},\mathbf{C}^n)$ as $\varepsilon\to 0$. For $u\in C_0^\infty(\mathbf{C},\mathbf{C}^n)$ we find a constant $M>0$ such that

$$|u(z_2)-u(z_1)|\leq M\,|z_2-z_1|$$

for all $z_1, z_2\in\mathbf{C}$. Using polar coordinates we obtain

$$\begin{aligned}|(A_\varepsilon u)(z_2)-(A_\varepsilon u)(z_1)| &\leq \frac{1}{2\pi}\int_{|\xi|\geq\varepsilon}\frac{|u(z_2+\xi)-u(z_1+\xi)|}{|\xi|}ds\,dt\\ &\leq \frac{1}{2\pi}\int_0^R\int_0^{2\pi} M\,|z_2-z_1|\,d\theta\,dr \qquad (A.1)\\ &\leq RM\,|z_2-z_1|\,,\end{aligned}$$

where $R>\operatorname{diam}(\operatorname{supp}(u))+\max\{|z_1|\,,|z_2|\}$. Similarly, we estimate

$$\begin{aligned}\left|(A_{\varepsilon_2}u)(z)-(A_{\varepsilon_1}u)(z)\right| &\leq \frac{1}{2\pi}\int_{\varepsilon_1\leq|\xi|\leq\varepsilon_2}\left|\frac{1}{\xi}u(z+\xi)\right|ds\,dt \qquad (A.2)\\ &\leq \|u\|_{C^0}\;|\varepsilon_2-\varepsilon_1|\,.\end{aligned}$$

The first estimate implies that $A_\varepsilon u$ is continuous and the second that $(A_\varepsilon u)$ converges in C^0 as $\varepsilon\to 0$.

Let $K\subset\mathbf{C}$ be a compact set and assume $u\in C_0^\infty(K,\mathbf{C}^n)$, i.e. $u\in C_0^\infty(\mathbf{C},\mathbf{C}^n)$ with $\operatorname{supp}(u)\subset\overset{\circ}{K}$. We estimate

$$\begin{aligned}|(A_\varepsilon u)(z)| &\leq \frac{1}{2\pi}\int_0^{2\pi}\int_\varepsilon^\infty |u(z+\xi)|\,dr\,d\theta\\ &\leq \|u\|_0\operatorname{diam}(\operatorname{supp}(u))\\ &\leq M\|u\|_0\end{aligned}$$

for a constant $M=M(K)>0$.

If we denote the directional derivative with respect to the vector e by D_e we find for fixed $z \in \mathbf{C}$

$$\begin{aligned} |(d_h Au)(z) - (AD_e u)(z)| &= |(Ad_h u)(z) - (AD_e u)(z)| \\ &\leq M \, \|d_h u - D_e u\|_0 \to 0 \end{aligned}$$

as $h \to 0$, where M only depends on $\mathrm{supp}(u)$. Hence

$$AD_e = D_e A.$$

Repeating the argument finally yields for every multi-index α and every partial derivative D^α:

$$D^\alpha A = AD^\alpha.$$

Hence A defines a linear map from $C_0^\infty(\mathbf{C}, \mathbf{C}^n)$ into $C^\infty(\mathbf{C}, \mathbf{C}^n)$ satisfying

$$A\bar{\partial} = \bar{\partial} A$$

on $C_0^\infty(\mathbf{C}, \mathbf{C}^n)$. We will show that

$$(\bar{\partial} \circ A)\, u = u$$

for all $u \in C_0^\infty(\mathbf{C}, \mathbf{C}^n)$, then $w := Ag$ will solve the problem $\bar{\partial} w = g$ for $g \in C_0^\infty(\mathbf{C}, \mathbf{C}^n)$. For $\xi = s + it$ and fixed $z \in \mathbf{C}$ define

$$v(s,t) := u(z + \xi).$$

Assuming $\xi \neq 0$, we compute

$$\begin{aligned} \frac{1}{\xi}(\bar{\partial} v) ds \wedge dt &= \frac{1}{\xi}\left(\frac{\partial v}{\partial s} ds + \frac{\partial v}{\partial t} dt\right) \wedge (dt - i ds) \\ &= -\frac{i}{\xi} dv \wedge (ds + i dt) \\ &= -i \; d\left(\frac{v}{\xi}\right) \wedge (ds + i dt), \end{aligned}$$

where we used

$$d\left(\frac{1}{\xi}\right) = -\frac{1}{\xi^2}(ds + i dt).$$

Invoking Stokes' theorem we deduce for $u \in C_0^\infty(\mathbf{C}, \mathbf{C}^n)$ and $R > 0$ large enough

$$\begin{aligned}(A_\varepsilon \bar\partial u)(z) &= -\frac{1}{2\pi}\int_{\varepsilon\le|\xi|\le R} \frac{1}{\xi}(\bar\partial u)(z+\xi) ds\, dt \\ &= \frac{i}{2\pi}\int_{\varepsilon\le|\xi|\le R} d\Big(\frac{u(z+\xi)}{\xi}(ds+i dt)\Big) \\ &= \frac{1}{2\pi i}\int_{|\xi|=\varepsilon} u(z+\xi)\frac{1}{\xi}(ds+i dt) \\ &= \frac{1}{2\pi i}\int_0^{2\pi} u(z+\varepsilon e^{i\theta})\frac{1}{\varepsilon e^{i\theta}}\, i\varepsilon e^{i\theta}\, d\theta \\ &= \frac{1}{2\pi}\int_0^{2\pi} u(z+\varepsilon e^{i\theta}) d\theta,\end{aligned}$$

but this tends to $u(z)$ as $\varepsilon \to 0$, hence we have proved

$$A\bar\partial = \mathrm{Id},$$

which is equivalent to

$$\bar\partial A = \mathrm{Id}\,.$$

We cannot expect that Au has compact support, but we have at least the following decay estimate:

Proposition A.1.1 *Let $u \in C_0^\infty(\mathbf{C}, \mathbf{C}^n)$. If $|\, z\,|$ is sufficiently large then*

$$|\,(Au)(z)\,| \le\ C\ |\, z\,|^{-1}$$

for some constant $C > 0$.

Proof Pick $R > 0$ so that $\mathrm{supp}(u) \subset B_R(0)$. We have

$$\begin{aligned}(Au)(z) &= \lim_{\varepsilon\to 0}\frac{-1}{2\pi}\int_{|\xi|\ge\varepsilon}\frac{u(z+\xi)}{\xi} ds\, dt \\ &= \lim_{\varepsilon\to 0}\frac{-1}{2\pi}\int_\varepsilon^\infty\int_0^{2\pi} u(z+re^{i\theta})e^{-i\theta} d\theta\, dr \\ &= -\frac{1}{2\pi}\int_0^\infty\int_0^{2\pi} u(z+re^{i\theta})e^{-i\theta} d\theta\, dr.\end{aligned}$$

Consider $|z| > R$ and write $v(r,\theta) := u(z+re^{i\theta})e^{-i\theta}$. Pick (r_0, θ_0) in such a way that

$$-z = r_0 e^{i\theta_0}$$

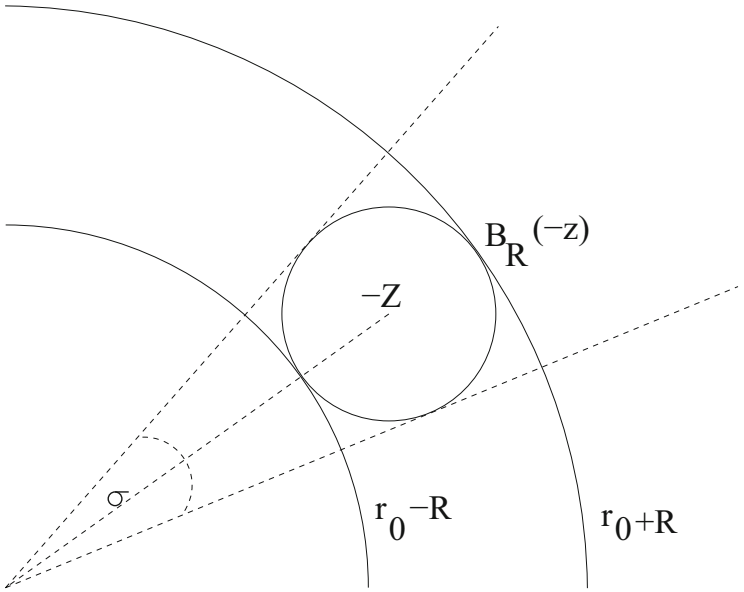


Fig. A.1 The cone C

Then $B_R(-z)$ contains $\mathrm{supp}(v)$ and lies in a cone C with vertex at 0 having angle σ so that

$$\frac{R}{r_0} = \sin\left(\frac{\sigma}{2}\right).$$

Moreover for $(r, \theta) \in C$ then $v(r, \theta) = 0$ if

$$r \in [0, r_0 - R] \cup [r_0 + R, +\infty)$$

(see Fig. A.1). As $r_0 \to +\infty$ we see that $\sigma \to 0$. More precisely, for large $|z|$

$$0 \leq \sigma \leq \frac{4R}{|z|}.$$

Using this we compute for large $|z|$

$$\begin{aligned} \mid (Au)(z) \mid &\leq \frac{1}{2\pi} \int_{|z|-R}^{|z|+R} \int_{\theta_0-\sigma/2}^{\theta_0+\sigma/2} \mid v(r, \theta) \mid d\theta\, dr \\ &\leq \frac{\|u\|_{C^0}}{2\pi} \cdot 2R\sigma \\ &\leq \frac{4R^2}{\pi} \|u\|_{C^0} \cdot \mid z \mid^{-1}, \end{aligned}$$

which completes the proof. □

We will use Proposition A.1.1 to show that Au may be viewed as a smooth function on the Riemann sphere $S^2 = \mathbf{C} \cup \{\infty\}$.

The Riemann sphere S^2 is the one-point compactification of $\mathbf{C}$, and equipped with the structure of a complex manifold by defining an atlas $\{\alpha, \beta\}$ by

$$\begin{aligned} U = \mathbf{C} &\xrightarrow{\alpha} \mathbf{C} \\ z &\longmapsto z \end{aligned} \tag{A.3}$$

$$\begin{aligned} V = (\mathbf{C} \setminus \{0\}) \cup \{+\infty\} &\xrightarrow{\beta} \mathbf{C} \\ z &\longmapsto \begin{cases} \frac{1}{z} \text{ for } z \neq +\infty \\ 0 \text{ for } z = +\infty \end{cases}. \end{aligned}$$

We recall the classical removable singularity theorem for a holomorphic map on the punctured ball (see Chapter 4.3.1. of [7]).

Proposition A.1.2 *Let $B_\varepsilon^* = \{z \in \mathbf{C} \mid 0 < |z| < \varepsilon\}$ be the punctured open ε-disk. Assume $f : B_\varepsilon^* \to \mathbf{C}$ is holomorphic and bounded. Then f extends in a unique way to a holomorphic map $\tilde{f} : B_\varepsilon \to \mathbf{C}$ with*

$$\tilde{f} \mid_{B_\varepsilon^*} = f.$$

□

Corollary A.1.3 *Let $u \in C_0^\infty(\mathbf{C}, \mathbf{C}^n)$. Then Au is the restriction of a smooth map $\tilde{u} : S^2 \to \mathbf{C}^n$ satisfying $\tilde{u}(\infty) = 0$.*

Proof By the preceding results we have

$$\begin{aligned} \bar{\partial}(Au) &= u \quad \text{on} \quad \mathbf{C} \\ (Au)(z) &\to 0 \quad \text{as} \quad |z| \to \infty. \end{aligned}$$

Since $u(z) = 0$ for $|z| \geq R$, where R is a sufficiently large constant, we see that $Au : \mathbf{C} \setminus B_R \to \mathbf{C}$ is holomorphic and bounded. Hence

$$z \longmapsto (Au)\left(\frac{1}{z}\right)$$

is holomorphic and bounded on some punctured ε-disk B_ε^*. By Proposition A.1.2 the map

$$z \longmapsto (Au)\left(\frac{1}{z}\right) = (Au) \circ \beta^{-1}(z)$$

is the restriction of a holomorphic map on B_ε. Hence, using the definition of the complex (smooth) structure on S^2 we see that

$$Au : S^2 \supset \mathbf{C} \to \mathbf{C}^n$$

is the restriction of a smooth map $S^2 \to \mathbf{C}^n$. Clearly this map takes the value 0 at ∞. This completes the proof of Corollary A.1.3. □

Summing up, we have proved

Theorem A.1.4 *A defines a linear map from* $C_0^\infty(\mathbf{C}, \mathbf{C}^n)$ *into* $C^\infty(S^2, \mathbf{C}^n)$ *with* $(Au)(\infty) = 0$. *Moreover on* $\mathbf{C}$

$$\bar\partial(Au) = u.$$

□

View $\mathbf{C}^n$ as the trivial vectorbundle $S^2 \times \mathbf{C}^n \to S^2$ over S^2. Moreover we introduce the vectorbundle $X \to S^2$, whose fiber over $z \in S^2$ consists of all real linear maps

$$\phi : T_z S^2 \to \{z\} \times \mathbf{C}^n$$

which anti-commute with i, i.e.

$$\phi i + i\phi = 0.$$

The previously introduced Cauchy-Riemann operator may be viewed as a linear map from $C^\infty(S^2 \times \mathbf{C}^n)$ into $C^\infty(X)$, where $C^\infty(E)$ denotes the space of smooth sections for some vector bundle E. Indeed, defining

$$\bar\partial : C^\infty(S^2 \times \mathbf{C}^n) \to C^\infty(X)$$

$$(\bar\partial u)(z) = (Tu)(z) + i \circ (Tu(z)) \circ i$$

with

$$i : T_z S^2 \longrightarrow T_z S^2 \ , \ i\left(\frac{\partial}{\partial s}\right) = \frac{\partial}{\partial t}$$

and $T_z\mathbf{C}^n = \mathbf{C}^n$, then the relation to the previously defined operator $\bar{\partial}$ is as follows. Consider $\mathbf{C} \subset S^2$ and the vector field $z \longmapsto \frac{\partial}{\partial s}\,|_z$ on $\mathbf{C}$. Then

$$\begin{aligned}(\bar{\partial}u)(z)(\frac{\partial}{\partial s}|_z) &= Tu(z)(\frac{\partial}{\partial s}) + i \circ (Tu(z)) \circ i\frac{\partial}{\partial s}|_z \\ &= \frac{\partial u}{\partial s}(z) + i\frac{\partial u}{\partial t}(z),\end{aligned}$$

where $z = s + it$. We obtain the following:

Theorem A.1.5 *$\bar{\partial} : C^\infty(S^2 \times \mathbf{C}^n) \to C^\infty(X)$ is a surjective Fredholm operator. The kernel consists of all constant maps. As a real linear operator the index ind$(\bar{\partial})$ is therefore $2n$.*

Proof Assume $\bar{\partial}u = 0$. Then $u_{|\mathbf{C}} : \mathbf{C} \to \mathbf{C}^n$ is a bounded holomorphic map and therefore constant by Liouville's theorem. This shows that $\ker(\bar{\partial})$ consists of the constant maps. Next assume $g \in C^\infty(X)$. Using a smooth partition of unity we can write g as

$$g = g_U + g_V$$

where $\mathrm{supp}(g_U) \subset U = \mathbf{C} \subset S^2$ and $\mathrm{supp}(g_V) \subset S^2 \setminus \{0\}$. The map $g_U(z) : \mathbf{C} \to \mathbf{C}^n$ is complex anti-linear for every $z \in U = \mathbf{C}$, and is compactly supported with respect to z. Therefore g^U defined by $g^U(z) = g_U(z)\frac{\partial}{\partial s}|_z$ belongs to $C_0^\infty(\mathbf{C}, \mathbf{C}^n)$. With $u = Ag^U \in C^\infty(S^2 \times \mathbf{C}^n)$ we have

$$\bar{\partial}u = g_U \quad \text{on} \quad \mathbf{C}$$

and

$$u(\infty) = 0.$$

By the same argument we find $v \in C^\infty(S^2 \times \mathbf{C}^n)$ with $v(0) = 0$ and

$$\bar{\partial}v = g_V.$$

Hence we have with $w = u + v \in C^\infty(S^2 \times \mathbf{C}^n)$

$$\bar{\partial}w = \bar{\partial}u + \bar{\partial}v = g_U + g_V = g.$$

□

A.2 Regularity Theory

Assume, J is any smooth almost complex structure on $\mathbf{C}^N$ such that $J(w) = i$ for $w \in \mathbf{R}^N$. Let $D^+ := \{z \in \mathbf{C} \,|\, |z| < 1\ ,\ \mathrm{Im}(z) \geq 0\}$. Assume that

$$u : D^+ \longrightarrow \mathbf{C}^N$$

solves the partial differential equation

$$\partial_s u + J(u)\partial_t u = 0 \text{ on } \overset{\circ}{D^+}$$

and the boundary condition

$$u((-1, +1)) \subset \mathbf{R}^N.$$

Let us assume that $u \in W^{1,p}_{loc}(D^+, \mathbf{C}^N)$, where $2 < p < \infty$, i.e. we understand the above partial differential equation in the sense of weak derivatives.

Theorem A.2.1 *Under the above hypothesis, we have* $u \in W^{2,p}_{loc}(D^+, \mathbf{C}^N)$.

This has the following corollary.

Corollary A.2.2 *We have* $u \in W^{k,p}_{loc}(D^+, \mathbf{C}^N)$ *for all* $k \in \mathbf{N}$. *In particular, u is smooth.*

Let us first prove the corollary assuming that Theorem A.2.1 is true. Assume that J is any smooth almost complex structure on $\mathbf{C}^n$ with $J(u) = i$ for $u \in \mathbf{R}^n$ (here $n \geq 1$ may be any integer). Define for $(z, u, v) \in \mathbf{C} \times \mathbf{C}^n \times \mathbf{C}^n$ a linear map

$$\tilde{J}(z, u, v) : \mathbf{C} \times \mathbf{C}^n \times \mathbf{C}^n \longrightarrow \mathbf{C} \times \mathbf{C}^n \times \mathbf{C}^n$$

by

$$\tilde{J}(z, u, v)(\delta, h, k) := \Big(i\delta\ ,\ J(u)h\ ,\ J(u)k + \mathrm{Im}(\delta)(DJ(u)v)(J(u)v) + \\ + \mathrm{Re}(\delta)J(u)(DJ(u)v)(J(u)v)\Big).$$

We observe that $\tilde{J}(z, u, v)^2(\delta, h, k) = -(\delta, h, k)$, hence $\tilde{J}$ is an almost complex structure on $\mathbf{C}^{2n+1}$. Moreover, by our assumption on J, we have $DJ(u)v = 0$ for $u, v \in \mathbf{R}^n$, which implies $\tilde{J}(z, u, v) = i$ if $u, v \in \mathbf{R}^n$. Let now $u \in W^{1,p}_{loc}(D^+, \mathbf{C}^n)$ be a solution of

$$\partial_s u + J(u)\partial_t u = 0 \text{ on } \overset{\circ}{D^+}$$

$$u((-1, +1)) \subset \mathbf{R}^n.$$

By Theorem A.2.1, the map u also belongs to the class $W^{2,p}_{loc}(D^+, \mathbf{C}^n)$. We define $w \in W^{1,p}_{loc}(D^+, \mathbf{C}^{2n+1})$ by

$$w(s,t) := (s+it, u(s,t), \partial_s u(s,t)).$$

We observe that

$$w((-1,+1)) \subset \mathbf{R}^{2n+1}.$$

Moreover, on D^+:

$$\begin{aligned}
&\partial_s w + \tilde{J}(w)\partial_t w \\
&= (1, u_s, u_{ss}) + \tilde{J}(w)(i, u_t, u_{st}) \\
&= (1+i^2, u_s + J(u)u_t, u_{ss} + J(u)u_{st} + (DJ(u)u_s)(J(u)u_s)) \\
&= (0, 0, u_{ss} + J(u)u_{st} + (DJ(u)u_s)u_t) \\
&= (0, 0, \partial_s(u_s + J(u)u_t)) \\
&= (0, 0, 0).
\end{aligned}$$

Applying again Theorem A.2.1 we deduce $w \in W^{2,p}_{loc}(D^+, \mathbf{C}^{2n+1})$. In particular, $\partial_s u \in W^{2,p}_{loc}(D^+, \mathbf{C}^n)$. Since

- $\partial_t u = J(u)\partial_s u$,
- $J(u) \in W^{2,p}_{loc}(D^+, \mathrm{End}(\mathbf{C}^n))$ by Proposition 4.3.2,
- $W^{k,p}_{loc}(D^+, \mathbf{C})$ is a Banach algebra for $2 < p < \infty$ and $k \geq 1$,

we see that $\partial_t u \in W^{2,p}_{loc}(D^+, \mathbf{C}^n)$. Therefore, $u \in W^{3,p}_{loc}(D^+, \mathbf{C}^n)$. Proceeding for w in the same way as we did with u, we find that $u, u_s, u_t, \frac{\partial^k u}{\partial s^k}$ belong to $W^{1,p}_{loc}(D^+, \mathbf{C}^n)$ for all $k \geq 1$.

Lemma A.2.3 *For every multi index $\alpha \in \mathbf{N}^2$ we have $D^\alpha u \in W^{1,p}_{loc}(D^+, \mathbf{C}^n)$.*

Proof We proceed by induction with respect to $|\alpha|$. We know already that the assertion is true for $|\alpha| \leq 1$. Moreover, we know that for $|\alpha| \leq 1$ there exist smooth maps $F^\alpha : (\mathbf{C}^n)^{|\alpha|+1} \to \mathbf{C}^n$ such that

$$D^\alpha u = F^\alpha(u, \partial_s u).$$

Indeed, for $|\alpha| = 0$ this is trivial. For $\alpha = (1,0)$ we have

$$\partial_s u = \mathrm{pr}_2(u, \partial_s u)$$

and for $\alpha = (0, 1)$

$$\partial_t u = J(u)\partial_s u,$$

hence $F^{(0,1)}(a, b) = J(a)b$. Assume now that for every $|\alpha| \leq k$, the maps $D^\alpha u$ belong to $W^{1,p}_{loc}(D^+, \mathbf{C}^n)$ and

$$D^\alpha u = F^\alpha(u, \partial_s u, \ldots, \frac{\partial^{|\alpha|} u}{\partial s^{|\alpha|}}) \tag{A.4}$$

for some smooth map F^α. We show now that this also holds for $|\alpha| = k + 1$. The right hand side of Eq. (A.4) belongs to $W^{1,p}_{loc}(D^+, \mathbf{C}^n)$, hence $\partial_s D^\alpha u$ belongs to $L^p_{loc}(D^+, \mathbf{C}^n)$ and satisfies

$$\begin{aligned}\partial_s D^\alpha u &= \sum_{j=0}^{|\alpha|} D_j F^\alpha(u, \ldots, \frac{\partial^{|\alpha|} u}{\partial s^{|\alpha|}}) \frac{\partial^{j+1} u}{\partial s^{j+1}} \\ &=: F^{\alpha+(1,0)}(u, \partial_s u, \ldots, \frac{\partial^{|\alpha|+1} u}{\partial s^{|\alpha|+1}}).\end{aligned}$$

The map $F^{\alpha+(1,0)}$ is clearly smooth in all its variables. Since all the variables belong to $W^{1,p}_{loc}(D^+, \mathbf{C}^n)$, we conclude the same for $\partial_s D^\alpha u$. Consider now $\partial_t D^\alpha u$, which belongs to $L^p_{loc}(D^+, \mathbf{C}^n)$ and satisfies

$$\begin{aligned}\partial_t D^\alpha u &= \sum_{j=0}^{|\alpha|} D_j F^\alpha(u, \ldots, \frac{\partial^{|\alpha|} u}{\partial s^{|\alpha|}}) \frac{\partial^j}{\partial s^j} \partial_t u \\ &= \sum_{j=0}^{|\alpha|} D_j F^\alpha(u, \ldots, \frac{\partial^{|\alpha|} u}{\partial s^{|\alpha|}}) \frac{\partial^j}{\partial s^j} (J(u)\partial_s u) \\ &=: F^{\alpha+(0,1)}(u, \ldots, \frac{\partial^{|\alpha|+1} u}{\partial s^{|\alpha|+1}}),\end{aligned}$$

where again $F^{\alpha+(0,1)}$ is a smooth map. Hence the assertion is proved also for $|\alpha| = k + 1$. Hence we have proved the lemma. Smoothness of u follows directly from the Sobolev embedding theorem, which completes also the proof of the corollary. □

In order to prove Theorem A.2.1, we use the difference quotient technique. Define for $h \in \mathbf{R}\backslash\{0\}$:

$$u^h(s, t) := \frac{1}{h}(u(s + h, t) - u(s, t)).$$

We note that $u((-1,+1)) \subset \mathbf{R}^n$ implies that

$$u^h(s,0) \in \mathbf{R}^n.$$

We will show that around every point $(s_0,t_0) \in D^+$ there is an open neighborhood U in D^+ so that

$$\|u^h\|_{W^{1,p}(U)} \leq \text{ const}$$

for all sufficiently small $|h|$ with $h \neq 0$. It is not difficult to show that

$$u^h \longrightarrow \partial_s u \text{ in } L^p(U, \mathbf{C}^n)$$

as $h \to 0$. For every sequence $h_k \to 0$ the sequence $(u^{h_k})_{k\in\mathbf{N}}$ must have a weakly convergent subsequence in $W^{1,p}(U,\mathbf{C}^n)$, which also has to converge weakly in $L^p(U,\mathbf{C}^n)$ to the same limit. Since convergence in the L^p–norm implies weak convergence, we see that u^h converges weakly to $\partial_s u$ in $W^{1,p}(U,\mathbf{C}^n)$ as $h \to 0$, therefore $\partial_s u$ belongs to $W^{1,p}(U,\mathbf{C}^n)$. For more details on the properties of difference quotients we refer the reader to the book by Avner Friedman [51].

We carry out the argument for the point $(s_0,t_0) = (0,0)$. Without loss of generality we may also assume that $u(0,0) = 0$. We have

$$\begin{aligned} 0 &= u_s^h + (J(u)u_t)^h \\ &= u_s^h + J(u)u_t^h + J(u)^h u_t(s+h,t) \\ &= u_s^h + iu_t^h + (J(u)-i)u_t^h + J(u)^h u_t(s+h,t). \end{aligned} \tag{A.5}$$

Pick a smooth map $\beta : \mathbf{R} \to [0,1]$ such that

$$\beta|_{[0,1/2]} \equiv 1\ ,\ \beta|_{[1,\infty)} \equiv 0$$

and $\beta'(s) < 0$ for $1/2 < s < 1$. For $\varepsilon > 0$ we define

$$\beta_\varepsilon(s,t) := \beta\left(\frac{\sqrt{s^2+t^2}}{\varepsilon}\right).$$

We now multiply Eq. (A.5) by β_ε and obtain

$$\beta_\varepsilon \overline{\partial}(u^h) = \beta_\varepsilon(i - J(u))u_t^h - \beta_\varepsilon J(u)^h u_t(s+h,t). \tag{A.6}$$

We rewrite this as

$$\begin{aligned} \overline{\partial}(\beta_\varepsilon u^h) = (\overline{\partial}\beta_\varepsilon)u^h + (i - J(u))\partial_t(\beta_\varepsilon u^h) - \\ -(i - J(u))(\partial_t\beta_\varepsilon\ u^h) - (\beta_\varepsilon J(u)^h)u_t(s+h,t). \end{aligned}$$

For $\varepsilon < 1$ and $|h|$ small enough, the map $\beta_\varepsilon u^h$ belongs to $W^{1,p}(D^+, \mathbf{C}^n)$ and has support in D^+. Hence with $D^+_\varepsilon := \{z \in D^+ \mid |z| < \varepsilon\}$

$$\begin{aligned} c\|\beta_\varepsilon u^h\|_{1,p,D^+} &\leq \|\overline{\partial}(\beta_\varepsilon u^h)\|_{0,p,D^+} \\ &\leq c_\varepsilon \|u^h\|_{0,p,D^+_\varepsilon} + \|i - J(u)\|_{L^\infty(D^+_\varepsilon)} \|\beta_\varepsilon u^h\|_{1,p,D^+} + \\ &\quad + c_\varepsilon \|u^h\|_{0,p,D^+_\varepsilon} + \|\beta_\varepsilon J(u)^h\|_{L^\infty(D^+)} \|u_t(s+h,t)\|_{0,p,D^+_\varepsilon}, \end{aligned} \tag{A.7}$$

where we used the L^p–estimate for the Cauchy Riemann operator. For $|h|$ small enough, more precisely for $\varepsilon + |h| < 1$ we have

$$\|u_t(s+h,t)\|_{0,p,D^+_\varepsilon} \leq \|u\|_{1,p,D^+}. \tag{A.8}$$

The embedding $W^{1,p}(D^+, \mathbf{C}^n) \hookrightarrow C^0(\overline{D^+}, \mathbf{C}^n)$ for $2 < p < \infty$ is compact. Hence given $\delta > 0$ we can find a constant c_δ such that

$$\|u\|_{L^\infty(D^+, \mathbf{C}^n)} \leq \delta \|u\|_{1,p,D^+} + c_\delta \|u\|_{0,p,D^+} \tag{A.9}$$

for all $u \in W^{1,p}(D^+, \mathbf{C}^n)$. Indeed, fixing δ and arguing indirectly, we find a sequence $(u_k) \subset W^{1,p}(D^+, \mathbf{C}^n)$ with $\|u_k\|_{1,p,D^+} = 1$ and

$$\|u_k\|_{L^\infty(D^+, \mathbf{C}^n)} > \delta \|u_k\|_{1,p,D^+} + k\|u_k\|_{0,p,D^+}.$$

The sequence (u_k) is bounded in the L^∞–norm by the Sobolev embedding theorem, hence $u_k \to 0$ in $L^p(D^+, \mathbf{C}^n)$. After passing to a suitable subsequence, we may assume also that $u_k \to u$ in L^∞ using the compact embedding. Then $u_k \to u$ in L^p as well, hence $u = 0$. Now this implies

$$0 = \lim_{k\to\infty} \|u_k\|_{L^\infty(D^+, \mathbf{C}^n)} \geq \delta + \lim \sup_{k\to\infty} (k\|u_k\|_{0,p,D^+}) \geq \delta,$$

which is a contradiction and proves (A.9). We now estimate for a suitable constant M depending on J

$$\begin{aligned} |\beta_\varepsilon J(u)^h(s,t)| &= \left| \beta_\varepsilon(s,t) \frac{J(u(s+h,t)) - J(u(s,t))}{h} \right| \\ &\leq M\, |\beta_\varepsilon(s,t)| \left| \frac{u(s+h,t) - u(s,t)}{h} \right| \\ &\leq M\, |\beta_\varepsilon u^h|. \end{aligned}$$

Therefore,

$$\|\beta_\varepsilon J(u)^h\|_{L^\infty(D^+, \mathbf{C}^n)} \leq M\, \|\beta_\varepsilon u^h\|_{L^\infty(D^+, \mathbf{C}^n)}. \tag{A.10}$$

Employing (A.9), we deduce from (A.10) for every $\delta > 0$

$$\|\beta_\varepsilon J(u)^h\|_{L^\infty(D^+,\mathbf{C}^n)} \leq M\delta\|\beta_\varepsilon u^h\|_{1,p,D^+} + Mc_\delta\|\beta_\varepsilon u^h\|_{0,p,D^+}. \tag{A.11}$$

Note that $\|i - J(u)\|_{L^\infty(D_\varepsilon^+)} \to 0$ for $\varepsilon \to 0$. Take $\varepsilon > 0$ so small that $\|i - J(u)\|_{L^\infty(D_\varepsilon^+)} < \frac{c}{2}$. Then we deduce from (A.7), (A.8) and (A.11)

$$\left(\frac{c}{2} - M\delta\|u\|_{1,p,D^+}\right)\|\beta_\varepsilon u^h\|_{1,p,D^+} \leq \left(2c_\varepsilon + Mc_\delta\|u\|_{1,p,D^+}\right)\|u^h\|_{0,p,D_\varepsilon^+}.$$

Pick now $\delta > 0$ smaller than $\frac{c}{4M\|u\|_{1,p,D^+}}$ so that

$$\frac{c}{4}\|\beta_\varepsilon u^h\|_{1,p,D^+} \leq c_{\varepsilon,\delta,u}\|u^h\|_{0,p,D_\varepsilon^+} \tag{A.12}$$

for all $|h| > 0$ small enough. Since $u \in W^{1,p}(D^+,\mathbf{C}^n)$, we know that $u^h \to \partial_s u$ in $L^p(D_\varepsilon^+,\mathbf{C}^n)$. Hence the right hand side of (A.12) is bounded (and well–defined) for $0 < |h| < h_0$ and h_0 sufficiently small. Consequently $\partial_s u$ belongs to $W^{1,p}(D_{\frac{\varepsilon}{2}}^+,\mathbf{C}^n)$. The same argument carried out at any boundary point of D^+ and any interior point shows that $\partial_s u \in W^{1,p}_{loc}(D^+,\mathbf{C}^n)$. Since $\partial_t u = J(u)\partial_s u$ and $p > 2$, the same is also true for $\partial_t u$. □

A.3 The Maslov Index for Loops of Totally Real Subspaces of $\mathbf{C}^n$

Denote by $\mathcal{T}(n)$ the set of totally real subspaces of $\mathbf{C}^n$. We will identify the set $\mathcal{T}(n)$ with $\mathrm{GL}(\mathbf{C}^n)/\mathrm{GL}(\mathbf{R}^n)$ via the map

$$\gamma : \mathrm{GL}(\mathbf{C}^n)/\mathrm{GL}(\mathbf{R}^n) \longrightarrow \mathcal{T}(n)$$

$$[A] \longmapsto A(\mathbf{R}^n).$$

We view $\mathrm{GL}(\mathbf{R}^n)$ as a subset of $\mathrm{GL}(\mathbf{C}^n)$. We note that it is actually a closed Lie subgroup of $\mathrm{GL}(\mathbf{C}^n)$ because it just consists of the invertible $n \times n$-matrices with real entries. Then the quotient $\mathrm{GL}(\mathbf{C}^n)/\mathrm{GL}(\mathbf{R}^n)$ is a smooth manifold and a map $f : \mathrm{GL}(\mathbf{C}^n)/\mathrm{GL}(\mathbf{R}^n) \longrightarrow M$ into some smooth manifold M is smooth if and only if $f \circ \pi : \mathrm{GL}(\mathbf{C}^n) \longrightarrow M$ is smooth, where π is the projection onto the quotient (see the chapter about homogeneous manifolds in [113]).

Lemma A.3.1

- *U(n) and O(n) are deformation retracts of* $\mathrm{GL}(\mathbf{C}^n)$ *and* $\mathrm{GL}(\mathbf{R}^n)$ *respectively*
- $\pi_1(\mathrm{GL}(\mathbf{C}^n), Id)$ *is isomorphic to* $\mathbf{Z}$ *generated by the homotopy class of*

$$t \longmapsto \begin{pmatrix} e^{2\pi i t} & 0 \\ 0 & Id \end{pmatrix} \; 0 \leq t \leq 1$$

Proof Choose $A \in \mathrm{GL}(\mathbf{C}^n)$. Then $A\bar{A}^T$ is hermitian and positive definite, hence the square root $\sqrt{A\bar{A}^T}$ is uniquely defined. Denote the set of hermitian positive definite $n \times n$-matrices by $\mathcal{P}$. One checks easily that the smooth map

$$\begin{aligned} \mathrm{GL}(\mathbf{C}^n) &\longrightarrow \mathcal{P} \times \mathrm{U}(n) \\ A &\longmapsto (\sqrt{A\bar{A}^T},\ (\sqrt{A\bar{A}^T})^{-1} \cdot A) \end{aligned}$$

is the inverse of

$$\begin{aligned} \mathcal{P} \times \mathrm{U}(n) &\longrightarrow \mathrm{GL}(\mathbf{C}^n) \\ (P, R) &\longmapsto PR, \end{aligned}$$

so $\mathrm{GL}(\mathbf{C}^n)$ is diffeomorphic to $\mathcal{P} \times \mathrm{U}(n)$. This proves the first part of the lemma because $\mathcal{P}$ is a convex subset of the vector space of hermitian $n \times n$-matrices (the proof for the real case $\mathrm{O}(n)$ is similar to the complex one).

Now $\mathrm{U}(n-1)$ can be viewed as a closed Lie subgroup of $\mathrm{U}(n)$ by

$$\begin{aligned} i : \mathrm{U}(n-1) &\longrightarrow \mathrm{U}(n) \\ A &\longmapsto \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned}$$

Then the quotient

$$\mathrm{U}(n)/\mathrm{U}(n-1)$$

is a homogeneous manifold and the projection

$$\mathrm{U}(n) \longrightarrow \mathrm{U}(n)/\mathrm{U}(n-1)$$

is a fibration with fiber $\mathrm{U}(n-1)$. We observe that the map

$$\begin{aligned} \varphi : \mathrm{U}(n)/\mathrm{U}(n-1) &\longrightarrow S^{2n-1} \\ A \cdot \mathrm{U}(n-1) = [A] &\longmapsto Ae_0 \end{aligned}$$

is a diffeomorphism, where $e_0 := (0, \ldots, 0, 1) \in \mathbf{R}^{2n} \approx \mathbf{C}^n$. Using this diffeomorphism we get an exact homotopy sequence (see [105], p.90)

$$\cdots \to \pi_2(S^{2n-1}, [\mathrm{Id}]) \to \pi_1(\mathrm{U}(n-1),\mathrm{Id}) \xrightarrow{i_*} \pi_1(\mathrm{U}(n),\mathrm{Id}) \to \pi_1(S^{2n-1}, [\mathrm{Id}]) \to \cdots .$$

So

$$i_* : \pi_1(\mathrm{U}(n-1),\mathrm{Id}) \longrightarrow \pi_1(\mathrm{U}(n),\mathrm{Id})$$

is an isomorphism for $n \geq 2$.

Now $\mathrm{U}(1) \approx S^1$ and $\pi_1(\mathrm{U}(1),\mathrm{Id}) \approx \mathbf{Z}$ generated by $\{t \longmapsto (e^{2\pi i t})\}$. Hence $\pi_1(\mathrm{U}(n),\mathrm{Id}) \approx \mathbf{Z}$ generated by

$$\left\{ t \longmapsto \begin{pmatrix} e^{2\pi i t} & 0 \\ 0 & \mathrm{Id} \end{pmatrix} \right\}.$$

□

Consider the following sequence

$$\pi_1(\mathrm{GL}(\mathbf{C}^n), \mathrm{Id}) \xrightarrow{\pi_*} \pi_1(\mathrm{GL}(\mathbf{C}^n)/\mathrm{GL}(\mathbf{R}^n), [\mathrm{Id}]) \xrightarrow{b} \{-1, 1\}$$

where π_* is the homomorphism induced by the quotient map

$$\pi : \mathrm{GL}(\mathbf{C}^n) \to \mathrm{GL}(\mathbf{C}^n)/\mathrm{GL}(\mathbf{R}^n).$$

The homomorphism b is defined as follows. Let ϕ be a loop in $\mathrm{GL}(\mathbf{C}^n)/\mathrm{GL}(\mathbf{R}^n)$ with base point [Id]. We take a continuous lift $\tilde{\phi} : [0, 1] \to \mathrm{GL}(\mathbf{C}^n)$ with

$$\begin{aligned} \tilde{\phi}(0) &= \mathrm{Id} \\ (\pi \circ \tilde{\phi})(t) &= \phi(t) \quad , \quad t \in [0, 1]. \end{aligned}$$

We must have $\tilde{\phi}(1) \in \mathrm{GL}(\mathbf{R}^n)$ and define

$$b([\phi]) := \operatorname{sign} \det(\tilde{\phi}(1))$$

If we choose a different lift $\hat{\phi} : [0, 1] \longrightarrow \mathrm{GL}(\mathbf{C}^n)$ of ϕ with $\hat{\phi}(0) = \mathrm{Id}$, then there is a continuous path $A : [0, 1] \longrightarrow \mathrm{GL}(\mathbf{R}^n)$ with $A(0) = \mathrm{Id}$ and $\tilde{\phi}(t) = \hat{\phi}(t) \cdot A(t)$. Then $\operatorname{sign} \det(\tilde{\phi}(1))$ does not depend on the particular choice of $\tilde{\phi}$ because of $\det A(1) > 0$. Moreover if we take a loop ψ in $\mathrm{GL}(\mathbf{C}^n)/\mathrm{GL}(\mathbf{R}^n)$ with base point [Id] homotopic to ϕ then we can lift the homotopy to $\mathrm{GL}(\mathbf{C}^n)$. This gives us a continuous path in $\mathrm{GL}(\mathbf{R}^n)$ connecting $\tilde{\phi}(1)$ and $\tilde{\psi}(1)$ so

$$\operatorname{sign} \det(\tilde{\phi}(1)) = \operatorname{sign} \det(\tilde{\psi}(1))$$

and b is well defined. It is an easy exercise to show that b is a group homomorphism.

Proposition A.3.2 *The sequence*

$$0 \to \pi_1(\mathrm{GL}(\mathbf{C}^n), Id) \stackrel{\pi_*}{\to} \pi_1(\mathrm{GL}(\mathbf{C}^n)/\mathrm{GL}(\mathbf{R}^n), [Id]) \stackrel{b}{\to} \{-1, 1\} \to 0$$

is exact.

Proof Assume $\pi_*(\{\tilde{\phi}\}) = 0$, where $\{\tilde{\phi}\}$ denotes the homotopy class of some loop $\tilde{\phi}$ in $\mathrm{GL}(\mathbf{C}^n)$ passing through Id. We have to show that $\{\tilde{\phi}\} = 0$. There is a homotopy

$$H : [0, 1] \times [0, 1] \longrightarrow \mathrm{GL}(\mathbf{C}^n)/\mathrm{GL}(\mathbf{R}^n)$$

connecting the loop $[\tilde{\phi}]$ in $\mathrm{GL}(\mathbf{C}^n)/\mathrm{GL}(\mathbf{R}^n)$ to the constant loop [Id]:

$$\begin{aligned} H(0, t) = \pi \circ \tilde{\phi} &= [\tilde{\phi}] \ ; \ t \in [0, 1] \\ H(s, 0) = H(s, 1) &= [\mathrm{Id}] \\ H(1, t) = [\mathrm{Id}] &\ ; \ t \in [0, 1] \end{aligned}$$

We can lift H to a homotopy $\tilde{H}$ on $\mathrm{GL}(\mathbf{C}^n)$:

$$\begin{aligned} \tilde{H} : [0, 1] \times [0, 1] &\to \mathrm{GL}(\mathbf{C}^n) \\ \tilde{H}(0, t) &= \tilde{\phi}(t) \ ; \ t \in [0, 1] \\ \tilde{H}(s, 0), \tilde{H}(s, 1), \tilde{H}(1, t) &\in \mathrm{GL}(\mathbf{R}^n) \ \ s, t \in [0, 1] \end{aligned}$$

Define $j(s) := \tilde{H}(s, 0)^{-1}$. Then $j(0) = \tilde{\phi}(0)^{-1} = \mathrm{Id}$. Replacing $\tilde{H}$ by

$$(s, t) \longmapsto \tilde{H}(s, t) j(s)$$

and using that $\tilde{H}(0, 0) = \tilde{\phi}(0) = \mathrm{Id}$, we may assume without loss of generality that $\tilde{H}(s, 0) = \mathrm{Id}$. Now define a homotopy

$$\tilde{G} : [0, 1] \times [0, 1] \longrightarrow \mathrm{GL}(\mathbf{C}^n)$$

by

$$(\sigma, t) \longmapsto \begin{cases} \tilde{H}(1 + 2t(\sigma - 1), 2t\sigma) & : \ 0 \le t \le \frac{1}{2} \\ \tilde{H}(2\sigma(1 - t), 1 + 2(\sigma - 1)(1 - t)) & : \ \frac{1}{2} \le t \le 1 \end{cases}$$

Then

$$\tilde{G}(0, t) = \begin{cases} \tilde{H}(1 - 2t, 0) = \mathrm{Id} & : \ 0 \le t \le \frac{1}{2} \\ \tilde{H}(0, 2t - 1) = \tilde{\phi}(2t - 1) & : \ \frac{1}{2} \le t \le 1 \end{cases}$$

is homotopic to $\tilde{\phi}$ and

$$\tilde{G}(1,t) \;=\; \begin{cases} \tilde{H}(1,2t) & : \; 0 \le t \le \frac{1}{2} \\ \tilde{H}(2(1-t),1) & : \; \frac{1}{2} \le t \le 1 \end{cases}$$

is a loop in $\mathrm{GL}(\mathbf{R}^n)$ passing through Id. Observe that

$$\begin{aligned} \tilde{G}(\sigma,0) = \tilde{H}(1,0) \;&=\; \mathrm{Id} \\ \tilde{G}(\sigma,1) = \tilde{H}(0,1) \;&=\; \phi(1) \;=\; \mathrm{Id}. \end{aligned}$$

If we can show that $\tilde{G}(1,\,.\,)$ is a contractible loop in $\mathrm{GL}(\mathbf{C}^n)$ then we are done. Because of Lemma A.3.1 we can homotope $\tilde{G}(1,\,.\,)$ to a loop $\hat{\psi}$ in $\mathrm{O}(n)\subset\mathrm{U}(n)$ passing through Id. Again by Lemma A.3.1 we find a homotopy

$$H \,:\, [0,1]\times[0,1] \longrightarrow \mathrm{U}(n)$$

with

$$\begin{aligned} H(0,t) &= \hat{\psi}(t) \\ H(1,t) &= \begin{pmatrix} e^{2\pi i k t} & 0 \\ 0 & \mathrm{Id} \end{pmatrix} \;\; ; \; k \in \mathbf{Z} \end{aligned}$$

and

$$H(s,0) \;=\; H(s,1) \;=\; \mathrm{Id}.$$

But then $h := \det \circ H : [0,1]\times[0,1] \longrightarrow S^1$ would be a homotopy between the constant loop

$$h(0,t) \;=\; \det \hat{\psi}(t) \;=\; \pm 1$$

and the loop

$$h(1,t) \;=\; e^{2\pi i k t}$$

which is only possible for $k = 0$, so $t \longmapsto \tilde{\psi}(t)$ is homotopic to the constant loop Id in $\mathrm{GL}(\mathbf{C}^n)$. Hence π_* is injective. Moreover b is surjective since obviously

$$b\left(\left\{\, t \longmapsto \begin{pmatrix} e^{\pi i t} & 0 \\ 0 & \mathrm{Id} \end{pmatrix}\right\}\right) \;=\; -1$$

Next assume $b\big(\,\{[\phi]\}\,\big) = 1$. Then $[\phi]$ is represented by an arc

$$\phi : [0,1] \to \mathrm{GL}(\mathbf{C}^n)$$

with $\phi(0) = \mathrm{Id}$, $\phi(1) \in \mathrm{GL}(\mathbf{R}^n)$ and $\det(\phi(1)) > 0$. Now we homotope ϕ to a loop $\hat{\phi}$ in $\mathrm{GL}(\mathbf{C}^n)$ with base point Id so that the end points of the arcs during the homotopy remain in $\mathrm{GL}(\mathbf{R}^n)$. Indeed take a path Γ in $\mathrm{GL}(\mathbf{R}^n)$ connecting $\phi(1)$ and Id. We get the desired homotopy by running along ϕ and then following some piece of Γ. Hence $\{[\hat{\phi}]\} = \{[\phi]\}$ and $\pi_*(\{\hat{\phi}\}) = \{[\phi]\}$. Obviously $b \circ \pi_* \equiv 1$ which completes the proof of the proposition. □

Combining the statements of Proposition A.3.2 and Lemma A.3.1, we obtain the following result.

Corollary A.3.3 *The following sequence is exact*

$$0 \to \mathbf{Z} \xrightarrow{\pi_*} \pi_1(\mathcal{T}(n), \mathbf{R}^n) \xrightarrow{b} \{-1, 1\} \to 0,$$

where

$$\left\{t \longmapsto e^{2\pi i t}\mathbf{R} \oplus \mathbf{R}^{n-1}\right\} = \pi_*(1)$$

$$b\left(\left\{t \longmapsto e^{\pi i t}\mathbf{R} \oplus \mathbf{R}^{n-1}\right\}\right) = -1$$

and $t \in [0, 1]$. □

Theorem A.3.4 *The fundamental group* $\pi_1(\mathcal{T}(n), \mathbf{R}^n)$ *is isomorphic to* $\mathbf{Z}$ *and generated by the loop*

$$\alpha : [0, 1] \longrightarrow \mathcal{T}(n)$$

$$t \longmapsto e^{\pi i t}\mathbf{R} \oplus \mathbf{R}^{n-1}.$$

Proof We have to show that the group $\pi_1(\mathcal{T}(n), \mathbf{R}^n)$ is generated by α. Then we are done because there is no number $k \neq 0$ that satisfies $\alpha^k = 1$. Let $\beta \in \pi_1(\mathcal{T}(n), \mathbf{R}^n)$. In the case $b(\beta) = 1$, i.e. $\beta \in \ker b = \mathrm{Im}(\pi_*)$, we have for some integer k

$$\{t \longmapsto e^{2\pi k t}\mathbf{R} \oplus \mathbf{R}^{n-1}\} = \beta,$$

so that $\beta = \alpha^{2k}$. In the case $b(\beta) = -1$ we have $b(\alpha\beta) = 1$ and as before $\alpha\beta = \alpha^{2k}$ for some k, i.e. $\beta = \alpha^{2k-1}$. □

Definition A.3.5 The class of α in $\pi_1(\mathcal{T}(n), \mathbf{R}^n)$ is called the *Maslov class*. If ϕ is a loop in $\mathcal{T}(n)$ with base point $\mathbf{R}^n$ then there is a number $\mu_n(\phi)$ so that $\{\phi\} = \alpha^{\mu_n(\phi)}$. This number is called the *Maslov index* of ϕ.

We remark that we have a natural inclusion

$$\begin{aligned} \mathcal{T}(n) \times \mathcal{T}(m) &\subset \mathcal{T}(n+m) \\ (L, K) &\longmapsto L \oplus K \end{aligned}$$

for $(m, n) \in \mathbf{N}^2$. Given $L \in \pi_1(\mathcal{T}(m), \mathbf{R}^m)$ and $K \in \pi_1(\mathcal{T}(n), \mathbf{R}^n)$ we obtain an element

$$L \oplus K \in \pi_1(\mathcal{T}(m+n), \mathbf{R}^{m+n}).$$

Since we may assume without loss of generality that

$$L(t) = e^{\pi i l t}\mathbf{R} \oplus \mathbf{R}^{m-1}$$
$$K(t) = e^{\pi i k t}\mathbf{R} \oplus \mathbf{R}^{n-1}$$

we have

$$(L \oplus K)(t) = e^{\pi i l t}\mathbf{R} \oplus \mathbf{R}^{m-1} \oplus e^{\pi i k t}\mathbf{R} \oplus \mathbf{R}^{n-1}.$$

Proposition A.3.6 *With the notation above we have*

$$\mu_{m+n}(L \oplus K) \;=\; \mu_m(L) \;+\; \mu_n(K).$$

Proof We only have to verify that the loops

$$t \longmapsto e^{\pi i l t}\mathbf{R} \oplus \mathbf{R}^{m-1} \oplus e^{\pi i k t}\mathbf{R} \oplus \mathbf{R}^{n-1}$$

and

$$t \longmapsto e^{\pi i (l+k) t}\mathbf{R} \oplus \mathbf{R}^{n+m-1}$$

represent the same element in $\pi_1(\mathcal{T}(n), \mathbf{R}^n)$. This is the case if we can homotope the matrix

$$\begin{pmatrix} e^{\pi i l t} & 0 \\ 0 & 1 \end{pmatrix}$$

into

$$\begin{pmatrix} 1 & 0 \\ 0 & e^{\pi i l t} \end{pmatrix}$$

in $\mathrm{GL}(\mathbf{C}^2)$. We may do this via

$$\tau \longmapsto \begin{pmatrix} \cos\tau & \sin\tau \\ -\sin\tau & \cos\tau \end{pmatrix} \begin{pmatrix} e^{\pi i l t} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \cos\tau & \sin\tau \\ -\sin\tau & \cos\tau \end{pmatrix}^{-1},$$

where $0 \leq \tau \leq \pi/2$. □

A.4 Fredholm Theory for Cauchy Riemann Operators on the Disk

Let $z \longmapsto L(z)$, $z \in \partial D$ be a smooth loop of totally real subspaces in $\mathbf{C}^n$, i.e.

$$L(z) \cap iL(z) = \{0\}$$
$$\dim_{\mathbf{R}} L(z) = n.$$

For $2 < p < \infty$ we denote by V_L^p the Banach space consisting of all maps $u \in W^{1,p}(D, \mathbf{C}^n)$ satisfying

$$u(z) \in L(z) \qquad \text{for } z \in \partial D.$$

We note that this definition makes sense because V_L^p can be embedded into $C^0(\bar{D}, \mathbf{C}^n)$ if $p > 2$. Assume that $A \in L^\infty(D, \mathcal{L}_{\mathbf{R}}(\mathbf{C}^n))$, where $\mathcal{L}_{\mathbf{R}}(\mathbf{C}^n)$ denotes the set of real linear maps from $\mathbf{C}^n$ into $\mathbf{C}^n$. We define a continuous linear operator

$$T : V_L^p \longrightarrow L^p(D, \mathbf{C}^n)$$

by

$$Tu = u_s + iu_t + Au = \bar{\partial} u + Au. \tag{A.13}$$

Our main result is the following:

Theorem A.4.1 *The operator T is Fredholm and its index is given by*

$$ind(T) = \mu_n(L) + n.$$

Let us recall the following notions: A bounded linear operator between two Banach spaces is called a *Fredholm operator* if its kernel and its cokernel are finite dimensional; the dimension of the kernel minus the dimension of the cokernel is then called the *index* of the Fredholm operator. The range of a Fredholm operator is always closed. If we only demand that the range is closed and the kernel is finite dimensional then the operator is called a *Semi-Fredholm operator*. It is a standard fact that the index of a continuous family of Fredholm operators remains constant while the dimensions of the kernels and cokernels in general do not.

The proof of Theorem A.4.1 consists of several parts. The main ingredients are the following estimates which we proved for the case $p = 2$. For the case $p > 2$ see [99] or [106].

Theorem A.4.2 *Given $p \in (2, \infty)$. Then there exists a constant $c > 0$ such that*

$$\|u_s + iu_t\|_{L^p(D^+)} \geq c\|u\|_{W^{1,p}(D^+)} \tag{A.14}$$

for all smooth $u\colon D^+ \longrightarrow \mathbf{C}^n$ *with compact support in* D^+*, where* $D^+ \subset \mathbf{C}$ *is the upper half of some open disk with radius* ε*, and*

$$u((-\varepsilon, \varepsilon)) \subset \mathbf{R}^n.$$

There exists a constant $\tilde{c} > 0$ *such that*

$$\|u_s + iu_t\|_{L^p(D)} \geq \tilde{c}\|u\|_{W^{1,p}(D)} \tag{A.15}$$

for all smooth $u\colon D \longrightarrow \mathbf{C}^n$ *with compact support in* D *where* $D \subset \mathbf{C}$ *is some open disk.* □

We will first show that our operator T is semi–Fredholm. The main tool is the following proposition, which is often called Weyl's lemma.

Proposition A.4.3 *Let* X, Y, Z *be Banach spaces. Assume* $T : X \to Y$ *is a continuous linear operator and let* $C : X \to Z$ *be a compact operator. Assume there exists a positive constant* c*, so that*

$$\|x\|_X \leq c(\|Tx\|_Y + \|Cx\|_Z)$$

for all $x \in X$*. Then* T *is semi–Fredholm.*

Proof If $x \in \ker T$ then we have by assumption

$$\|x\|_X \leq c\|Cx\|_Z. \tag{A.16}$$

We will show that every bounded sequence in $\ker T$ has a convergent subsequence. The only Banach spaces with this property are finite dimensional spaces. This would prove that $\ker T$ is finite dimensional.

Let $(x_k)_{k\in\mathbf{N}}$ be any bounded sequence in $\ker T$. Then the sequence $(Cx_k)_{k\in\mathbf{N}}$ has a convergent subsequence (which we denote again by (Cx_k)), since C is a compact operator. Inequality (A.16) implies that the sequence $(x_k)_{k\in\mathbf{N}}$ is a Cauchy sequence and therefore converges.

Now consider $\tilde{X} := X/\ker T$ which is a Banach space because $\ker T$ is closed and X is Banach. Denote the induced operator by $\tilde{T} : \tilde{X} \to Y$. If we can show that there is a positive constant d so that

$$\|\tilde{x}\|_{\tilde{X}} \leq d\|\tilde{T}\tilde{x}\|_Y$$

for all $\tilde{x} \in \tilde{X}$ then the range $R(\tilde{T})$ of $\tilde{T}$ is closed. Since $R(T) = R(\tilde{T})$ this would imply that T is semi–Fredholm.

Arguing indirectly, we assume that there is a sequence $(\tilde{x}_k)_{k\in\mathbf{N}} \subset \tilde{X}$ with $\|\tilde{x}_k\|_{\tilde{X}} = 1$ and

$$\|\tilde{T}\tilde{x}_k\|_Y \longrightarrow 0$$

as $k \to \infty$. Recalling that $\tilde{x}_k = x_k + \ker T$ for some $x_k \in X$ and

$$\|\tilde{x}_k\|_{\tilde{X}} = \inf_{v \in \ker T} \|x_k - v\|_X = \operatorname{dist}(x_k, \ker T),$$

we pick sequences $\varepsilon_k \searrow 0$ and $x_k \in \tilde{x}_k$ so that

$$1 \le \|x_k\|_X \le 1 + \varepsilon_k.$$

Since C is a compact operator we may assume that Cx_k converges in Z after passing to a suitable subsequence. Then

$$\begin{aligned} \|x_k - x_l\|_X &\le c\|T(x_k - x_l)\|_Y + c\|C(x_k - x_l)\|_Z \\ &\le c\|\tilde{T}\tilde{x}_k\|_Y + c\|\tilde{T}\tilde{x}_l\|_Y + c\|Cx_k - Cx_l\|_Z \\ &\to 0 \end{aligned}$$

as $k, l \to \infty$, hence the sequence x_k converges to some $x \in X$ and $Tx = 0$. This implies $\|\tilde{x}\|_{\tilde{X}} = 0$ in contradiction to $\|\tilde{x}\|_{\tilde{X}} = 1$. □

We will apply this result to the Cauchy Riemann type operator T in (A.13).

Proposition A.4.4 *The operator T from Theorem A.4.1 satisfies the following estimate: There exists a constant $c > 0$ such that*

$$\|u\|_{W^{1,p}(D)} \le c\,(\|Tu\|_{L^p(D)} + \|u\|_{L^p(D)}) \tag{A.17}$$

for all $u \in V_L^p$. Moreover, T is semi–Fredholm.

Before we start with the proof of Proposition A.4.4 we will simplify the situation. Since

$$\|Au\|_{L^p(D)} \le d\|u\|_{L^p(D)}$$

for some constant $d > 0$, it follows that

$$\|Tu\|_{L^p(D)} + d\|u\|_{L^p(D)} \ge \|\bar{\partial}u\|_{L^p(D)}.$$

Hence it suffices to prove Proposition A.4.4 for $T = \bar{\partial}$.

Proof of Proposition A.4.4 We will prove the estimate for smooth functions only. The general estimate then follows easily because smooth functions are dense in $W^{1,p}(D)$ and T is continuous. We leave this as an exercise for the reader. If u has compact support in D we have nothing to do, because of the L^p estimate for the Cauchy Riemann operator. We have to choose a suitable open covering of the (closed) disk D, take a subordinate partition of unity and prove the estimate for the separate 'pieces' of u. The situation near boundary points $z_0 \in \partial D$ requires a special treatment.

Consider the conformal map

$$\psi_\varepsilon : H^+ \longrightarrow D\backslash\{-z_0\}\ ,\ H^+ := \{s+it \in \mathbf{C}\,|\,t \geq 0\}$$

$$\psi_\varepsilon(z) := \frac{i-\varepsilon z}{i+\varepsilon z},$$

which maps an upper half disk D^+ centered at 0 onto a small neighborhood $U(z_0) \subset D$ of z_0. Assume, u has compact support in $U(z_0)$, then $u \circ \psi_\varepsilon$ has compact support in D^+ and $(u \circ \psi_\varepsilon)(s,0) \in L(\psi_\varepsilon(s,0))$. Since $L(\psi_\varepsilon(s,0)) \subset \mathbf{C}^n$ is totally real, we can find a smooth map

$$A : D^+ \longrightarrow \mathrm{GL}(\mathbf{C}^n)$$

so that

$$A(s,0)\, L(\psi_\varepsilon(s,0)) = \mathbf{R}^n.$$

Consider now the map

$$v : D^+ \longrightarrow \mathbf{C}^n$$

$$v(s,t) := A(s,t)(u \circ \psi_\varepsilon)(s,t),$$

which satisfies the boundary L^p–estimate (A.14). Writing $(\psi_\varepsilon^1, \psi_\varepsilon^2)$ for the components of ψ_ε and using $\overline{\partial}\psi_\varepsilon = 0$, we obtain

$$\overline{\partial}(u \circ \psi_\varepsilon) = (\partial_s \psi_\varepsilon^1 - i\,\partial_s \psi_\varepsilon^2)\overline{\partial} u(\psi_\varepsilon).$$

We then estimate using $Ai = iA$

$$\begin{aligned}
\|u\|_{W^{1,p}(U(z_0))} &\leq c\,\|v\|_{W^{1,p}(D^+)} \\
&\leq c\|\overline{\partial} v\|_{L^p(D^+)} \\
&= c\,\|\overline{\partial} A \circ (u \circ \psi_\varepsilon) + (\partial_s \psi_\varepsilon^1 - i\,\partial_s \psi_\varepsilon^2) A \circ \overline{\partial} u \circ \psi_\varepsilon\|_{L^p(D^+)} \\
&\leq c(\|u\|_{L^p(U(z_0))} + \|\overline{\partial} u\|_{L^p(U(z_0))}). \qquad \text{(A.18)}
\end{aligned}$$

Remark If the spaces $L(\psi_\varepsilon(s,0))$ are not totally real then A is not complex linear and $\overline{\partial}$ transforms into a different differential operator when applied to u.

We now cover D by a ball $B_\varepsilon \subset \overset{\circ}{D}$ and the sets $U(z_0)$, $z_0 \in \partial D$, so that we have the estimates (A.15) and (A.18) respectively. Denote this cover by $U_1, \ldots, U_m$ and choose a subordinate partition of unity $\beta_1, \ldots, \beta_m$. We observe that the norms $\|u\|_{L^p(D)}$ and $\sum_{i=1}^m \|\beta_i u\|_{L^p(U_i)}$ are equivalent. The same applies to the

corresponding Sobolev norms of type $W^{1,p}$. We remark at this point that we used the Banach algebra structure of $W^{1,p}$ for $p > 2$ in order to prove equivalence of these two norms.

Hence we compute, where $d > 0$ is a generic constant

$$\begin{aligned}
\|\bar{\partial}u\|_{L^p(D)} &\geq d \sum_{i=1}^{m} \|\beta_i \bar{\partial}u\|_{L^p(U_i)} \\
&= d \sum_{i=1}^{m} \|\bar{\partial}(\beta_i u) - (\bar{\partial}\beta_i)u\|_{L^p(U_i)} \\
&\geq d \sum_{i=1}^{m} \left(\|\bar{\partial}(\beta_i u)\|_{L^p(U_i)} - \|(\bar{\partial}\beta_i)u\|_{L^p(U_i)} \right) \\
&\geq d \left(\sum_{i=1}^{m} \|\beta_i u\|_{W^{1,p}(U_i)} \right) - d\|u\|_{L^p(D)} \\
&\geq d\|u\|_{W^{1,p}(D)} - d\|u\|_{L^p(D)}.
\end{aligned}$$

This completes the proof of inequality (A.17). Since the embedding $W^{1,p}$ into L^p is compact, Proposition A.4.4 finally follows from Proposition A.4.3. □

Summarizing the discussion above, we know now that all the operators $T : V_L^p \to L^p$ with $T = \bar{\partial} + A$ are semi-Fredholm.

Next we would like to find a continuous homotopy of semi-Fredholm operators connecting T with some 'simpler' operator T_0 where we can show that it is Fredholm. Then we want to conclude that T must be Fredholm as well. For this reason we will need the following result (see T. Kato's book [76]).

Proposition A.4.5 *Let E, F be Banach spaces and $\phi : [0,1] \to \mathcal{L}(E,F)$ a continuous path of bounded linear operators such that every $\phi(s)$ is a semi-Fredholm operator. If $\phi(0)$ is a Fredholm operator then every $\phi(s)$ is Fredholm and*

$$ind(\phi(s)) = ind(\phi(0))$$

for all $s \in [0,1]$, where 'ind' denotes the Fredholm index.

Proof Let

$$s_0 := \sup\{s \in [0,1] \mid \phi(\tau) \in \mathrm{Fred}(E,F) \text{ for all } \tau \in [0,s)\}.$$

We have $s_0 > 0$ because the set of Fredholm operators in $\mathcal{L}(E,F)$ is open (which we will not prove here) and the path ϕ is continuous. If we can show that $\phi(s_0)$ is Fredholm as well then we are done. Indeed, if $\phi(s_0)$ were Fredholm and $s_0 < 1$ then $\phi(s_0 + t)$ would have to be Fredholm for small $t > 0$ in contradiction to the

definition of s_0. By the continuity property of the index the map

$$\begin{aligned} [0, s_0) &\longrightarrow \mathbf{Z} \\ s &\longmapsto \operatorname{ind}(\phi(s)) \end{aligned}$$

would then be constant. Let $S := \phi(s_0)$. By assumption $\ker(S)$ is finite dimensional. Fix any topological complement X of $\ker(S)$ in E, i.e. $E = \ker(S) \oplus X$ and X is closed so that the projections onto $\ker(S)$ and X are continuous linear maps. We claim that the operators $\phi(s)|_X : X \to F$ are injective as well, provided $|s - s_0|$ is sufficiently small.

The operator $S|_X : X \to R(S)$ is a bijective bounded linear operator between Banach spaces, hence $(S|_X)^{-1}$ is also continuous by the open mapping theorem. Therefore there exists a positive constant c so that

$$\|Sx\|_F \geq c \, \|x\|_E \tag{A.19}$$

for all $x \in X$. If $|s - s_0| < \varepsilon$ with $\varepsilon > 0$ sufficiently small then

$$\|\phi(s) - S\| \leq \frac{c}{2}.$$

Combining these two inequalities, we obtain for $x \in X$:

$$\begin{aligned} \|\phi(s)x\|_F &\geq \|Sx\|_F - \|(\phi(s) - S)x\|_F \\ &\geq c\|x\|_E - \frac{c}{2}\|x\|_E \\ &= \frac{c}{2}\|x\|_E. \end{aligned}$$

proving the claim.

The operators $\phi(s)|_X$ are actually Fredholm if $s < s_0$ and $|s - s_0| < \varepsilon$ because the inclusion $i : X \hookrightarrow E$ is Fredholm (note that $\ker i$ is trivial and $\operatorname{coker} i = \ker(S)$) and

$$\operatorname{ind}(\phi(s)|_X) = \operatorname{ind}(\phi(s)) + \operatorname{ind}(i)$$

(we do not prove this here), which implies

$$\begin{aligned} \dim(\operatorname{coker}(\phi(s)|_X)) &= -\operatorname{ind}(\phi(s)|_X) \\ &= -\operatorname{ind}(\phi(s)) - \operatorname{ind}(i) \\ &= -\operatorname{ind}(\phi(s)) + \dim(\ker(S)). \end{aligned}$$

Hence the codimension of $R(\phi(s)|_X)$ remains constant for $s \in (s_0 - \varepsilon, s_0)$, say m. We claim that the codimension of $R(S) = R(S|_X)$ cannot exceed m either, showing

that S is Fredholm and hence completing the proof. Arguing indirectly, we pick an $(m+1)$–dimensional subspace Y of F so that $Y \cap R(S) = \{0\}$. Then $R(S) + Y$ is a closed subspace of F and we consider the operators

$$T_s : X \times Y \longrightarrow F$$

$$T_s(x, y) := \phi(s)x + y$$

for $|s - s_0| < \varepsilon$ and $s \in [0, 1]$. The operator

$$T_{s_0} : X \times Y \longrightarrow R(S) + Y$$

is bounded and bijective, hence its inverse is continuous (open mapping theorem). Therefore there exists a positive constant d, so that

$$\|T_{s_0}(x, y)\|_F \geq d(\|x\|_E + \|y\|_F)$$

for all $(x, y) \in X \times Y$. Then we have

$$\|T_s(x, y)\|_F \geq \frac{d}{2}(\|x\|_E + \|y\|_F)$$

if ε is sufficiently small, so the operators T_s are injective as well. Since $R(\phi(s)|_X)$ has codimension m and Y has dimension $m+1$, the two spaces must have a nontrivial intersection. But this contradicts the fact that T_s is injective. □

Let us return to our original operator T in (A.13). Using Proposition A.4.5, we may remove the zero order term Au by a homotopy without affecting the Fredholm property or the index of the operator. We would like to simplify the boundary condition L by a homotopy between L and a simpler loop L' having the same Maslov–index. The problem here is that we are dealing now with the operator $\overline{\partial}$ defined on different Banach spaces V_L^p. Proposition A.4.5 does not accommodate this situation directly, but there is a way to formulate the problem in the right context.

Proposition A.4.6 *Write* $\bar{\partial}_{L_t} : V_{L_t}^p \to L^p(D, \mathbf{C}^n)$ *for* $u \longmapsto u_s + iu_t$ *where* (L_t) , $0 \leq t \leq 1$ *is a continuous family of smooth loops* $L_t : \partial D \to \mathcal{T}(n)$ *of totally real subspaces in* $\mathbf{C}^n$.

Then $\bar{\partial}_{L_t}$ *is Fredholm if and only if* $\bar{\partial}_{L_0}$ *is Fredholm. Moreover, we have*

$$ind\, \bar{\partial}_{L_t} = ind\, \bar{\partial}_{L_0}.$$

Proof We have to get rid of the different domains of definition $V_{L_t}^p$ of the operators $\overline{\partial}_{L_t}$. Since the spaces $L_t(e^{2\pi is}) \subset \mathbf{C}^n$ are totally real, we may find matrices $A_t(s) \in \mathrm{GL}(\mathbf{C}^n)$ depending continuously on t and smoothly on $s \in [0, 1]$ so that

$$L_t(e^{2\pi is}) = A_t(s)L_0(e^{2\pi is})$$

and

$$A_0(s) \equiv \mathrm{Id}_{\mathbf{C}^n}.$$

Then $t \longmapsto A_t(1)^{-1}A_t(0)$ is a continuous path of matrices starting at the identity and mapping $L_0(1)$ onto itself. Since the general (real) linear group of $L_0(1)$ consists of two connected components, we may find a continuous family of smooth paths

$$\gamma_t : [0,1] \longrightarrow \mathrm{GL}(L_0(1))$$

with the properties

$$\gamma_t(0) = \mathrm{Id}\ ,\ \gamma_t(1) = A_t(1)^{-1}A_t(0)\ ,\ \gamma_0 \equiv \mathrm{Id}.$$

Since $L_0(1)$ is totally real, we may extend each $\gamma_t(s)$ to a map in $\mathrm{GL}(\mathbf{C}^n)$. We define now

$$B_t(s) := A_t(s)\gamma_t(s)$$

so that still $B_t(s)L_0(e^{2\pi is}) = L_t(e^{2\pi is})$ and in addition $B_t(0) = B_t(1)$, $B_0 \equiv \mathrm{Id}$. Hence B_t can be viewed as a contractible smooth loop in $\mathrm{GL}(\mathbf{C}^n)$ defined on ∂D. Therefore we have smooth extensions

$$\hat{B}_t : D \longrightarrow \mathrm{GL}(\mathbf{C}^n).$$

We define continuous linear isomorphisms as follows:

$$\Phi_t : V^p_{L_0} \longrightarrow V^p_{L_t}$$

$$(\Phi_t u)(z) := \hat{B}_t(z)u(z)$$

and

$$\Psi_t : L^p(D, \mathbf{C}^n) \longrightarrow L^p(D, \mathbf{C}^n)$$

$$(\Psi_t u)(z) := \hat{B}_t(z)u(z).$$

The operator $\overline{\partial} : V^p_{L_t} \to L^p(D, \mathbf{C}^n)$ is Fredholm if and only if the operator $\Psi_t^{-1} \circ \overline{\partial} \circ \Phi_t : V^p_{L_0} \to L^p(D, \mathbf{C}^n)$ is Fredholm and their indices are equal. We compute

$$(\Psi_t^{-1} \circ \overline{\partial} \circ \Phi_t)(u) = \overline{\partial}u + B_t^{-1}\overline{\partial}B_t \cdot u$$

and consider the homotopy

$$\overline{\partial} u + \tau B_t^{-1} \overline{\partial} B_t \cdot u : V_{L_0}^p \longrightarrow L^p(D, \mathbf{C}^n)$$

for $0 \leq \tau \leq 1$, which consists of semi–Fredholm operators by Proposition A.4.4. Using Proposition A.4.5 we may therefore drop the term $u \longmapsto B_t^{-1} \overline{\partial} B_t \cdot u$ without affecting the Fredholm property or the index, thus proving the proposition. □

Remark In our situation, we are removing the zero–order term

$$K(u) := B_t^{-1} \overline{\partial} B_t \cdot u.$$

We remark that K maps $W^{1,p}(D, \mathbf{C}^n)$ into itself. Composing with the compact embedding $W^{1,p}(D, \mathbf{C}^n) \hookrightarrow L^p(D, \mathbf{C}^n)$, we see that K is a compact operator and perturbing a Fredholm operator with a compact operator does not do any harm. Hence Proposition A.4.5 is much stronger than we actually needed. In situations where the domain is unbounded, Rellich's compactness theorem is false and zero order perturbations need not be compact anymore. In these cases Proposition A.4.5 is extremely useful.

We will write T_μ^d for the following Cauchy Riemann operator with "standard" boundary conditions

$$\begin{aligned} T_\mu^d : V_{L_\mu^d}^p &\longrightarrow L^p(D, \mathbf{C}^d) \\ T_\mu^d u &:= u_s + i u_t \end{aligned}$$

where

$$V_{L_\mu^d}^p := \{u \in W^{1,p}(D, \mathbf{C}^d) \mid u(z) \in L_\mu^d(z) \text{ for } z \in \partial D\}$$

and

$$L_\mu^d(e^{2\pi i t}) := e^{\pi i \mu t} \mathbf{R} \oplus \mathbf{R}^{d-1}$$

so that the Maslov–index of L_μ^d is given by

$$\mu_d(L_\mu^d) = \mu.$$

Consider the operator

$$\begin{aligned} T : V_L^p &\longrightarrow L^p \\ u &\longmapsto \bar{\partial} u + Au, \end{aligned}$$

where L has Maslov index $\mu_n(L) = k$. Define a homotopy by

$$\bar{\partial}_t := \bar{\partial} + tA \; : \; V_L^p \longrightarrow L^p \; ; \; 0 \leq t \leq 1.$$

Because of Proposition A.4.4 all these operators are semi-Fredholm. So we only need to discuss $\bar{\partial}$ with boundary condition L. Now we can find a homotopy in $\mathcal{T}(n)$ connecting L with L_k^n. By Proposition A.4.6 it is sufficient to investigate the operator T_k^n. We would like to compare the following operators:

1. $T_\mu^d \oplus T_\tau^l$ on
   $\left\{(u, v) \in W^{1,p}(D, \mathbf{C}^d \oplus \mathbf{C}^l) \mid (u, v)(z) \in (L_\mu^d \oplus L_\tau^l)(z) \text{ for all } z \in \partial D \right\}$
   and
2. $T_{\mu+\tau}^{d+l}$ on
   $\left\{w \in W^{1,p}(D, \mathbf{C}^{d+l}) \mid w(z) \in (L_{\mu+\tau}^{d+l})(z) \text{ for all } z \in \partial D \right\}$.

Both operators have image in $L^p(D, \mathbf{C}^{d+l})$. In Appendix A.3, Proposition A.3.6, we saw that $L_{\mu+\tau}^{d+l}$ and $L_\mu^d \oplus L_\tau^l$ are homotopic. Again by Proposition A.4.6 we only need to consider direct sums of Cauchy Riemann operators. Clearly, if two of the operators T_μ^d, T_τ^l and $T_\mu^d \oplus T_\tau^l$ are (semi)- Fredholm then it applies also to the third one. Moreover

$$\operatorname{ind}(T_\mu^d \oplus T_\tau^l) \; = \; \operatorname{ind}(T_\mu^d) + \operatorname{ind}(T_\tau^l)$$

if they are Fredholm. The following propositions show that T_0^1 and T_1^1 are Fredholm operators of index 1 and 2 respectively.

Proposition A.4.7 *T_0^1 is Fredholm and its index is one.*

Proof Clearly $\ker(T_0^1) = \mathbf{R}$, i.e. the constant real-valued maps. We know already that $R(T_0^1)$ is closed since T_0^1 is semi–Fredholm. If we can show that $R(T_0^1)$ is also dense then we are done. We will actually show that $R(T_0^1) \supset C_0^\infty(D, \mathbf{C})$, where $C_0^\infty(D, \mathbf{C})$ denotes the set of smooth complex valued functions on D with compact support in $\overset{\circ}{D}$; which is dense in $L^p(D, \mathbf{C})$. We start with some preparations:

Take any smooth compactly supported map $f : \mathbf{C} \longrightarrow \mathbf{C}$ which has the symmetry

$$f(z) \; = \; \overline{f(\bar{z})}.$$

By Theorem A.1.5 there exists $u \in C^\infty(S^2, \mathbf{C})$ with $u(\infty) = 0$ such that on $\mathbf{C}$

$$\bar{\partial} u \; = \; f.$$

Moreover, such a map u is uniquely determined by the condition $u(\infty) = 0$ by Liouville's theorem. We view here $\mathbf{C}$ as a subset of $S^2 = \mathbf{C} \cup \{\infty\}$. We would like

to show that such a solution u has to satisfy $u(\mathbf{R}) \subset \mathbf{R}$. Defining

$$v(z) \ := \ \overline{u(\bar{z})}$$

we compute with $z = s + it = (s, t)$

$$\begin{aligned}(\bar{\partial} v)(z) &= \overline{u_s(s,-t)} \ - \ i(\overline{u_t(s,-t)}) \\ &= \overline{(\bar{\partial} u)(\bar{z})} \\ &= \overline{f(\bar{z})} \\ &= f(z).\end{aligned}$$

Hence

$$\bar{\partial}(u - v) \ = \ 0$$

and also $(u - v)(\infty) = 0$. So that $u = v$. By definition of v this implies

$$u(\mathbf{R}) \subset \mathbf{R},$$

Now take the biholomorphic map

$$\phi : \ S^2 \longrightarrow S^2$$

$$\phi(z) := \begin{cases} i\frac{1+z}{1-z} : z \neq \ \infty, +1 \\ \ \ \infty \ \ : \ \ z = \ +1 \\ \ -i \ \ : \ \ z = \ \infty \end{cases}$$

mapping the closed unit disk D onto the extended upper half plane

$$\begin{aligned}\overline{H}^+ &:= H^+ \cup \{+\infty\} \\ &:= \{ \, z \in \mathbf{C} \ \mid \ \mathrm{Im}(z) \geq 0\} \cup \{+\infty\}.\end{aligned}$$

Then $u \circ \phi : \overset{\circ}{D} \longrightarrow \mathbf{C}$ is smooth and $(u \circ \phi)(\partial D) \subset \mathbf{R}$. We introduce $\phi = \alpha + i\beta$ and compute using $\overline{\partial}\phi = 0$ and $\overline{\partial} u = f$:

$$\begin{aligned}\bar{\partial}(u \circ \phi)(z) &= f(\phi(z)) \cdot (\alpha_s - i\beta_s)(z) \\ &= f(\phi(z)) \, \overline{\partial_s \phi(z)}.\end{aligned}$$

Observe that $\partial_s \phi(z)$ is nonzero everywhere.

Given any map $g \in C_0^\infty(D, \mathbf{C})$, we define now f on the upper half plane H^+ by

$$f(w) := \left(\overline{\partial_s \phi(\phi^{-1}(w))}\right)^{-1} g(\phi^{-1}(w)).$$

This map has compact support inside $\overset{\circ}{H}{}^+$ and therefore vanishes near $\mathbf{R}$. We extend it to $\mathbf{C}$ by the rule

$$f(\bar{z}) \;=\; \overline{f(z)}.$$

Starting with this function f, we may find now some $u \in C^\infty(S^2, \mathbf{C})$ with $u(\infty) = 0$, $u(\mathbf{R}) \subset \mathbf{R}$ and $\overline{\partial} u = f$. We have arranged the function f in such a way that

$$\overline{\partial}(u \circ \phi|_D) = g.$$

Since $u \circ \phi$ is a smooth map defined on S^2, we have $u \circ \phi|_D \in W^{1,p}(D, \mathbf{C})$. □

Proposition A.4.8 *T_1^1 is Fredholm and its index is* 2.

Proof It suffices to show that T_2^1 is Fredholm and $\operatorname{ind}(T_2^1) = 3$. Indeed, if T_2^1 is Fredholm then $T_2^1 \oplus T_0^1$ is also Fredholm. But $T_2^1 \oplus T_0^1$ and $T_1^1 \oplus T_1^1$ are homotopic, hence T_1^1 must be Fredholm as well and

$$\begin{aligned} 2\operatorname{ind}(T_1^1) &= \operatorname{ind}(T_1^1 \oplus T_1^1) \\ &= \operatorname{ind}(T_2^1 \oplus T_0^1) \\ &= \operatorname{ind}(T_2^1) + 1 \\ &= 4 \end{aligned}$$

Let $L(z) := z\,\mathbf{R}$, $z \in \partial D$, which is a loop in $\mathcal{T}(1)$ with Maslov–index 2. We consider

$$\bar{\partial} : V_L^p \longrightarrow L^p$$

and we observe that the set of smooth maps $f : D \to \mathbf{C}$, which vanish on a neighborhood $U = U(f)$ of 0 and near ∂D are dense in $L^p(D, \mathbf{C})$, $1 < p < \infty$. We know from the proof of Proposition A.4.7 that the Cauchy Riemann operator T_0^1 is onto. Take any smooth f as described above and observe that

$$g(z) := \frac{f(z)}{z}$$

defines a smooth map on D vanishing near 0 and near ∂D. Pick $u \in V^p_{L^1_0}$ with $\bar\partial u = g$ and let $v(z) := z\,u(z)$. Then

$$\begin{aligned}(\bar\partial v)(z) &= z\,(\bar\partial u)(z)\\ &= z\,g(z)\\ &= f(z).\end{aligned}$$

Moreover, $v(z) \in z\,\mathbf{R}$ for $z \in \partial D$. Hence

$$\bar\partial : V^p_L \longrightarrow L^p$$

is onto. Let $\bar\partial u = 0$ and $u \in V^p_L$. Then u is holomorphic and smooth up to the boundary by the results in Appendix A.2, hence u can be expanded in a power series

$$u(z) = \sum_{k=0}^{\infty} a_k z^k.$$

We claim that this series converges for all $z \in \bar D$ to $u(z)$. The argument is as follows: Because u is continuous on $\bar D$, the coefficients of the power series are given by

$$a_k = \frac{1}{2\pi i}\int_{\partial D}\frac{u(\zeta)}{\zeta^{k+1}}d\zeta = \frac{1}{2\pi i}\int_0^{2\pi} u(e^{i\phi})ie^{-ik\phi}d\phi$$

with integration over ∂D (not just a circle with smaller radius). Since $u|_{\partial D}$ is continuous, we may expand it in a Fourier series (identifying $\partial D \approx \mathbf{R}/2\pi\mathbf{Z}$):

$$u(e^{i\phi}) = \sum_{k=-\infty}^{\infty} c_k e^{ik\phi}\ ,\ 0 \le \phi \le 2\pi,$$

where the coefficients are given by

$$c_k = \frac{1}{2\pi}\int_0^{2\pi} u(e^{i\phi})e^{-ik\phi}d\phi$$

which are the same as the coefficients a_k of the power series above for $k \ge 0$. Since the Fourier series converges, the power series above also converges for $|z| = 1$ and they are actually the same, i.e. $c_k = 0$ whenever $k < 0$. We saw that the Fourier series

$$\sum_{k=0}^{\infty} a_k e^{ikt}$$

converges. Since for all $t \in [0, 2\pi]$

$$\sum_{k=0}^{\infty} a_k e^{ikt} \in e^{it}\,\mathbf{R}$$

we deduce

$$\sum_{k=0}^{\infty} a_k e^{i(k-1)t} \in \mathbf{R}$$

for all $t \in [0, 2\pi]$. Hence

$$\sum_{k=0}^{\infty} \bar{a}_k e^{i(1-k)t} \;=\; \sum_{k=0}^{\infty} a_k e^{i(k-1)t}.$$

Comparing the coefficients yields

$$\begin{aligned} a_0 &= \bar{a}_2, \\ a_1 &= \bar{a}_1, \\ a_3 = a_4 \;&=\; \ldots \;= 0. \end{aligned}$$

Hence the general form of an element in the kernel of $\bar{\partial} : V_L^p \to L^p$ is

$$u(z) \;=\; a_0 + a_1 z + \bar{a}_0 z^2$$

with $a_1 \in \mathbf{R}$ and $a_0 \in \mathbf{C}$. Consequently

$$\dim(\ker(\bar{\partial} : V_L^p \longrightarrow L^p)) \;=\; 3$$

and the index is also three. □

Proof of Theorem A.4.1 Consider T_k^n and assume first that $k > 0$. We have

$$T_k^n \;\simeq\; T_k^1 \oplus (T_0^1)^{n-1},$$

where $\simeq$ means 'homotopic'. So T_k^n is Fredholm if and only if T_k^1 is. On the other hand

$$\left(\bigoplus_{k-1} T_0^1\right) \oplus T_k^1 \simeq \bigoplus_{k} T_1^1.$$

Hence T_k^1 is Fredholm and we compute

$$\begin{aligned} \operatorname{ind}(T_k^n) &= \operatorname{ind}(T_k^1 \oplus (T_0^1)^{n-1}) \\ &= \operatorname{ind}(T_k^1) + (n-1)\operatorname{ind}(T_0^1) \\ &= \operatorname{ind}(T_k^1) + n - 1 \end{aligned} \tag{A.20}$$

and

$$\begin{aligned} 2k &= k \cdot \operatorname{ind}(T_1^1) \\ &= \operatorname{ind}\left(\bigoplus_k T_1^1\right) \\ &= \operatorname{ind}\left(\bigoplus_k T_0^1\right) + \operatorname{ind}(T_k^1) \\ &= k - 1 + \operatorname{ind}(T_k^1). \end{aligned} \tag{A.21}$$

From (A.21) we conclude that $\operatorname{ind}(T_k^1) = k + 1$ and (A.20) provides

$$\operatorname{ind}(T_k^n) = k + 1 + n - 1 = k + n = \mu_n(L_k^n) + n.$$

So the theorem is true for $k > 0$. Because of

$$T_0^n \simeq \bigoplus_n T_0^1$$

it is also true for $k = 0$. If $k < 0$ we have to consider

$$T_{-k}^n \oplus T_k^n \simeq T_0^n \oplus T_0^n.$$

This shows that T_k^n is Fredholm and

$$\begin{aligned} -k + n + \operatorname{ind}(T_k^n) &= \operatorname{ind}(T_{-k}^n) + \operatorname{ind}(T_k^n) \\ &= \operatorname{ind}(T_{-k}^n \oplus T_k^n) \\ &= \operatorname{ind}(T_0^n \oplus T_0^n) \\ &= 2\operatorname{ind}(T_0^n) \\ &= 2n. \end{aligned}$$

Hence $\operatorname{ind}(T_k^n) = n + k = \mu_n(L_k^n) + n$ and the proof of Theorem A.4.1 is complete. □

A.5 The Similarity Principle and Applications

For $2 < p < \infty$ we denote by $V^p = V^p_{L^n_0}$ the Banach space consisting of all $u \in W^{1,p}(D, \mathbf{C}^n)$ satisfying $u(\partial D) \subset \mathbf{R}^n$, where $D \subset \mathbf{C}$ is the unit disk and let

$$\bar\partial : V^p \to L^p$$

be the standard Cauchy Riemann operator

$$u \longmapsto \frac{\partial u}{\partial s} + i\frac{\partial u}{\partial t}.$$

By the results in Appendix A.4 we know that $\bar\partial : V^p \to L^p(D, \mathbf{C}^n)$ is a surjective Fredholm operator of index n with kernel being the constants in $\mathbf{R}^n$.

Theorem A.5.1 *Assume $A \in L^\infty(D, \mathcal{L}_{\mathbf{R}}(\mathbf{C}^n))$ and $w \in W^{1,p}_{loc}(\overset{\circ}{D}, \mathbf{C}^n)$, where $2 < p < \infty$. Let w be a solution of*

$$\begin{aligned} \bar\partial w + Aw &= 0 \qquad \text{in} \qquad \overset{\circ}{D} \\ w(0) &= 0. \end{aligned}$$

Then there exists

$$\Phi \in \bigcap_{2<q<\infty} W^{1,q}(D, \mathcal{L}_{\mathbf{C}}(\mathbf{C}^n))$$

with

$$\Phi(0) = \mathrm{Id}\ , \quad \Phi(z) \in \mathrm{GL}(\mathbf{C}^n)$$

and a map $f : D \to \mathbf{C}^n$ with $f(0) = 0$ such that for $z \in D$

$$w(z) = \Phi(z) f(z),$$

and for some $0 < \varepsilon \leq 1$, a sufficiently small number and $D_\varepsilon \subset \mathbf{C}$ the disk of radius ε, f is holomorphic on D_ε.

Before we start with the actual proof, we point out some facts as a preparation: Denote by χ_ε the characteristic function of the closed ε-disk $D_\varepsilon \subset D$, $0 < \varepsilon \leq 1$. The Sobolev embedding result provides the estimate

$$\|u\|_{C^0(D)} \leq c_p \|u\|_{W^{1,p}(D)} \quad \text{for } p > 2. \tag{A.22}$$

We compute for $p > 2$, $u \in V^p$ and $B \in L^\infty(D, \mathcal{L}_{\mathbf{R}}(\mathbf{C}^n))$

$$\begin{aligned} \|\chi_\varepsilon Bu\|_{L^p(D)} &\leq \|B\|_{L^\infty(D)} \|u\|_{C^0(D)} \|\chi_\varepsilon\|_{L^p(D)} \\ &\leq c\varepsilon^{2/p} \|u\|_{W^{1,p}(D)} \end{aligned} \tag{A.23}$$

for a suitable constant $c = c(p, B) > 0$ not depending on ε.

Consider the operator $T_0 : V^p \longrightarrow L^p(D, \mathbf{C}^n) \times \mathbf{R}^n$ for $2 < p < \infty$ defined by

$$T_0 u := (\bar{\partial} u, u(1)).$$

It follows from the proof of Proposition A.4.7 that the operator T_0 is bijective. We estimate with (A.22)

$$\begin{aligned} \|\bar{\partial} u\|_{L^p(D)} + \mid u(1) \mid &\leq \|u\|_{W^{1,p}(D)} + \|u\|_{C^0(D)} \\ &\leq C(p) \cdot \|u\|_{W^{1,p}(D)}. \end{aligned}$$

Hence T_0 is a continuous bijective linear operator and T_0^{-1} is also continuous because of the open mapping theorem. Define for $\varepsilon \in (0, 1)$

$$T_\varepsilon : V^p \longrightarrow L^p(D, \mathbf{C}^n) \times \mathbf{R}^n$$

by

$$T_\varepsilon u := (\bar{\partial} u + \chi_\varepsilon Bu, u(1)). \tag{A.24}$$

If $\varepsilon \in (0, 1)$ is small enough then T_ε is still a continuous linear isomorphism with T_ε^{-1} being continuous too. By elliptic regularity theory $T_\varepsilon : V^p \to L^p(D, \mathbf{C}^n) \times \mathbf{R}^n$ is an isomorphism for all $p > 2$ provided it is for at least one p (see Appendix A.2). This means that the same $\varepsilon > 0$ is good for all $p > 2$.

Proof of Theorem A.5.1 Let us review first the idea of the proof. We would like to write w as Φf, where $\Phi(z) : \mathbf{C}^n \to \mathbf{C}^n$ is complex linear and f is holomorphic. Let us find out what kind of differential equation Φ has to satisfy. We view Φ as a map from the disk into $\mathcal{L}_{\mathbf{C}}(\mathbf{C}^n)$, the set of complex linear maps which we identify with $\mathbf{C}^{n^2}$. Using that $\overline{\partial} w + Aw = 0$ and inserting naively $w = \Phi f$ we obtain $0 = \overline{\partial}\Phi + A\Phi$. This would fit perfectly into our framework for Cauchy–Riemann type operators if $\Phi \longmapsto A\Phi$ was a linear map from $\mathbf{C}^{n^2}$ into $\mathbf{C}^{n^2}$, i.e. if $A\Phi$ was complex linear. Hence we would like to replace the real linear map A by a complex linear map $\hat{A}$ so that still $\overline{\partial} w + \hat{A} w = 0$. After having done this, we will then solve the equation $\overline{\partial}\Phi + \hat{A}\Phi = 0$.

Denote the complex conjugation map by

$$\begin{aligned} \Gamma : \mathbf{C}^n &\longrightarrow \mathbf{C}^n \\ z &\longmapsto \bar{z}. \end{aligned}$$

We split $A(z) \in \mathcal{L}_{\mathbf{R}}(\mathbf{C}^n)$ into its complex-linear and complex-anti linear parts

$$A(z) = A_l(z) + A_a(z),$$

where

$$A_l(z) := \frac{1}{2}(A(z) - iA(z)i),$$
$$A_a(z) := \frac{1}{2}(A(z) + iA(z)i).$$

We pick a map $C \in L^\infty(D, \mathcal{L}_{\mathbf{C}}(\mathbf{C}^n))$ satisfying

$$C(z)w(z) = \overline{w(z)} = \Gamma(w(z)),$$

where w is the map given by the assumptions of the theorem. We define

$$\hat{A}(z) = A_l(z) + A_a(z) \circ \Gamma \circ C(z)$$

and note that

$$\hat{A} \in L^\infty(D, \mathcal{L}_{\mathbf{C}}(\mathbf{C}^n)),$$

i.e. $\hat{A}(z)$ is complex linear. Moreover

$$\hat{A}(z)w(z) = A(z)w(z).$$

View $\mathcal{L}_{\mathbf{C}}(\mathbf{C}^n)$ as $\mathbf{C}^N$ with $N = n^2$ and $\mathcal{L}(\mathbf{R}^n) \subset \mathcal{L}_{\mathbf{C}}(\mathbf{C}^n)$ as $\mathbf{R}^N$. We consider now the operator

$$\begin{aligned} T_\varepsilon : V^p &\longrightarrow L^p \times \mathcal{L}(\mathbf{R}^n) \\ T_\varepsilon \Phi &= (\bar{\partial}\Phi + \chi_\varepsilon \hat{A}\Phi, \Phi(1)) \end{aligned}$$

with

$$V^p = \{\Phi \in W^{1,p}(D, \mathcal{L}_{\mathbf{C}}(\mathbf{C}^n)) \mid \Phi(\partial D) \subset \mathcal{L}(\mathbf{R}^n)\}$$
$$L^p = L^p(D, \mathcal{L}_{\mathbf{C}}(\mathbf{C}^n)).$$

If $\varepsilon > 0$ is small enough we find unique solutions $\tilde{\Phi}_\varepsilon$ of

$$\begin{aligned} \bar{\partial}\tilde{\Phi}_\varepsilon(z) + \chi_\varepsilon(z)\hat{A}(z)\tilde{\Phi}_\varepsilon(z) = 0 \quad \text{for } z \in \overset{\circ}{D}, \\ \tilde{\Phi}_\varepsilon(\partial D) \subset \mathcal{L}(\mathbf{R}^n), \\ \tilde{\Phi}_\varepsilon(1) = \mathrm{Id}\,. \end{aligned}$$

The solutions $\tilde{\Phi}_\varepsilon$ are contained in V^p for all $2 < p < \infty$ because T_ε is an isomorphism for all $p > 2$. Moreover

$$\begin{aligned}\|\tilde{\Phi}_\varepsilon - \mathrm{Id}\|_{C^0(D)} &\leq c_p \|\tilde{\Phi}_\varepsilon - \mathrm{Id}\|_{W^{1,p}(D)} \\ &\leq \tilde{c}_p \|T_\varepsilon \tilde{\Phi}_\varepsilon - T_\varepsilon(\mathrm{Id})\|_{L^p(D)} \\ &= \tilde{c}_p \|\chi_\varepsilon(\hat{A} \circ \mathrm{Id})\|_{L^p(D)} \\ &= \tilde{c}_p \|\hat{A}\|_{L^p(D_\varepsilon)} \longrightarrow 0 \text{ for } \varepsilon \to 0.\end{aligned}$$

Note that we needed $\tilde{\Phi}_\varepsilon(1) = \mathrm{Id}$ because then $T_\varepsilon(\tilde{\Phi}_\varepsilon - \mathrm{id}) = (\chi_\varepsilon(\hat{A} \circ \mathrm{Id}), 0)$. In particular, $\tilde{\Phi}_\varepsilon(z) \in \mathrm{GL}(\mathbf{C}^n)$ for $z \in D$ provided $\varepsilon > 0$ is small enough. Fix a sufficiently small $\varepsilon_0 \in (0, 1)$ and put $\tilde{\Phi} := \tilde{\Phi}_{\varepsilon_0}$. We compute on D_{ε_0} with $\tilde{f}(z) = \tilde{\Phi}^{-1}(z) w(z)$ and recalling that $\tilde{\Phi}$ commutes with i:

$$\begin{aligned}0 &= \bar{\partial} w + A w \\ &= \bar{\partial} w + \hat{A} w \\ &= (\bar{\partial}\tilde{\Phi} + \hat{A}\tilde{\Phi})\tilde{f} + \tilde{\Phi}(\bar{\partial}\tilde{f}) \\ &= \tilde{\Phi}(\bar{\partial}\tilde{f}).\end{aligned}$$

Since $\tilde{\Phi}(z) \in \mathrm{GL}(\mathbf{C}^n)$ we deduce $\bar{\partial}\tilde{f} = 0$ on D_{ε_0}, so $\tilde{f}$ is holomorphic on D_{ε_0}. Define

$$\begin{aligned}\Phi(z) &:= \tilde{\Phi}(z)\tilde{\Phi}^{-1}(0) \\ f(z) &:= \tilde{\Phi}(0)\tilde{f}(z).\end{aligned}$$

Then f is holomorphic on D_{ε_0}, $f(0) = 0$ because of $w(0) = 0$, $\Phi(0) = \mathrm{Id}$ and

$$w(z) = \Phi(z) f(z).$$

□

Next we consider a version of the theorem above where we have real boundary values. Take

$$D^+ := \{z \in D \mid \mathrm{Im}(z) \geq 0\}$$

Theorem A.5.2 *Assume $A \in L^\infty(D^+, \mathcal{L}_{\mathbf{R}}(\mathbf{C}^n))$ and $w \in W^{1,p}_{loc}(D^+, \mathbf{C}^n)$, $2 < p < \infty$, satisfying*

$$\begin{aligned}\bar{\partial} w + A w = 0 \qquad &\text{on} \qquad \overset{\circ}{D}{}^+ \\ w((-1, 1)) \subset \mathbf{R}^n, \qquad &w(0) = 0.\end{aligned}$$

Then there exists $\Phi \in \bigcap_{2<q<\infty} W^{1,q}(D^+, \mathcal{L}_{\mathbf{C}}(\mathbf{C}^n))$ *with*

$$\Phi(z) \in \mathrm{GL}(\mathbf{C}^n) \ , \quad \Phi(0) = Id$$

$$\Phi(z) \in \mathrm{GL}(\mathbf{R}^n) \subset \mathcal{L}(\mathbf{R}^n) \qquad for \quad z \in (-1, 1)$$

and a map $f : D^+ \to \mathbf{C}^n$ *with*

$$f(z) \in \mathbf{R}^n \qquad for \quad z \in (-1, 1) \ , \ f(0) = 0,$$

holomorphic on some smaller half–disk D^+_ε*, such that*

$$w(z) = \Phi(z) f(z).$$

Proof This result can be reduced to Theorem A.5.1. Extend A to a map in $L^\infty(D, \mathcal{L}_{\mathbf{R}}(\mathbf{C}^n))$ by

$$A(z) = \overline{A(\bar{z})},$$

where " $\overline{}$ " means replacing all coefficients by the complex conjugate ones. Extend w by

$$w(z) = \overline{w(\bar{z})}.$$

Then $w \in W^{1,p}(D, \mathbf{C}^n)$ as one verifies easily. Now apply Theorem A.5.1 and find

$$w(z) = \Phi(z) f(z),$$

where z lies in some disk D_ε and Φ is obtained as described in Theorem A.5.1. The corresponding map $\tilde{\Phi}$ satisfies

$$\begin{aligned} \bar{\partial}\tilde{\Phi} + \chi_\varepsilon \hat{A}\tilde{\Phi} = 0 \quad \text{on} \quad \overset{\circ}{D} \\ \tilde{\Phi}(1) = \mathrm{Id}, \quad \tilde{\Phi}(\partial D) \subset \mathcal{L}(\mathbf{R}^n). \end{aligned} \tag{A.25}$$

We note that $\overline{\tilde{\Phi}(\bar{z})}$ also satisfies (A.25). By uniqueness, we have

$$\tilde{\Phi}(z) = \overline{\tilde{\Phi}(\bar{z})}.$$

This implies $\tilde{\Phi}(t) \in \mathcal{L}(\mathbf{R}^n)$ for $t \in (-1, 1)$. Therefore also

$$\Phi((-1, 1)) \subset \mathcal{L}(\mathbf{R}^n)$$

and consequently

$$f((-\varepsilon, \varepsilon)) \subset \mathbf{R}^n.$$

□

Assume L is a loop in $\mathcal{T}(n)$ with Maslov index $\mu_n(L)$. Assume that $2 < p < \infty$ and $u \in W^{1,p}_{loc}(D, \mathbf{C}^n)$ is a solution of

$$\begin{aligned} u_s + iu_t + Au &= 0 \qquad \text{on } \overset{\circ}{D} \\ u(z) &\in L(z) \qquad \text{for } z \in \partial D. \end{aligned}$$

If u is not identically zero then an application of Theorem A.5.1 shows that the zeros of u in $\overset{\circ}{D}$ are isolated. Indeed, if $z_0 \in \overset{\circ}{D}$ and $u(z_0) = 0$ then we have

$$u(z) = \Phi(z)\sigma(z), \tag{A.26}$$

where σ is holomorphic near z_0, $\sigma(z_0) = 0$ and

$$\Phi \in \bigcap_{2<p<\infty} W^{1,p}$$

near z_0. If z_0 is a cluster point of zeros of u, we see that it is also a cluster point of zeros for σ. This implies however that σ vanishes identically near z_0. Denote by Γ the subset of $\overset{\circ}{D}$ consisting of all z_1 having an open neighborhood on which u vanishes. By definition Γ is open and contains z_0. If z_2 belongs to the closure of Γ in $\overset{\circ}{D}$ we apply Theorem A.5.1 to z_2 and find that $z_2 \in \Gamma$. Hence $\Gamma = \overset{\circ}{D}$ implying by continuity that $u = 0$ on D. This shows that interior zeros are isolated unless u vanishes on D.

Assume next that $z_0 \in \partial D$ is a zero of u. Let

$$D^+ = \{z \in \mathbf{C} \mid |z| < 1, \quad \operatorname{Im}(z) \geq 0\}.$$

For a suitable open neighborhood U of z_0 in D and $\varepsilon > 0$ the following map is biholomorphic with $\phi(0) = z_0$ and $\phi((-1, 1)) = \partial D \cap U$.

$$\phi\colon D^+ \xrightarrow{\sim} U$$

$$\phi(z) := z_0 \frac{i - \varepsilon z}{i + \varepsilon z}.$$

Define $v := u \circ \phi$ and compute

$$\begin{aligned} v_s + i\, v_t &= Du(\phi)\phi_s + i\, Du(\phi)\phi_t \\ &= [Du(\phi) + i\, Du(\phi)\, i]\phi_s \\ &= [(u_s + iu_t)(\phi)\, ds - i(u_s + iu_t)(\phi)\, dt]\phi_s \\ &= [(-A(\phi)v)ds + (iA(\phi)v)dt]\phi_s \\ &=: -\tilde{A}v. \end{aligned}$$

Hence v satisfies on D^+ an equation of the form

$$v_s + i\, v_t + \tilde{A}v = 0$$

and

$$v(\theta) \in L(\phi(\theta))$$

for $\theta \in (-1, 1)$. Next we pick a smooth $\mathrm{GL}(\mathbf{C}^n)$-valued map Ψ on D^+ satisfying

$$\Psi(\theta)(L(\phi(\theta))) = \mathbf{R}^n$$

for $\theta \in (-1, 1)$. Define $w\colon D^+ \to \mathbf{C}^n$ by

$$w(z) := \Psi(z)v(z) \tag{A.27}$$

so that $w((-1, 1)) \subset \mathbf{R}^n$. We then compute

$$\begin{aligned} w_s + i\, w_t &= \Psi_s v + \Psi v_s + i\, \Psi_t v + i\, \Psi v_t \\ &= \Psi[v_s + i\, v_t] + (\Psi_s + i\, \Psi_t)v \\ &= \Psi[-\tilde{A}v + \Psi^{-1}(\Psi_s + i\, \Psi_t)v] \\ &=: -\hat{A}w, \end{aligned}$$

hence $w\colon D^+ \to \mathbf{C}^n$ satisfies

$$\begin{aligned} w((-1, 1)) &\subset \mathbf{R}^n \\ w_s + i\, w_t + \hat{A}w &= 0 \quad \text{on } D^+. \end{aligned} \tag{A.28}$$

Since $w(0) = 0$ we deduce from Theorem A.5.2 that

$$w(z) = \Phi(z)\sigma(z)$$

on

$$D_\varepsilon^+ := \{z \in \mathbf{C} \mid |\, z \,| < \varepsilon\ ,\ \mathrm{Im}(z) \geq 0\}$$

for some $\varepsilon \in (0, 1)$ and $\sigma((-\varepsilon, \varepsilon)) \subset \mathbf{R}^n$,

$$\Phi(z) \in \mathrm{GL}(\mathbf{R}^n) \ \text{ for } z \in \partial D_\varepsilon^+$$

and

$$\Phi \in \bigcap_{2<p<\infty} W^{1,p}.$$

In the case where 0 is a cluster point of zeros, the function w must vanish identically near zero. Hence we obtain also a cluster point $z_1 \in \overset{\circ}{D}$ of zeros and are back in the case of interior zeros. Summarizing, we have proved the following proposition:

Proposition A.5.3 *Any $W^{1,p}(D, \mathbf{C}^n)$-solution of*

$$\begin{aligned} u_s + iu_t + Au &= 0 \qquad \text{on } \overset{\circ}{D} \\ u(z) &\in L(z) \qquad \text{for } z \in \partial D \end{aligned}$$

(where $2 < p < \infty$) has only a finite number of zeros unless it is identically zero. □

Now let us consider the case of complex dimension one, i.e. $n = 1$. We will see that the Maslov index of L is already determined by the zeros of u. Let $z_0 \in D$ be a zero of u, where we assume that u does not vanish identically. If $z_0 \in \overset{\circ}{D}$ we have $u(z) = \Phi(z)\sigma(z)$ with

$$\sigma(z) = \sum_{k=k_0}^{\infty} a_k (z - z_0)^k$$

where $a_{k_0} \neq 0$. We note that the number $k_0 \geq 1$ does not depend on the representation $\Phi\sigma$ for u because it equals the winding number $w(u|_{\partial B_\rho(z_0)}, 0)$ of $u|_{\partial B_\rho(z_0)}$ around 0, where $\rho > 0$ is some small number. We define the local order $o(z_0, u)$ by

$$k_0 := o(z_0, u).$$

If $z_0 \in \partial D$ then we can apply Theorem A.5.2 and we define similarly the local order $o(z_0, u)$ for a boundary point.

We have the following useful theorem.

Theorem A.5.4 *Assume $u\colon D \to \mathbf{C}$ is not identically zero and satisfies*

$$\begin{aligned} u_s + iu_t + Au &= 0 \qquad \text{on } \overset{\circ}{D} \\ u(z) &\in L(z) \qquad \text{for } z \in \partial D, \end{aligned} \tag{A.29}$$

where $A \in L^\infty(D, \mathcal{L}_{\mathbf{R}}(\mathbf{C}))$. Then

$$\mu_1(L) = \sum_{z \in N_{bd}} o(z,u) + 2 \sum_{z \in N_{int}} o(z,u) \geq 0, \tag{A.30}$$

with

$$N_{bd} := \{z \in \partial D \mid u(z) = 0\}$$

and

$$N_{int} := \{z \in \overset{\circ}{D} \mid u(z) = 0\}.$$

Proof Let $r_1 := \sup\{|z| \mid z \in \overset{\circ}{D}, u(z) = 0\}$. Then $r_1 \in [0,1)$ since the zeros of u are isolated. Denote by D_r the closed disk of radius r. For $r \in (r_1, 1)$ we have $0 \notin u(\partial D_r)$. Hence the winding number $w(u|_{\partial D_r}, 0)$ is well defined and equals the sum of the local winding numbers for points $z \in \overset{\circ}{D}_r$ with $u(z) = 0$. Then we obtain for $r \in (r_1, 1)$:

$$w(u|_{\partial D_r}, 0) = \sum_{z \in N_{int}} o(z,u).$$

Let $\varepsilon_0 > 0$ be the minimal distance of two zeros of u on the boundary of D. Let $\varepsilon_1 = \min\{\frac{1}{2}\varepsilon_0, 1 - r_1\}$ and pick $\varepsilon \in (0, \varepsilon_1)$. Define a closed subdomain of D by

$$D_\varepsilon^* := D \setminus \bigcup_{z \in N_{bd}} B_\varepsilon(z)$$

(see Fig. A.2). Since u has no zeros in $\overline{D}_\varepsilon^* \setminus D_r$, we have

$$w(u|_{\partial D_\varepsilon^*}, 0) = w(u|_{\partial D_r}, 0).$$

Consider the loop $L_\varepsilon^*(z) := \frac{u(z)}{|u(z)|}\mathbf{R}$, $z \in \partial D_\varepsilon^*$ in $\mathcal{T}(1)$. Clearly, the winding number $w(u|_{\partial D_\varepsilon^*}, 0)$ and the Maslov index $\mu_1(L_\varepsilon^*)$ are related by

$$\mu_1(L_\varepsilon^*) = 2\, w(u|_{\partial D_\varepsilon^*}, 0),$$

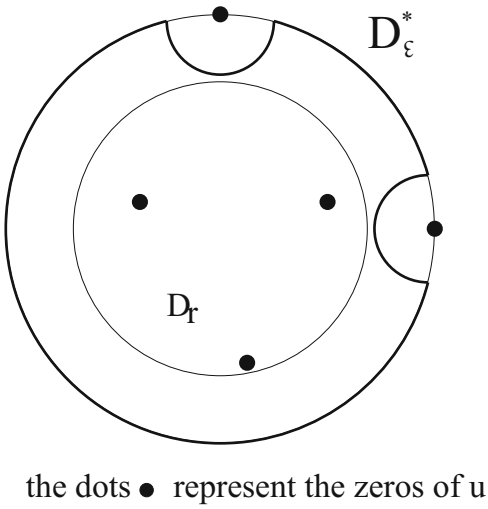


Fig. A.2 The domain D_ε^*

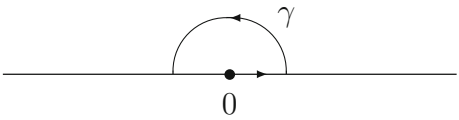
hence

$$\mu_1(L_\varepsilon^*) = 2 \sum_{z \in N_{int}} o(z, u).$$

It remains to compare $\mu_1(L)$ and $\mu_1(L_\varepsilon^*)$. We observe that

$$L_\varepsilon^*(z) = L(z) \quad \text{for } z \in \partial D \cap \partial D_\varepsilon^*.$$

Hence any difference between $\mu_1(L_\varepsilon^*)$ and $\mu_1(L)$ is contributed by the zeros of u on ∂D. Fix a zero $z_0 \in \partial D$. Since we are only concerned now with the map u on a small neighborhood of z_0, we may assume that u is defined on some small half–disk D_δ^+, $u(0) = 0$ is an isolated zero and $u((-\delta, \delta)) \subset \mathbf{R}$. Moreover, we may assume that $u(z) = \Phi(z)\sigma(z)$ with some holomorphic function σ. All this follows from the similarity principle, Theorem A.5.2, and its proof (in particular Eqs. (A.27) and (A.28)). We compute the Maslov index for the loop $\hat{L}_\delta$ sketched in the following figure by running once along the boundary of the half disk D_δ^+.



On $(-\delta, \delta)$ $\hat{L}_\delta$ equals $\mathbf{R}$, and on γ we have

$$\hat{L}_\delta(z) = u(z)\mathbf{R}$$

We claim that we may replace u by the holomorphic function σ without changing the Maslov index. Indeed, since $u(z) = \Phi(z)\sigma(z)$ we can take the homotopy

$$t \longmapsto \Phi(tz)^{-1}\hat{L}_\delta(z)$$

for $t \in [0, 1]$. Hence without loss of generality the loop is given by

$$\begin{aligned}\hat{L}_\delta(z) &= \mathbf{R} \quad \text{for } z \in [-\delta, \delta]\\ \hat{L}_\delta(z) &= \sigma(z)\mathbf{R} \quad \text{for } |z| = \delta\ ,\ \mathrm{Im}(z) > 0.\end{aligned}$$

If $\delta > 0$ is small enough we might replace σ by $a_{k_0}z^{k_0}$, where $k_0 = o(0, w) = o(z_0, u)$. We can achieve this by considering the homotopy

$$\begin{aligned}\hat{L}^t_\delta(z) &:= \sigma_t(z)\mathbf{R}\\ &:= \left(a_{k_0}z^{k_0} + t\sum_{l=1}^{\infty} a_{k_0+l}\, z^{k_0+l}\right)\mathbf{R}\end{aligned}$$

for $t \in [0, 1]$ and $|\ z\ | = \delta$, $\mathrm{Im}(z) > 0$. We observe that $\sigma_t(z) \neq 0$ for all $t \in [0, 1]$ and all z as above.

Now the Maslov index of $\hat{L}_\delta$ is given by

$$\mu_1(\hat{L}_\delta) = o(z_0, u).$$

The difference $\mu_1(L) - \mu_1(L^*_\varepsilon)$ equals the sum of all this boundary contributions. Therefore

$$\begin{aligned}\mu_1(L) &= \mu_1(L^*_\varepsilon) + \sum_{z\in N_{bd}} o(z, u)\\ &= 2\sum_{z\in N_{int}} o(z, u) + \sum_{z\in N_{bd}} o(z, u).\end{aligned}$$

This completes the proof of Theorem A.5.4. □

Remark There are some (trivial) corollaries: First, if u solves Eq. (A.29) and $\mu(L) < 0$, then u has to vanish identically. If $\mu(L) = 0$ then u cannot have any zeros.

Formula (A.30) has some extremely useful applications. Let $2 < p < \infty$ and $A \in L^\infty(D, \mathcal{L}_{\mathbf{R}}(\mathbf{C}))$. We consider the operator

$$\begin{aligned}T\colon V^p_L &\longrightarrow L^p\\ Tu &= u_s + iu_t + Au\end{aligned}$$

where

$$V_L^p = \{u \in W^{1,p}(D, \mathbf{C}) \mid u(z) \in L(z) \quad \text{for } z \in \partial D\}.$$

We know from Appendix A.4 that the Fredholm index $\mathrm{ind}(T)$ of T is given by

$$\mathrm{ind}(T) = \mu_1(L) + 1.$$

Theorem A.5.4 allows us to derive an upper bound for the dimension of the kernel of T. As a consequence we will see that T is necessarily surjective if $\mu_1(L) \geq -1$, and injective otherwise. This is the contents of Theorem A.5.5 below. We want to emphasize that it is a special feature of embedded pseudoholomorphic curves in dimension four to prove surjectivity of the linearized Cauchy Riemann operator without having to make a generic choice for the almost complex structure. This is especially important in contact geometry since we can only consider almost complex structures $\tilde{J}$ on the symplectisation which map X_λ onto $\frac{\partial}{\partial t}$, thus restricting severely the freedom to perturb $\tilde{J}$. Hence the following theorem is the heart of the implicit function theorem in Sect. 7.5.

Theorem A.5.5 *Let T be as described above. Then*

$$\dim(\ker(T)) = \max\{0, \mu_1(L) + 1\}.$$

Proof Since $\mathrm{ind}(T) \leq \dim(\ker(T))$ we infer from the index formula

$$\mu_1(L) + 1 \leq \dim(\ker(T)).$$

Since always $\dim(\ker(T)) \geq 0$ we obtain

$$\dim(\ker(T)) \geq \max\{0, \mu_1(L) + 1\}.$$

In order to show the opposite inequality assume first that $\mu_1(L) \leq -1$. Then Theorem A.5.4 shows us that any $u \in \ker(T)$ must vanish identically. Consequently $\mu_1(L) \leq -1$ implies $\dim(\ker(T)) = 0$.

Now assume $\mu_1(L) \geq 0$. Since $\mathrm{ind}(T) = \mu_1(L) + 1$, we see that $l := \dim(\ker(T)) \geq 1$. Let $u_1, \ldots, u_l$ be a basis for $\ker(T)$. Arguing indirectly, we assume that

$$l > \mu_1(L) + 1.$$

The number of zeros of each u_k is bounded by the Maslov index, hence we may pick points $z_1, \ldots, z_{\mu_1(L)+1}$ in ∂D such that

$$u_k(z_j) \neq 0$$

for $k \in \{1, \ldots, l\}$ and $j \in \{1, \ldots, \mu_1(L) + 1\}$. We define a linear map

$$\wedge \colon \mathbf{R}^l \longrightarrow \bigoplus_{j=1}^{\mu_1(L)+1} L(z_j)$$

by

$$\wedge(\lambda_1, \ldots, \lambda_l) := \left(\sum_{k=1}^{l} \lambda_k u_k(z_1), \ldots, \sum_{k=1}^{l} \lambda_k u_k(z_{\mu_1(L)+1}) \right).$$

The equation $\wedge(\lambda_1, \ldots, \lambda_l) = 0$ is a system of $\mu_1(L)+1$ equations with l unknown variables over the real numbers: Indeed, if e_j is a generator of $L(z_j)$, then we may write $u_k(z_j) = \mu_k^{(j)} e_j$, with a nonzero real number $\mu_k^{(j)}$ and the equations

$$0 = \sum_{k=1}^{l} \lambda_k u_k(z_j) \text{ and } 0 = \sum_{k=1}^{l} \lambda_k \mu_k^{(j)}$$

are equivalent. Since by assumption $l > \mu_1(L) + 1$ there exists a nonzero vector $(\lambda_1, \ldots, \lambda_l)$ in $\mathbf{R}^l$ with $\wedge(\lambda_1, \ldots, \lambda_l) = 0$. Let $u := \sum_{k=1}^{l} \lambda_k u_k$. Then $u \in \ker(T)$ and has at least $\mu_1(L) + 1$ zeros. By Theorem A.5.4 it must be identically zero. Since however $u_1, \ldots, u_l$ was a basis for $\ker(T)$ we obtain a contradiction. This shows that

$$\dim(\ker(T)) \leq \mu_1(L) + 1$$

if $\mu_1(L) \geq 0$ and the proof of Theorem A.5.5 is complete. □

Corollary A.5.6 *If T is as described above then we have the following implications:*

- $\mu_1(L) < -1 \Longrightarrow$ *T is injective but not surjective*
- $\mu_1(L) = -1 \Longrightarrow$ *T is bijective*
- $\mu_1(L) \geq 0 \Longrightarrow$ *T is surjective but not injective*

Proof This is a trivial consequence of Theorem A.5.5 and the index formula $\operatorname{ind}(T) = \mu_1(L) + 1$. □

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